Engineering Mathematics - I · Unit_01 · Unit 1

DRAFT — UE26MA141A · Unit_01 — Unit 1 · Complete Unit Master

Source-complete searchable master · transcript, slide text, page visuals and archive references

Hugging Face PESU_Materials source tree, official units.json catalogue, transcript sidecars, slide text/PDF/PPT sources and the selected First_Year_8enterprise archive; see the provenance datasets in the release ZIP. Generated 18 October 2026.

· How to use this pack

Search this single unit master by lesson, timestamp, slide text or source page. Read the transcript in timestamp order, then compare extracted slide text with its page-faithful source figure. Formula candidates and archive question mappings are explicitly flagged for review. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Unit_01 · Unit 1

1 UE26MA141A · Unit_01: Unit 1

This unit master organizes source material for Engineering Mathematics - I, Unit_01, officially titled Unit 1. It includes 17 video transcripts and 11 source slide decks, with page-level text and rendered images where available. This draft preserves source wording and page order; it does not claim a complete human subject review or full audiovisual review. EV0000001

SHORT-NOTE RECAP

Official catalogue entry: UE26MA141A, Unit_01 (Unit 1). Bucket folder present: yes. Videos: 17; slide decks: 11.

EV0000001

2 Continuity and differentiability of a function of two independent variable

This lesson is indexed from the source video titled Continuity and differentiability of a function of two independent variable. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence.

EV0000002 · EV0000003

Graphs and diagrams

Figure G000001. Introduction to Partial differentiation · source slide page 1.

EV0000259

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

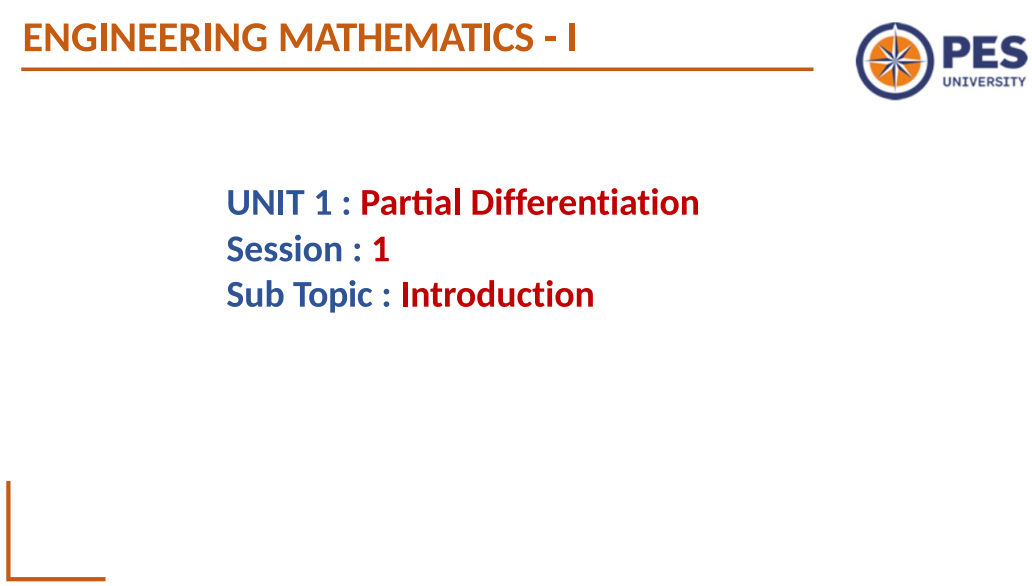

Figure G000002. Introduction to Partial differentiation · source slide page 2.

EV0000262

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

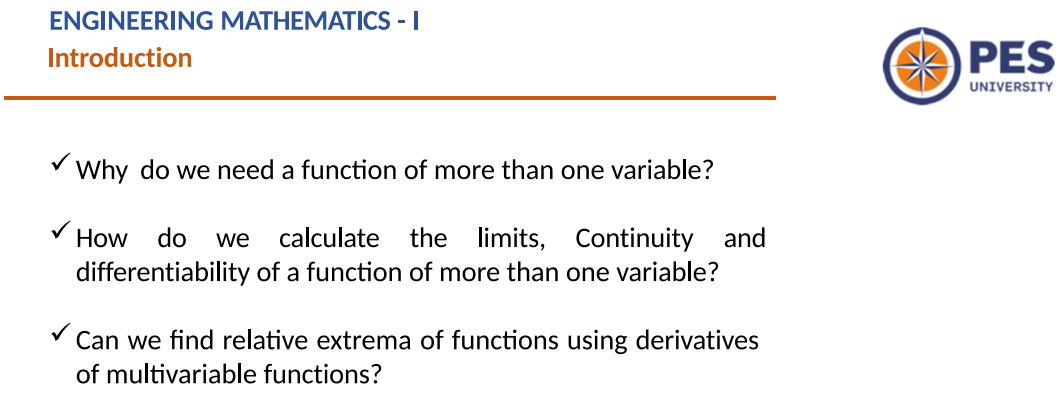

Figure G000003. Introduction to Partial differentiation · source slide page 3.

EV0000265

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Introduction

Why do we need multivariable functions?

Quantities which depend on two or more variables are represented using functions with two or more variables.

Examples for functions of several variables:

1. The area (A) of a rectangle depends on its length (l) and breadth (b).

Figure G000004. Introduction to Partial differentiation · source slide page 4.

EV0000268

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Introduction

2. The volume (v) of a cylinder depends on its radius (r) and height (h).

3. Electrical potential V within a cathode ray tube vary with spatial positions and time.
i.e. $v = f(x, y, z, t)$

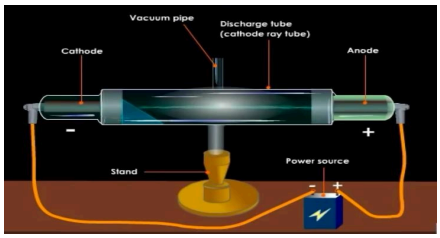

Figure G000005. Introduction to Partial differentiation · source slide page 5.

EV0000271

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Introduction

PES UNIVERSITY

4. The Propagation of Electromagnetic wave depends on the Magnetic field, Electric field, Wavelength and the direction of propagation.

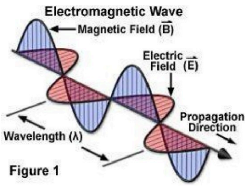

Figure 1

5. The Weather(v) of a region depends on Topography of the place, pressure, humidity and temperature distribution.

Figure G000006. Introduction to Partial differentiation · source slide page 6.

EV0000274

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Introduction

PES UNIVERSITY

6. To study the rate of spreading of a pandemic like COVID 19 which again depends on the area, population, awareness amongst the people in that area, immunity level of the people, distribution of the population based on their age group, gender and other classification.

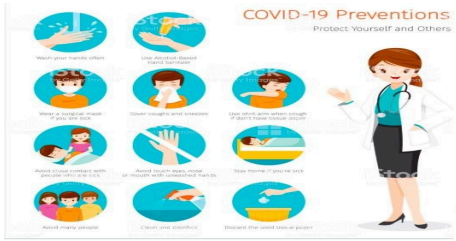

Figure G000007. Introduction to Partial differentiation · source slide page 7.

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How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Functions of Two Independent Variables

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- A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes on. $\{f(x, y), (x, y) \in D\}$

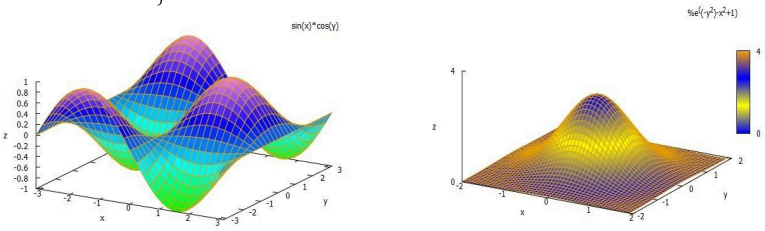

Figure G000008. Introduction to Partial differentiation · source slide page 8.

EV0000280

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Limit of a Function of Two Variables

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A function $f(x, y)$ is said to approach the limit L as (x, y) approaches to (a, b), if the limit remains the same along any path of (x, y) approaching to (a, b).

$$\lim_{(x,y)\to(a,b)} f(x, y) = L$$

There is potentially an infinite number of paths to consider.

PROBLEMS ON FINDING THE LIMITS

1. Evaluate the limit of the following function

$$\lim_{(x,y)\to(2,1)} (x^2 - 2xy + 3y^2 - 4x + 3y - 6)$$

Figure G000009. Introduction to Partial differentiation · source slide page 9.

EV0000283

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

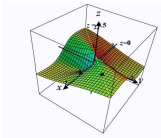
ENGINEERING MATHEMATICS - I

Problems on Limits

PES UNIVERSITY

2. Show that for the following function limit does not exist.

$$\lim_{(x,y)\to(0,0)} \frac{2xy}{3x^2 + y^2}$$



Along the line $y = 0$, the value of the limit Is zero, whereas along $y = x$ the values is ½

Figure G000010. Introduction to Partial differentiation · source slide page 10.

EV0000286

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Continuity of a Function of Two Variables

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- Remember with functions of one variable, if the function was continuous then $\lim_{x \to c} f(x) = f(c)$
- The same is true for functions of two variables. If the function is continuous then $\lim_{(x,y)\to(a,b)} f(x, y) = f(a,b)$
- A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful.
- If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths.

Figure G000011. Introduction to Partial differentiation · source slide page 11.

EV0000289

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Problems on Continuity of a function of two Variables

PES UNIVERSITY

1. Consider the function

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

The function is not continuous at (0,0) as f(0,0) is not defined. However we can easily verify that limit (x,y) approching to (0,0) exists.

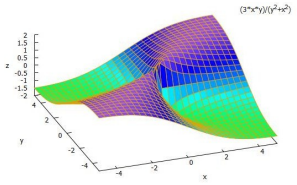

Figure G000012. Introduction to Partial differentiation · source slide page 12.

EV0000292

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Problems on Continuity of a function of two Variables

PES UNIVERSITY

2. Consider the function

$$f(x, y) = \frac{2xy}{3x^2 + y^2}$$

The function is not continuous at the origin as the limit of the function as (x,y) approaches to (0,0) does not exists.

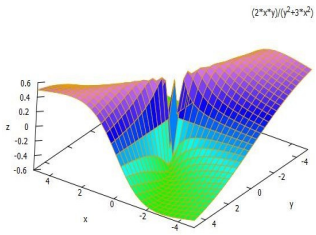

Figure G000013. Introduction to Partial differentiation · source slide page 13.

EV0000295

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

History of Partial Differentiation

- Partial differentiation occurs even in ordinary processes of the calculus. The simplest example of partial differentiation is used by Leibnitz for the first time while defining product rule of differentiation.

- The Modern Partial derivative symbol was created by Adrien-Marie Legendre (1786) but later abandoned it.
- Carl Gustav Jacob Jacobi re introduced the symbol in 1841.

Figure G000014. Introduction to Partial differentiation · source slide page 14.

EV0000298

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Definition of Partial Differentiation

- A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant.
- Let $z = f(x, y)$ be a function of two variables x& y then the partial derivative of z with respect to x treating y as constant is denoted by $\frac{\partial z}{\partial x}$ or z_x or $\frac{\partial f}{\partial x}$ or f_x and with respect to y by treating x as constant is denoted by $\frac{\partial z}{\partial y}$ or z_y or $\frac{\partial f}{\partial y}$ or f_y and is defined as

$$\frac{\partial z}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h}$$

$$\frac{\partial z}{\partial y} = \lim_{h \to 0} \frac{f(x, y + h) - f(x, y)}{h}$$

Figure G000015. Introduction to Partial differentiation · source slide page 15.

EV0000301

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

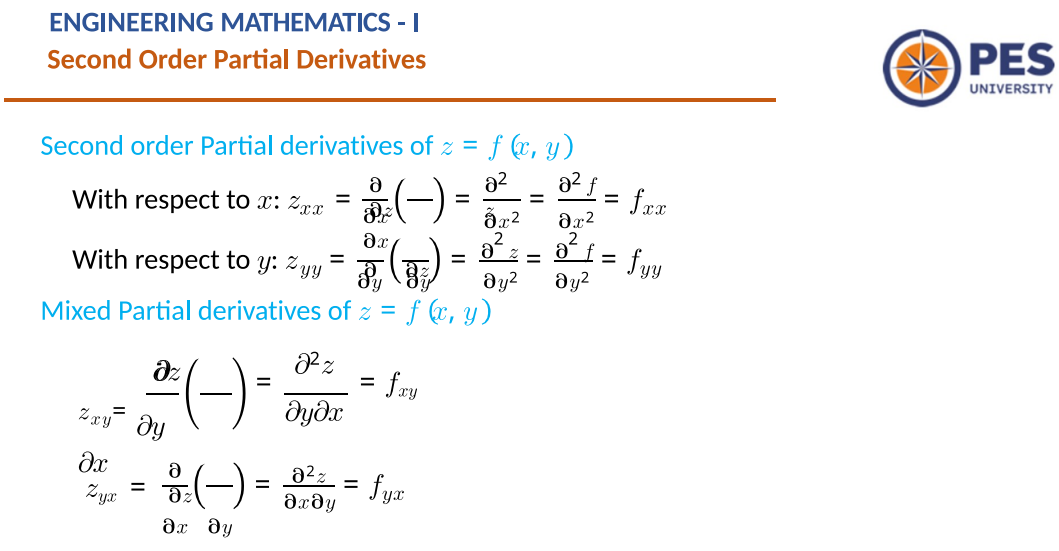

Figure G000016. Introduction to Partial differentiation · source slide page 16.

EV0000304

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

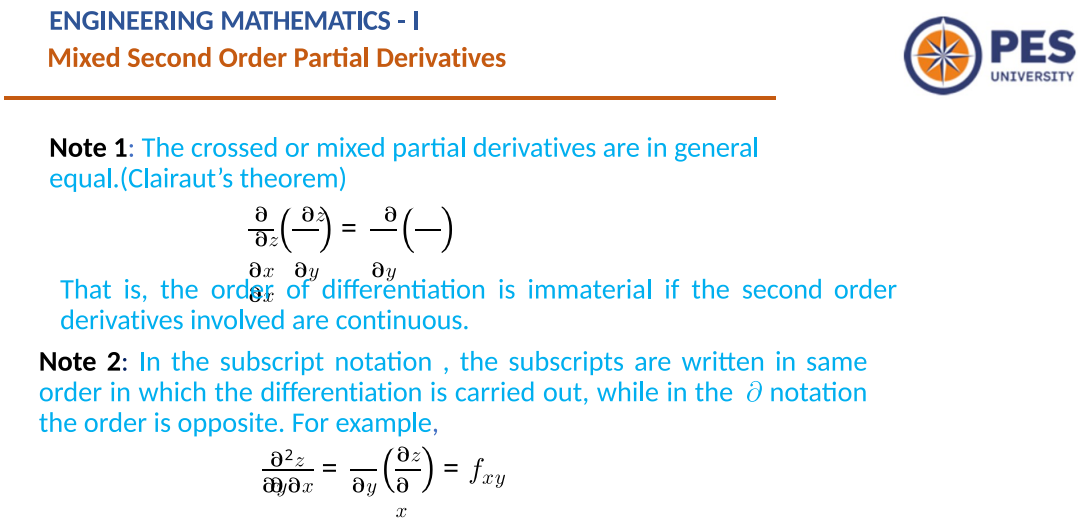

Figure G000017. Introduction to Partial differentiation · source slide page 17.

EV0000307

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000018. Introduction to Partial differentiation · source slide page 18.

EV0000310

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000258 · EV0000259 · EV0000260
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 1 Sub Topic : Introduction EV0000261 · EV0000262 · EV0000263
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Introduction Why do we need a function of more than one variable? How do we calculate the limits, Continuity and differentiability of a function of more than one variable? Can we find relative extrema of functions using derivatives of multivariable functions? EV0000264 · EV0000265 · EV0000266
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Introduction Why do we need multivariable functions? Quantities which depend on two or more variables are represented using functions with two or more variables. Examples for functions of several variables: 1. The area (A) of a rectangle depends on its length (l) and breadth (b). EV0000267 · EV0000268 · EV0000269
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Introduction The volume (v) of a cylinder depends on its radius (r) and height (h). Electrical potential V within a cathode ray tube vary with spatial positions and time. i.e. $v = f(x, y, z, t)$ [Image text, OCR] Vacuumpipe Dischargetube (cathoderaytube) Cathode Anode Powersource Stand EV0000270 · EV0000271 · EV0000272

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Introduction 4. The Propagation of Electromagnetic wave depends on the Magnetic field, Electric field, Wavelength and the direction of propagation. [Image text, OCR] ATMOSPHEREFACTORS AIRPRESSURE snuueersrtck EMPERATURE HUMIDITY CLOUDINESS WIND 5. The Weather(v) of a region depends on Topography of the place, pressure, humidity and temperature distribution. EV0000273 · EV0000274 · EV0000275

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Introduction 6. To study the rate of spreading of a pandemic like COVID 19 which again depends on the area, population, awareness amongst the people in that area, immunity level of the people, distribution of the population based on their age group, gender and other classification. [Image text, OCR] COVID-19Preventions Protect Yourself and Others Wash your hands often Use Alcohol-Based Hand Sanitizer Ge Wear asurgcalmask Cover coughs and sneezes Use shirt arm when cough) f you are sick f donthave tssue aper Avod clsecontat wih peopieaae sck Avoid toucheyes.nose or mouthwih nwashed hands Stay homeyou're sick Aid many people Clen snd csifect Dscart the used tisse poper EV0000276 · EV0000277 · EV0000278

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Functions of Two Independent Variables A function of two variables denoted by $z = f(x, y)$ is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the domain of f and its range is the set of values that f takes ⊠f (x, y),(x, y) ⊠ D⊠ on. [Image text, OCR] Gnuplot(windowid:0) □ %e(-y²)-x²+1) 60.000030.0000 scale:1.00000.1.00000 [Image text, OCR] Gnuplot(windowid:0) □ × sin(x)*cos(y) 0.8 0.6 0.4 0.2 0 -0.2 -0.4 -0.6 -0.8 -1 ew:60.0000,30.0000 scale:1.00000,1.00000 EV0000279 · EV0000280 · EV0000281

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Limit of a Function of Two Variables A function $f(x, y)$ is said to approach the limit L as (x, y) approaches to (a, b), if the limit remains the same along any path of (x, y) approaching to (a, b). lim f (x, y) ⊠ L (x, y)⊠(a,b) There is potentially an infinite number of paths to consider. PROBLEMS ON FINDING THE LIMITS 1. Evaluate the limit of the following function lim $(x2 - 2xy + 3y2 - 4x + 3y - 6)$ $(x,y)\to(2,1)$ EV0000282 · EV0000283 · EV0000284

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Problems on Limits 2. Show that for the following function limit does not exist. $2xy$ lim $(x,y)\to(0,0)$ $3x$ 2 2 + y Along the line $y = 0$, the value of the limit Is zero, whereas along $y = x$ the values is ½ EV0000285 · EV0000286 · EV0000287

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Continuity of a Function of Two Variables Remember with functions of one variable, if the function was continuous then lim f (x) ⊠ f (c) x⊠c The same is true for functions of two variables. If the function is continuous then lim f (x, y) ⊠ f (a,b) (x, y)⊠(a,b) A function is continuous at a point if the graph doesn't have any holes or breaks at that point. Graphing the function is helpful. If a function is not continuous at (a,b), this does not mean the limit does not exist. We will need to consider the approaching (a,b) from different paths. EV0000288 · EV0000289 · EV0000290

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Problems on Continuity of a function of two Variables 1. Consider the function $3x2y = x2 + y2$ f x, y The function is not continuous at (0,0) as f(0,0) is not defined. However we can easily verify that limit (x,y) approching to (0,0) exists. [Image text, OCR] (3*x*y)/ (y²+x²) Z 0 4 2 2 X EV0000291 · EV0000292 · EV0000293

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Problems on Continuity of a function of two Variables 2. Consider the function $f\ x,\ y\ 2xy = 3x2 + y2$ The function is not continuous at the origin as the limit of the function as (x,y) approaches to (0,0) does not exists. [Image text, OCR] (2*x*y) / (y²+3*x²) 0.6 0.4 0.2 Z 0 -0.2 -0.4 -0.6 2 X 2 4 EV0000294 · EV0000295 · EV0000296

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I History of Partial Differentiation Partial differentiation occurs even in ordinary processes of the calculus. The simplest example of partial differentiation is used by Leibnitz for the first time while defining product rule of differentiation. The Modern Partial derivative symbol was created by Adrien-Marie Legendre (1786) but later abandoned it. Carl Gustav Jacob Jacobi re introduced the symbol in 1841. EV0000297 · EV0000298 · EV0000299

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Definition of Partial Differentiation A partial derivative of a function of several variables is the derivative with respect to one of those variables, with the others held constant. Let $z = f(x,\ y)$ be a function of two variables x& y then the partial derivative of z with respect to x treating y as $\partial z\ \partial f$ with by constant is denoted by $or\ zx\ or\ or\ fx$ and $\partial x\ \partial x$ respect to y by treating x as constant is denoted $or\ \partial f\ \partial z\ or\ z\ or\ fy$ and is defined as $y\ \partial y\ \partial y\ \partial z\ f\ x + h,\ y - f(x,\ y) = \lim h\to 0\ \partial x\ \partial z\ h\ f\ x,\ y + h - f(x,\ y) = lim\ h\to 0\ \partial y\ h$ EV0000300 · EV0000301 · EV0000302

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Second Order Partial Derivatives Second order Partial derivatives of $z = f\ x,\ y\ 2\ 2\ \partial\ \partial z\ \partial\ z\ \partial\ f$ With respect to x: $z = = = = f\ xx\ xx\ \partial x\ \partial x\ \partial\ \partial z\ \partial x2\ \partial\ z\ \partial x2\ \partial\ f\ 2\ 2$ With respect to y: $z = = = = f\ yy\ yy\ \partial y\ \partial y\ \partial y2\ \partial y2$ Mixed Partial derivatives of $z = f\ x,\ y\ \partial\ \partial z\ zxy = \partial y\ \partial x\ \partial 2z = = fxy\ \partial y\partial x\ \partial\ \partial z\ \partial x\ \partial y\ \partial 2z\ zyx = = = f\ yx\ \partial x\partial y$ EV0000303 · EV0000304 · EV0000305

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Mixed Second Order Partial Derivatives Note 1: The crossed or mixed partial derivatives are in general equal.(Clairaut's theorem) $\partial\ \partial z\ \partial\ \partial z\ \partial x\ \partial y\ \partial y\ \partial x =$ That is, the order of differentiation is immaterial if the second order derivatives involved are continuous. Note 2: In the subscript notation , the subscripts are written in same order in which the differentiation is carried out, while in the ∂ notation the order is opposite. For example, $\partial 2z\ \partial\ \partial z = = f\ xy\ \partial y\partial x\ \partial y\ \partial x$ EV0000306 · EV0000307 · EV0000308

18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000309 · EV0000310 · EV0000311

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.932] Hello, everyone. Welcome back to class two of partial differentiation. Today we'll be discussing continuity, history of partial derivative, definition of partial derivative, higher order partial derivative, and its geometric interpretation. In the previous session, [00:00:19.940] We defined what is the continuity of a function of single variable. So we defined the three necessary conditions, and then we said that a function f of x is said to be continuous if limit as x approaches to c of f of x is equal to f of c. [00:00:38.340] A similar thing can be done for a function of two independent variables. So I have a function f of x y and I say that the function f of x y is continuous if limit as xy approaches to ab of f of x y is f of ab. [00:00:57.060] We'll solve some examples so that the definition will be a little

more clear to you. A function is continuous at a point. [00:01:05.028] So geometrically it means that there is no hole or break in the graph. Unlike see for a function of single variable, what we said is a function is said to be discontinuous if there is a gap. [00:01:19.076] Right. So if a function looks like this. But for a function of bivariate, I need to be in three-dimensional plane. So let me just mark a three-dimensional plane. [00:01:33.220] So I'll have something like this. I have x, y, and z axis. So my my surface is here. Right? So something like this. My surface would look something like this. But there is a hole. So somewhere here, there is a hole in the surface. [00:01:51.012] I'll show you how it looks. So or there is a break in the surface. I can't call that as a break. A break can be considered only for a curve. So there is a discontinuity. So if there is a hole or gap or dent or whatever you want to call in the surface, then we say that the function is not continuous. [00:02:10.404] But you need to be careful about something. A function is uh not continuous doesn't imply that the limit does exist. So what do I mean by that? So if see this is the limit does not exist. [00:02:27.940] is not a condition, right? So uh that means even if the limit exists, the function need not be continuous. I know things are not that clear. So let's take an example. [00:02:41.060] So here is the curve. So as you see, there is a curve that is 3x square y by x square plus y square. So the curve looks like this. [00:02:50.000] If you observe carefully, as I go closer to 0, see if I come closer to 0, there is a folding that is happening. So that means there is actually a gap that is there at 0, 0. [00:03:02.980] Correct? So let's see uh what part of the definition of continuity fails for this? I need to check if this function is continuous or the curve must be very smooth. That's what I mean by continuous, right? So the surface has to be a very smooth surface. [00:03:17.540] How do I check if the uh function is continuous? I need to check the limit as xy. [00:03:25.668] x y approaching to zero zero of this function 3x square y divided by x square plus y square this must be equal to f of 0 comma 0. That's the condition. [00:03:41.732] But now is it satisfied is the question. Before we attempt to find the left hand side, let's check the value of f of 00. For this function, what is the value of f of 0,0? [00:03:55.552] If I substitute, I will get 3 into 0 into 0 divided by 0 plus 0, which is obviously in the form of 0 by 0, which is an indeterminate form. And no matter whatever the value I get in the left hand side, I cannot compare this with an indeterminate form. [00:04:14.820] So this function is not continuous, whatever the function I have here is not continuous because the function is not defined at the origin. Limit might exist. [00:04:27.296] Okay, so if I take the limit from along the x axis, I'll get zero along the y axis also, it might be zero along any other line y is equal to x or parabola, it can be zero. But no matter whatever the value of this limit is, I cannot equate this to an indeterminate value. [00:04:45.764] So function is not continuous because the value of the function is not defined at the origin. Is it clear? So I hope you understood the first example. Let's take another example. It's the same thing that we did yesterday. Okay? So

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:05.508] Let's see, what is this example now? You look at the graph carefully. So suppose I have a function 2xy divided by 3x square plus y square. This is the same function we took yesterday to explain the limits in the previous session, right? So uh suppose I want to calculate the [00:05:23.364] Limit for this. As you know, the limit does not exist, right? So we took the limit as uh XY approaching to 0, 0. [00:05:31.428] of this function, which is 2xy divided by 3x square plus y square. And we calculated along different paths we took y equal to 0 and then we took y is equal to x. [00:05:44.400] x equal to 0. You take any path, the value that we get is different. So the limit does not exist. So this limit

does not exist. Also the function, what is the value of the function at 0 comma 0? It will be 0 by 0, which is again an indeterminate form. Okay. [00:06:01.636] So the limit does not exist as also the value of the function is not defined. So this graph here will tell you what is happening to the function, right? [00:06:13.892] So we'll see. Look at the graph a little bit carefully. At zero, there is actually a a crunch or whatever you want to call it. Um so it's there is a break in the graph or it's not smooth. [00:06:26.212] It's like a boundary and then it suddenly drops down at 0, 0, and then of course at every other point the function exists. So these two examples will help you to understand that a function is continuous if the surface is smooth. I took an example of a bivariate function. [00:06:45.604] And as as usual, as I mentioned in the previous lecture, it is very easy for us to check if the function is not continuous. If you want to check if the function is continuous, the best way to do that is look at the graph. [00:06:59.728] So as and when you move to a higher dimension, it's always better that you visualize what is happening rather than just calculating analytically. Alright? Okay. So [00:07:13.444] Next, I want to um go a little bit or talk a little bit about the history of partial differentiation. So most of us will be interested to know. [00:07:24.292] Who invented all these things? At least if not interested. Sometimes you want to curse people, right? You want to say why did he invent? Why did Newton invent so that I have to learn? Why did Jacobi do this? Why did Leibniz do this? Uh but at least we should be uh grateful to few people who made [00:07:41.344] our life much easier by inventing something that is very, very useful. Okay? So uh partial differentiation is a concept that is not very new to you. It's only that you're hearing or it's been named differently, right? [00:07:56.640] So when you did learn differentiation of a function of single variable, you did something called as the product rule, right? I hope you remember product rule. And all of you were uh you know very very comfortable with product rule because things were very simple. What does this product rule say is suppose I want to differentiate [00:08:15.876] uh function, which is actually product of two functions. Say I have um so I have um say x q [00:08:28.324] Sorry. Okay. So I have a function which is something like x cube into sin x. Let's take this. So can you tell me how do you calculate the uh derivative for this? The product rule says that you keep the first function as it is, and you differentiate the second. [00:08:47.972] Right? Plus, you keep the second function as it is, and then you differentiate the first. And all of you were very comfortable when they define product rule. But did you question yourself that when I'm differentiating this, why am I not doing anything to this function here? [00:09:08.708] So basically what you did in product rule is when you were differentiating one function, you treated the other one as constant, right? Suppose you treat this as a constant. Let's say that I'll I'll cut this out and I'll call this as k. [00:09:24.368] So what is the derivative of k sin x now? It is just k cos x. You will not worry about whatever that constant is. The next time what you did, so let me just erase it. So the next time what you did is so what I did derivative of x cube into sine x. [00:09:43.044] Right? So in the first case I took this as K. [00:09:47.524] And I treated that as k. So I got this as k into derivative of cos x, I mean derivative of sin x will be cos x. So for the next term, that is the second term, what you did? You took sine x to be some constant. So let's say call that as k1.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.592] And then so you treated that as K1 and you differentiated XQ which is 3x square. Is it clear whatever I'm talking? So partial differentiation also means the same thing. [00:10:18.308] It's not something new. So Leibniz when he did define product rule, he actually took one of the variables. So let me leave uh this with

this. D of uv is u d v plus v d u, right? [00:10:38.276] So I could do this by treating one of the function as constant while differentiating the other. So same thing was extended when it was a case of a function of more than one independent variable. [00:10:53.380] So they were um suggesting many symbols. So Leibniz was the first person to introduce partial differentiation. In fact uh he in his letter to um Lobithal he used the symbol delta. [00:11:08.900] for partial derivative with respect to x. So he didn't knew that uh to use the same symbol and he used gamma for partial derivative with respect to y. But later uh he didn't continue his work on partial derivatives. So legendary [00:11:23.552] Legendary what he did is he actually defined the modern partial derivative. So how do we define that symbol? The partial derivative of a function. Suppose I have a function z is equal to f of x y. Then the partial derivative of this function is denoted by double [00:11:41.380] The symbol is not delta, this is dou. So dou z by dou x. [00:11:48.228] That's a notation. So what does this mean? You have z, which is a function of more than one variable, and I'm going to differentiate that with respect to x by treating all other variables that are there with me as constant. That's what I'm going to do. [00:12:07.844] Is the definition clear? So uh the re the very new uh concept of partial differentiation or the symbol that you're using is because of Jacobi. Jacobi actually reintroduced that symbol of Do, and then even today we use it. But there are other notations also for this. [00:12:26.864] We will see uh how it is defined and you know, uh what it is and all. But the concept of partial differentiation was introduced by Leibniz, and then again it's like differentiating like it's like ordinary derivative. [00:12:40.292] But you have to treat all other variables as constant and differentiate only the variable with respect to which we are finding the derivative. Okay. [00:12:52.324] So the proper definition goes like this. A partial derivative of a function of several variables is a derivative with respect to one of those variables with the other's held constant. [00:13:05.700] Okay. And of course there are different notations. So as I was talking, see we have z is equal to f of x y and this is the notation I was talking about, dou z by dou x, or you can take z x. This is the most easier one for problems we preferably use this. [00:13:20.544] Okay. So if Z is a function of say XY and another variable, say t, okay, then you can find three partial derivatives. You can find Zx, you can find ZY, and also you can find Z T. [00:13:39.588] So when you differentiate this zx, what is the meaning of zx? I am differentiating z like you differentiate any other function with respect to x, but with the constraint that you have to treat y and t as constant. [00:13:57.956] As the definition uh is similar to the definition of the total derivative or derivative of uh a single variable function, all the properties, rules, and all those things that you used for the derivative of a function of single variable holds good here also. [00:14:16.196] like product rule, quotient rule, chain rule, derivative of an implicit function, composite functions. All those things are applicable to partial derivatives also. [00:14:26.596] Okay. So hopefully the definition is clear, right? So let me take a simple example so that things will be a little more clear to you. Let's take uh a function which looks like say z is equal to [00:14:40.400] I am taking bivariate, maybe I can do another one which is a multivariate function. Okay. So I'll take the same example which is x cube sine x, but now this time I'll take x cube sine 1. So [00:14:53.476] Now, as soon as I ask you to differentiate, the first question that comes to your mind is with respect to what? Right? I need to specify that. So I'll ask you to find the partial derivative of Z with respect to X.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.580] Now this is very simple. So whenever you have x, you just look at the function that has x in it. [00:15:16.068] don't see any

other thing, no matter how many variables you had here. Okay? So maybe I can add another maybe C can square t to just to make it a little more complicated. Or I can add an e to the power of W to this and an Ln of uh say um some P. [00:15:34.820] Okay. So this Z is a function of five independent variables. So I have X, Y, T, W, P, no matter, it could be N number of variables. But when I ask you to find ZX, [00:15:48.964] All these things doesn't matter for me because all of them will become constant. And I just need to differentiate x cubed, which is 3x square, and you retain all these things as it is because they are constant. Okay. [00:16:06.660] Now, let's say I want you to calculate ZP. So, can you tell me? Let me choose another color so that you can visualize properly. Okay. So suppose I want ZP. What will be ZP now? What all will be constant? You don't need to worry about any of these functions. All these functions are constant. [00:16:26.064] And uh you just need to differentiate Ln of P, which will be 1 by P and the remaining things will come as it is. Right. The next thing is let me choose another color. So now suppose I want to differentiate ZY. So you have to look at the term that has y in it. [00:16:44.804] Right, which is sine y. So derivative of sine y will be cos y. The other terms will remain as it is. As I said, all the rules of differentiation uh hold good for partial derivative also. But I'll discuss it more when we uh do problems. So uh [00:17:04.996] Again, when I say all the rules of differentiation holds good for partial derivative, you know that we found the first derivative and then we extend it to find the second derivative, third derivative. You can continue that. [00:17:18.468] So, even in partial derivative, you can find the second order partial derivative, which are denoted by zxx and zy. But there is one thing that is different in partial derivative. Suppose I have uh say I have zx. [00:17:36.740] And Zx might look like you know sine y into 3x square, something like that. I'll continue with the same example that I did in the previous page. Now don't you think that I can differentiate Z with respect to Y again, right? [00:17:52.992] So I can differentiate Zx with respect to Y again. [00:17:57.924] And I'll call that as Z Y. So you need to be careful about the notation. This so what will what will I get? I have to treat 3x square as constant, but I can differentiate sin y which will be cos y 4. [00:18:14.340] So what I did is I had Zx and then ZY and I can differentiate this Zx and ZY with respect to the other parameters now. [00:18:24.944] With respect to itself is higher order, that's what we saw now x, zx and z y. But I can differentiate zx with respect to y and then z y with respect to x. And they are called the mixed partial derivatives. So zxy and z. [00:18:41.680] Y x. Okay? So just be careful about the notation. So ZxY suppose you have uh ZXY. [00:18:51.300] What does this mean is first you go in the same order as we are prefixing. Okay. First you calculate the derivative of z with respect to x and then with respect to y. So if I'm doing z y x, then that means I will differentiate first with respect to y and then with respect to [00:19:08.448] X. So you move from inside to outside. So um suppose let's take an example. Suppose I'll ask you to calculate Z XY X. [00:19:21.924] And then another x. Okay. So what is the order of derivative? So you have z. The first thing you need to do is find zx. [00:19:30.688] Okay. So from ZX you find ZXY. So differentiate that with respect to Y. Zxy you differentiate with respect to X again. And then you finally you differentiate with respect to uh X again. So it is ZY XX. [00:19:48.388] Okay. So the partial derivative, mixed order partial derivative, you need to be careful about the notation. [00:19:57.892] So, uh there is a Clydoids theorem which says that the mixed order partial derivatives are equal. That is, if the order of differentiation is uh immaterial.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:10.768] Okay so this is a very important theorem. I would like to leave a note for you here to prove this Clydoids theorem. So try to prove this theorem. [00:20:21.092] if possible. Okay? And the subscript notation, the proof is very simple. You don't need to do too much. Use the definition of uh limit definition of derivative and you can show that uh those two limits are equal. So that's how so in the subscript notation the subscripts are written in the same order, this we discussed already. [00:20:38.704] Um so next class we will see what is the geometry of uh this partial derivative. Why [00:20:44.752] What geometrically what does the first derivative represent, second derivative represent. For a function of single variable, you already know that, right? The first derivative represents velocity, second derivative represents acceleration. But now we are in a higher dimension and we want to check what it represents. So that we'll take up in the next session. [00:21:03.876] Okay, thank you.

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Full transcript: 3363 normalized words, 86 timestamped segments, 1266.112 seconds duration. Low-confidence segments flagged: 0.

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3 Introduction to Partial Differentiation

This lesson is indexed from the source video titled Introduction to Partial Differentiation. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000022 · EV0000023

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.028] Hello everyone, welcome to Engineering Mathematics One Unit Two, which is partial differentiation. Myself, Dr. Anita from Department of Science and Humanities. As the name itself suggests partial differentiation is uh differentiating something partially. [00:00:16.912] So we'll see what is that that we are differentiating partially. Okay? So the topics that are included in the module are definition of partial derivative, total derivative and chain rule, partial differentiation of composite and implicit functions, Euler's theorem. [00:00:33.440] Maxima and minima for functions of two variables, Lagrange's method of undetermined multipliers, and as an application of all the Ebo concepts you will be learning errors and approximations. [00:00:44.804] I know most of you might not like math that much, right? The only reason uh that you're here is maybe by force or uh you want to I hardly doubt that many of you want to learn things uh by choice, right? Uh so [00:01:01.408] To make things a little more interesting, whenever we teach a topic, it's very essential that we ask ourselves questions. That is, why I need to study the topic, where exactly I'm going to apply the topic, and how can I apply the topic. [00:01:18.436] Right? So even for this topic, we have three fundamental questions that are why do we need a function of more than one variable? How do we calculate the limit, continuity, and differentiability for a function of more than one variable? And [00:01:34.948] When we find the derivative, how can I apply that to find the relative extrema of

functions? By the end of this unit, you will get the answers to all these questions. So before we uh define, let's see why do we need a multivariable function? [00:01:56.004] On an uh larger scale, we can define a multivariable question as a function as quantities which depends on two or more variables are represented using functions of two or more variables, for example. [00:02:11.268] Suppose you have a function of single variable. Let's say you have a function y is equal to f of x. [00:02:18.724] If we write like this, we know that y is a dependent variable which depends on x. But now we want to define a function which is a function of two independent variables. This is called a bivariate. [00:02:33.092] Because both x and y are independent, we'll assign this function to another variable called z. [00:02:40.484] Suppose similarly if you want to define a function of three independent variables, let's say a function of x, y and z, where all three x y and z are independent variables, then w will be the dependent variable. [00:02:54.756] Not just this, suppose you want to define a function of n independent variable, we can define it as x1, x2, etcetera, xn. So this is a function of n independent variable. [00:03:10.628] Uh whatever we do in this unit, we'll restrict ourselves to a bivariate function, but same thing can be extended to a multivariate function. Now let's see why do we need a function of more than one independent variable. [00:03:26.500] The first case would be when you uh find the area of a rectangle. You know that area will be a function of just length and width. [00:03:36.740] So it's a function of two independent variables. Suppose you similarly if you want to find the volume of a cylinder, then we it's a function of radius as well as its height. [00:03:47.428] Electric potential if you want to find uh in within a cathode ratio, we know that it's a multivariable function with uh spatial coordinates as the variables as well as time t. [00:04:00.964] The propagation of electromagnetic wave, which depends on the magnetic field, electric field, wavelength, and the direction of propagation. As you observe, this is a function of four independent variables. Not just that. Suppose I'll ask you to predict the weather in your area today. [00:04:19.716] you would need some parameters before you say what could be the weather in the next twenty four hours, right? So the variables would be topol o topography of the place, pressure, and then humidity and the temperature distribution, etcetera. [00:04:33.860] But if there is a cyclone or any other kind of major change in the weather, then you might have to include other parameters to decide the or to predict the weather conditions. [00:04:47.492] You can also give an example of the recent pandemic like COVID nineteen. Suppose you want to study how uh the pandemic is spreading, in that case, you need to consider many parameters, which may include like um

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:01.444] the area, the population in that area, awareness among the people, the immunity level of the people who are living in that area, and the distribution of population based on the age group, gender, and other classifications. Okay? So [00:05:16.208] As you see, the Ebo examples make one thing clear that we are surrounded by function of more than one variable. So if we study uh just a function of single variable, it is only for us to understand the concept. [00:05:30.928] But if you want to conclude or make a a decision or draw a conclusion about any given situation, we have to consider always a function of more than one variable. Now let's see the definition. [00:05:46.468] So we define again a bivariate function and same thing you can extend it to n number of variables. A function of two variables denoted by so here we are denoting it as z is equal to f of x y. [00:05:59.312] is a rule that assigns to each ordered pair of real numbers XY in a set D a unique real number denoted by f of x y. [00:06:09.540] The set D is the domain of f and its range is the set of all the values that it takes. What I mean by this in

simple terms is suppose you have a function z is equal to f of x of single variable. When do we say that this is a function [00:06:26.020] you will have a domain which is a set of all real numbers. So this is the domain for us. And then you have another set of real numbers called the range. [00:06:39.396] Right? And then we we will map every element in the domain to a unique element in the range by this function f. So for every element in the domain you have an image which is in the range. [00:06:55.556] Similarly, we can define a function of bivariable or function of two variables. [00:07:02.852] So even here it's a mapping from the set of all real numbers. So this is again R, which is the domain, to the set of all real numbers again, which is the range. Okay. [00:07:16.048] But instead of having a real number here, I'll have an ordered pair here. So I'll have an ordered pair x y which will be mapped on to a real number z in using the function f. For example, let's take suppose I have z is equal to x cube y. [00:07:34.180] Okay and then you want to plot let's say an ordered pair like 2 comma 1. So where can we map this ordered pair 2 comma 1? When I put x is equal to 2 and y is equal to 1 in this expression, this will be mapped on to 8. [00:07:50.212] Okay? So a function of two variables is nothing but a set of ordered pair which is mapped onto a unique real number. I hope uh the concept is clear, right? The next question is, how does a function of more than one variable look like? [00:08:09.860] So it was easy so far to plot any function. Suppose y is equal to x square is given, you know how it looks, right? It's just a parable which is symmetric about the y axis. So it would look something like that. [00:08:23.712] But now if I ask you to plot a function z is equal to say sin x into cos y, it you can observe that it's not that easy for us to visualize how it looks. [00:08:35.236] So we have luckily we have tools now which will help us to plot functions of n variables, at least two variables, because as I in increase the number of independent variables, the coordinate system has to increase. [00:08:50.020] For example, for a bivariate function I need three coordinate system. For a function with three independent variables, I have to add another dimension, which is practically not possible for us to demonstrate. So we will restrict ourselves for a function of two independent variables. [00:09:07.140] Now this is the graph, the first graph that you see is for the function sin x into cos y. So it's beautifully done with the a range of values. And this graph here is for a bivariate function again, which looks like e to the power of 1 minus x square minus y square. [00:09:26.052] Okay. So we have plotted this function and this is the graph of the function. So hopefully you understood what is the definition of a function of two independent variable. How does the graph of the uh multivariable function look like? Specifically bivariate. [00:09:44.324] The next thing would be we'll see uh how to find the limit of a function of two independent variables. [00:09:54.788] So let's recollect before we go what is the limit of a function of single variable. Suppose I have a function of one independent variable, which is like y is equal to f of x. Then we say that limit of f of x will go to a value L.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:14.244] As x approaches to A, if the only condition that I need to worry is as the distance between X and A reduces, the distance between f of x and l reduces. Right? [00:10:27.940] In fact, what you did is suppose this is the line where I have my point A, then you can let X approach to A either from the left or from the right. And then you calculate what happens to the value of the function. [00:10:46.148] In both the cases, if it approaches to the value L, we say that the limit of the function exists and it is equal to L. We can do a similar thing for a function of two variables now. [00:11:01.060] So suppose I have a function f of x y, so let's denote it by z is equal to f of x y. I say that the function f of x, comma y will tend to a limit L as [00:11:17.156] X comma y approaches to a point, let's say x comma y

approaches to the point, a comma b, if and only if this function f of x y takes the same value L. [00:11:32.964] When xy approaches to a b along any given path. What do I mean by that? As I was mentioning, in case of single variable function, x will approach a either from the left or from the right. [00:11:52.292] But now that I am in two dimension, XY, suppose this is my point x, y, can approach the point a comma b along different paths. So it can be along a straight line. [00:12:07.332] Let me choose a different color to show different path. Or it could be along let's take an ellipse. So it could be on an ellipse. Okay. Or [00:12:19.908] It could be on a parabola. So which my which might look something like that. So you have a parabola and on which it goes. So this XY can approach to this AB along different paths. [00:12:34.340] So when we find the limit of a two variable function, then we need to be careful that as xy approaches to ab along any given path, the function f of x, y approaches to the same value L. [00:12:54.244] So that's one thing that you have to be careful about while calculating the limit. [00:12:59.460] Let's take an example so that things will be a little more clear. Okay. So I have a a problem here, that is I have taken limit of this function and I need to calculate what happens to the limit as xy approaches to 2, minus 1. [00:13:15.072] Let's plug in the value and check. Suppose if we put x is equal to 2 and y is equal to minus 1, I get 2x into minus 1 plus 3 minus 1 the whole square, minus 4 into 2 plus 3 into minus 1 and then minus 6. [00:13:35.300] If we simplify this, we get 4 plus 4 minus 3 minus 8, this would be plus 3, and then minus 3 and then minus 6. [00:13:47.300] So if you observe carefully, we can do some simplification for this. I can cancel this, I can cancel this, and the final answer would be minus six. So what is the meaning of this minus six? [00:13:59.556] As xy approaches to 2 comma minus 1 along any given path, this function here will take the value minus 6, irrespective of the path taken by x y to reach the point 2 comma minus 1. Okay. But we'll let's take another example. [00:14:20.004] Suppose I need to calculate the value of the limit of this function, 2xy divided by 3x square plus y square. This is the graph of this function. [00:14:31.940] Now as soon as you look at the function, you can conclude that as x y goes very close to 00, this function takes a value zero by zero, which is an indeterminate form. [00:14:44.068] Right. So I just want to make sure that this limit does not exist. How am I going to prove that it does not exist? So the only way to prove that a limit does not exist is I will let this point X, Y approach to 0, 0 along different paths.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.260] Okay what do I mean by different path? So let's say that it approaches let's say that x y approaches to 0 0 so maybe case 1 I can take along the line y is equal to 0. And you know that this line y is equal to 0 is nothing but x axis. [00:15:20.996] Right. So what I'll do is I'm going to let this x y approachy approaches to 00 along this line y is equal to 0. In that case, what happens to the limit? [00:15:33.860] I have to calculate limit as y is 0, I will get only x approaching to 0, 2x into 0 divided by 3x square plus 0. So I don't have to simplify this. This will be anyway 0. Right? [00:15:50.468] Let's take uh along a different path. So let me change the color. Now let's say if uh xy approaches to 00 along uh along the path you can choose y is equal to x square, okay, which is a parable. [00:16:07.268] What happens if uh it approaches along y is equal to x square? So I'll get limit extend to 0. Wherever I have [00:16:16.100] a y I need to replace it with x square, right? So I'll be replacing it with x square and the denominator will be 3x square plus y square will be x to the power of 4. [00:16:29.572] And if I simplify this, I'll get 2x cube divided by x square. If I take out x square common, I will get 3 plus x square. Of course I have limit. Okay. So if I simplify, you can see that this will be 0. [00:16:48.452] So luckily along two paths I got the limit to be same. But what

happens if I take the value that xy approaches to 00 along [00:17:03.684] The line y is equal to x. So I know that y is equal to x is a line that passes through the origin like this. So if it approaches, then I'll get limit extending to 0 of 2x. Wherever I have y, I'll replace it with x divided by 3x square plus x square. [00:17:24.356] And if you observe carefully, this will be just 2x square divided by 4x square, and of course this is just half. So we just proved that these three values are not same. That means the limit of this function does not exist. [00:17:42.596] As you saw in the previous example, it won't be uh easy for us to check if the limit exists. But it's fairly easier for us to check if the limit does not exist, because we can take two random paths and show that the value is not equal. [00:17:59.492] Okay. So a better way to understand if the graph is uh does have a limit or if it's continuous is based on the graphing. [00:18:07.760] So it's very important that you graph a bivariate or a multivariable function so that you can decide whether the limit exists or the graph is continuous, at is there a discontinuity at any point or something like that. Let's define now continuity for a function of [00:18:25.316] a single variable. So before we go for a bivariate function, you recollect when what what is the definition for continuity of a single variable function? We say that a function y is equal to f of x is continuous if it satisfies three properties. [00:18:46.084] The first one is limit as extending to A of F of X must exist. [00:18:54.820] The second thing would be f of a must exist. And the third one is limit as extending to A of F of X must be F of A. [00:19:08.964] So if these three conditions are satisfied, then we say that the function f of x is continuous at x is equal to c. Right. So let's see what does the graph of that function look like? Suppose if I uh so a function is said to be continuous, if there is no [00:19:25.328] break in the graph. So that's a smooth curve. And suppose I have any break in the graph, something like this, then you know that the function is not continuous. Similarly, in the next class we are going to define well how how to define [00:19:40.016] uh the continuity for a bivariate function or in general maybe for a multivariable function. So I'll see you all soon in the next class. Thank you.

EV0000034 · EV0000035 · EV0000036 · EV0000037

Full transcript: 3066 normalized words, 80 timestamped segments, 1190.848 seconds duration. Low-confidence segments flagged: 0.

EV0000022 · EV0000023 · EV0000024 · EV0000025

4 Geometrical interpretation of Partial derivatives

This lesson is indexed from the source video titled Geometrical interpretation of Partial derivatives. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000038 · EV0000039

Graphs and diagrams

Figure G000019. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 1.

EV0000313

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

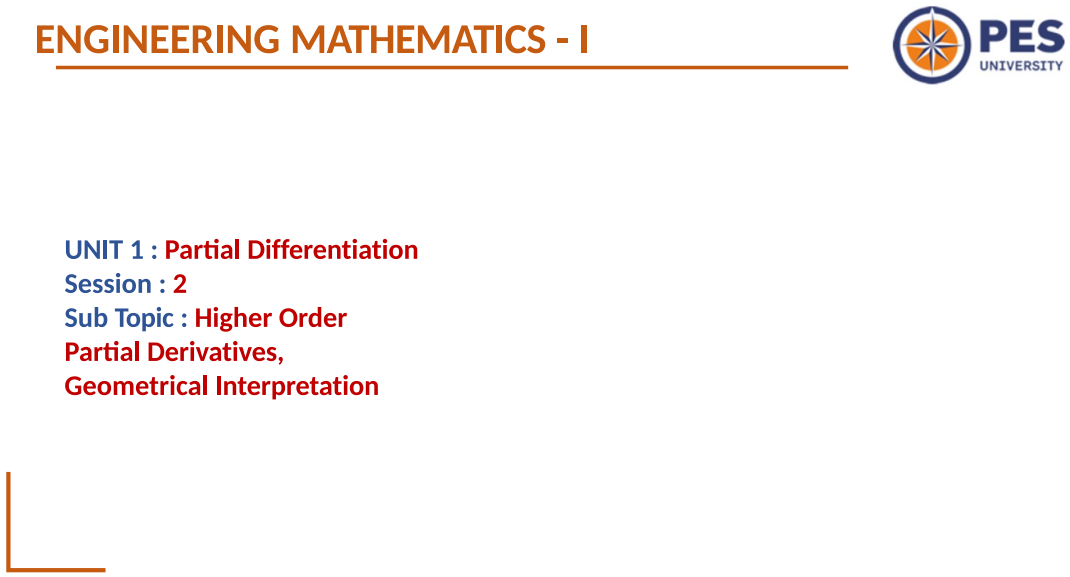

Figure G000020. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 2.

EV0000316

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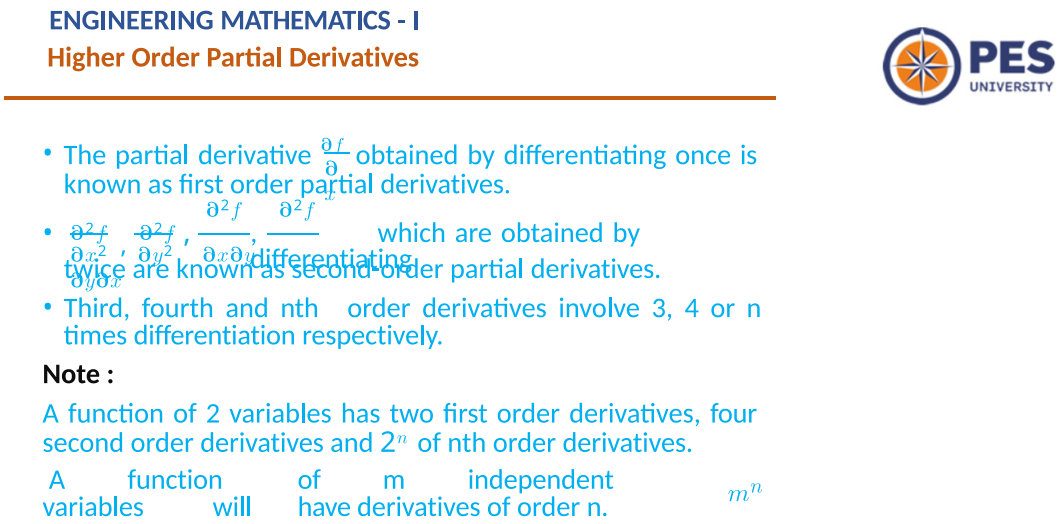

Figure G000021. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 3.

EV0000319

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ENGINEERING MATHEMATICS - I

Geometrical Interpretation of Partial differentiation

PES UNIVERSITY

We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point.

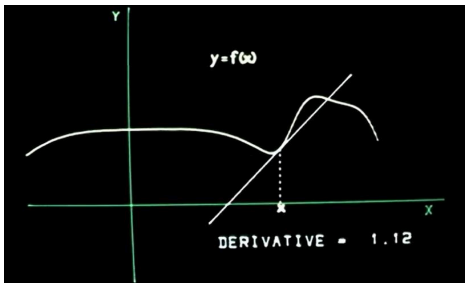

IMAGE SOURCE: https://www.youtube.com/watch?v=Q8mbXy0oJj8

Figure G000022. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 4.

EV0000322

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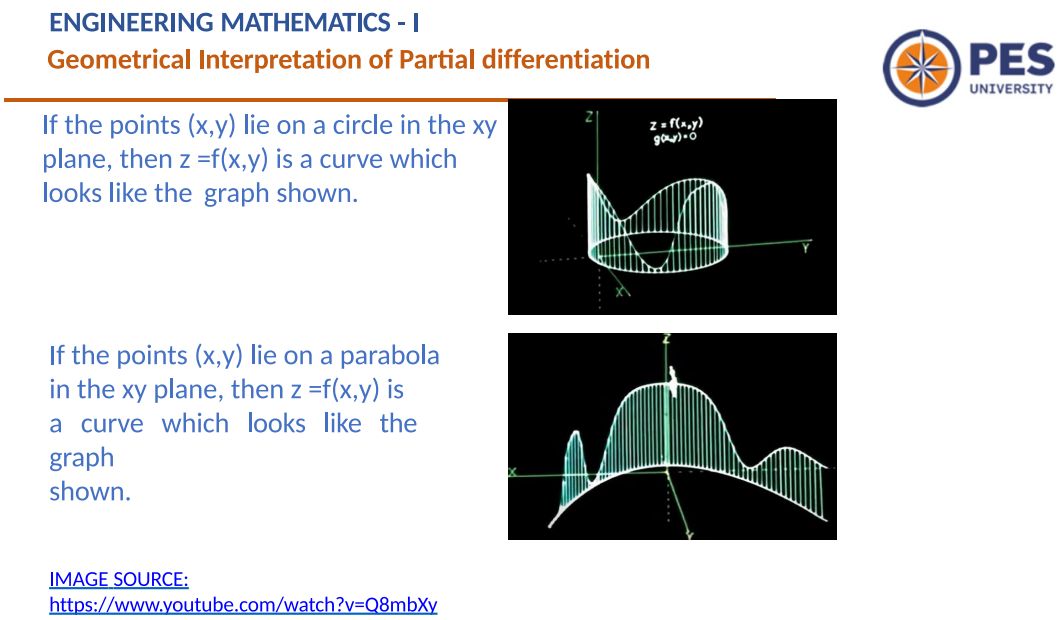

Figure G000023. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 5.

EV0000325

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ENGINEERING MATHEMATICS - I

Geometrical Interpretation of Partial differentiation

PES UNIVERSITY

Partial derivative of the function f(x,y) at the point (x_0, y_0) can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function.

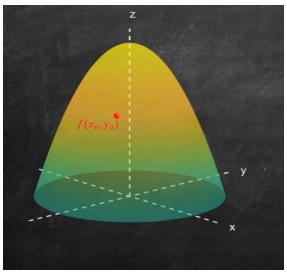

Figure G000024. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 6.

EV0000328

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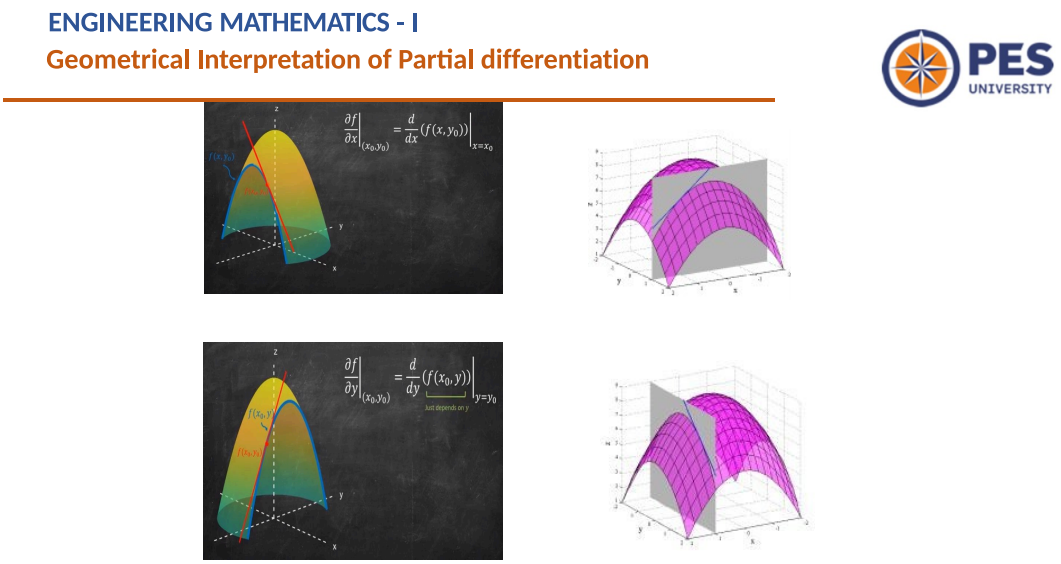

Figure G000025. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 7.

EV0000331

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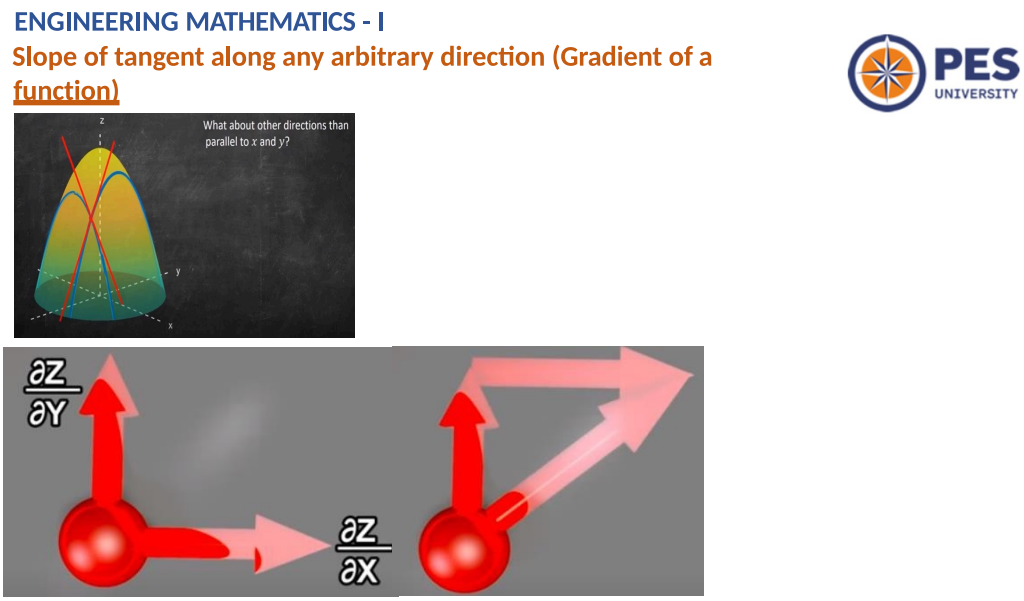

Figure G000026. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 8.

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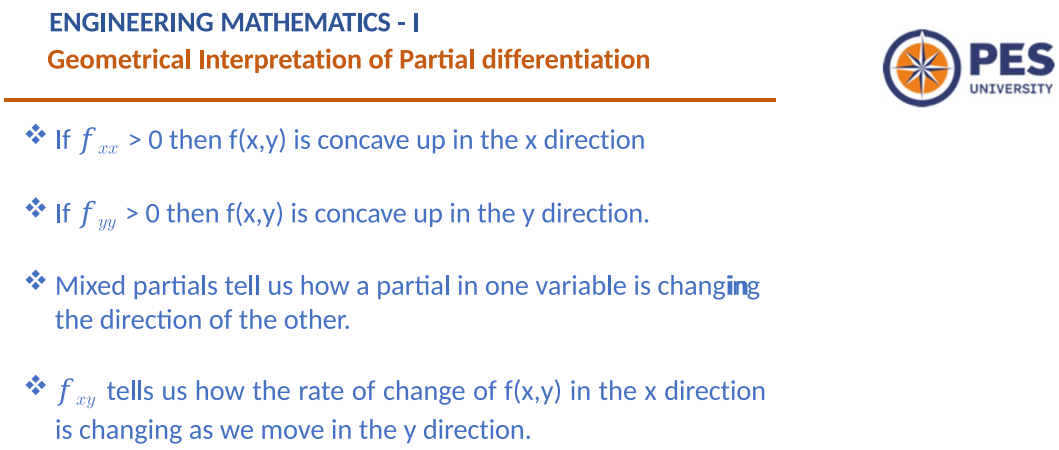

Figure G000027. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 9.

EV0000337

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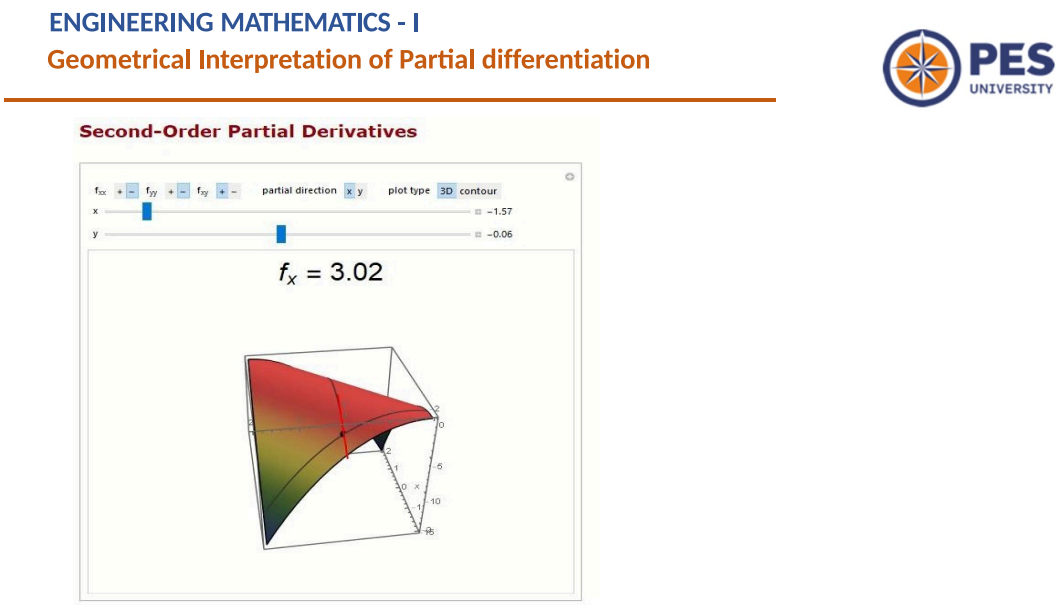

Figure G000028. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 10.

EV0000340

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Figure G000029. Geometrical interpretation of Partial derivatives Problems on partial derivatives · source slide page 11.

EV0000343

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000312 · EV0000313 · EV0000314
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 2 Sub Topic : Higher Order Partial Derivatives, Geometrical Interpretation EV0000315 · EV0000316 · EV0000317
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Higher Order Partial Derivatives The partial derivative ∂f obtained by differentiating once is ∂x known as first order partial derivatives. $\partial 2 f$, $\partial 2 f$ $\partial 2 f$ $\partial 2 f$, which are obtained by differentiating • , $\partial x 2$ $\partial y 2$ $\partial x \partial y$ $\partial y \partial x$ twice are known as second-order partial derivatives. Third, fourth and nth order derivatives involve 3, 4 or n times differentiation respectively. Note : A function of 2 variables has two first order derivatives, four second order derivatives and $2n$ of nth order derivatives. mn A function of m independent variables will have derivatives of order n. EV0000318 · EV0000319 · EV0000320
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation We know that the derivative of a single valued function gives the slope of the tangent line to the curve at the given point. [Image text, OCR] y=f(cx) X DERIVATIVE1.12 IMAGE SOURCE: https://www.youtube.com/watch?v=Q8mbXy0oJj8 EV0000321 · EV0000322 · EV0000323

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation [Image text, OCR] z=f(x.y) ga). If the points (x,y) lie on a circle in the xy plane, then z =f(x,y) is a curve which looks like the graph shown. If the points (x,y) lie on a parabola in the xy plane, then z =f(x,y) is a curve which looks like the graph shown. IMAGE SOURCE: https://www.youtube.com/watch?v=Q8mbXy0oJj8 EV0000324 · EV0000325 · EV0000326
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation Partial derivative of the function f(x,y) at the point $(x0, y0)$ can be visualized as the slope of the tangent line along the direction of the independent variable with respect to which we would differentiate the function. [Image text, OCR] f(xo,yo) EV0000327 · EV0000328 · EV0000329
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation [Image text, OCR] af (f(x, yo)) ax dx (xo,yo) x=xo f(x,yo) [Image text, OCR] Z of d dy (f(xo, y)) dyl(xo,yo) y=yo f(xo.y) Just dependsony f(xo,yo) EV0000330 · EV0000331 · EV0000332
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Slope of tangent along any arbitrary direction (Gradient of a function) [Image text, OCR] Z Whataboutotherdirectionsthan paralleltoxandy? [Image text, OCR] 2 Z ax [Image text, OCR] ← EV0000333 · EV0000334 · EV0000335
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation If $f_{xx} > 0$ then f(x,y) is concave up in the x direction If $f_{yy} > 0$ then f(x,y) is concave up in the y direction. Mixed partials tell us how a partial in one variable is changing in the direction of the other. f_{xy} tells us how the rate of change of f(x,y) in the x direction is changing as we move in the y direction. EV0000336 · EV0000337 · EV0000338
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Geometrical Interpretation of Partial differentiation [Image text, OCR] Second-OrderPartialDerivatives + partial direction plot type 3D contour -1.57 -0.06 f=3.02 EV0000339 · EV0000340 · EV0000341
11. Source slide page 11 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000342 · EV0000343 · EV0000344

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.444] Welcome back. We were discussing about the differentiability [00:00:06.656] of a multivariable function and we call that as a partial derivatives. We also defined what is a higher order derivatives and what is the mixed order partial derivative. So what basically we did is suppose z is equal to f of x y was a bivariate function, we just define [00:00:24.336] How to find ZX, how to find ZY, and uh if you had observed we uh actually did these many second order partial derivatives, right? But of course these were uh equal, but still. So what one thing we observed is [00:00:44.292] A function of two independent variables, suppose you have a function of two independent variables, which is a bivariate function, it has two first order derivatives, which are zx and zy, and it has four derivative. [00:00:58.980] second order derivative, which are Zx, ZYY, ZXY, and ZYX. That means, suppose I have a function f [00:01:08.996] which is uh a function okay let me call z as a function of n independent variable. So say let's say x1, x2, etc, x uh n, then the mth [00:01:23.876] Um okay, so here let me

make this as m. So suppose I have m a function of m independent variable, then the nth order derivative. So let me just erase this to give you an idea what I'm talking about. [00:01:37.700] So I have a function z, which is a function of m independent variables. So I have x1, x2, etc., xm, then I can differentiate this. [00:01:50.020] in M to the power of N order. That means you the first order derivative will have two terms. The second order will have [00:01:59.712] Again four terms for a two variable and so like that it continues. Suppose you have a function of m independent variable, then you will have m to the power of n derivatives of order n. Okay. So [00:02:13.572] Uh suppose n is equal to 2, suppose you want the second derivative, then there are m square second derivative uh functions. Okay But the most basic question is what is the geometry of this partial derivative? [00:02:32.004] As you know, suppose I have a function of one independent variable. So I have a function which is y is equal to f of x. Let's say the graph that you see here is the graph of that function, y is equal to f of x. Right? Then what is the dy by dx represents? [00:02:50.244] You know that the derivative always gives me the change, right? So derivative always represents change. But change in what? [00:02:59.332] So I here I'm studying the change in the position that is happening corresponding to this x. So as I change my position on the x axis, what is the change that is happening on the curve? That's given by dy by dx. [00:03:13.252] So it's basically geometrically what you are doing is finding the slope of the tangent at every point. Correct? So the derivative at a particular point is nothing but the slope of the tangent. That's for a single variable function. Now if I have a bivariate function. [00:03:32.452] Let's say I have this. So, okay. So what is this graph represents? I have a points which lie on the circle. You see a circle in the bottom? So for every x and y which lie on a circle, I'll calculate the corresponding value of z. [00:03:47.792] So I am calculating the value of set. And that lie on this curve. Right? That means this function here, x x y, suppose x y lie. So there's another example here. So in this case the points x comma y lie on a parabola. [00:04:05.316] Okay? So for every point on the parabola, these are the points that corresponds to value of Z. So Z changes like this, so it goes on the surface. [00:04:16.132] So as we saw yesterday, a function of two independent variables always gives me a surface in three dimensions. Correct? So for every value of x and y, you have a corresponding value of z. So what is this partial derivative of a function at any point gives you? [00:04:36.004] It gives you the slope of the tangent. So can you imagine? So I have a surface like this, okay? So if I take a point like this, how many tangents can I draw to that surface here? So let's say I have one and then I have another tangent, I have another tangent, I have another tangent. So as you know, if you have a surface, [00:04:55.664] Something like this, and if you have a point, at this point, you please uh don't think that I am intersecting, this is a tangent because it's in 3D, it looks like that. So uh every color will represent a different tangent, okay.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:10.864] So I might not be that perfect, but let's let's draw a few tangents to that. So I I'm just sketching some of the tangents to that graph at that particular point. [00:05:24.740] So because it's a surface, at any given point you will have infinitely many tangents to the surface. Right? Are you following what I'm trying to say? So [00:05:35.652] Let me erase all these things. So I have a surface here, and then what I'm trying to find is what is the value of the tangent or the slope of the tangents at this particular point. There are infinitely many such tangent. [00:05:49.540] Right. I just want to find the slope of one such tangent. What is the tangent? That tangent is the variable with respect to which you are going to differentiate. [00:06:01.124] Right? So suppose I have Z is equal to F of XY. Let's say that

this is the surface which represents Z is equal to F of XY. Then Zx [00:06:10.500] will give me, so I am calculating Z x a at X0 Y0. This derivative at this point will give me the slope of the tangent to the curve at x0 y naught, which is parallel to x axis. [00:06:27.300] Are you following? So zx at x0 y naught gives me the slope of the tangent to this surface at the point x0 y naught which is parallel to the x axis. [00:06:41.604] So then what will ZY be? So to for you to better understand this is how it looks like. Okay? So f of x naught and it has infinitely many tangents. So look at this picture, maybe that gives you a better idea. [00:06:54.096] So I have uh suppose this is x axis for me, horizontal axis, and at this particular point if you see then I have taken the tangent line which is parallel to the x axis. Similarly, if I take another curve, then [00:07:10.692] The partial derivative with respect to y gives me the slope of the tangent. So this is here dou z by dou y at x0 y naught. This gives me the slope of the tangent to this surface at this point. [00:07:27.908] And what is the tangent? That tangent which is parallel to the y-axis. So that might look something like this. Okay. So let me go back to this. So yeah. So this is what we get. [00:07:45.252] Okay. So geometally the first order partial derivatives, suppose uh let me write this as zx and this is zy. [00:07:54.436] So the per first order partial derivative with respect to x gives me the slope of the tangent parallel to x axis and z y gives me the slope of the tangent parallel to the y axis. [00:08:07.024] Okay? So there are many tangents, there are infinitely many tangents, but I'm taking only the tangents that are parallel to X and parallel to Y axis. But the general question that would arise is [00:08:22.084] What if I want to find the slope of a tangent that is neither parallel to x axis nor parallel to y axis? Right? So that means what I if I had the my coordinate system like this and you had a surface. [00:08:38.756] And you know the tangents that you got now is either it was parallel to x axis or it was parallel to y axis, something like that. Okay, so we got that parallel to y axis. [00:08:51.780] Now what I want to do is I don't want these two. Okay, I got this already. But I want something something the that is different. So something like that. What how do I get the slope of this or what will this represent? [00:09:06.224] Okay. So it's basically if you look at that carefully, I have this is ZX and then this is Z1, right? This is ZX and this is Z1. So you know that any given line for me in a system can be resolved into its components, right? [00:09:23.600] I can resolve that into the component along x axis and the component along y axis. So if I treat them as vectors, then by vector addition I get another vector which is the resultant of these two. And this vector also gives me the slope of the tangent and this is called the gradient. [00:09:44.932] Right. So that's the gradient. Let's see how does it look like. So this is how the gradient look like. [00:09:54.212] So you're going to learn more about the gradient uh when we study vector calculus. Okay. So I have x axis and then y axis, and then I just took the resultant. So if you look at this line, this is the tangent to the surface where I want to find the slope of the tangent.

EV0000042 · EV0000043 · EV0000044 · EV0000045

Transcript block 3 (source interval 000600-000899 seconds): [00:10:11.792] Alright. So this will give me the resultant. Uh now we'll see what is the geometry of the second order partial derivative. [00:10:23.396] As you know that like the first order uh derivative in the for one variable gives you velocity. The second derivative so suppose uh what I mean is suppose you have y is equal to f of x, then uh dy by dx gives you velocity. [00:10:40.676] Right? And then D square y by dx square gives you acceleration. Do you know what do they call for the higher order derivative? T cube y by dx cube or um d to the power of four y by d x power four. So uh this is [00:10:57.520] uh velocity, this is acceleration, this is uh jowl, this is jerk, and then [00:11:03.920] uh you have pop and other things. So up to fifth or sixth derivative I think there is a name given. But you also know that

suppose you want to find whether the curve is concave up or concave down, you use usually the second derivative, right? When you did maxima minima. [00:11:19.936] So see even the impartial derivatives it's the same thing. Suppose I have the second order partial derivative which is fxx and if this is greater than zero, that means the function is concave up in the x direction. It's not that easy for me to plot in 2D, but I'll show you a video. [00:11:38.564] And then if f y y is greater than zero, then it is concave up in the y direction. [00:11:45.604] Same thing with the mixed order partial derivative. So what is this fxy or f y x will give you? Fxy will tell me how the partial derivative of f x is changing in y direction. Right? [00:12:04.420] So what is the change in fx along the y direction? So what will f yx will tell you? What is the change in f y along the x direction? So how much of f y is changing along the x direction? So that's what is given by the mixed second order partial derivative. [00:12:24.324] Okay. Um so let's see. I mean it's um again, it's not that easy for you to uh visualize. So here w you can try this. This is uh just from Ultra Am uh lab. Um what I have done is I have [00:12:40.144] animated that so that you can visualize for different values of fxx. So I have taken the cases where um uh for the first case as you can see I have taken f of xx to be uh sorry negative. So let me just uh erase that. [00:12:55.040] So in the first case I have taken FXX to be negative and then FYY also to be negative and then I have taken fx y to be positive. So this is the graph of that. [00:13:07.440] Okay so you just follow the red line there which is the tangent line which should tell you that it is um whether it is concave up in the x direction or concave down in the y direction or whatever is happening. Okay. So uh the as and when the video moves you can watch what [00:13:25.104] Is happening. So I have rotated so that you can visualize it in a better way. So this is when um both are negative. [00:13:32.576] And then I have increased the value of x. Now I have changed fx to be fxx to be positive. And you can see that it is concave up along the uh y axis or x axis, sorry. And then I have changed now fx is positive, but I have changed x fxy to be negative. [00:13:51.616] Yeah, it's it's nice to see these uh visuals. Um you can try by yourself by changing different values. Um just to check what happens. Okay. So you take different cases, like um you can make this positive. [00:14:05.828] and this negative and this positive. So you have so many combinations that you can or you can make all positive, all negative. [00:14:13.252] So and try what is happening to the graph, right? And you can keep on changing the value of X and changing the value of Y. That will help you to visualize the tangent. Do you see the tangent here? So I want you to observe the movement of this tangent, how it is moving when I change the value of [00:14:30.336] um x and y okay corresponding to the x axis or corresponding to y axis how what is the transition that is happening okay so uh I can also take partial derivative with respect to y and then partial derivative with respect to x and other things okay. [00:14:48.260] Um, so this these things will be better if you um try by yourself. So that things will be that much uh clearer to you. [00:14:59.268] So let me erase. Uh in the next class what we are going to do is we'll uh try try and solve some problems on uh partial derivatives. We'll try to find uh mixed order partial derivatives, we'll try to verify the Claret theorem and all those things. So but I'm not going to solve too many problems.

EV0000046 · EV0000047 · EV0000048 · EV0000049

Transcript block 4 (source interval 000900-001199 seconds): [00:15:16.816] I want you to explore. But um I have given a link uh in my next slide. [00:15:22.340] uh I'll give you a link in the end of the uh class wherein you can try all these things. You can try um uh how to plot a three dimensional graph, how to do this animation, and you know, how to check the partial derivative

and his geometrical interpretation and other things. [00:15:39.472] So uh I'll see you when I solve some problems on partial derivative. Thank you.

EV0000050 · EV0000051 · EV0000052 · EV0000053

Full transcript: 2575 normalized words, 66 timestamped segments, 945.301333 seconds duration. Low-confidence segments flagged: 0.

EV0000038 · EV0000039 · EV0000040 · EV0000041

5 Problems on Total derivatives, Chain rule

This lesson is indexed from the source video titled Problems on Total derivatives, Chain rule. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000054 · EV0000055

Graphs and diagrams

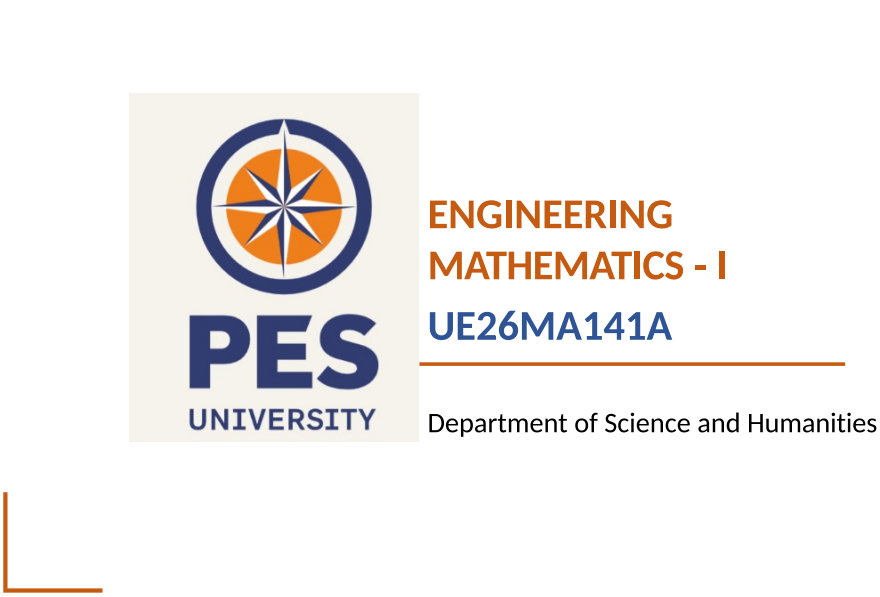

Figure G000030. Total derivative Chain rule problems · source slide page 1.

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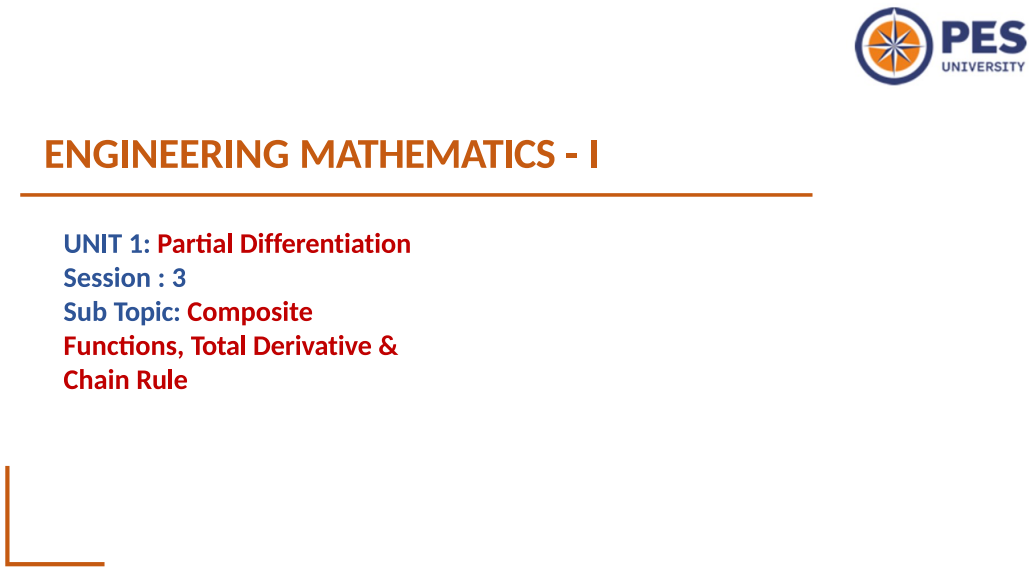

Figure G000031. Total derivative Chain rule problems · source slide page 2.

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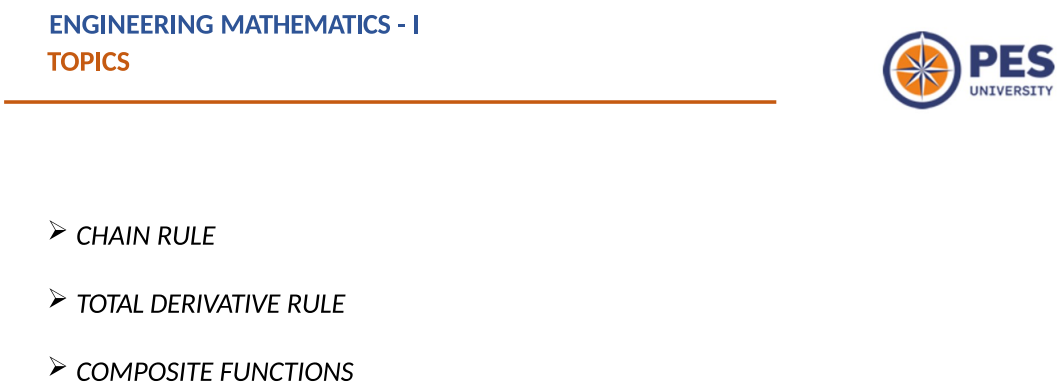

Figure G000032. Total derivative Chain rule problems · source slide page 3.

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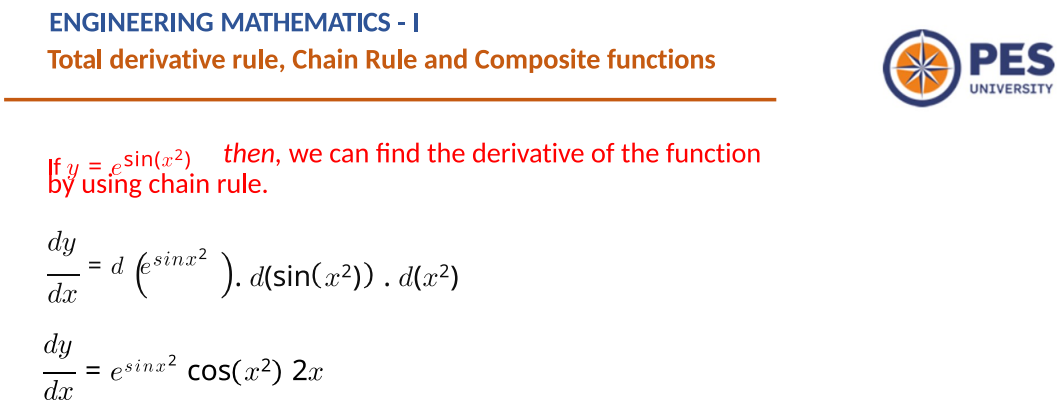

Figure G000033. Total derivative Chain rule problems · source slide page 4.

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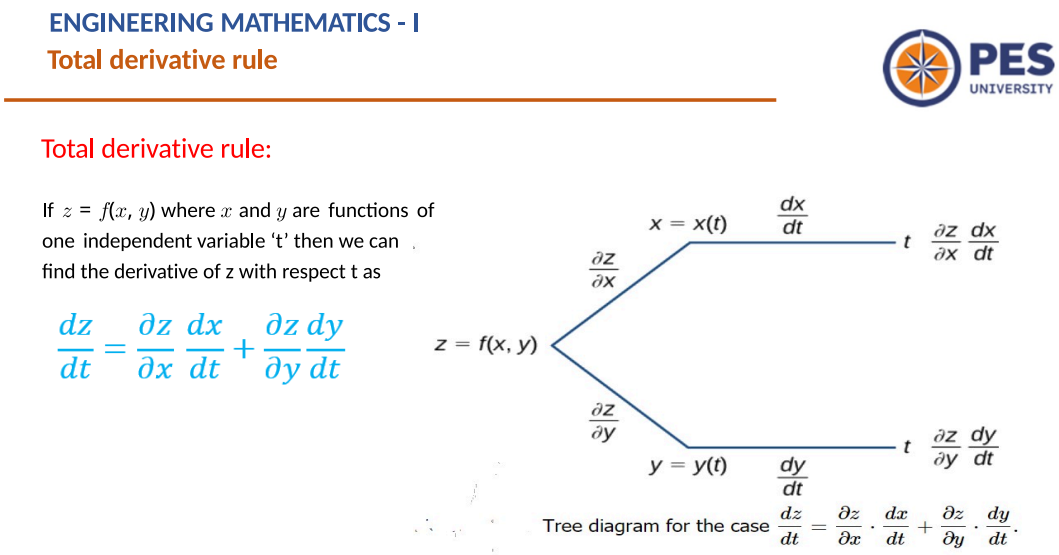

Figure G000034. Total derivative Chain rule problems · source slide page 5.

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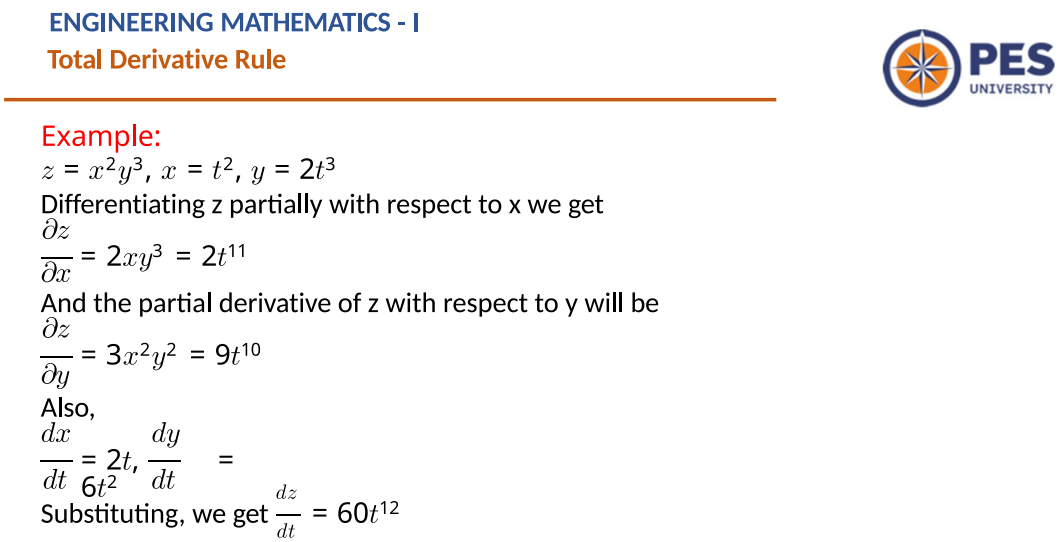

Figure G000035. Total derivative Chain rule problems · source slide page 6.

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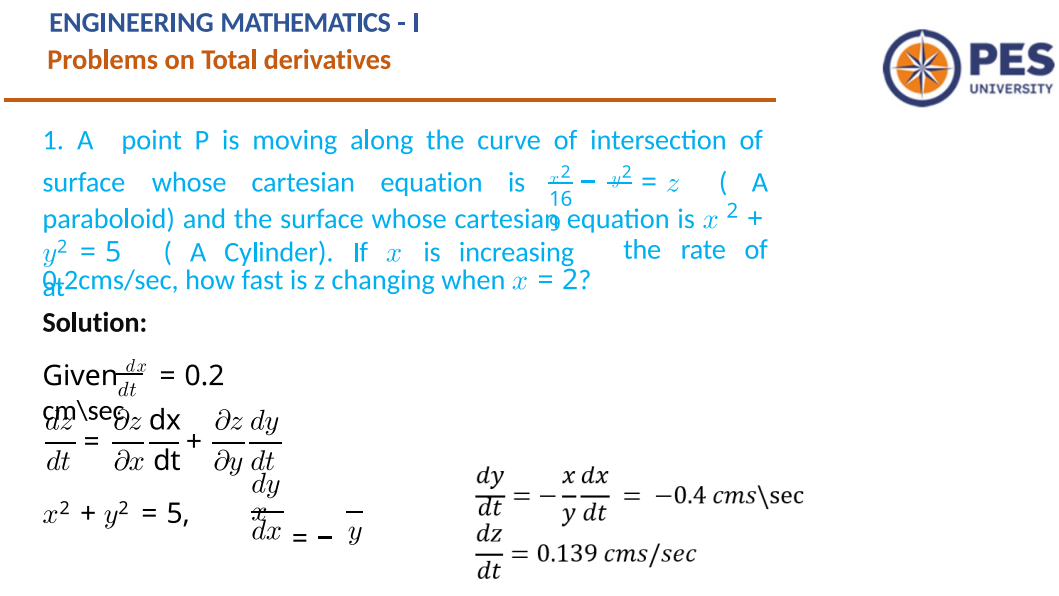

Figure G000036. Total derivative Chain rule problems · source slide page 7.

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ENGINEERING MATHEMATICS - I

Problems on Total derivatives

2. Given that $f(2,-3) = 6$, $f_x(2,\ ) = 1.3$ and $f_y(2,-3) = -3$ $= -0.6$, approximate $f(2.1,-3.03)$.

Solution:

The total differential approximates how much f changes from the point $(2,-3)$ to $(2.1,-3.03)$. With $dx = 0.1$ and $dy = 0.03$, we have

$dz = f_x(2,-3)\,dx + f_y(2,-3)\,dy$

$= 1.3(0.1) + (-0.6)(-0.03)$

$= 0.148$

The change in z is approximately 0.148, so we approximate

$f\ 2(1,-3.03\ ) \approx 6.148$

Figure G000037. Total derivative Chain rule problems · source slide page 8.

EV0000367

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ENGINEERING MATHEMATICS - I

Problems on Total Derivatives

3. One side of a triangle is increasing at a rate of 3cm/s and second side is decreasing at a rate of 2cm/s. If the area of a triangle remains constant, at what rate does the angle between the sides change when the first side is 20cms long, the second side is 30cms and the angle is $\frac{\pi}{6}$.

Solution:

A= $\frac{1}{2}$(base *height)

$= \frac{1}{2}xy\sin\theta$

$\frac{dA}{dt} = \frac{\partial A}{\partial x}\frac{dx}{dt} + \frac{\partial A}{\partial y}\frac{dy}{dt} + \frac{\partial A}{\partial \theta}\frac{d\theta}{dt}$

Figure G000038. Total derivative Chain rule problems · source slide page 9.

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ENGINEERING MATHEMATICS - I

Problems on Total Derivatives

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Contd...

$$\frac{dA}{dt} = -\left(y\sin\theta(3) + x\sin\theta(-2) + xy\cos\theta\ \frac{d\theta}{dt}\right)$$

Here $x = 20cm$; $y = 30cm$; $\theta = \frac{\pi}{6}$

$$\frac{d\theta}{dt} = \frac{-25}{300} = 0.04831$$

θ should decrease at the rate of 0.0481 rad/sec

Figure G000039. Total derivative Chain rule problems · source slide page 10.

EV0000373

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ENGINEERING MATHEMATICS - I

Problems on Total derivatives

PES UNIVERSITY

4. A cylindrical steel storage tank is to be built that is 10ft tall and 4ft across in diameter. It is known that the steel will expand/contract with temperature changes; is the overall volume of the tank more sensitive to changes in the diameter or in the height of the tank?

Solution:

A cylindrical solid with height h and radius r has volume $V = \pi r^2 h$.

$\frac{\partial V}{\partial r} = V_r(r, h) = 2\pi rh$ and $\frac{\partial V}{\partial h} = V_h(r, h) = \pi r^2$

$dV = (2\pi rh)\, dr + (\pi r^2) dh$.

When $h = 10$ and $r = 2$, we have

$dV = 40\pi dr + 4\pi dh$.

Figure G000040. Total derivative Chain rule problems · source slide page 11.

EV0000376

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ENGINEERING MATHEMATICS - I

Problems on Total derivatives

Contd....

Note that the coefficient of dr is $40\pi \approx 125.7$; the coefficient of dh is a tenth of that, approximately 12.57. A small change in radius will be multiplied by 125.7, whereas a small change in height will be multiplied by 12.57. Thus the volume of the tank is more sensitive to changes in radius than in height.

Figure G000041. Total derivative Chain rule problems · source slide page 12.

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ENGINEERING MATHEMATICS - I

Chain Rule-Composite Functions

We can define chain rule for a function of two or more independent variable:

If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v, given by

$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial u} \quad \text{and}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\frac{\partial y}{\partial v}$$

Figure G000042. Total derivative Chain rule problems · source slide page 13.

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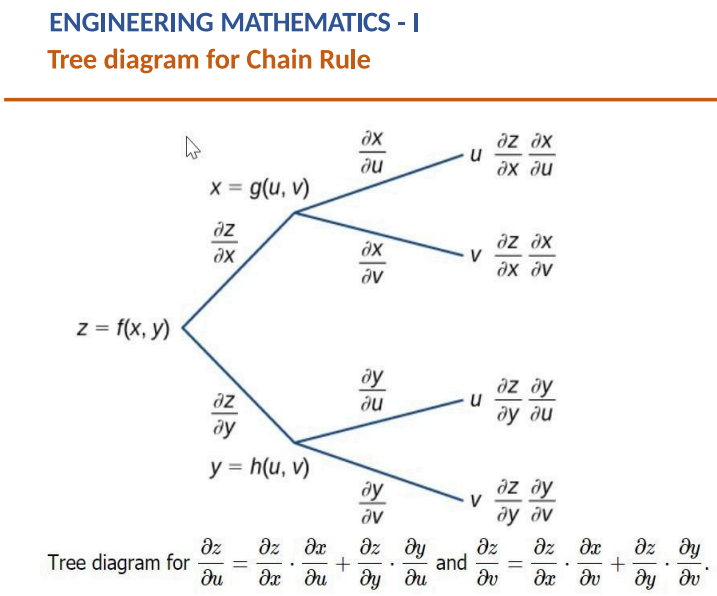

Figure G000043. Total derivative Chain rule problems · source slide page 14.

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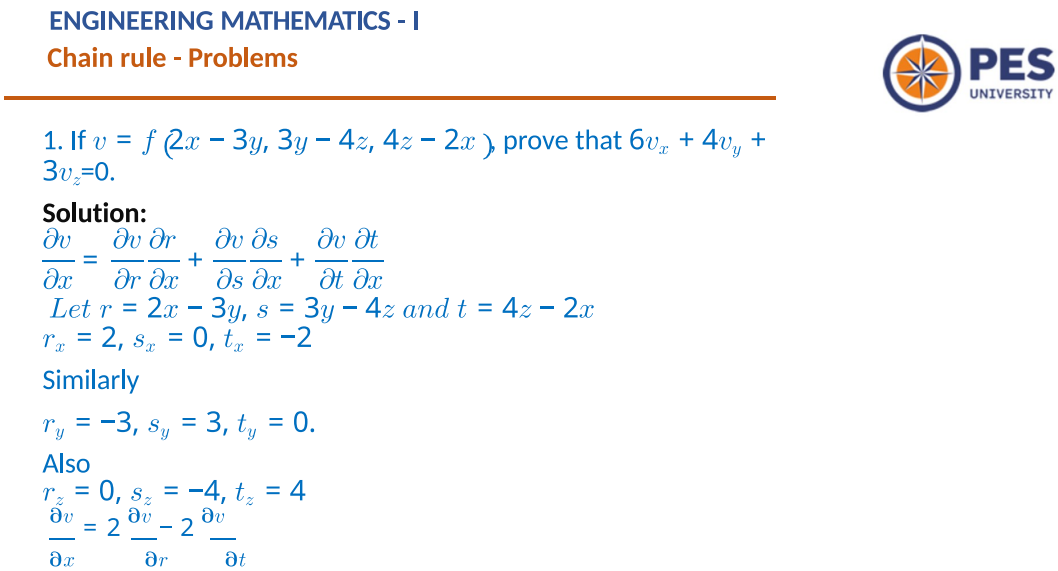

Figure G000044. Total derivative Chain rule problems · source slide page 15.

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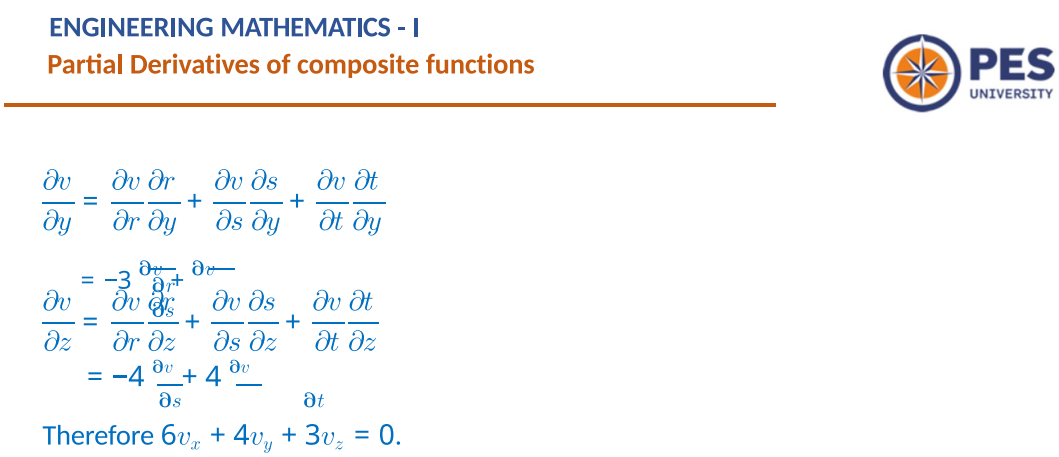

Figure G000045. Total derivative Chain rule problems · source slide page 16.

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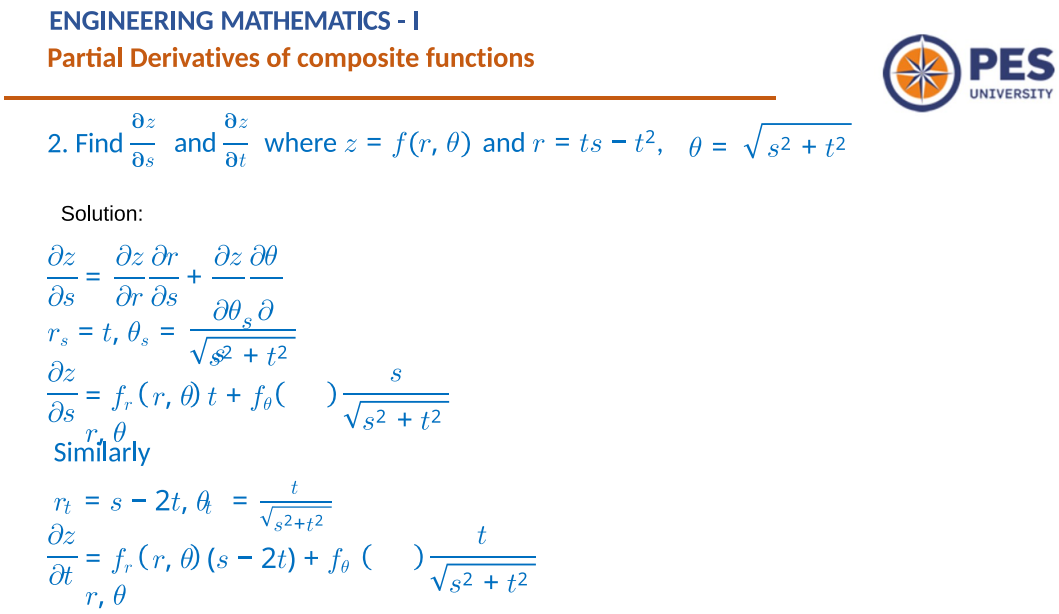

Figure G000046. Total derivative Chain rule problems · source slide page 17.

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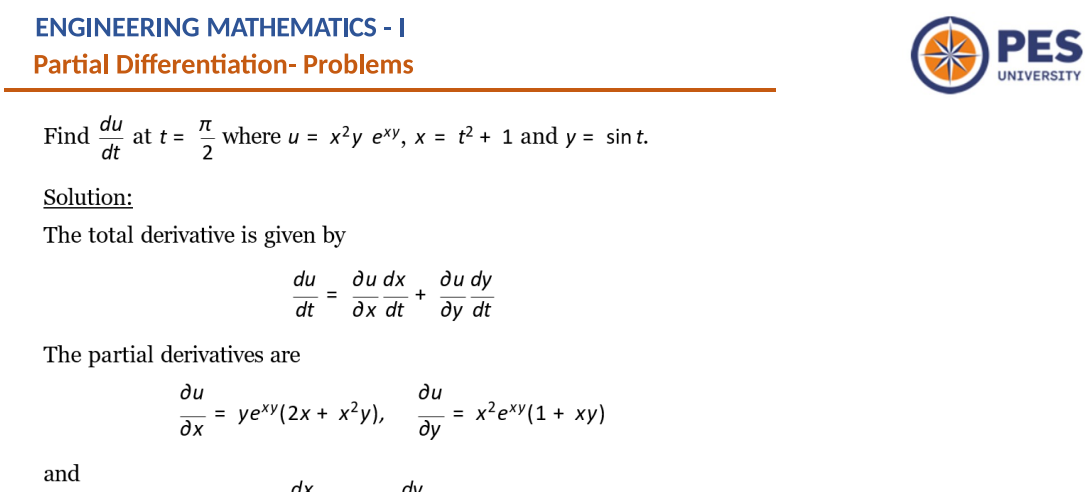

Figure G000047. Total derivative Chain rule problems · source slide page 18.

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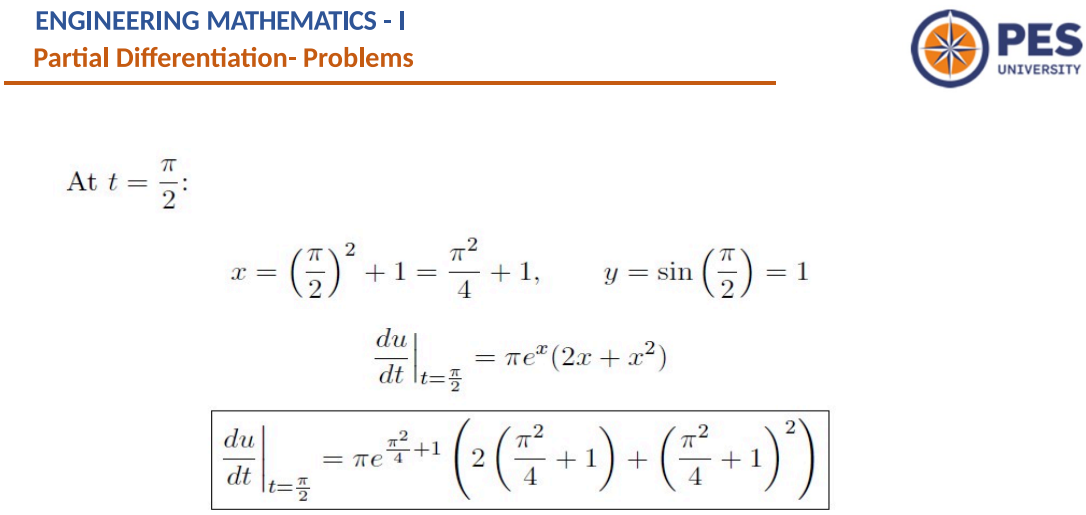

Figure G000048. Total derivative Chain rule problems · source slide page 19.

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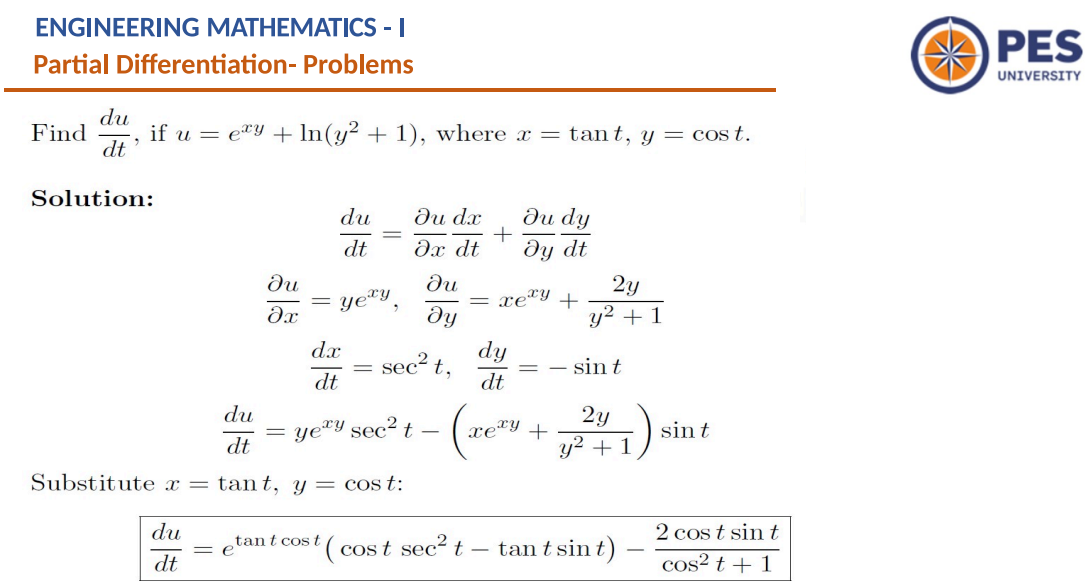

Figure G000049. Total derivative Chain rule problems · source slide page 20.

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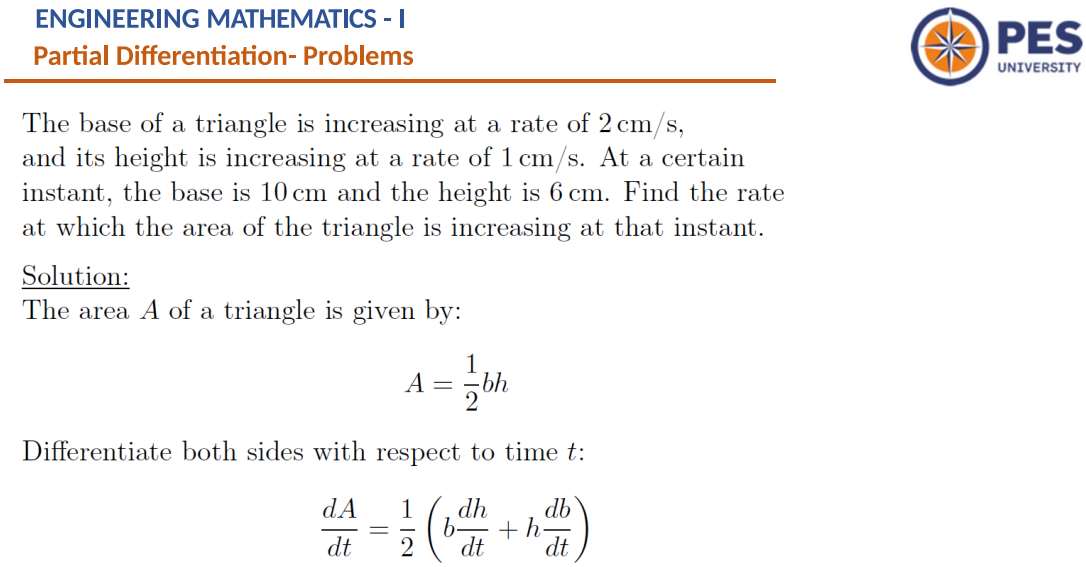

Figure G000050. Total derivative Chain rule problems · source slide page 21.

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

Substituting the known values:

$b = 10\,\text{cm}\ h = 6\,\text{cm}\ \frac{db}{dt} = 2\,\text{cm/s}\ \frac{dh}{dt} = 1\,\text{cm/s}$

$$\frac{dA}{dt} = \frac{1}{2}(10 \cdot 1 + 6 \cdot 2) = \frac{1}{2}(10 + 12) = \frac{22}{2} = 11\,\text{cm}^2/\text{s}$$

Figure G000051. Total derivative Chain rule problems · source slide page 22.

EV0000409

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

Water is being poured into a conical tank at a rate of $9\,\text{cm}^3/\text{s}$. The height and radius of the cone are always equal. Find the rate at which the water level is rising when the height is 6 cm.

Solution:
The volume of a cone is given by:

$$V = \frac{1}{3}\pi r^2 h$$

Since the height and radius are always equal ($r = h$), substitute

$$V = \frac{1}{3}\pi h^2 h = \frac{1}{3}\pi h^3$$

Differentiate both sides with respect to t:

$$\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$$

Figure G000052. Total derivative Chain rule problems · source slide page 23.

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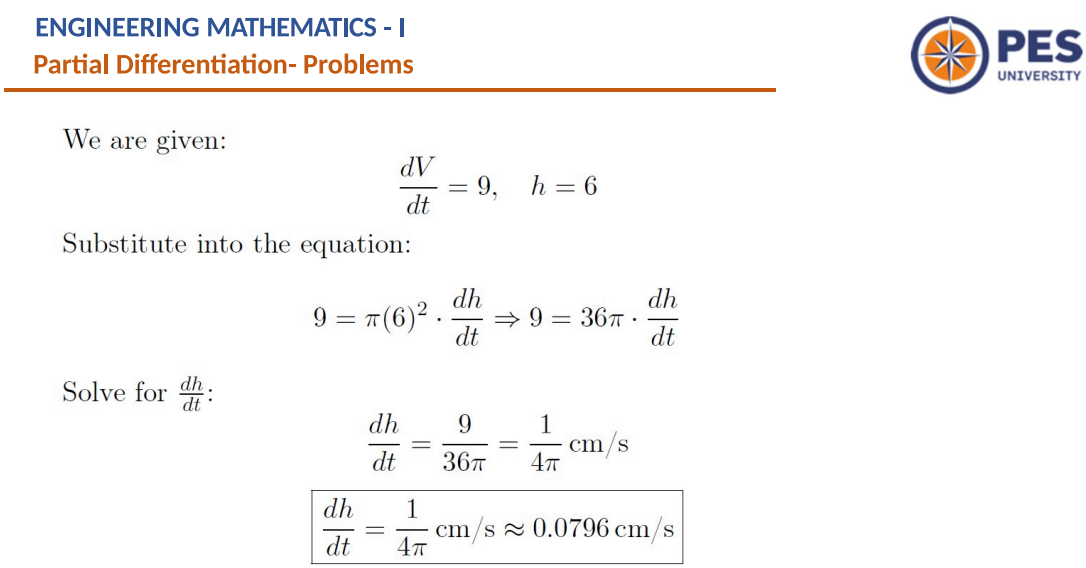

Figure G000053. Total derivative Chain rule problems · source slide page 24.

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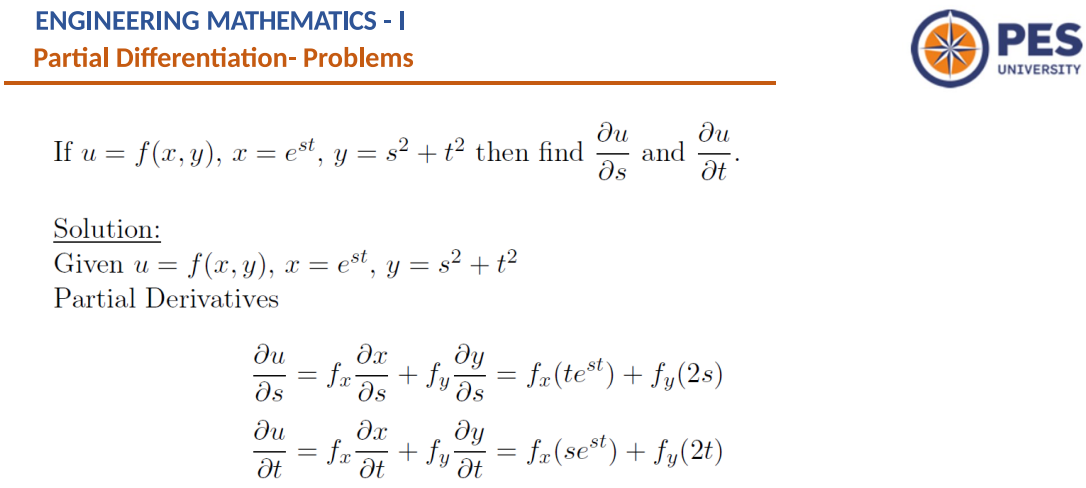

Figure G000054. Total derivative Chain rule problems · source slide page 25.

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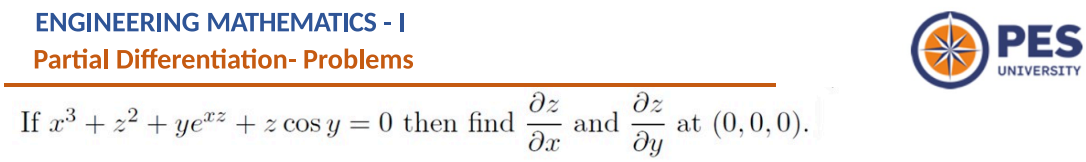

Figure G000055. Total derivative Chain rule problems · source slide page 26.

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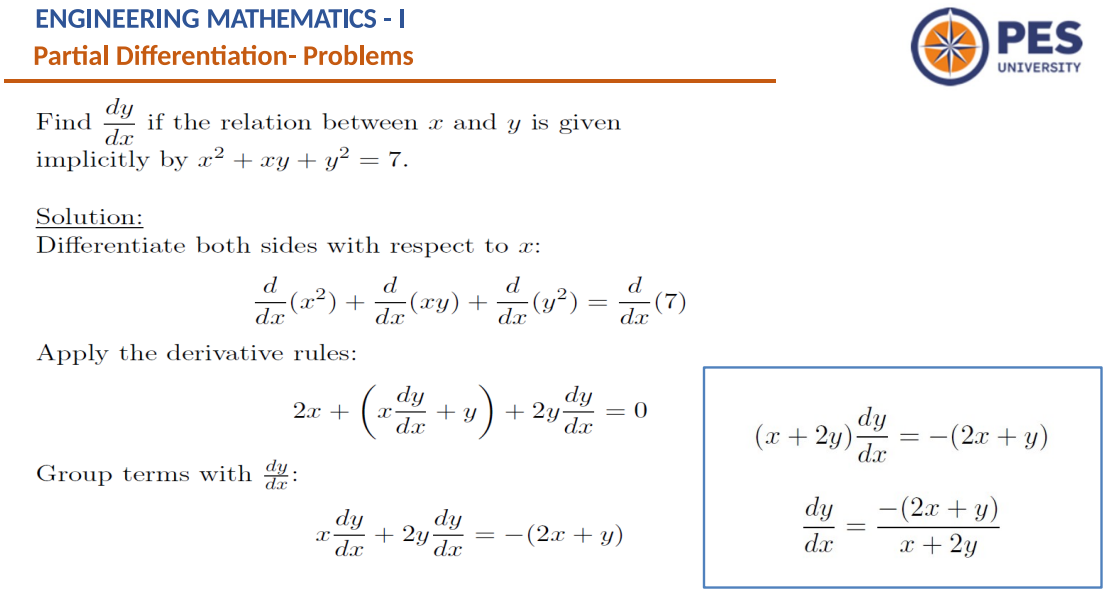

Figure G000056. Total derivative Chain rule problems · source slide page 27.

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Figure G000057. Total derivative Chain rule problems · source slide page 28.

EV0000427

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000345 · EV0000346 · EV0000347
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1: Partial Differentiation Session : 3 Sub Topic: Composite Functions, Total Derivative & Chain Rule EV0000348 · EV0000349 · EV0000350
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I TOPICS CHAIN RULE TOTAL DERIVATIVE RULE COMPOSITE FUNCTIONS EV0000351 · EV0000352 · EV0000353
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Total derivative rule, Chain Rule and Composite functions If $y = e\sin(x2)$ then, we can find the derivative of the function by using chain rule. $dy = d\ esinx2\ .\ d(\sin\ x2)\ .\ d(x2)\ dx\ dy = esinx2\ \cos\ x2\ 2x\ dx$ EV0000354 · EV0000355 · EV0000356
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Total derivative rule Total derivative rule: If $z = f(x, y)$ where x and y are functions of one independent variable ‘t’ then we can find the derivative of z with respect t as [Image text, OCR] dx X=x(t) dt z dx dxdt dx z = f(x, y) dy y = y(t) dy dy dt dt dz 0z dx 0z dy Tree diagram for the case dt 0c dt oy dt EV0000357 · EV0000358 · EV0000359

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Total Derivative Rule Example: $z = x2y3$, $x = t2$, $y = 2t3$ Differentiating z partially with respect to x we get $\partial z = 2xy3 = 2t11\ \partial x$ And the partial derivative of z with respect to y will be $\partial z = 3x2y2 = 9t10\ \partial y$ Also, $dx\ dy = 2t$, $= 6t2\ dt\ dt\ dz$ Substituting, we get $= 60t12\ dt$ EV0000360 · EV0000361 · EV0000362

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total derivatives 1. A point P is moving along the curve of intersection of $x2\ y2$ surface whose cartesian equation is $- = z$ (A 16 9 2 paraboloid) and the surface whose cartesian equation is $x + y2 = 5$ (A Cylinder). If x is increasing at the rate of 0.2cms/sec, how fast is z changing when $x = 2$? Solution: Given $dx = 0.2$ cm\sec $dt\ \partial z$ dx $\partial z\ dy\ dz = + dt\ \partial x$ dt $\partial y\ dt\ dy\ x = - x2 + y2 = 5$, $dx\ y$ EV0000363 · EV0000364 · EV0000365

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total derivatives 2. Given that $f(2, -3) = 6$, $fx\ 2, -3 = -0.6$, approximate $f(2.1, -3.03)$. Solution: = 1.3 and $fy\ 2, -3$ The total differential approximates how much f changes from the point $(2, -3)$ to $(2.1, -3.03)$. With $dx = 0.1$ and $= 0.03$, we have $dz = fx\ 2, -3\ dx + fy\ 2, -3\ dy = 1.3(0.1) + (-0.6)(-0.03) = 0.148$ The change in z is approximately 0.148, so we approximate $f\ 2.1, -3.03 \approx 6.148\ dy$ EV0000366 · EV0000367 · EV0000368

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total Derivatives 3. One side of a triangle is increasing at a rate of 3cm/s and second side is decreasing at a rate of 2cm/s. If the area of a triangle remains constant, at what rate does the angle between the sides change when the first side is 20cms long, π the second side is 30cms and the angle is . Solution: 1 6 A= (base *height) 2 1 = $xysin\theta\ 2 = dA\ \partial A\ dx\ \partial A\ dy\ \partial A\ d\theta + + dt\ \partial x\ dt\ \partial y\ dt\ \partial\theta\ dt$ EV0000369 · EV0000370 · EV0000371

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total Derivatives Contd... $dA\ 1\ dt\ 2\ d\theta = ysin\theta\ 3 + xsin\theta\ -2 + xycos\theta\ dt$ Here $x = 20cm$; $y = 30cm$; $\theta = \pi\ 6 = 0.04831\ d\theta\ -25 = dt\ 300\sqrt{3}\ \theta$ should decrease at the rate of 0.0481 rad/sec EV0000372 · EV0000373 · EV0000374

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total derivatives 4. A cylindrical steel storage tank is to be built that is 10ft tall and 4ft across in diameter. It is known that the steel will expand/contract with temperature changes; is the overall volume of the tank more sensitive to changes in the diameter or in the height of the tank? Solution: A cylindrical solid with height h and radius r has volume $V = \pi r2h$. $\partial V = V\ \partial V = V\ r, h = 2\pi rh$ and $r, h = \pi r2\ r\ h\ \partial r\ \partial h\ dV = (2\pi rh)\ dr + (\pi r2)dh$. When $h = 10$ and $r = 2$, we have $dV = 40\pi dr + 4\pi dh$. EV0000375 · EV0000376 · EV0000377

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Problems on Total derivatives Contd.... Note that the coefficient of dr is $40\pi \approx 125.7$; the coefficient of dh is a tenth of that, approximately 12.57. A small change in radius will be multiplied by 125.7, whereas a small change in height will be multiplied by 12.57. Thus the volume of the tank is more sensitive to changes in radius than in height. EV0000378 · EV0000379 · EV0000380

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Chain Rule-Composite Functions We can define chain rule for a function of two or more independent variable: If $z = f(x, y)$ is a bivariate function and x, y are functions of two or more independent variables say u, v then we can find the partial derivative of z with respect to u and v, given by $\partial z = \partial z\ \partial x + \partial z\ \partial y$ and $\partial y\ \partial u\ \partial z\ \partial y\ \partial u\ \partial x\ \partial u\ \partial z = \partial z\ \partial x + \partial v\ \partial x\ \partial v\ \partial y\ \partial v$ EV0000381 · EV0000382 · EV0000383

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Tree diagram for Chain Rule [Image text, OCR] ax du u dxau x=g(u,v) ax ax dv dxdv z=f(x, y) y au u oyau ay y=h (u, v) ay az ay av ay dv 02 dy dy Tree diagram for + and dy + du du du dy EV0000384 · EV0000385 · EV0000386

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Chain rule - Problems 1. If $v = f\ 2x - 3y, 3y - 4z, 4z - 2x$, prove that $6vx + 4vy + 3vz$=0. Solution: $\partial v\ \partial v\ \partial r\ \partial v\ \partial s\ \partial v\ \partial t = + + \partial x\ \partial r\ \partial x\ \partial s\ \partial x\ \partial t\ \partial x\ Let\ r = 2x - 3y, s = 3y - 4z\ and\ t = 4z - 2x\ rx = 2, sx = 0, tx = -2$ Similarly $ry = -3, sy = 3, ty = 0$. Also $rz = 0, sz = -4, tz = 4\ \partial v = 2\ \partial v - 2\ \partial v\ \partial x\ \partial r\ \partial t$ EV0000387 · EV0000388 · EV0000389

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Partial Derivatives of composite functions $\partial v\ \partial v\ \partial r\ \partial v\ \partial s\ \partial v\ \partial t = + + \partial y\ \partial r\ \partial y\ \partial s\ \partial y\ \partial t\ \partial y = -3\ \partial v + \partial v\ \partial r\ \partial s\ \partial v\ \partial v\ \partial r\ \partial v\ \partial s\ \partial v\ \partial t = + + \partial z\ \partial r\ \partial z\ \partial s\ \partial z\ \partial t\ \partial z = -4\ \partial v + 4\ \partial v\ \partial s\ \partial t$ Therefore $6vx + 4vy + 3vz = 0$. EV0000390 · EV0000391 · EV0000392

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Partial Derivatives of composite functions ∂z 2. Find ∂z and where $z = f\ r, \theta$ and $r = ts - t2, \theta = s2 + t2\ \partial s\ \partial t$ Solution: $\partial z\ \partial z\ \partial r\ \partial z\ \partial \theta = + \partial s\ \partial r\ \partial s\ \partial \theta s \partial s\ rs = t, \theta s = s2 + t2\ \partial z\ s = fr\ r, \theta\ t + f\theta\ r, \theta\ \partial s\ s2 + t2$ Similarly $t\ r = s - 2t, \theta = t\ t\ \partial z\ s2+t2\ t = fr\ r, \theta\ (s - 2t) + f\theta\ r, \theta\ \partial t\ s2 + t2$ EV0000393 · EV0000394 · EV0000395

18. Source slide page 18 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du Find at t = dt 2 Solution: The total derivative is given by du du dx . du dy dt dx dt dy dt The partial derivatives are du du = yexy(2x + x²y), = x²exy(1 + xy) dx dy and dx dy = 2t, = cost dt dt EV0000396 · EV0000397 · EV0000398

19. Source slide page 19 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] T At t = 2 72 +1= +1 y = sin 2 4 du = πe"(2x +x² dt\t= du π 72 72 =πe4 2 +1 +1 dt It= 4 EV0000399 · EV0000400 · EV0000401

20. Source slide page 20 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du Find if u = ey + ln(y² + 1), where = tant, y = cost. dt Solution: PES du dx Ou dy du UNIVERSITY dt Oc dt Oy dt du du 2y = cery + = yery. 0c dy y2 +1 dc dy - sin t dt dt du 2y = yery sec² t — sin t Cedy dt y2 + 1 Substitute = tant, y = cos t: du etantcost(cost sec² t - tan t sint) 2 cos t sin t ☰e dt cOs2 t + 1 EV0000402 · EV0000403 · EV0000404

21. Source slide page 21 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] The base of a triangle is increasing at a rate of 2 cm/s, and its height is increasing at a rate of 1 cm/s. At a certain instant, the base is 10 cm and the height is 6 cm. Find the rate at which the area of the triangle is increasing at that instant. Solution: The area A of a triangle is given by: 1 A = bh 2 Differentiate both sides with respect to time t: dA dh db +h dt 2 dt dt EV0000405 · EV0000406 · EV0000407

22. Source slide page 22 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Substituting the known values: b = 10 cm h = 6 cm = 2cm/s dh = 1cm/ dt dA 22 (10·1+6·2) (10 + 12) = 11 cm 2 — dt EV0000408 · EV0000409 · EV0000410

23. Source slide page 23 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Water is being poured into a conical tank at a rate of 9 cm? /s. The height and radius of the cone are always equal. Find the rate at which the water level is rising when the height is 6 cm. Solution: The volume of a cone is given by: 1 V= Since the height and radius are always equal (r = h), substitute 1 1 V =÷πh²h = πh3 3 3 Differentiate both sides with respect to t: dV dh dt dt EV0000411 · EV0000412 · EV0000413

24. Source slide page 24 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] We are given: dV = 9, h = 6 dt Substitute into the equation: dh dh 9 = π(6)2 → 9 = 36π· dt dt Solve for h: dt dh 9 cm/s dt 36π 4π dh 1 cm/s 0.0796 cm/s dt 4π EV0000414 · EV0000415 · EV0000416

25. Source slide page 25 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du Ou If u = f(x,y), = est, y = s² + t² then find and 0s at Solution: Given u = f(c,y), = est, y = s² + t2 Partial Derivatives Ou 0c Oy = fc = fx(test) + fy(2s) 0s 0s du 0c ay +fyat = fx = fx(sest) + fy(2t) at ϑt EV0000417 · EV0000418 · EV0000419

26. Source slide page 26 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] ϑ2 0z If x² + z2 + ye + z cos y = 0 then find and at (0, 0, 0). dy Solution: Let F(x, y, z) = x² + x2 + ye"z + z cos y. Then, Fx = 3x² + zyeαz, Fz = 2z + xyerz + cos y. ϑz Fc 3x2 + zyerz F 2z + cyez + cos y EV0000420 · EV0000421 · EV0000422

27. Source slide page 27 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] dy Find if the relation between c and y is given dc implicitly by 2 + cy + y² = 7. Solution: Differentiate both sides with respect to : d d d d -(α²) + - (xy) +(y²) (7) dx dc dc Apply the derivative rules: dy 2c + +y +2y =0 dc dx Group terms with dy dy +2y- = -(2c +y) C dc dx [Image text, OCR] dy (c+ 2y) = -(2c + y) dc dy -(2c +y) dc c+ 2y EV0000423 · EV0000424 · EV0000425

28. Source slide page 28 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000426 · EV0000427 · EV0000428

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.852] In this session, we will solve problems on total derivative, chain rule, and implicit function. [00:00:17.156] Let's begin with total derivatives. [00:00:23.972] Point du by dt at a point t equals to pi by 2 where u equals x square y times a power x y and x equals t square plus 1 and y equals sine t. [00:00:43.588] From the data, we can see that u is a function of x and y, where x and y are the functions of p. [00:00:52.612] So we can find the ordinary derivative of u with respect to t, which is called the total derivative of u with respect to t and given by the formula du by dt equals dou by doux times dx by dt plus [00:01:10.788] dou by dou y times d y by dt. Now for the given u let us compute dou by dou x and dou by dou y so u by dou x is given by y times e power x y [00:01:30.148] of 2x plus x square y. Similarly, dou by dou y equals x square times e power x y of one plus x y. Also [00:01:45.092] For the given x and y, d x by dt as well as dy by dt becomes 2t as well as cos t

respectively. Now because we want du by dt at the point t equals to pi by 2, we need to find [00:02:03.044] X and Y at the point. So [00:02:10.948] At t equals to pi by 2, x becomes pi square by 4 plus 1 and y becomes 1 because of sine t. [00:02:22.468] Also, du by dt becomes because y is 1, it becomes only e power x times 2x plus x square. [00:02:34.212] We also see that when t is pi by 2, the second term doesn't exist because d by by dt is cost t and the cost t is 0 at t equals to pi by 2. So finally at t equals to pi by 2, we see that d u by dt becomes pi times e power. [00:02:53.572] Pi square by four plus one of two times pi square by four plus one plus pi square by four plus one multiply. So this is the answer for the problem. This is the value of d u by dt at the point d equals to pi by two. [00:03:16.132] Let us look at a similar problem. Find du by dt. If u is given by e power x y plus log of y square plus one, where x equals t and y equals cos t. We use a same approach what we we have used for the previous problem. [00:03:36.452] So this is the formula for the total derivative of u with respect to d. Now for the given u, dou u by dou x becomes y times e power x y. Plus two y divided by y square plus one. [00:03:55.844] Also, for given x and y, d x by dt becomes second square d and dy by dt becomes minus sine d. So when you subject all this back to one, so t u by dt becomes y e power x y. [00:04:13.764] times second square t minus x e power x y plus two y divided by y square plus one into sin t. [00:04:24.324] Now because u is a function of t, we express x and y in terms of t. So when we replace x and y by tan t and cos respectively, the final answer for du by dt is going to be this much. [00:04:50.052] Let us solve an application problem on total derivative. [00:04:57.860] The base of a triangle is increasing at a rate of 2 centimeter per second, and its height is increasing at a rate of 1 centimeter per second. At a certain instant, the base is 10 centimeters, the height is 6 centimeters.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.404] Find the rate at which the area of a triangle is increasing at that instant. So for this problem we need to compute the rate at which the area of the triangle is increasing. So [00:05:35.972] the area of a triangle is given by this formula so we differentiate with respect to t using product rule so d a by dt becomes half d times d h by dt plus h into db by dt [00:05:52.836] In the problem, it is given that the base of a triangle is increasing at a rate of 2 centimeters per second. So, so dB by dt becomes 2. [00:06:03.364] And the height of a triangle is increasing at a rate of 1 centimeter per second. So dh by dt is 1. Also, p and h are respectively 10 and 6. [00:06:15.652] So when we subject all this into this expression, we see that dA by dt equals 11 square centimeters per second. It means the rate at which the area of the triangle is increasing is 11 square centimeters per second. [00:06:41.188] Let us look at one more application problem on total derivative. Water is being poured into a conical tank at a rate of 9 cubic centimeter per second. It is given that the height and radius of the cone are equal. [00:06:58.852] Find the rate at which the water level is rising when the height is 6 centimeter. So for this problem, we need to compute dh by dt from volume of a cone. So let us start with volume of a cone, which is given by 1 by 3 times. [00:07:17.796] pi r square h. Because height and radius are equal when we replace r equals h, v becomes 1 by 3 pi h power 3. Let us differentiate with respect to d [00:07:34.820] So we see that 1 by 3 pi is a constant and h power 3 becomes 3 h square into d h by dt. After calcation 3 and 3, we get dv by dt to be pi h square dh by dt. Now let us use all the values and compute d h by dt. It is given that [00:07:54.884] P V by D T is 9 and H is 6. So when you subpute these two values in this expression, we see that d H by T T becomes 0.0796 centimeter per second. [00:08:11.556] So these are two application problems on total derivative. [00:08:18.692] Now let us move on to a chain rule. It is given that u equals f of x y.

It means u is a function of x and y, where x and y are the functions of s and t. [00:08:32.228] then u becomes composite function of s and t. So we can compute dou u by dou s as well as dou u by dou t. Now this is a chain rule for partial derivativity what we know from theory. Now for the given x and y [00:08:48.452] Do x by dou s is going to be t times e power s t. dou x by dou t becomes s times e power s t. becomes two s dou y by dou t becomes two t. [00:09:05.988] So these are the values of do u buy do s as well as do u buy do t for the given data. [00:09:15.684] Let us solve a problem on implicit function with three variables. It is given that if x power 3 plus z square plus y times e power x z plus z cos by 0. [00:09:29.764] Then find a value of douz by doux as well as douz by dou y at the point zero zero zero. We know from theory the formula for dou z by doux is minus of f x divided by f z. [00:09:45.508] And similarly, dou z by dou y equals minus f y divided by f z. It means we take this as some function of xy and z, then we find only partial derivative of f with respect to x y.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:04.388] as well as z. Then using the formula we can compute douz by doux and dou z by dou y at the point zero zero zero. So for this problem we see that dou z by doux is zero and dou z by dou y is minus one [00:10:23.460] Let us solve another problem on implicit function with two variables. So phi dy by dx. If the relation between x and y is given by [00:10:33.828] x square plus x pi plus y square equals 7. Of course there are two ways to solve this problem. Another method what we can use uh we can use dy by dx equals [00:10:45.984] Minus of partial derivative of f with respect to x divided by f with respect to y. It means we can take this as some function of x and y, then use the partial derivative and use this formula, we can get the answer. So another method which is called direct method, which is shown as follows. So for the given [00:11:05.892] Data x square plus xy plus y square equal to 7. Differentiate both the sets with respect to x. So we get uh 2x plus x into dy by dx plus y plus 2y time dy by dx is 0. [00:11:21.604] Now from this particular uh data we solve for dy by dx. So we see that the answer for this problem is minus of two x plus y divided by x plus two y. [00:11:38.052] Thank you.

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Full transcript: 1545 normalized words, 45 timestamped segments, 707.989 seconds duration. Low-confidence segments flagged: 0.

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6 Total derivative, Chain rule and Implicit functions

This lesson is indexed from the source video titled Total derivative, Chain rule and Implicit functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000066 · EV0000067

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.284] Hello, everyone. Let's welcome you all to another session of partial differentiation. Today we'll be learning chain rule, composite functions, and total derivative of implicit function. It's like an extension of what you already did in the case of a function of single variable. [00:00:21.572] For example, suppose uh I asked you to differentiate this function, which is y is equal to e to the power of sine of x square. So how would you do the derivative of that function? [00:00:33.252] There are three functions which you need to take it. The first one is exponential. So derivative of exponential would be e to the power of sine of x square. Next you have to differentiate sine, which will be cos of x square. Next you differentiate x square, which will be 2x. [00:00:51.108] So the final answer for this would be e to the power of sine of x square into cos of x square into 2x. [00:00:59.108] This was just an example of a function which is a composition of two more functions because it's of single variable, right? So you had f composition g composition h of x. So that means you had f of g of you had f of g of h of x. [00:01:19.748] And when you differentiate it, you followed the order of differentiating f first, then g, and then in the end with respect to h. So we can do a similar thing for a function of multivariable. [00:01:35.876] For example, suppose I have a function z, which is a function of two independent variables x and y. Okay, I have a function here. [00:01:45.828] Now suppose I want to differentiate, as you all know, I if I say differentiate z, the first question that you would ask is with respect to what we should differentiate, because there are two independent variables. Now that I say along with uh z, suppose I'll take z as say x square y q. [00:02:05.860] Okay. So it's a function of two independent variables. And then if x and y again are functions of one some independent variable, in this case it could be t. So I'll take x is equal to t and y is equal to 2t squared. [00:02:23.908] So basically what happens if I know that z is a function of two independent variables and in turn x and y are functions of some other variable t, then I need to find the total derivative of z with respect to t. [00:02:39.876] Right. So one way to do that is you can write z as substitute x is equal to t, so you will get t square. If you substitute y as 2t square, you will get 2t square, the whole q. So this eventually will become 8 t to the power of 8. [00:02:57.284] So now I reduce z to be a function of single variable t. So you can differentiate this z with respect to t now like you did the function of single variable. So dz by dt would be 64 t to the power of 7. [00:03:13.348] But always it might not be that easier for me to substitute the independent variables and reduce it to a single variable and differentiate. So what we can do is we have another alternate method. So let me erase this. [00:03:26.784] So we can define it's as good as so what I did now is I had let me go back to that same example. I had z is equal to x square y cube. I took x is equal to t and then I took y is equal to some 2t square. Right. So now [00:03:45.508] What happens if I differentiate x with respect to t, I can get the total derivative because it's a function of single variable. So I'll get dx by dt. So dx by dt is 1 and then dy by dt will be 4 t squared. [00:04:03.620] I already know the partial derivative of z with respect to x and y. So I can write dz by dt as dou z by doux, which is 2x and keeping y cube as it is into dx by dt, which is 1. [00:04:20.548] Plus dou z by dou y, which will be x square, I am keeping it as it is. You differentiate y cube with respect to y. So I will get 3y square into dy by dt, which is 4 t square. [00:04:35.556] Now we'll substitute for x and y and then we'll see if we get the same answer like the previous case. So I can write dz by dt as I will get 2 and then value of x is t and then y cube will be again 8. [00:04:52.164] T to the power of six. I think I took t square if I'm not wrong. So let me make x as t square. So this will be 2t for me. Okay. Oh, anyway. So plus

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:06.084] You can substitute x square again, so this will be t power 4, 3 into 2 t square, the whole square into 4 t square. And we can simplify this. So this is an alternate method of [00:05:20.228] Finding the total derivative of a function of two or more independent variables when each of the independent variables is a function of another variable. Okay? So single variable, for example. So that means whenever z is a function of xy. [00:05:40.740] And x and y are functions of some parameter say t. So here y could be a function of t and uh x is also another function of t. In that case, I can write [00:05:54.128] I can actually find the total derivative of z with respect to a single variable t. I can do that by finding the partial derivative of z with respect to x into the total derivative of x with respect to whatever the variable you have t. [00:06:10.852] Plus, I can also do z y dy by d. So just keep keep in mind this formula. Okay. You can apply this formula when uh the independent variables are functions of a single variable. [00:06:27.524] Now let's move on. Suppose I have a I have a problem now, okay? A point P is moving along the curve of intersection whose Cartesian equation is given. So a paraboloid. A paraboloid you can imagine like a cup, okay? So it's like a three-dimensional. So it's like a like this with the [00:06:46.852] With the base, of course. So it goes like that. Something like that. Okay, you can imagine like a cup, a cut. So a paraboloid and the surface whose Cartesian equation that means I have a cylinder and a paraboloid. So a point P is moving on the surface of a cylinder and a paraboloid. [00:07:06.916] They have given that if x incre is increasing at the rate of 0.2 centimeters per second, how fast is z changing when x is equal to 2? Okay. So how do we calculate the solution for this particular problem? Let's write down first what all is given to me. [00:07:25.636] Okay. So I have uh x square by 16 minus y square by 9 equal to z, which is a paraboloid. And then I have a surface whose Cartesian form is cylinder. They have given me dx by dt. So let's write down what all is given. [00:07:41.380] dx by dt is given to me as 0.2 centimeters per second. [00:07:49.412] I need to calculate how fast Z is changing when X is equal to two. So the only constraint I have is [00:07:59.748] Um I don't know what is dy by dt is, so I'm not worried about that. I need to calculate what is the value of z. Do I know z already? Yes, z is x square by 16 minus y square by 9. So what is the partial derivative of z with respect to x? [00:08:19.300] with respect to x will be x by eight. Partial derivative of z with respect to y will be minus 2y by 9. So I got these two partial derivatives. And they have already given me dx by dt. [00:08:35.620] So the question asks you to find how fast the change in Z is happening along the direction of T. So maybe I need to find what is dz by dt. Right? [00:08:49.796] So uh can you find the only okay what is the formula for dz by dt now? You know that this is dou z by dou x as z is a function of xy into t x by dt plus. [00:09:03.972] Do z by dou y into t y by d. This is what I have. Okay. [00:09:11.876] Now, in this what all we know I need this at x is equal to 2. So that's fine. We can substitute x is equal to 2, that's not a problem. I can find this, I already know this, I know this, I know this, but I don't know this. So how do I find dy by dt? [00:09:29.860] For that I have another example, another third, right? So let me erase that first. So I have an equation of the cylinder x square plus y square equal to zero or five. [00:09:41.572] You know that this is an implicit function, right? So you can differentiate this. If y is a function of x, so I can definitely differentiate this, I'll get 2x plus 2y into dy by dx actually equal to 0.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:00.676] Can I do this? Now, what do I get from this? I'll get dy by dx equal to minus x by y. [00:10:10.852] But I don't want dy by dx. What I want is dy by dt. Now again you can write dy by dx as dy by dt into dt by dx.

These things are known to you when you did the first order partial derivatives. Minus x by y. [00:10:29.508] You can also find what is the value of y when x is equal to 2. So, so that you can substitute. But first, let's find out what is dy by dt is. So, dy by dt will be minus x by y into dx by dt. [00:10:44.832] So I'm just moving this term to the right hand side in this particular equation. [00:10:51.268] Okay. So I have an expression for dy by dt. So let me erase the other things and bring back the uh things that we need now. Okay. So um uh all that is required I have now. So I need to study the change that is happening in said. [00:11:08.292] when x is equal to 2. Okay? So let's see. Suppose now uh what is required is dz by dt, correct? And what is the formula for dz by dt now? You have zx into dx by dt. [00:11:26.916] plus ZY into TY by D. [00:11:32.644] Now you know everything that you want. So you know zx. Zx we got it as x by 8 into dx by dt. They have already given it as 0.2. Z y also we know which is minus 2y by 9. [00:11:47.940] dy by dt, I just now got that to be minus x by y into dx by dt which is again 0.2. Now the one good thing is I got this y cancel so I don't need to worry about calculating the value of y. Just put x is equal to 2, you will get the answer. [00:12:06.564] Okay. So if dz by d t is positive you can say that z is increasing when x is equal to 2. If dz by dt is negative you can say that z is decreasing at say x is equal to 2. [00:12:18.884] So if I need to calculate dz by dt, let's finish this. At x is equal to 2, I'll get 0.2 divided by 4, and then plus or minus or plus. This will become plus and x is again 2, I have to substitute. So this is 4 into 0.2 divided by 9. [00:12:37.956] So you can uh definitely simplify that and uh for sure this is greater than zero. All I'm interested in to know the behavior of the function. So I just want to know at x is equal to 2 what is happening to z. So that will give me where exactly my point is. [00:12:55.392] Okay. So I hope you understood the when you have a function of uh two independent variables and each one of them depends on another variable, how do you study the change, right? [00:13:09.348] Now another application of this uh would be to find the value of a function. [00:13:16.484] at any given point, because I don't know the function at all. So all I know is f of 2 comma minus 3 is 6, and I know the partial derivative of f with respect to x, and I know partial derivative with respect to y. [00:13:29.920] They want you to calculate the value of the function at a different point. So here the point that they have given is 2.1 and minus 3.03. [00:13:40.228] So, if you observe the given point and the point at which we need the value of the function, you see that x has incremented to 0.1. So that means I x is there is a small change in x which is 0.1 and a small change in y which is 0 point. [00:13:56.944] 03. Again, you just apply the definition of dz. So dz will be zx into dx. If I don't want to restrict myself to any particular variable, I can write the equation like this also. [00:14:11.140] So it's like a total derivative. So dz will be zx dx plus z y d by. And you substitute the values that are there and you can find that approximately it is equal to 6.418. 6.148. Okay that's another application of this uh derivative. [00:14:31.204] I have another problem. Suppose one side of a triangle is increasing at the rate of 3 centimeters per second, and the second side is decreasing at the rate of 2 centimeters per second. If the area of a triangle remains constant, [00:14:44.720] So one side is increasing, another one is decreasing, but area remains constant obviously. So uh what is at what rate does the angle between the side should change when the first side is 20 centimeter and the second is 30 centimeter.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:00.676] Okay. So they have given suppose I have a triangle. So you understood the problem? [00:15:07.396] They have given me this side and this side. So one of them will increase, another one will decrease. I want to know at what rate this theta should change, whether theta should increase or theta should decrease, so that the

area of the triangle remains constant. [00:15:25.252] So I'll make a small construction for you to understand why I'm using that. Suppose I'll draw a perpendicular because I need the height. And you know the area is half into base into height. Let me uh take the base as uh [00:15:38.244] x and suppose this side is y, then you know that sine theta, right? Uh sine theta will be opposite side by hypotenius. So if I write that sine theta [00:15:51.588] will be opposite side by hypotenous which is h by y or this height will be y sin theta. That's how I got this expression half x y sin theta. So I have area the formula for area I have is area is equal to half into x y into sin theta. [00:16:10.340] So basically this area is a function of three independent variables x, y, and th. [00:16:16.964] Now, how do you calculate the rate? So, because there are three independent variables, you calculate the partial derivative with respect to x, dx by dt, with respect to y, dy by dt, with respect to theta, d theta by t t. Okay. [00:16:32.260] And uh what is required is this, they say that area remains constant. So what is the meaning of that uh area remains constant means dA by dt is 0. [00:16:43.300] I don't want any change to happen to this area. So there if the area of the triangle remains constant, so the meaning of this is d a by dt will be 0. [00:16:55.460] So if dA by dt is 0, use this equation to get d theta by t t, which will be 0.04831, which is not uh too much of a change, but definitely angle is changing. Okay. So I hope you appreciate this example so that things will be that much clearer to you. [00:17:14.116] Again, I have another wherein you have to conclude certain uh assertions. A cylindrical steel storage tank is to be built that is 10 feet tall, and so uh let's let's imagine a picture. So you have to build a cylinder, then what all you need to consider is uh how high you want. So height of the cylinder is say 10 feet. [00:17:35.428] And four feet across the diameter. So diameter is four feet. [00:17:40.068] It is known that the steel will expand or contract with temperature changes, right? So maybe the cylinder is heated or you know or is cooled or whatever it is, whatever the purpose of the cylinder is the overall volume of the tank more sensitive to the changes in the diameter. [00:17:59.652] or in the height of the tank. So that's the question. You need to tell me whether the change in the diameter or change in height will affect the volume. Okay? [00:18:10.084] This is also simple. So you know already the volume of a cylinder is V is equal to pi r square h, and then you know all the partial derivatives, and then you calculate what is d v at the given h and r. And we conclude that dv is equal to 40 pi dr plus 4 pi dh. [00:18:30.372] If you observe carefully, so I got an expression that the volume is changing as 40 pi dr plus 4 pi dh. [00:18:45.828] So if I take out a 4 pi d common from these two terms, what you can observe is this is 10 r plus h. Because the coefficient of r is 10, definitely the radius would contribute more towards the change in the volume than the height. [00:19:05.284] Okay. So sometimes you don't need to solve the problem completely. Just by doing a little bit of differentiation, you can conclude what is the change that is happening, what is contributing to the maximum change. Okay. Now let's move on. [00:19:22.788] So so far what we did, we had z as a function of x and y. So let me let me just draw a tree diagram so that things will be that much clearer. Okay. So I had z as a function of two independent variables, say x and y. [00:19:42.500] And then each of them were function of t. Correct. This is what I had in the previous case. So z was a function of two independent variables, and each of them was function of t. So I could find total derivative of z with respect to t. This is what I did. But now

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:02.916] Let's change little bit. So again I have a bivariate function Z. Z is a function of two independent variables, say x as well as y. But both x and y are functions of again two independent variables u and v. [00:20:21.092] So x is a

function of u and y is also a function of u and v. Now ultimately if I substitute x in terms of uv and y in terms of uv, I'll get z as a function of two independent variables u and v. [00:20:40.644] That means what I have to differentiate Z partially. Now it's not possible for me to differentiate Z totally. So either I can find Z u or I can find Z V. [00:20:54.436] But Z is a function of XY. So how am I going to find Z u now? Z u will be Zx into XU plus [00:21:07.172] Z y or ZY into Y U. So both with respect to U. I need to calculate this Z U, so I'll calculate Z with respect to X and then with respect to Y and then I'll go like this. [00:21:25.220] So I covered all u that is there in the equation. Now if I want Z V, so you can you know that it's already Zx into XV plus ZY into YV. [00:21:40.228] So you don't have to restrict yourself only to a function of two variables, right? It can be a function of n variables. So if z is a function of say again two independent variables x and y and each one of them is a function of three independent variables. [00:21:57.348] Then you can find the partial derivative of z with respect to all these three independent variables. But you don't need to restrict to only two independent variables, you can move on and find out what is it for a multivariable function as well. Let's take an example now. [00:22:14.372] Okay. They have given V to be a function of this. So V is a function of 2x minus 3y, 3y minus 4z, and 4z minus 2x. Then I need to prove that 6vx plus 4v plus 3vz is 0. [00:22:33.444] So, to make things little simpler, we call this 2x minus 3y to be r, 3y minus 4z to be some s, and 4z minus 2x to be some t. [00:22:46.308] So basically what I have is V is a function of three independent variables r, s, as well as t. [00:22:56.932] And each of them R is a function of XY, S is a function of X and Y, and T is also a function of X and Y. [00:23:10.628] That means I can calculate what is Vx. What will be Vx? Vx would be VR Rx V S S X V T T X. [00:23:23.780] And it's easy for you to calculate them. So I have taken S as 3y minus 4z. So can you tell me what is Sx and S y? S X is just minus 4 and S y is just 3. [00:23:38.084] Similarly, I have R as 2x minus 3y. So Rx will be 2 and then Ry will be minus 3. And T we have taken it as 4z minus 2x. So tz will be 4 and then sorry t [00:23:57.412] Um okay, so this is basically this is a function of okay, let me just change a little bit here. Let me use an eraser. So we uh r is a function of xy, but this is a function of y and z. Okay, so that's okay. [00:24:18.340] So S is a function of Yz and T is a function of phz. [00:24:26.084] Let's just move a little bit. Okay. Uh this is ZX. Okay. So we want Vx. How do we get Vx? Vx is VR R X. Then again V S, there is no X there, but you have to write. And then V T T X. So [00:24:44.260] the middle term will become 0. So this term here is 0 because I don't have s as a function of x. So you can calculate uh all of them and it's easy to check that if you substitute and uh if you simplify you will get the answer to be zero. Okay.

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Transcript block 6 (source interval 001500-001799 seconds): [00:25:02.276] I uh I have another example. So you need to find the z is a function of r and theta. So z is a function of r and theta. It's always better if you write the tree diagram every time you have a function of more than one independent variable, so that you won't do any mistake in calculating the partial derivative. [00:25:21.092] R is a function of T and S and theta is also a function of T and S. So how do you calculate ZS? I want Zs. So Z will be ZR R S plus Z theta theta S. [00:25:40.228] So what is ZR now? I have to keep it as it is because I don't know what the function is. So this is ZR, but I know what is R S is. See if here what will be Rs, Rs will be just T and R T will be S minus 2T. So Rs is T. [00:25:58.820] And uh z theta, I don't know, I have to keep it as it is, as I don't know what the function is. But theta s of course I know. So theta is uh given as root of s square plus t square. So theta s is one pi. [00:26:14.992] 2 root of s square plus t square into 2 s. I can get rid of this. So I

know what is theta s is okay which is s by root of s square plus t square. [00:26:28.772] Similarly, you can find the other uh variable. But the problem with this problem is you don't know what is this function. So I cannot substitute for zr and z theta. I need to retain them as it is in the solution as well. Okay. Alright. So

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Full transcript: 4234 normalized words, 104 timestamped segments, 1609.301333 seconds duration. Low-confidence segments flagged: 0.

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7 Implicit functions and problems

This lesson is indexed from the source video titled Implicit functions and problems. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000090 · EV0000091

Graphs and diagrams

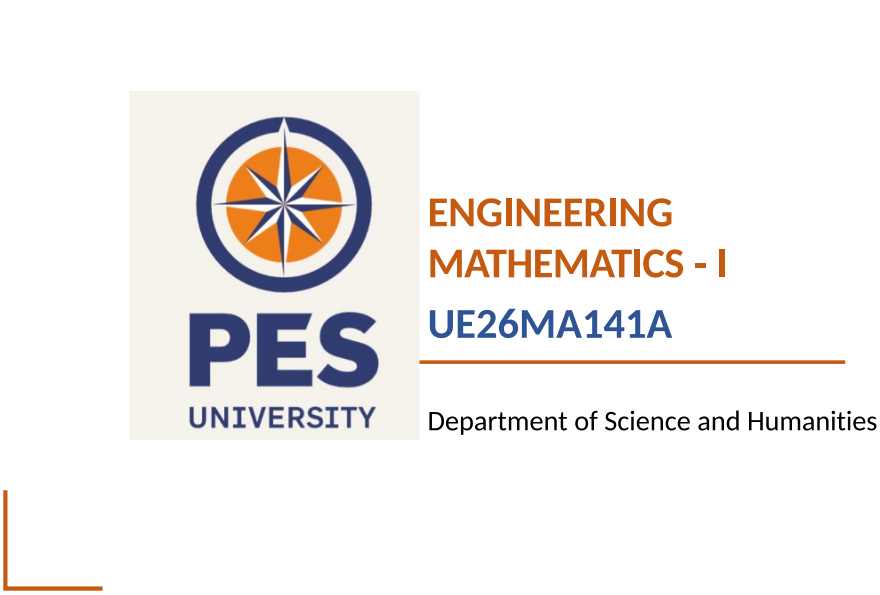

Figure G000058. Implicit functions Problems on Implicit and Composite functions · source slide page 1.

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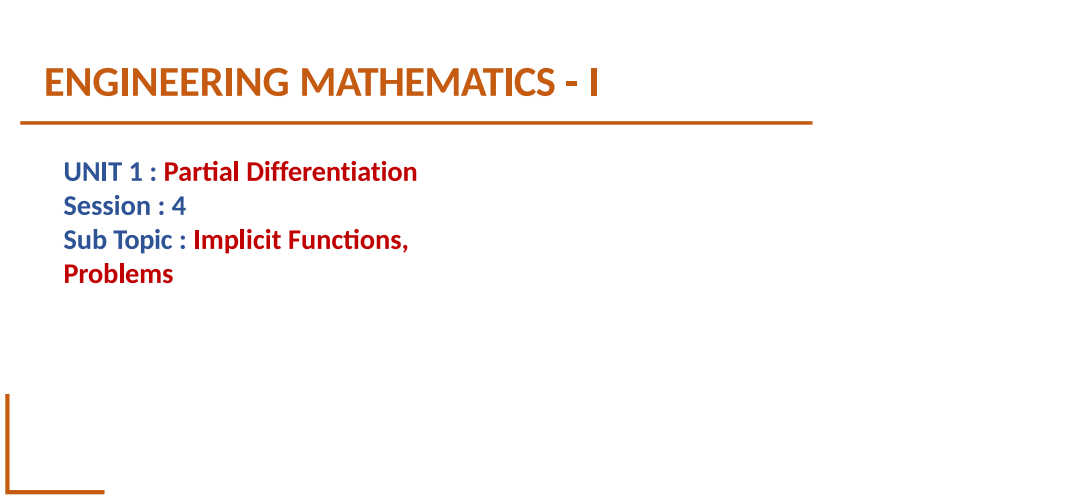

Figure G000059. Implicit functions Problems on Implicit and Composite functions · source slide page 2.

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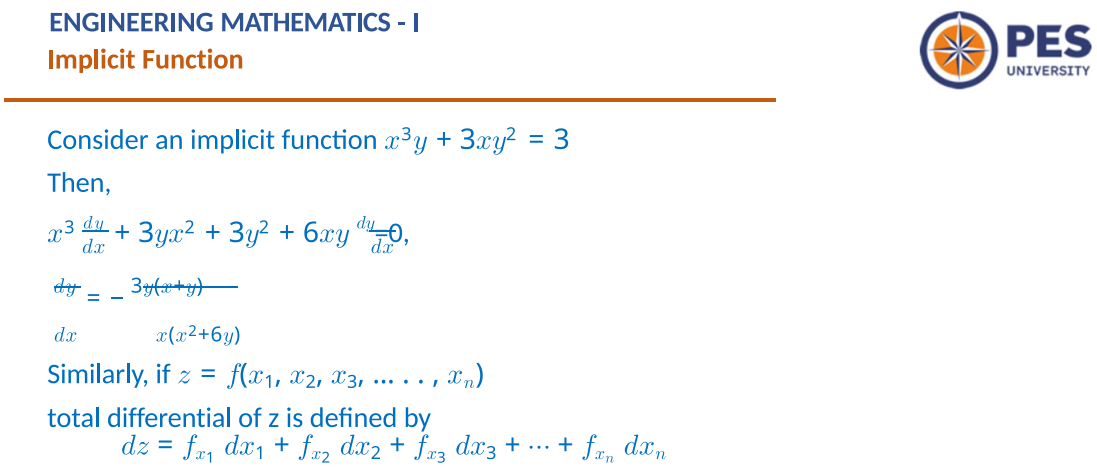

Figure G000060. Implicit functions Problems on Implicit and Composite functions · source slide page 3.

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ENGINEERING MATHEMATICS - I

Implicit Function

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If $z = f(x, y)$ is a function of two independent variables, then

$$dz = f_x(x, y)\,dx + f_y(x, y)\,dy$$

If z represents a level surface, i.e. if $z = f(x, y) = c$, then

$$dz = 0, \quad \frac{dy}{dx} = -\frac{f_x(x, y)}{f_y(x, y)}$$

The total derivative is also called the differential.

The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation.

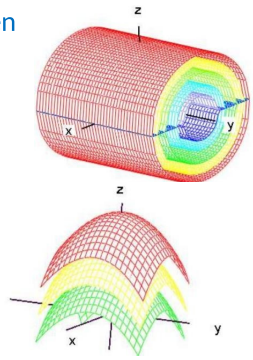

Image Source:http://calculus7.com/sitebuildercontent/sitebuilderfiles/lvlsurf3.pdf

Figure G000061. Implicit functions Problems on Implicit and Composite functions · source slide page 4.

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ENGINEERING MATHEMATICS - I

Implicit Function

Branch diagram for differentiating $z = f(x, y)$ w.r.t x.

Setting $\frac{dz}{dx} = 0$ leads to a simple computation formula for implicit differentiation

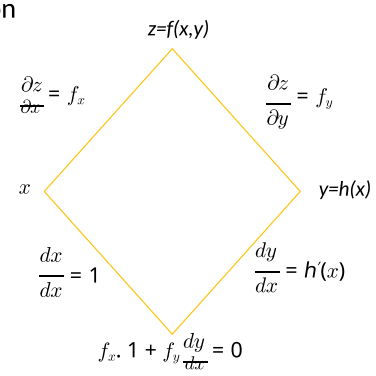

Figure G000062. Implicit functions Problems on Implicit and Composite functions · source slide page 5.

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ENGINEERING MATHEMATICS - I

Implicit function of three variables

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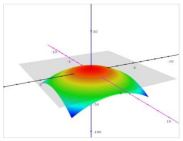
Let $f(x, y, z) = 0$ be the equation of an implicit function of three variables x, y, z. Suppose y and z are functions of x, then f is a function of one independent variable x and y , z are intermediate variables.



https://www.intmath.com/vectors/3d-grapher.php

Keeping z constant, differentiating w.r.t. x, we get

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial x} = 0$$

Solving, $\frac{\partial y}{\partial x} = -\frac{f_x}{f_y}$ provided $f_y \neq 0$.

Similarly, differentiating w.r.t. x, holding y constant

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial x} = 0,$$

Solving, $\frac{\partial z}{\partial x} = -\frac{f_x}{f_z}$, if $f_z \neq 0$.

Figure G000063. Implicit functions Problems on Implicit and Composite functions · source slide page 6.

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ENGINEERING MATHEMATICS - I

Implicit Functions -Problems

1. For the curve $xe^y+ye^x=0$, find the equation of the tangent line at the origin.

Solution:

$xe^y+ye^x=0$……(1)

Differentiating w.r.t x,

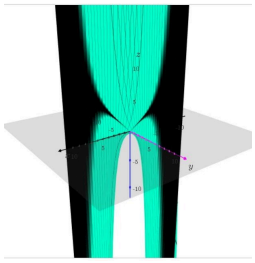
$$\frac{dy}{dx} = -\frac{ye^x + e^y}{xe^y + e^x}$$

Slope of the tangent at origin

$\frac{dy}{dx} at\ (0,0) = -1$

Equation of the tangent line at(0,0),

$y + x = 0$



https://www.intmath.com/vectors/3d-grapher.php

Figure G000064. Implicit functions Problems on Implicit and Composite functions · source slide page 7.

EV0000448

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

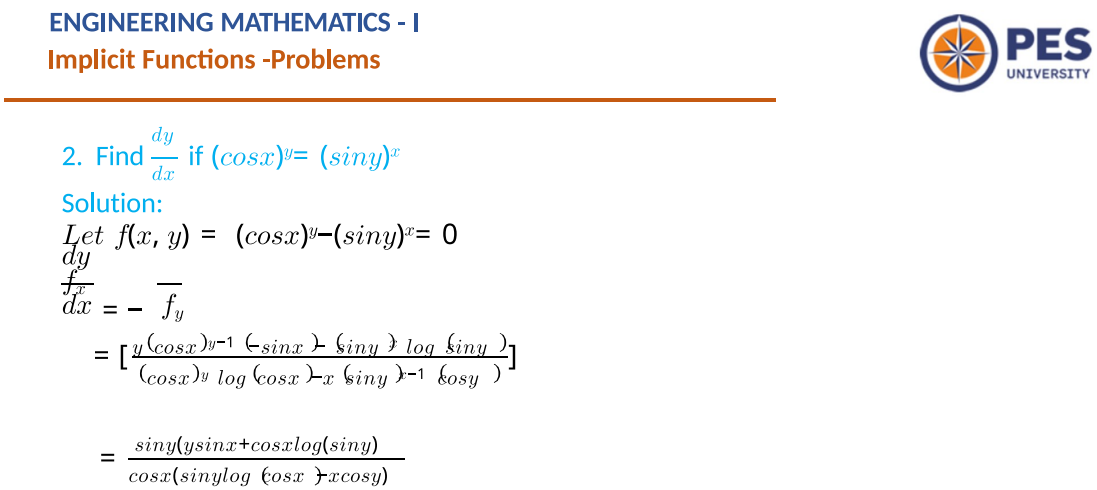


Figure G000065. Implicit functions Problems on Implicit and Composite functions · source slide page 8.

EV0000451

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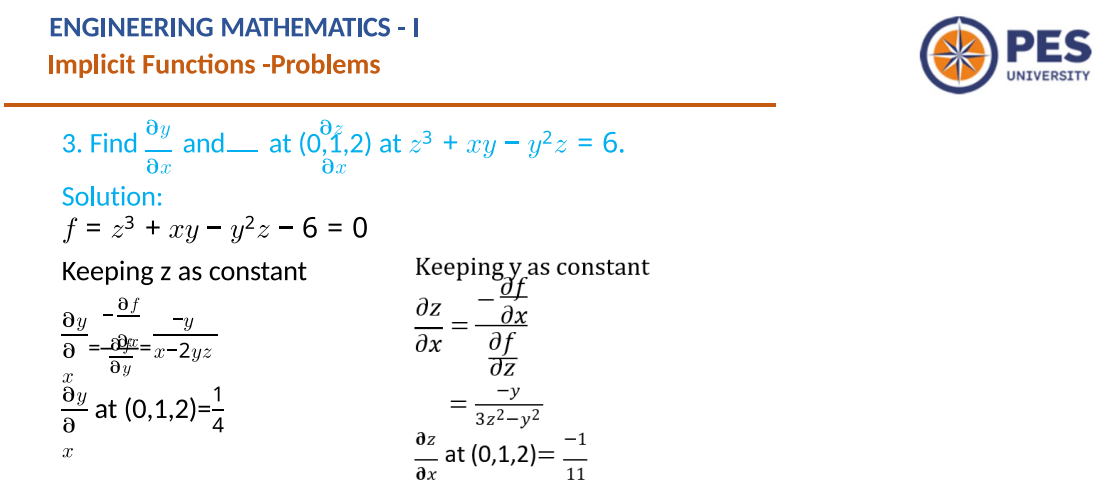

Figure G000066. Implicit functions Problems on Implicit and Composite functions · source slide page 9.

EV0000454

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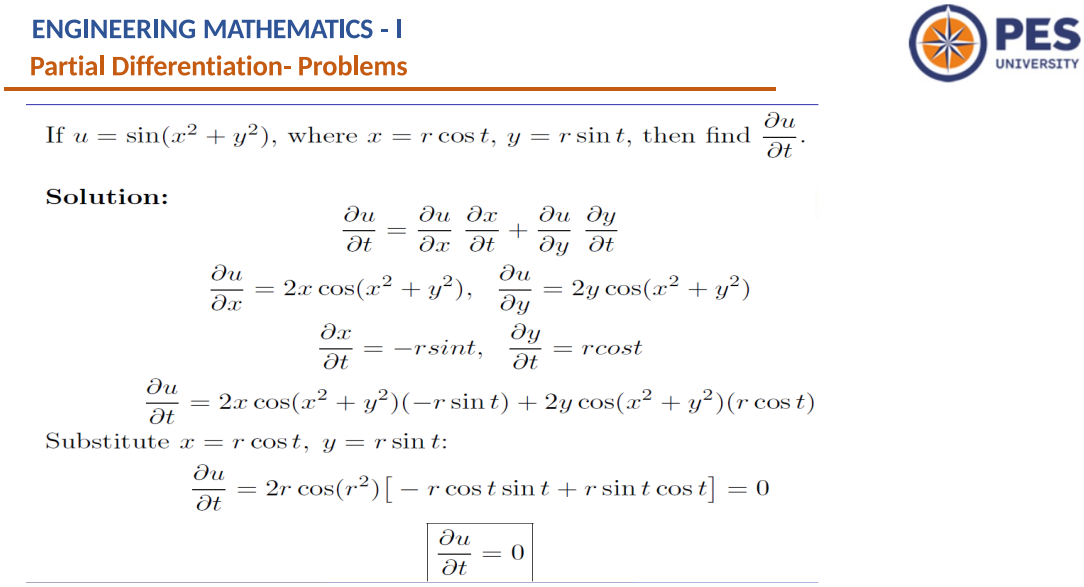

Figure G000067. Implicit functions Problems on Implicit and Composite functions · source slide page 10.

EV0000457

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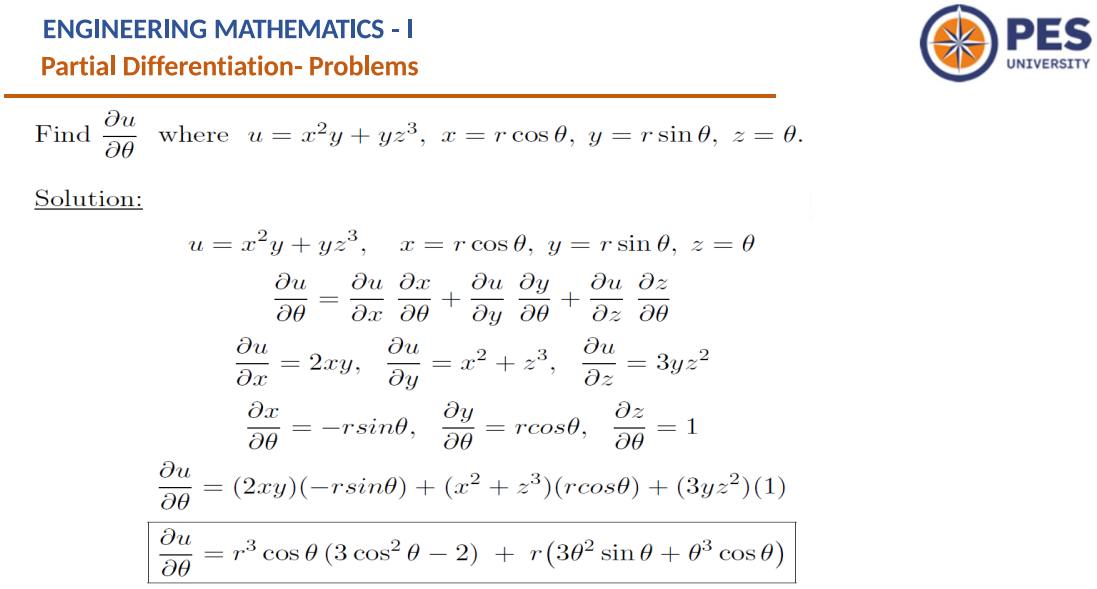

Figure G000068. Implicit functions Problems on Implicit and Composite functions · source slide page 11.

EV0000460

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

PES UNIVERSITY

If $u = f(r)$ where $r = \sqrt{x^2 + y^2 + z^2}$, then express $\dfrac{\partial u}{\partial x}$ in terms of $f'(r)$.

Solution:

$$\frac{\partial u}{\partial x} = \frac{du}{dr}\,\frac{\partial r}{\partial x},$$

$$\implies \frac{\partial u}{\partial x} = f'(r)\,\frac{\partial r}{\partial x},$$

$$\frac{\partial r}{\partial x} = \frac{x}{\sqrt{x^2 + y^2 + z^2}} = \frac{x}{r}$$

$$\therefore \boxed{\frac{\partial u}{\partial x} = f'(r)\frac{x}{r}}$$

Figure G000069. Implicit functions Problems on Implicit and Composite functions · source slide page 12.

EV0000463

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

If $u = e^{x^2+y^2}$, $x = r\cos\theta$, $y = r\sin\theta$, then find $\dfrac{\partial u}{\partial \theta}$.

Solution:

$$\frac{\partial u}{\partial \theta} = \frac{\partial u}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial u}{\partial y}\frac{\partial y}{\partial \theta}$$

$$\frac{\partial u}{\partial x} = 2xe^{x^2+y^2}, \quad \frac{\partial u}{\partial y} = 2ye^{x^2+y^2}$$

$$\frac{\partial x}{\partial \theta} = -r\sin\theta, \quad \frac{\partial y}{\partial \theta} = r\cos\theta$$

Thus,

$$\frac{\partial u}{\partial \theta} = 2xe^{x^2+y^2}(-r\sin\theta) + 2ye^{x^2+y^2}(r\cos\theta)$$

Figure G000070. Implicit functions Problems on Implicit and Composite functions · source slide page 13.

EV0000466

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

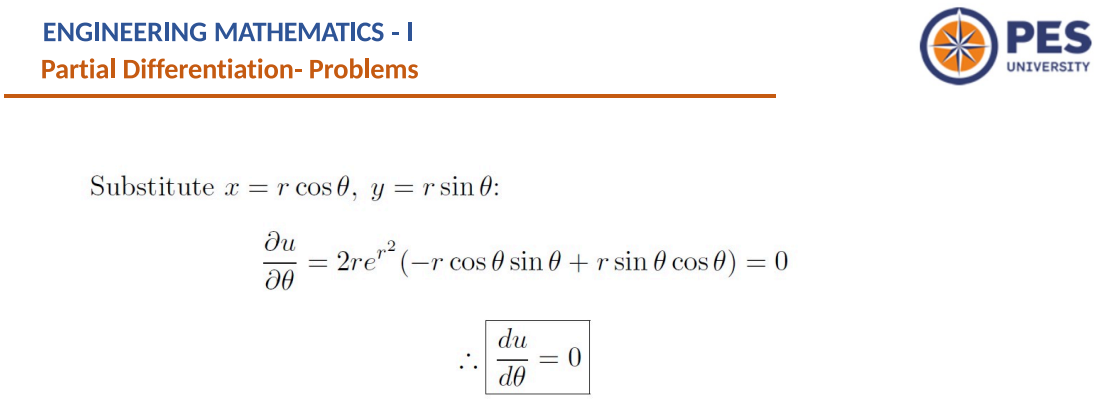

Figure G000071. Implicit functions Problems on Implicit and Composite functions · source slide page 14.

EV0000469

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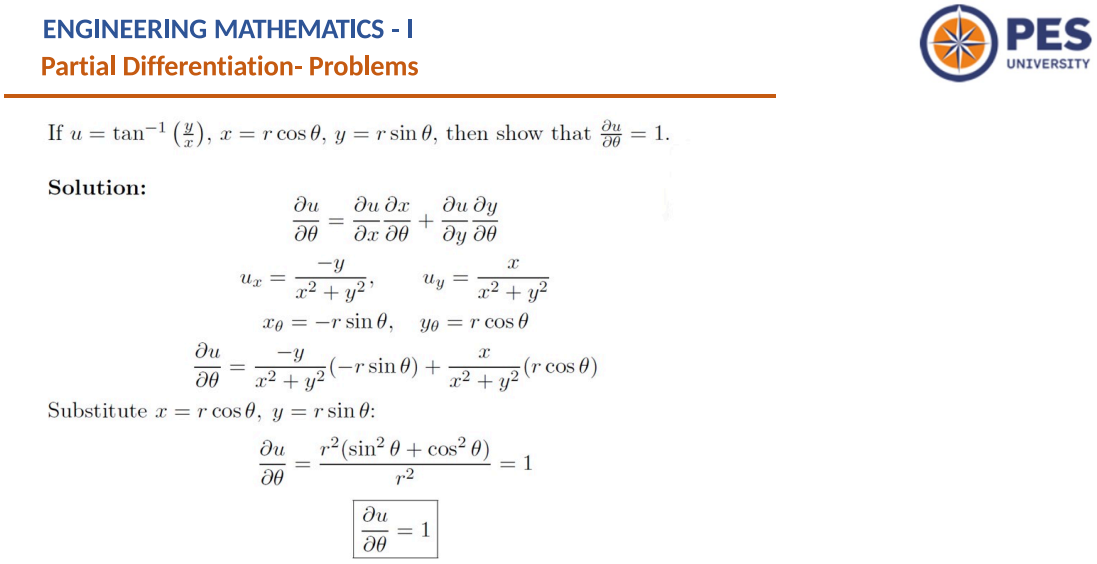

Figure G000072. Implicit functions Problems on Implicit and Composite functions · source slide page 15.

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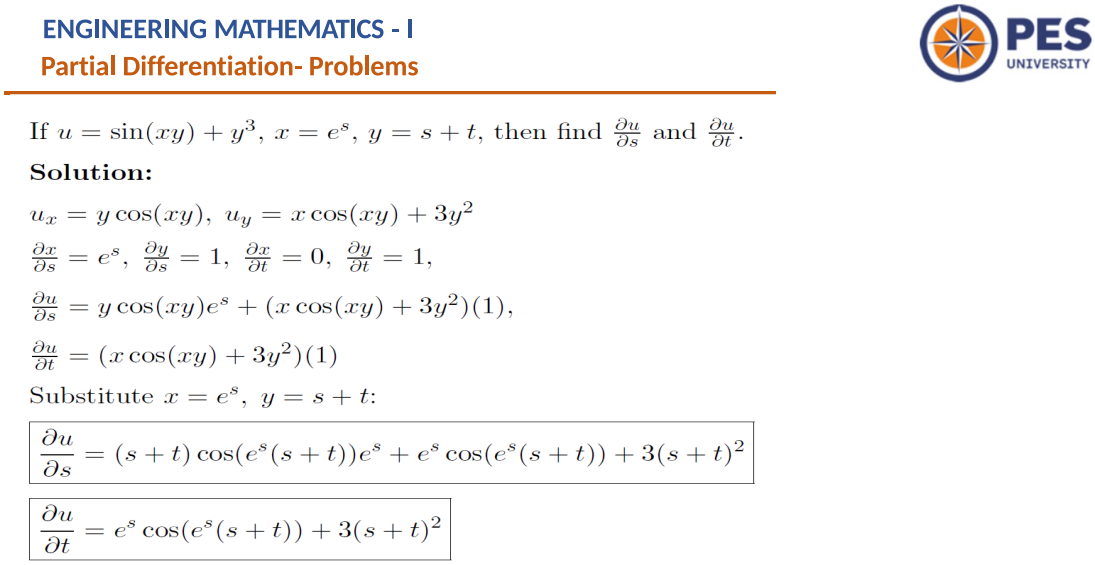

Figure G000073. Implicit functions Problems on Implicit and Composite functions · source slide page 16.

EV0000475

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

ENGINEERING MATHEMATICS - I

Partial Differentiation- Problems

PES UNIVERSITY

If $u = f(r, \theta)$, $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}(y/x)$, then find $\dfrac{\partial u}{\partial x}$ and $\dfrac{\partial u}{\partial y}$.

Solution:

$$\frac{\partial u}{\partial x} = f_r \frac{\partial r}{\partial x} + f_\theta \frac{\partial \theta}{\partial x}, \quad \frac{\partial u}{\partial y} = f_r \frac{\partial r}{\partial y} + f_\theta \frac{\partial \theta}{\partial y}$$

Derivatives:

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \ \frac{\partial r}{\partial y} = \frac{y}{r}, \ \frac{\partial \theta}{\partial x} = -\frac{y}{r^2}, \ \frac{\partial \theta}{\partial y} = \frac{x}{r^2}$$

Hence:

$$\boxed{\frac{\partial u}{\partial x} = f_r \frac{x}{r} - f_\theta \frac{y}{r^2}} \qquad \boxed{\frac{\partial u}{\partial y} = f_r \frac{y}{r} + f_\theta \frac{x}{r^2}}$$

Figure G000074. Implicit functions Problems on Implicit and Composite functions · source slide page 17.

EV0000478

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000075. Implicit functions Problems on Implicit and Composite functions · source slide page 18.

EV0000481

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000429 · EV0000430 · EV0000431
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 4 Sub Topic : Implicit Functions, Problems EV0000432 · EV0000433 · EV0000434
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Implicit Function Consider an implicit function $x3y + 3xy2 = 3$ Then, $x3\,dy + 3yx2 + 3y2 + 6xy\,dy$=0, dx $dx\,dy = -3y(x+y)\,dx\,x(x2+6y)$ Similarly, if $z = f(x1, x2, x3, \ldots, xn)$ total differential of z is defined by $dz = fx1\,dx1 + fx2\,dx2 + fx3\,dx3 + \cdots + fxn\,dxn$ EV0000435 · EV0000436 · EV0000437
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Implicit Function If $z = f(x, y)$ is a function of two independent variables, then $dz = fx\ x, y\,dx + fy\ x, y\,dy$ If z represents a level surface, i.e. if $z = f(x, y) = c$, then $dy\ fy(x, y)\,dz = 0, ' = -dx\ fx(x, y)$ The total derivative is also called the differential. The differential is used in integration by substitution and in approximation. It is also useful to calculate the Errors in approximation. Image Source:http://calculus7.com/sitebuildercontent/sitebuilderfiles/lvlsurf3.pdf EV0000438 · EV0000439 · EV0000440

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Implicit Function Branch diagram for differentiating $z = f(x, y)$ w.r.t x. dz Setting = 0 leads to a simple computation formula for dx implicit differentiation z=f(x,y) $\partial z = f y \partial y \partial z \partial x = f x x$ y=h(x) $d y = h'(x)$ $dx = 1$ dx dx dy fx. $1 + f y dx = 0$ EV0000441 · EV0000442 · EV0000443

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Implicit function of three variables Let $f(x, y, z) = 0$ be the equation of an implicit function of three variables x, y, z. Suppose y and z are functions of x, then f is a function of one independent variable x and y , z are intermediate variables. Keeping z constant, differentiating w.r.t. x, we get https://www.intmath.com /vectors/3d-grapher.php ∂f ∂f ∂y + = 0 ∂x ∂y ∂x ∂y f x Solving, = − provided $f \neq 0$. y ∂x $f y$ Similarly, differentiating w.r.t. x, holding y constant ∂f + ∂f ∂z = 0, ∂x ∂z ∂x ∂z fx Solving, = − , if $fz \neq 0$. ∂x f z EV0000444 · EV0000445 · EV0000446

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Implicit Functions - Problems 1. For the curve $xey+yex=0$, find the equation of the tangent line at the origin. Solution: $xey+yex=0$.....(1) Differentiating w.r.t x, [Image text, OCR] -5 y -10 dy yex + ey = − dx xey + ex Slope of the tangent at origin dy at (0,0) = −1 https://www.intmath.com/ dx Equation of the tangent line at(0,0), $y + x = 0$ vectors/3d-grapher.php EV0000447 · EV0000448 · EV0000449

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Implicit Functions - Problems dy 2. Find if $(cosx)y = (siny)x$ dx Solution: Let $f(x, y) = (cosx)y-(siny)x = 0$ dy fx = − dx fy y $cosx$ $y-1$ $-sinx - siny$ x log $siny$ = | | $cosx$ y log $cosx$ $-x$ $siny$ $x-1$ $cosy$ $siny(ysinx+cosxlog(siny) = cosx(sinylog$ $cosx$ $-xcosy)$ EV0000450 · EV0000451 · EV0000452

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Implicit Functions - Problems ∂y ∂z 3. Find and at (0,1,2) at $z3 + xy - y2z = 6$. ∂x ∂x Solution: $f = z3 + xy - y2z - 6 = 0$ Keeping z as constant ∂f [Image text, OCR] Keepingy.as constant df 0z 0x dx of dz - y 3z2-y2 az -1 at (0,1,2)= ax 11 − = ∂x= ∂y $-y$ ∂f ∂y ∂x $x-2yz$ 1 4 ∂y at (0,1,2)= ∂x EV0000453 · EV0000454 · EV0000455

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Ou If u = sin(c² + y²), where = rcost, y = r sint, then find at Solution: du Ou ϑc Ou Oy at ϑc at Oy at Ou du = 2x cos(x² + y²) = 2y cos(x² + y²) 0c Oy 0c dy = -rsint, = rcost at at du = 2x cos(x² + y²)(-r sint) + 2y cos(x² + y²)(r cost) at Substitute = r cost, y = r sin t: du = 2r cos(r²)[- r cost sint + r sint cost] = 0 at Ou =0 at EV0000456 · EV0000457 · EV0000458

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du Find where u = ²y + yz², = rcos ϑ, y = rsinϑ, ≈ = ϑ. 00 Solution: = rcosϑ, y = rsinϑ, = ϑ du Ou Or du Oy Ou ϑz + 0x0ϑ Oy 00 0z 00 du du du = 2cy, = 3yx2 0c Oy ϑz 0c dy 0z = -rsinϑ, = rcosϑ, =1 00 00 00 Ou (2cy)(-rsinϑ) + (x² +z")(rcosϑ) +(3yz²)(1) 00 du = r3 cos ϑ(3 cos² ϑ - 2) + r(302 sin ϑ + 03 cos ϑ) 00 EV0000459 · EV0000460 · EV0000461

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du If u = f(r) where r = √2 + y2 + z2, then express dc in terms of f'(r). Solution: du du Or Ox-dr0x Ou Or = f'(r) 0c 0c 0r C C /αc2 + y2 + 22 r du =f(r) 0c EV0000462 · EV0000463 · EV0000464

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Ou If u = ex²+y², = rcos ϑ, y = rsinϑ, then find 00 Solution: Ou du dc Ou Oy ϑc 00 0y 00 du du = 2rex²+y2 = 2yec2+y2 Oy 0c dy -r sin 0, r cos ϑ 00 00 Thus, du = 2cet2+y²(r sin 0) + 2yer+s²(r cos 0) 00 EV0000465 · EV0000466 · EV0000467

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Substitute = r cosϑ, y = rsin ϑ: au -r cos ϑsin ϑ + r sin ϑ cosϑ) = 0 du dϑ EV0000468 · EV0000469 · EV0000470

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] If u = tan-1 (), = rcos ϑ, y = rsin ϑ, then show that = 1. Solution: Ou Ou Oc du Oy 00 0x 00 dy 00 -y C Uc Uy x2+ y2' x2 + y2 Cϑ = -r sin ϑ, ye = r cos ϑ Ou -y (-r sin 0) + - C 00 x2 + y2 Substitute = r cos ϑ, y = r sin ϑ: du r2(sin2 ϑ + cos² ϑ) ∶ 1 00 r2 du 00 EV0000471 · EV0000472 · EV0000473

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Solution: Ux = y cos(cy), uy = ccos(cy) + 3y2 =1,=0,=1, = ycos(cy)e + (ccos(cy) + 3y²)(1), du = (ccos(cy) + 3y²)(1) at Substitute = e", y = s + t: Ou = (s + t) cos(es(s + t))e" + es cos(es(s + t)) + 3(s + t)² 0s du = es cos(es(s + t)) + 3(s + t)2 at EV0000474 · EV0000475 · EV0000476

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] If u = f(r,ϑ), r = √x2 + y², ϑ = tan-1(y/α), then du du find and Oy Solution: du 0r 00 du Or +fy 00 = fr +fe- = fr 0c 0c 0c dy oy Derivatives: Or Or 00 00 y y C 0c Oy ϑc r2 Oy r2 Hence: du du C y C =fr- -+fe- Oy r r EV0000477 · EV0000478 · EV0000479

18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000480 · EV0000481 · EV0000482

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.868] Now let's continue our uh lecture on implicit functions. It will be a continuation of chain rule and composite functions and other things that we saw in the previous video. Now you know that suppose you have a function. [00:00:18.852] which is like f of x y equal to c some constant, then we call this as an implicit function. For example, you can see here I have an problem x cube y plus 3xy square equal to 3. This is an implicit function. [00:00:35.204] Now if you remember, how did you differentiate this function? I have x cube y plus 3xy square equal to 3. How do we differentiate this? [00:00:48.452] You differentiate this using product rule, keep the first function as it is. Derivative of y is dy by dx, plus, keep the second function as it is, and you differentiate the first, which is 3x square, plus. [00:01:02.372] You keep the first function as it is, you differentiate the second again, 2y dy by dx plus I'll keep the second function as it is and I differentiate. The advantage of this is the right hand side will become 0. [00:01:17.540] Because it's a constant. And from this expression, it's easy for me to find what is dy by tx. Right? So if I compare this with my function of two independent variables, suppose I'll call this a z, this is as good as having z is equal to some constant. [00:01:37.348] or f of x y equal to some constant. So I can extend the same thing to a function of more than one independent variable. [00:01:49.380] So this we call the total derivative. So dy by dx is was nothing but the derivative. But now if z is a function of xy, but with a restriction. That

restriction you have to remember. That z must be a constant. [00:02:04.100] So here if z is a function of n independent variable and if it's equal to constant, I can always find the total derivative of z which is defined by [00:02:15.172] DZ is equal to, so you know this, dz is equal to partial derivative with respect to x, dx1 with respect to just now we discussed what is the total derivative, right? Now if z becomes constant, this is what we get. If z becomes constant, then it it looks like f of x comma y equal to c. [00:02:35.396] And there is a special name given to this. This is called a level surface. So Z is equal to constant will always represent a level surface. And it's not new to you, right? The level surface, for example, you take equipotential surfaces. So a surface on which at every point the potential is constant. [00:02:55.396] Right? Or isotherms where at every point the temperature remains the same. So this shows you that there is a cylinder here and every color indicates that at that particular point the temperature remains constant. [00:03:10.756] That means what suppose I have the equation of a cylinder as x square plus y square equal to some constant c. And for every constant you get different lay of the cylinder, and for every constant, the temperature, or suppose I take the temperature distribution, it remains the same. [00:03:29.828] If I'd want to consider a function of let's say three independent variables, suppose I'll give you x square plus y square plus z square equal to some constant c1, and what will this represent for you? [00:03:42.596] You know that this is nothing but a sphere centered at the origin, right? So I couldn't show you the complete sphere, so I have taken only the upper part of it. Now for every C1 you get different lay of that sphere. [00:03:56.708] So if C1 becomes 0, I will get the origin. If C1 become 1, I'll get the unit sphere. C1 become 89, I'll get a bigger sphere. So for every layer of the sphere, I'll get the value of x square plus y square plus z square to be constant on the surface of the sphere. [00:04:16.580] So here that's what I have shown. Every color would represent a different value of C, and it shows that for every color the temperature distribution or whatever you want to call it, the pressure or the potential or uh distribution of current, everything was remains constant. [00:04:34.148] So if you have a multivariable function which is of the form f of x1 comma x2 comma, I have n variable equal to c, this would represent a level surface. [00:04:50.340] Of course, it's difficult to plot because I don't have that many dimensions to plot, but definitely it would represent a level surface. And in case of level surface, you can find the total derivative. So how did I find the total derivative?

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.260] Let me go back to my previous slide so that it will be easier for you. [00:05:12.868] So how did I find the total derivative? I differentiate it given expression with respect to X and Y, and then this is the expression I got. But is there another way that I can somehow make use of a partial derivative to this? [00:05:30.020] Right? If you observe carefully, you have this, right? Let me take this as some set. Can you find out what is ZX from the given expression? [00:05:41.892] Isn't it ZX 3x square y plus 3y square? I am differentiating z partially with respect to x. Okay. And what is z y? Z y will be. Again I get x q plus. [00:06:01.028] 6x y and if you observe carefully I got this dy by dx as minus zx by sec y. This will be useful if your function is too complicated. [00:06:19.440] the see see the uh tedious task that we did here I have to use product rule maybe quotient rule or chain rule or whatever it is the function might be as complicated as possible but it's not possible for me to find the d by by dx very easily or very conveniently. [00:06:36.164] But now if I need to find the total derivative, you already know z is a constant, so that's not going to change. So if z is equal to f of x y is equal to c if it represents a level surface, then you can get dy by dx to be minus zx by z y.

That was a small observation. [00:06:56.068] And that was a good observation. So this total differential that you get. So maybe if you don't want to call D Web DS you can just call it as DF. [00:07:05.572] So if z is equal to f of x y is equal to c is a level surface, then df is nothing but minus zx divided by z y. And this df is called the differential. [00:07:18.692] In fact, you have already used this uh in your course. Okay. So let me just show you like where you have uh used it. So let me just go back to this slide. [00:07:31.812] Okay, so if I have a function of two independent variables x and y, then the total derivative [00:07:39.744] d dz is fx of x y dx plus f y of x y dy. So d z equal to 0, which is dy by dx is equal to minus f y x y by f x x y. So the total derivative as I said it's also called the differential. [00:07:57.316] And where where did we use this? So this is just a tree diagram to show you how we got this expression, which I showed you already. Okay. Now suppose I'll ask you to evaluate this integral, e to the power of x square, and then let's say I have x dx. Right? [00:08:14.340] How did you uh do this integration? As soon as you look at the integrand, it's clear that you have to make a substitution. So when you substitute x square equal to t, you got 2x dx to be 3. [00:08:29.408] So x dx would have been dt divided by 2. That is where I can apply the total differential. So the applic one of the applications of total differential is you can do integration by substitution. That's mean basically I'm I'm in x plane, now I'll be in dt plane. [00:08:47.120] All my x square x gets transformed and I'll get only a simple function as z to the power of t which you can integrate directly. Okay. Another appro uh application would be in errors and approximations that I'll discuss by the end of this unit when we do applications. Okay. [00:09:05.156] So this I have a three dimensional. Now suppose I have f of xyz equal to 0, uh be a function of implicit function of three independent variables xy and z. And again, I need to consider that z is a function of xy, otherwise it's difficult for me to express it. Okay then. [00:09:23.428] Now the problem is in the previous case I had uh y to be a function of x and z was a function of xy and this was constant. So I could find the total differential of z. [00:09:38.116] Now what I have is f is a uh or okay, z is a function of x and y definitely. Okay. So I cannot find dz as y is not a function of x. These two are independent. [00:09:51.524] So what I can find is I can find what is uh uh ZX if you want, and then what is ZY. Okay. Um those things you can find. So depending on the variable. So you can find ZX and

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.920] Z y so this is like an extension of what we just discussed to another variable. One more variable is added. Again it represents a level surface. So the graph here will tell you what is the distribution of temperature that has happening. So I got um dy by dx to be minus f x by f y, which is [00:10:23.568] obvious from the previous case and dz by dou z by dou x is minus f x by f z. So provided this is not 0 of course partial derivative of f with respect to z is not equal to 0. Okay, so that's a new thing for you to observe. [00:10:39.296] Uh what I want you to concentrate is Zx. Suppose you have f of x y z equal to c, then zx is minus f x by f z. So look at the advantage. Okay, now suppose I'll give you uh z. [00:10:55.876] uh maybe I'll give you x cube y square uh plus three x y z plus some sine of e to the power of z this is equal to some c just imagine how much effort you need to put to differentiate this with respect to [00:11:11.760] X partially, right? You have to separate Z and then or because X and Y are both independent. You'll get you don't know how what to do with uh you know Y X and keep treat Y as a constant and retain that and everything. But here this formula will tell you that [00:11:27.920] dou z by dou x is minus f x by f z. Similarly we can uh write another way of writing dou z by dou y if you want. You can extend it to n number of variables. Okay. So keep in

mind and uh we'll solve some problems so that things will be little more clear to you. [00:11:48.420] Now let's try some problems on implicit functions. So the first question is for the curve x e power y plus y e power x equal to 0, find the equivalent of the tangent line. [00:12:01.284] Okay. So uh we know that the equation of the tangent line you need slope, right? So if I call the equation uh of the tangent line to be y is equal to mx, then I need to calculate the slope. [00:12:15.748] So the slope of any line is nothing but its derivative, t y by dx. So let's use the definition of total derivative. So if this function here represents in [00:12:29.008] level surface. That means it is of the form z is equal to some function of xy or y x equal to zero. Now that's another constant zero. So [00:12:40.592] I know that the total derivative for this is given by so in this case if f of x y is equal to 0, then dy by dx is nothing but minus f x divided by f. [00:12:53.540] That's from the definition of total derivative. Right? So what is f of x y now for me for this problem? F of x comma y is x into e to the power of y plus y into e to the power of x. [00:13:09.860] So can you find a partial derivative of this with respect to x and with respect to y? So f x will be e to the power of y plus y into e to the power of x. That's f x. Now what will be f y then? [00:13:26.052] F y will be x into e to the power of y plus e to the power of x. So I got fx and f y, so I can find the slope. So dy by dx will be minus e to the power of y plus y e to the power of x. [00:13:43.012] divided by X e to the power of Y plus e to the power of X. [00:13:48.932] Now once I know what is the slope, so this is M, you can calculate the equation of the line passing through any point. Okay so if I need to calculate the slope passing through the origin, let's say because this is a line which passes through the origin. So at 00 M [00:14:07.876] which is dy by dx will become if I put y 0 I will get 1 here. This term will vanish. Similarly in the denominator this term will vanish and this will be 1. So the slope will be minus 1. So one of the equations of the tangent line for the given curve will look like y [00:14:25.648] is equal to minus x or y plus x is equal to 0. If you need to find the tangent in any other point, let's say I want to find at the point x0 y naught, then you can write this as y minus y naught is equal to m into x minus x naught. [00:14:43.792] Okay, so any point is fine, but for a uh for an example I have taken at the point zero comma zero. So that's the equation of a tangent line for that particular curve. Now let's move on to the second problem. [00:14:57.540] The question says, find dy by dx if cos x par y is equal to sine y to the power of x. Let's see if we can write this also in the form of f of x y equal to zero.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:12.548] Can we rewrite this? So I can write this as cos x to the power of y minus sin y to the power of x equal to f of x y which is equal to z. [00:15:27.012] That means this represents an implicit function for me now. Right? See, just imagine if you differentiate this directly, you need to differentiate like first cos x and then apply chain rule to this, it's to the power of y, and then sine y to the power of x. Things will become more complicated, right? So um [00:15:47.268] But if we use partial derivatives, things will be much simpler. The formula for dy by dx will be minus fx divided by f y. So what is the partial derivative of this function with respect to x? [00:16:02.596] When I differentiate this with respect to x, y is a constant. So I don't need to worry about that y. So this will be y minus y into cos x to the power of y minus 1 into derivative of cos x will be minus psi x. [00:16:22.276] Is this clear what I did? So I had this as the first [00:16:26.080] First this is just the first term. So I'm differentiating with respect to X, so Y as a constant. So this is cos x to the power of four or five or c whatever you want to call it. So let's differentiate the exponent first, which is y cos x to the power of y minus one. Ah derivative of cos x is [00:16:43.728] minus sin x. Now here this is like this is a

constant for me. So this is like a to the power of x. What is the derivative of a power x for you? If you remember derivative of a power x will be a to the power of x into ln of a [00:17:02.724] So I'll get sine y to the power of x into ln of y. That's my numerator. That's the numerator, okay? So let me put this inside the brackets so that you know what is the numerator is. Now I need to find the partial derivative of s with respect to y. [00:17:22.564] So that means cos x will become a constant for me now. So this is like a to the power of y. And derivative of a power y will be cos x to the power of y into ln of cos x. [00:17:36.612] Right? So this is like a to the power of y now. Minus same thing I have to do. This is a variable for me, but x is a constant. So I'll get x into sine y to the power of x minus 1 into derivative of sine y will be cos y. [00:17:56.132] Of course, not much simplific simplification can be done for this, but uh this is the expression that you get for dy by dx. If any particular point is given, then uh we may continue and substitute that point and find out. But you need to be careful. [00:18:12.128] that x cannot take the value 0, neither y can take the value 0 because ln of 0 is not defined. So that's a value of expression for dy by dx at x not equal to 0 and y not equal to [00:18:28.176] Zero. So if I apply these conditions, this derivative always good. So we'll try the next problem, which is again on uh uh implicit functions and uh we'll see how it goes. [00:18:39.716] Okay, let's try another problem now. The question is find dou y by dou x and double at the point one comma zero comma two for this particular function. [00:18:51.972] So this is a function of three independent variables x, y, and z, but also you can see that this is an implicit function. [00:19:00.324] So when I differentiate y partially with respect to x, I need to treat z as constant. That's what is expected. And when I differentiate z partially with respect to x, I need to treat y as constant. So I can find y x, which is nothing but partial derivative. [00:19:18.096] derivative of y with respect to x by taking f minus of f x by f y. This is using the formula for total derivative. Where what is f? [00:19:29.732] F is Z cube plus XY minus Y square Z. So that this is of the form f of XYZ equal to C. [00:19:41.220] So if f is this, then what is F x now? So Yx will be what is the partial derivative of F with respect to X? Please Z is constant. So you don't need to worry about Z. So uh it will be uh I have differentiating with respect to X. So Y is constant. [00:19:58.916] Uh right. So y minus uh uh two y.

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Transcript block 5 (source interval 001200-001499 seconds): [00:20:05.024] So that term also will vanish. So when I differentiate f with respect to x, this is the only term that exists. So this term is not there. So you can just take y divided by what is the partial derivative of this with respect to y. I have two terms with y in it. So I'll get x. [00:20:25.540] Minus two Y Z. [00:20:29.988] So y x, y x will be y divided by x minus 2 y z. This is what I get. But I need to calculate y x at the point 1 comma 0 comma 2. So if I need to calculate dou y by dou x. [00:20:47.664] At the point 1 comma 0 comet the value of y, y is 0 divided by x, x is 1 minus 2 into y is again 0 and z is 2. [00:21:01.828] Okay. So that means I'll get this value as zero because the numerator is zero for me. Okay. Now they want you to calculate dou z by dou x. [00:21:15.620] So similar to this, can you recollect and tell me what could be the formula for dou z by doux? Dose by dou x will be minus f x divided by f z. [00:21:28.356] So when you calculate the partial derivative of f with respect to x, we need to treat z as constant and y also as constant. So f x will be f x partial derivative of f with respect to x will be [00:21:45.316] uh I'll get y with a minus uh and so that's it. There's only one y term and f z will be three z square uh minus y square. [00:21:57.764] I am differentiating f partial b with respect to z. So dou z by dou x will be minus y by three z square minus y square. But again I need to calculate this at the point. [00:22:11.908] One comma, zero, comma two. So because y is zero, this

is also zero for me. So if the partial derivatives are zero at that particular point, it implies that the there is [00:22:24.336] Tangent of the slope or the slope of the tangent at this point parallel to x axis is a horizontal line and even at this point it's a horizontal line. Okay. In our next session we will try some problems on Taylor's and McLaren series.

EV0000106 · EV0000107 · EV0000108 · EV0000109

Full transcript: 3737 normalized words, 90 timestamped segments, 1360.213333 seconds duration. Low-confidence segments flagged: 0.

EV0000090 · EV0000091 · EV0000092 · EV0000093

8 Problems on Composite and Implicit functions

This lesson is indexed from the source video titled Problems on Composite and Implicit functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000110 · EV0000111

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.860] We will solve problems on composite functions. [00:00:10.628] Let us look at the first problem. It is given that u equals sin of x square plus y square where x equal to r cos t and y equal to r sin t. Then find the value of do u by dou t. [00:00:31.300] It is given that u is a function of x and y, where x and y are the functions of r and t. It means u becomes the composite function of r and t. So by using chain rule we get the formula. [00:00:49.476] dou u by dou t equals dou u by dou x into dou x by dou t plus dou u by dou y into dou y by dou t. Now for the given u [00:01:02.820] dou u by dou x becomes two x into cos of x square plus y square similarly dou by dou y becomes two y into cos of x square plus y square similarly for the given x and y [00:01:20.580] 2 x by dou t becomes minus r sin t, dou y by dou t becomes r cos t. So let's substitute all these values into this expansion that is 1. Then dou u by dou t becomes [00:01:37.700] 2x into cos of x square plus y square into minus r sin t plus 2y cos of x square plus y square into r cos t. Now because u is a composite function of r and t, we eliminate [00:01:57.252] Let is x and y. In the problem it is given that x equal to r cos t and y equal to r sin t. So when we use that, so du u by do t becomes two r cos of r square into minus r cos t times sin t plus. [00:02:16.548] R sin t into cos c. So after that it becomes zero. So w by do t becomes zero. So this is the answer for this problem. [00:02:31.428] So whenever we solve a problem on composite function, it is essentially to know the chain given in the problem. Now what is the meaning of chain? Absolutely problem here. U is a function of x and y. [00:02:50.148] X and Y are the functions of R and T. So this implies U becomes composive function of R and T. So based on this idea we can easily remember the chain rule involved in the problem. [00:03:09.412] Let us look at the second problem. Find dou u by dou theta where u equals x square y plus y z q, x equal to r cos theta, y equal to r sin theta, and z d equals to theta. In this problem, u is a function of x y z. [00:03:29.796] where xy z are the functions of r and theta it means u becomes the composite function of r and theta so in this problem we want only one particular case that is dou u by dou

theta so when we use chain rule [00:03:47.556] So dou by dou theta becomes this expression. Now for the given u, dou u by dou x becomes two x y. W by dou y is x square plus z power three. W u by dou z equals three y z square. [00:04:04.932] Similarly for the given xy and z, dou x by dou theta equals minus r sin theta, dou y by dou theta is r cos theta and dou z by dou theta is one because that is given as theta. So when we use all these substitutions back to one. [00:04:25.348] So one reduces to two x y of minus r sin theta plus x square plus z power 3 into r cos theta plus 3y z square. So since u is a composite function of r and theta [00:04:43.524] So we make use of the substitution x and y to be r cos t and r sin theta. So then the final answer for the problem is this r power 3 cos theta of 3 cos square theta minus 2 plus r three theta square sin theta plus theta power 3 into

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.140] [ASR low confidence; check audio] Gosteep. [00:05:07.364] Let us look at this particular problem. If u equal to a function of r where r equal to square root of x square plus y square plus z square, then express dou u by dou x in terms of f dash of r. [00:05:27.076] Again, when we look at the chain, it is given that u is a function of r. When r is a function of x, y, and z. So u become composite function of x, y, and z. So when we write the chain rule for dou u by dou x, it be it becomes [00:05:45.092] dou u by dou x equals dou by dou r into dou r by dou x so because u is f of r so dou u by dou r becomes f dash of r into dou r by dou x so let us find the value of dou r by dou x [00:06:02.660] Because r is square root of x square plus y square plus z square, so the partial derivative of r with respect to x is going to be x divided by square root of x square plus y square plus square is after calculation up to 12. So this is nothing but x y. [00:06:22.692] So when you substitute this into this expression, so do u by do x becomes f dash of r into x by r. So this is the answer for this particular problem. [00:06:38.628] Let us look at the fourth problem. It is given that u equals e power x square plus y square where x equal to r cos theta, y equal to r sin theta, then find a value of w by dou theta. [00:06:58.084] So in this problem it is given that u is a function of x and y. And x and y are the functions of r and theta, then u becomes the composite function of r and theta. So again we want only one particular case, so that is uh 2 by 2 theta. [00:07:17.316] So by using chain rule, we get dou u by two theta to be this. Let us call expression number one. Now for given u, dou u by two x is two x times e power x square plus y square. [00:07:32.708] and dou u by dou y becomes two y times e power x square plus y square then dou x by dou theta is minus r sin theta dou y by dou theta is r cos theta and substitute these four into expression one we get dou u by dou theta equal [00:07:52.804] 2x e power x square y square of minus r sine theta plus 2y e power x square plus y square into r cos theta. So what would be the next step in this particular problem? Can you stop here? So the answer is no [00:08:10.532] Because u is a composite function of r and theta, we need to eliminate this x and y. So by using even theta x equal to r cos theta and y equal to r sine theta, the given answer, the the answer becomes [00:08:27.876] Zero because after cancellation of this and this. [00:08:33.988] I hope this concept is clear to you. [00:08:40.740] Let us look at one more problem. If u equal to tan in us y by x, when x and y are the functions of r and theta, then show that 2 u by 2 theta equals 1. Again after writing the particular chain. [00:08:59.396] So this is a chain rule that is dou u by dou theta equals this. Then when we compute dou u by do x, dou u by dou y, then dou x by dou theta and dou y by dou theta and finally upon using x and y to be r cos and r sign respectively. [00:09:17.508] We see that two by dou theta becomes one. [00:09:32.580] So let us solve this problem. It is given that u equals sine of xy plus y q where x equals e power s and

y equals s plus t then find the value of dou u by dou s and dou u by dou d. [00:09:51.972] Now what is a chain for this particular problem? U is a function of x and y. When x and y are the functions of s and t then u becomes the composite function of s and t.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.652] So we can expect dou by douce and dou by dou t using chain rule. So upon writing the formula for chain rule and after finding all the partial derivatives for dou by douce and dou by dou t the final answer for this problem is going to be this as well as this. [00:10:28.516] And you must know why did we use x equal to e power s and y equal to s plus t after this particular step. So because u is a composite function of s and t. So we eliminate this x and y in the answer. [00:10:47.332] So let's solve this problem. It is given that u equals f of r theta, where r equals square root of x square plus y square and theta equals that inverse y by x. Then find the value of dou by dou x and dou by dou y. [00:11:07.012] So what is the chain involving problem? U is a function of r and theta where r and theta functions of x and y. Then u becomes composite function of x and y. Then when when we use chain rule, we get [00:11:22.820] dou u by dou x has this and dou u by dou y has this so after solving for dou r by dou x dou dita by dou x dou y and dou dita by dou y we get this and this as the answer for this particular problem [00:11:43.716] Thank you.

EV0000118 · EV0000119 · EV0000120 · EV0000121

Full transcript: 1644 normalized words, 42 timestamped segments, 711.573 seconds duration. Low-confidence segments flagged: 1. Review locators: 00:05:03.140

EV0000110 · EV0000111 · EV0000112 · EV0000113

9 Homogeneous functions and Euler's theorem

This lesson is indexed from the source video titled Homogeneous functions and Euler's theorem. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000122 · EV0000123

Graphs and diagrams

Figure G000076. Homogeneous functions and Euler s theorem · source slide page 1.

EV0000484

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

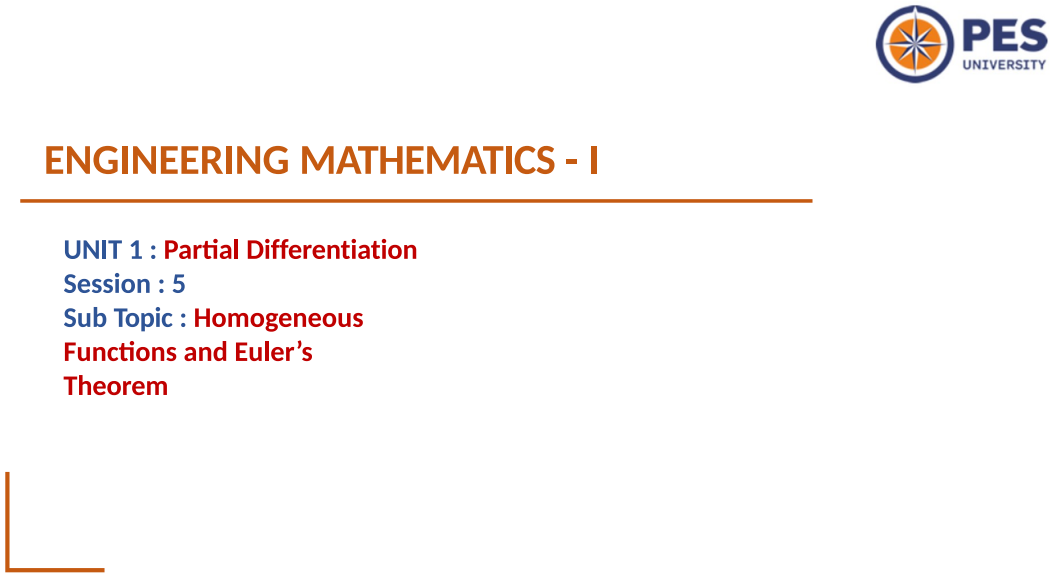

Figure G000077. Homogeneous functions and Euler s theorem · source slide page 2.

EV0000487

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

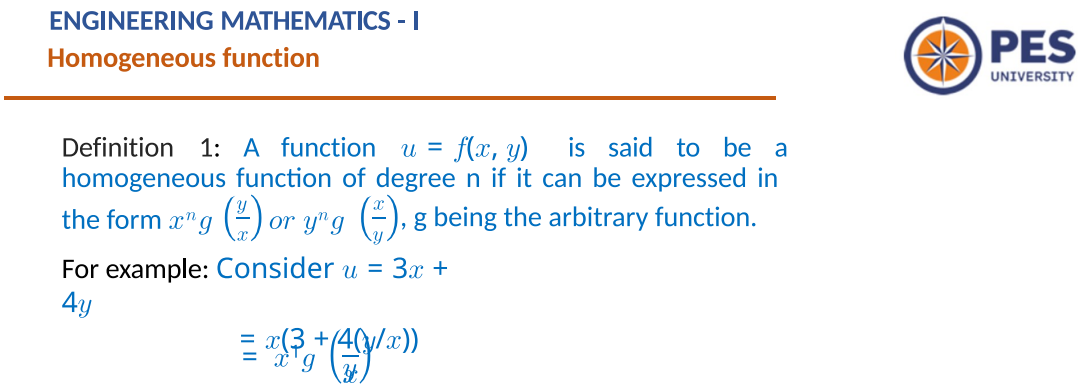

Figure G000078. Homogeneous functions and Euler s theorem · source slide page 3.

EV0000490

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ENGINEERING MATHEMATICS - I

Homogeneous function

Definition 2: A function $u = f(x, y, z)$ is said to be a homogeneous function of degree n if it can be expressed in the form $x^n g\left(\frac{y}{x}, \frac{z}{x}\right)$ or $y^n g\left(\frac{x}{y}, \frac{z}{y}\right)$, or $z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$, g being arbitrary function.

For example: Consider $u = x^3 + y^3 + z^3 + 3xyz$

$$= x^3\left(1+\left(\frac{y}{x}\right)^2 + \left(\frac{z}{x}\right)^3 + 3\left(\frac{y}{x}\right)\left(\frac{z}{x}\right)\right)$$

$$= x^3 g\left(\frac{y}{x}, \frac{z}{x}\right)$$

Therefore u is a homogeneous function of degree 3.

Figure G000079. Homogeneous functions and Euler s theorem · source slide page 4.

EV0000493

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ENGINEERING MATHEMATICS - I

Homogeneous function

- A constant function is homogeneous of degree 0.
- Linear functions are homogeneous of degree 1.
- The degree of the homogeneous function can be positive, negative or zero.
- How to relate homogeneous functions with their partial derivatives.(Euler's Theorem).
- Scaling of the given function.
- Extensive applications in Economics like production function and Marginal productivities, Elasticity of demand etc.,
- In Thermodynamics, Extensive variables are homogeneous of degree 1.
- Intensive variables are homogeneous of degree 0.

Figure G000080. Homogeneous functions and Euler s theorem · source slide page 5.

EV0000496

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

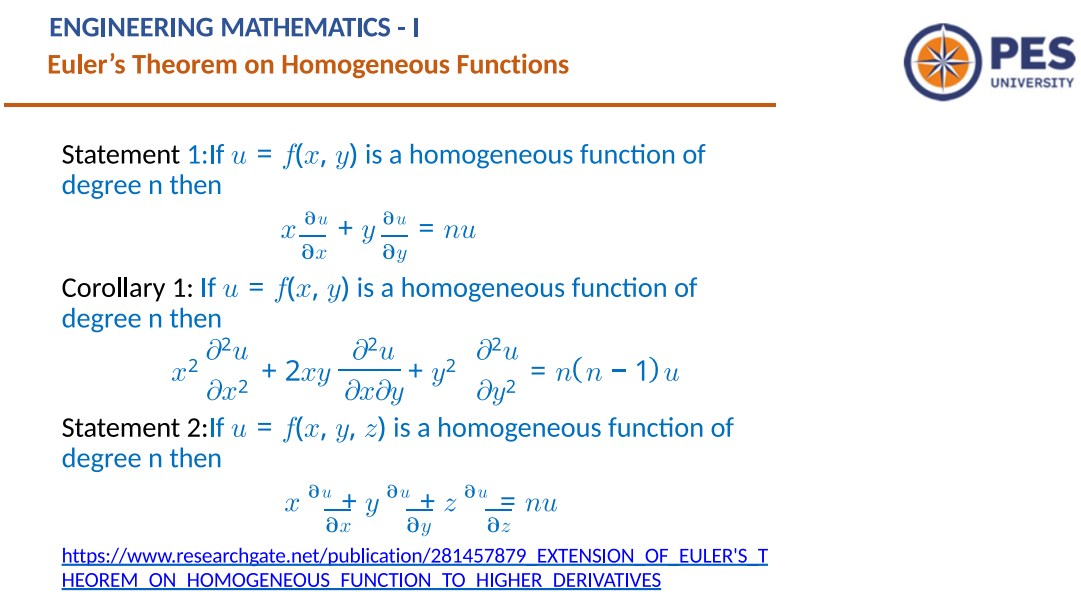

Figure G000081. Homogeneous functions and Euler s theorem · source slide page 6.

EV0000499

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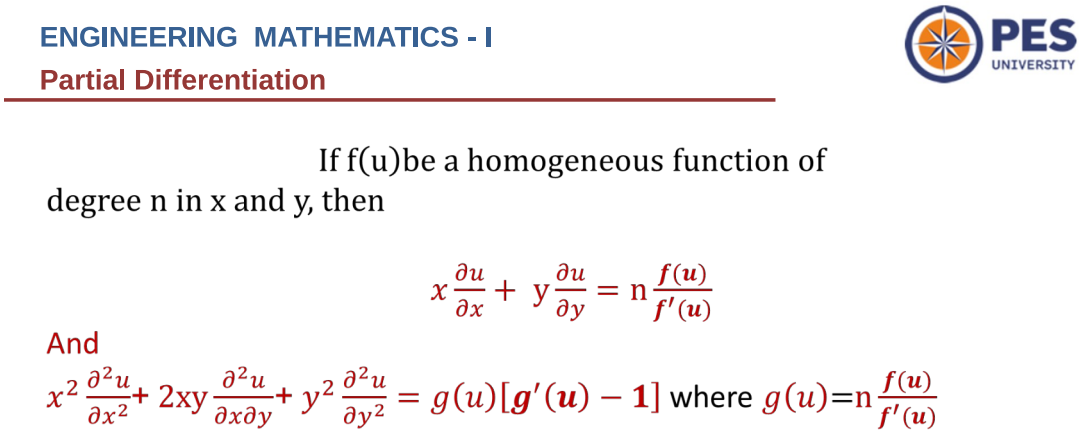

Figure G000082. Homogeneous functions and Euler s theorem · source slide page 7.

EV0000502

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000083. Homogeneous functions and Euler s theorem · source slide page 8.

EV0000505

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000483 · EV0000484 · EV0000485
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 5 Sub Topic : Homogeneous Functions and Euler's Theorem EV0000486 · EV0000487 · EV0000488
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Homogeneous function Definition 1: A function $u = f(x, y)$ is said to be a homogeneous function of degree n if it can be expressed in $y\ x\ x\ y$ the form $xng\ or\ yng$, g being the arbitrary function. For example: Consider $u = 3x + 4y = x(3 + 4(y/x))\ y = x1g\ x$ Therefore u is a homogeneous function of degree 1. EV0000489 · EV0000490 · EV0000491
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Homogeneous function Definition 2: A function $u = f(x, y, z)$ is said to be a homogeneous function of degree n if it can be expressed in y , $z\ x$, $z\ x$, y the form $xng\ or\ yng$, or zng , g being the $x\ x\ y\ y\ z\ z$ arbitrary function. For example: Consider $u = x3 + y3 + z3 + 3xyz\ 2 + 3 + 3\ y\ x\ z\ x\ y\ z\ x\ x = x3\ 1 + y$, $z = x3g\ x\ x$ Therefore u is a homogeneous function of degree 3. EV0000492 · EV0000493 · EV0000494

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Homogeneous function A constant function is homogeneous of degree 0. Linear functions are homogeneous of degree 1. The degree of the homogeneous function can be positive, negative or zero. How to relate homogeneous functions with their partial derivatives.(Euler's Theorem). Scaling of the given function. Extensive applications in Economics like production function and Marginal productivities, Elasticity of demand etc., In Thermodynamics, Extensive variables are homogeneous of degree 1. Intensive variables are homogeneous of degree 0. EV0000495 · EV0000496 · EV0000497
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Euler's Theorem on Homogeneous Functions Statement 1:If $u = f(x, y)$ is a homogeneous function of degree n then $x \partial u + y \partial u = nu \partial x \partial y$ Corollary 1: If $u = f(x, y)$ is a homogeneous function of degree n then $\partial 2u \partial 2u \partial 2u x2 + 2xy + y2 = n n - 1 u \partial x2 \partial x\partial y \partial y2$ Statement 2:If $u = f(x, y, z)$ is a homogeneous function of degree n then $x \partial u + y \partial u + z \partial u = nu \partial x \partial y \partial z$ https://www.researchgate.net/publication/281457879_EXTENSION_OF_EULER'S_THEOREM_ON_HOMOGENEOUS_FUNCTION_TO_HIGHER_DERIVATIVES EV0000498 · EV0000499 · EV0000500
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation EV0000501 · EV0000502 · EV0000503
8. Source slide page 8 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000504 · EV0000505 · EV0000506

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.996] Hello everyone. Welcome to another session of partial differentiation. Today we'll be learning something called as Euler's theorem, which is a wonderful observation made by Euler for a certain class of functions. The advantage and what are the uses for for this Euler's theorem we can discuss later. [00:00:22.788] Before we go to Euler's theorem, we need to uh refresh our memory about homogeneous functions. Uh you might have heard uh of homogeneous function when you solved differential equations. And that was the most easiest one because any homogeneous function you could reduce into a [00:00:40.452] variable separable form, right? I hope you remember this, right? So suppose you had a differential equation which was like dy by dx equal to x square plus y square divided by um say two x y. [00:00:56.196] So you know that all the terms here are of degree two. So that means I was uh in a comfortable position to reduce this to variable separable by making the substitution y is equal to bx. But now what happens is we will discuss [00:01:15.652] We will see how to check a homogeneous function. I have a function x square plus y square. Or I may take a function which looks like x cube plus three x square y plus y cube plus three x y square. [00:01:31.044] So this function here, you know that this is also homogeneous of degree 3, as every term here is of degree 3. So in case of a polynomial, it's easy for us to visualize how does the function look like, whether it is homogeneous and what is the degree. [00:01:47.556] But let me give you another example. Suppose I'll write a function which looks like sine inverse of x plus y plus z divided by something like x cube plus y cube plus z cube, or it could be tan inverse of something or e to the power of something. [00:02:04.192] So it could be e to the power of say x square plus y square, something like that. I know that this is homogeneous, but I'm not sure whether e whether e po of that is homogeneous. [00:02:15.396] Right? Similarly, this function. I don't know how to check this

function to be homogeneous. So let me erase all these things so that uh we'll go back to the definition. So now I need a shortcut to check whether a given function is homogeneous. If at all it is homogeneous, what is the degree of that function? [00:02:36.068] Right. So we are interested in finding two things. One is is the function homogeneous? And if it is homogeneous, what is its degree? [00:02:45.476] Let's say the definition says that a function u is equal to f of x y. Again, I'm restricting myself to a bivariate in the beginning. But if you want to go for a multivariable function, definitely you can do it by extending the definition. Okay. So this is said to be homogeneous function of degree n. [00:03:04.164] If it can be expressed in the form of either x to the power of n into some function of y by x, or if I can write it as y to the power of some m into some other function of x by y. [00:03:21.508] Then we say that the function is homogeneous of degree n in this case and m in this case. What do I mean by that? See, suppose I I'll take this example, x square plus y square. [00:03:34.436] How do I check if this is homogeneous? If I take out x square common, can I write this function as 1 plus y square pi x square. So here this is a function of x y or y bytes. So I can write this expression as x square into some function of y bytes. [00:03:54.692] That means this function is homogeneous of degree. [00:04:01.348] So is the definition clear? A function f of x y is said to be homogeneous of degree n. If it can be expressed in the form of x to the power of n into function of y by x or y par n into function of x by y. Whichever is convenient, you can choose according to the problem. [00:04:20.868] So in this case, uh in this example u is equal to 3x plus y, it's a homogeneous function of degree 1. Now suppose I have a function of three independent variables. So I have say u is a function of x, y, and z. [00:04:38.596] Then how am I going to uh find if it is homogeneous or not? Again we'll use the same definition, but now it's not just y by x, but it is z by x also. Correct? So like you see in this example. [00:04:52.608] you have x q plus y q plus z q plus three x y z I can take out x cube common and if I take out x cube common I can g get the remainder remaining terms in functions of y by x as well as z by x. Right.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:09.060] So if you can write a function in the form of x to the power of n into function of y by x, comma z by x, or y to the power of n into function of x by y, comma z by y. [00:05:22.596] Or it could be other way also. It could be using z power n. Z power n into function of x by z and y by z. We call that that is a homogeneous function of degree n negative. So when we solve problems, it will be more clear for you to how to check a function is homogeneous. [00:05:42.212] So basically what you need to do take out x to the power of n comma and express the remaining function in terms of y by x and z by x. Let's say you had another variable w, it should be w by x. So you can continue this, right? [00:05:57.988] Or you can write this as y to the power of n into some other function of now x by y and z by y and w by y and that continues. So whichever is convenient to check, you can do that. So hopefully the definition of homogeneous function is clear, right? [00:06:16.388] Now, there is a vast application of this homogeneous functions and Euler's theorem. [00:06:22.500] So before I go to Euler's theorem, um, by the way, he is Euler, as you can see in all his photographs, his one eye is shorter because he was uh blind by one eye. Um, in spite of that, um [00:06:36.676] he was an extraordinarily intelligent uh mathematician. He didn't had any former schooling or anything, but um of course uh uh he was with the family of uh Legrange or Barnooli, I don't remember, but [00:06:52.420] That's how uh if you're interested you can read more about uh who Euler was and what all his contributions are. So these are the few observations we can make uh before we define

what what's Euler's theorem is. As you can see, a constant function definitely is a homogeneous function of degree zero. [00:07:11.876] And of course the converse also works. So suppose you have a function which is of degree 0, then it should be a constant function. And linear functions as we saw in the previous example, so we took 3x plus 4y and we saw that this linear function is of degree 1 always. [00:07:29.220] So we know up till uh you know quintiple so suppose you can get a uh polynomial which is of degree five, then we can write it as a homogeneous if it is homogeneous, we can call the degrees five. [00:07:42.468] And it need not be just positive integer. Okay. Degree can be positive, negative, zero, could be anything. [00:07:50.660] So just to show you an example, suppose I had uh say root x plus root y divided by x plus y, if suppose this was my function, you know that I can take out root x common from the numerator and write this as 1 plus root of y by x. [00:08:06.416] And I can take out an x common from the denominator, right? This is 1 plus y by x. [00:08:12.452] So if I simplify this little bit, I can write this as x to the power of half minus 1 into some function of y by x. So this will be x to the power of minus half. That means this function here is a homogeneous function of degree minus half. [00:08:29.540] Right? So you don't need to worry about what the degree has to be. It's not always positive, negative. It could be rational, it could be positive, negative integer, or even zero. It's okay. [00:08:41.700] Ah so the remaining results I'll uh uh come back to these things when I define Euler's theorem. So let's go uh further and define. So what is a statement of Euler's theorem is if you have a homogeneous function. [00:08:57.732] Suppose u is a homogeneous function of degree n. This is the most important one. Then when I multiply x with the partial derivative of u with respect to x, y with the partial derivative of u with respect to y, it is same as n multiplied by the function itself. [00:09:17.252] That's what I said. So if you have a homogeneous function and if you have an expression like this, you don't have to simplify anything at all. So all you need to do is check if this is homogeneous. [00:09:30.564] Check the degree, then you can say directly that x u x plus y u y. So another form to write this is x into u x plus y into u y is equal to n into u. Okay. So this is a very very interesting result. [00:09:46.340] No wow, what is it happening geometrically? I'm finding the slope of the tangent parallel to x axis, but I'm multiplying that with x. So there is a scaling happening. I'm finding the slope or change along the y direction multiplied by y.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.080] So when I combine all the changes that I study, what I will get is just the scaling of the function itself. So basically you can think that there is a change that is happening along the x direction. [00:10:18.596] There is a change that is happening along the y direction for a surface, right? You study all those changes, then what you get is stacking of the functions one above the other end times. That's what you get. [00:10:31.300] So maybe um if I take the equation of a cylinder, you know that x square plus y square equal to c will represent a cylinder for me, right? Now you know that when I uh differentiate, what do I get if I differentiate this partially with respect to x? So let's say uh this is your u. [00:10:49.316] Okay. So what is UX? Ux is 2x and uh uy will be 2y again. So if I substitute in this equation I'll get x into 2x plus uh y into 2y, that will be just 2x square. [00:11:06.372] Plus 2 y square, or this is just 2 into the function itself. And this is a homogeneous function of degree 2. So basically what Euler observed is for a homogeneous function, homogeneous means which is smooth everywhere. For a homogeneous function, [00:11:24.292] The change along the x direction when scaled with x. Change along the y direction scaled with y will give you the entire function itself scaled by the factor of the degree. [00:11:38.532] Okay. So I might be uh saying too many

words, but it is not that uh complicated. Okay. Uh now there is very uh very many interesting uh results about this Euler's theorem. Uh but let's see what is the extension of Euler's theorem. This is for the first derivative. [00:11:56.516] Suppose because I have only ux and u y. What if if I want to extend it to a function of second variable? [00:12:04.612] So how many second order partial derivative exist? We discussed this in the previous lectures. So you know that uxx exists and then uy, and then you have two mixed order partial derivatives. You have ux y and uyx. But definitely they are equal. [00:12:21.380] Okay, because first order partial derivatives are continuous. So what Euler says is because this is second order, you scale it by 2. So when you multiply x square with u x, 2 x y with u x y square with u y, then it will be n into n minus 1 into u. [00:12:43.588] So it is scaled by a factor of n into n minus 1 into e. Similarly, suppose I want to go for an one more higher order derivative. So uh suppose I want to take the third order derivatives. How many third order partial derivatives can exist? [00:13:00.676] It is U XXX, right? Uy and then U XY. U X Y X. And then U X Y. [00:13:16.484] And then U Y Y X and then U Y X Y. Right? [00:13:25.860] So yesterday we just checked how many uh partial derivatives you can have. It's of order um three, right? So and uh there are uh two independent variable, so I can have another one. So basically I can have one more, so there'll be eight partial derivatives. [00:13:44.452] So it will be u y x y is there x y x y y x y is there anything that is left? So maybe xyx is missing. No x y x I did. Okay. So I'll take uh uh y. [00:14:04.644] X Y is also done. [00:14:12.100] Okay. So there is um there is one more partial derivative that comes. But anyway, so you can check the mixed order partial derivative and what happens to the right hand side will be n into n minus one into n minus two into u. That's what you're going to get. As in when you add. [00:14:28.400] one more variable to it. Okay. Um so let's see now what is the uh suppose I have a function of three independent variables. So u is a function of x, y, and z. [00:14:42.244] Then if it is homogeneous, so u is equal to f of x y z is a homogeneous function of degree n, then I get x u x plus y u y plus z u z equal to u. Okay. At this point I want you to explore [00:14:59.428] What is the corollary one that we did? So what is the higher order Euler's theorem?

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:06.148] So you can check what is a higher order Euler's theorem for a function of three independent variables. So I have given you a paper here if somebody is interested in what are the applications of Euler's theorem and its extension of Euler's theorem, you can go through this paper. [00:15:22.976] and see what is the least recent research that is happening in this field. Okay. [00:15:27.972] So now let's see uh what is uh what are the applications of this uh Euler's theorem. As I was telling, basically Euler's theorem is to relate homogeneous function with their partial derivative. See, that's what Euler observed. [00:15:45.092] But if I have a uh partial derivatives of a function of two independent variables, how am I going to relate them uh with the degree of their homogeneity? [00:15:57.060] So basically it also gives the scaling of the given function, that's because of n into u, that's what we discussed just now. Uh and then uh it has extensive applications in economics. [00:16:09.124] So it it can be like uh used in production function. Actual form of oiless theorem they use in production function, marginal productivity, elasticity of demand. I didn't put the equations in detail because uh, you know, most of you are engineers and uh you might want an engineering application to Euless theorem. [00:16:26.816] Uh so one uh thing that you can observe is in thermodynamics, they use oiless theorem. So all the extensive variables are homogeneous of degree one. So whenever you say it is a homogeneous function, you can apply oiless theory. And intensive variables are homogeneous

functions of degree. [00:16:46.820] Zero. Okay. So what do I mean by intensive and extensive is intensive variables may be like mass, which is all the factors depends on that, and extensive can be like the pressure and temperature and other things. [00:17:00.612] Um so anyway, you you might have learnt about these variables when you did uh thermodynamics. Maybe in your physics course you can uh mm uh better understand what is happening and uh what is the Euler's theorem. [00:17:13.860] what does it say and other things, right? So uh that's the definition and applications. Now what we'll do is we'll solve some problems on um uh Euler's theorem so that uh things will be a little more clear. So l let's do a quick recap on um uh what [00:17:32.836] we did today, uh in today's session. Because the Euler's theorem is the most important one wherein um you can reduce the problem. Actually you don't need to find the partial derivatives at all, right? So what I mean is suppose uh [00:17:48.484] The first thing that we did is we defined a homogeneous function. So these are the things that we did today. We defined what's a homogeneous function. [00:17:59.460] Basically, uh along with the degree. So if you know how to check if a function is homogeneous, you will definitely know the degree. And if you know the degree of the homogeneous function, then Euler's theorem says that x into ux [00:18:14.624] plus y into u y is equal to n into u. So basically the change that is happening along the x axis, along the y axis, it gets combined and it just scales the function itself. Okay. [00:18:28.292] And then we also observe the extension of this for the second order. So if you have a function of two independent variables, then x square u x plus 2xy u x y uh plus y square uy is also n into n minus 1 into [00:18:48.100] So I would like you to uh um do a little bit of research and see if you can extend Euler's theorem, as I was mentioning, this paper will help you, but uh see if you can extend this Euler's theorem uh for N variables. [00:19:04.772] N independent variables. So here I have taken three independent variables and I have given first order. So try to extend this to higher order uh partial derivatives or extend Euler's theorem to higher order variables. So I took three, you can try for four, five, and other things. [00:19:22.084] Okay. So in the next session we will solve problems on Euler's theorem.

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Full transcript: 3184 normalized words, 77 timestamped segments, 1167.936 seconds duration. Low-confidence segments flagged: 0.

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10 Problems on Euler's Theorem

This lesson is indexed from the source video titled Problems on Euler's Theorem. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000138 · EV0000139

Graphs and diagrams

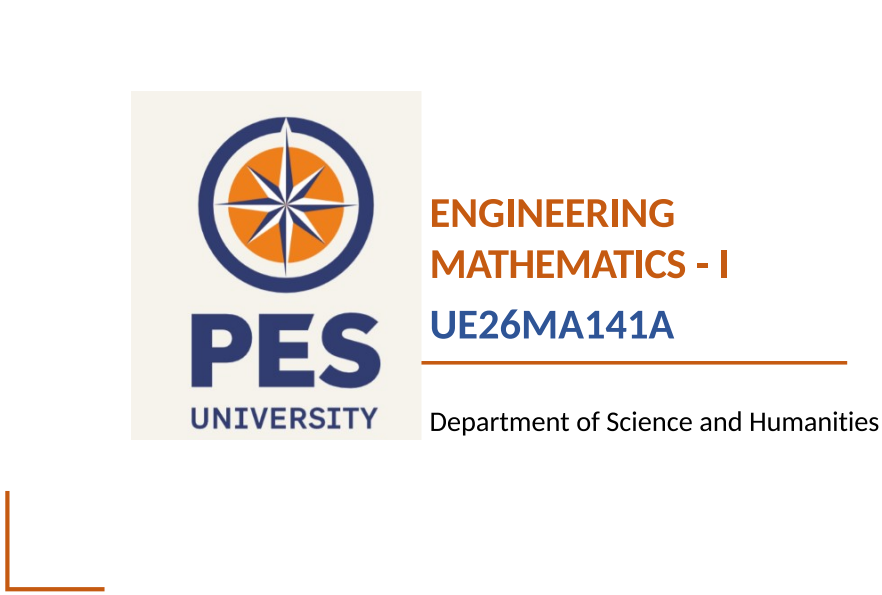

Figure G000084. Problems on Euler s theorem · source slide page 1.

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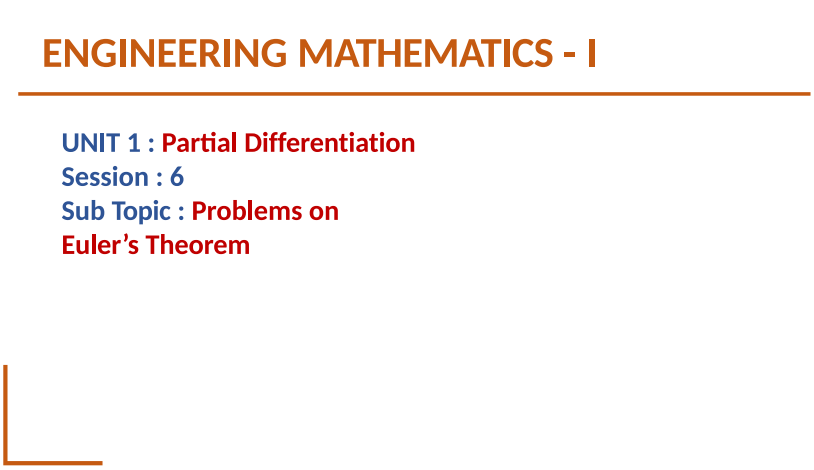

Figure G000085. Problems on Euler s theorem · source slide page 2.

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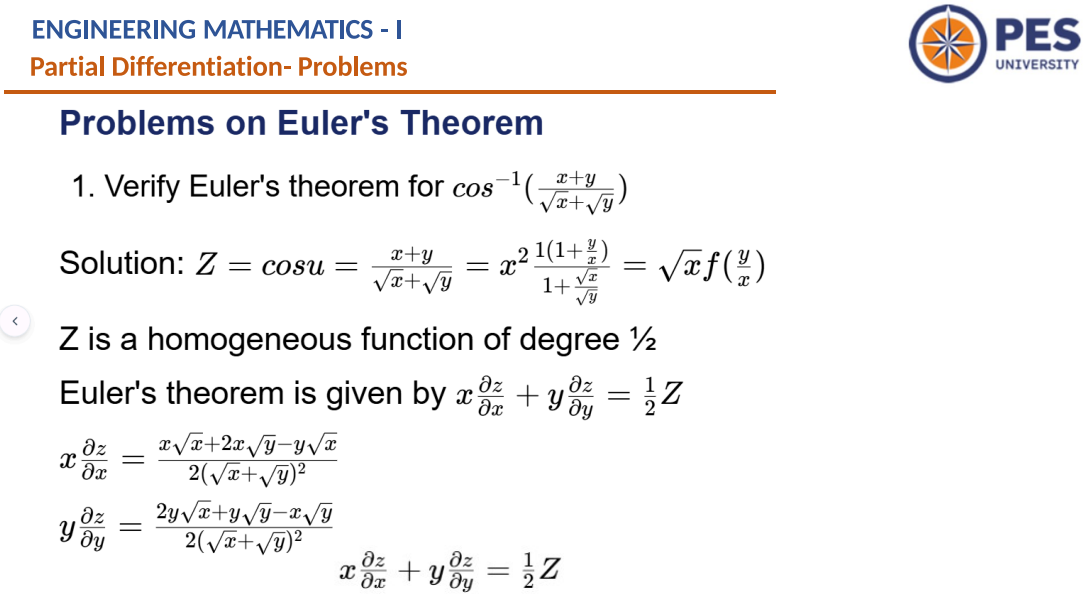

Figure G000086. Problems on Euler s theorem · source slide page 3.

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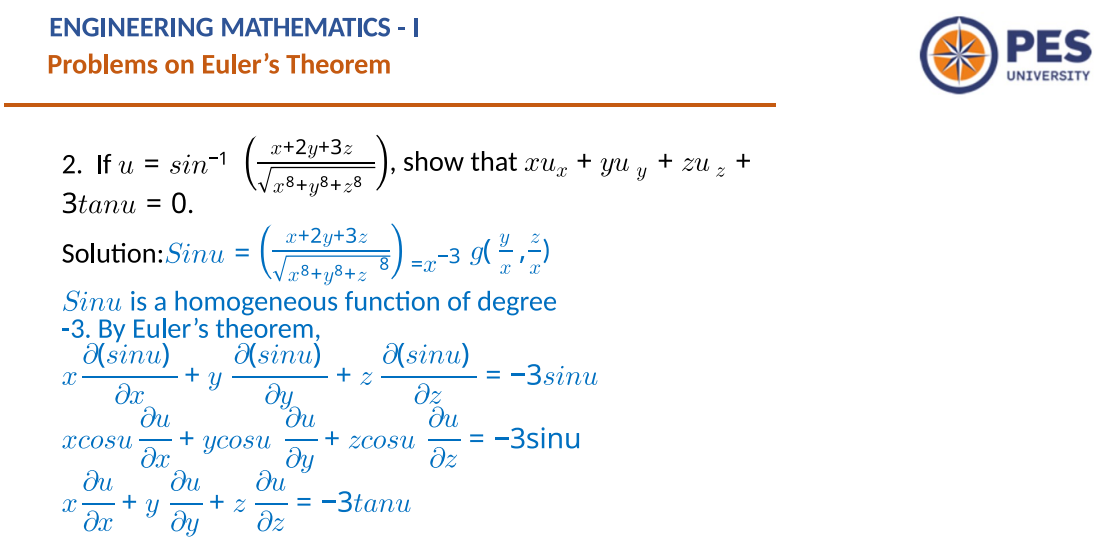

Figure G000087. Problems on Euler s theorem · source slide page 4.

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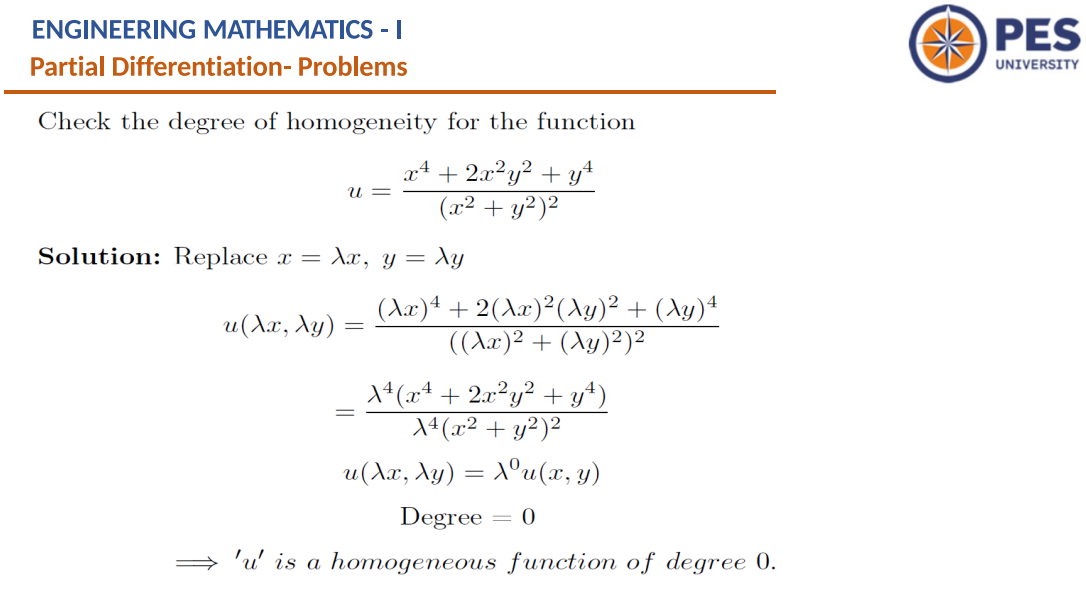

Figure G000088. Problems on Euler s theorem · source slide page 5.

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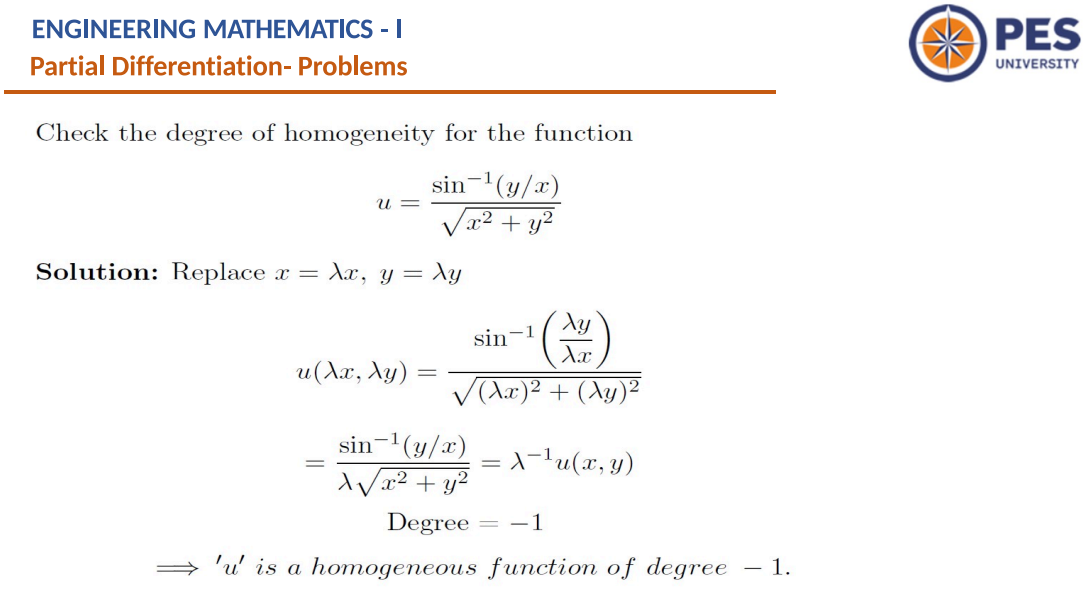

Figure G000089. Problems on Euler s theorem · source slide page 6.

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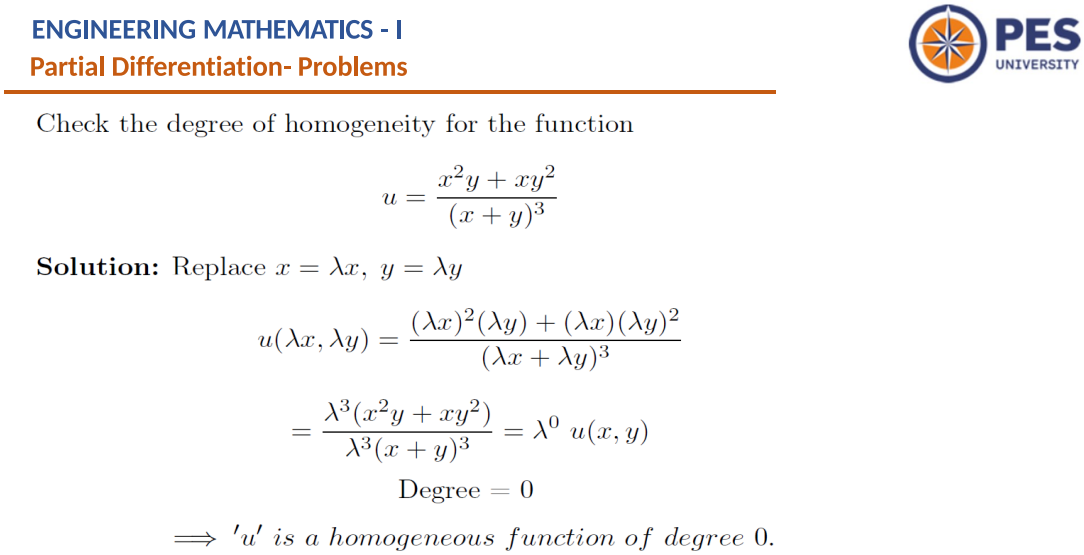

Figure G000090. Problems on Euler s theorem · source slide page 7.

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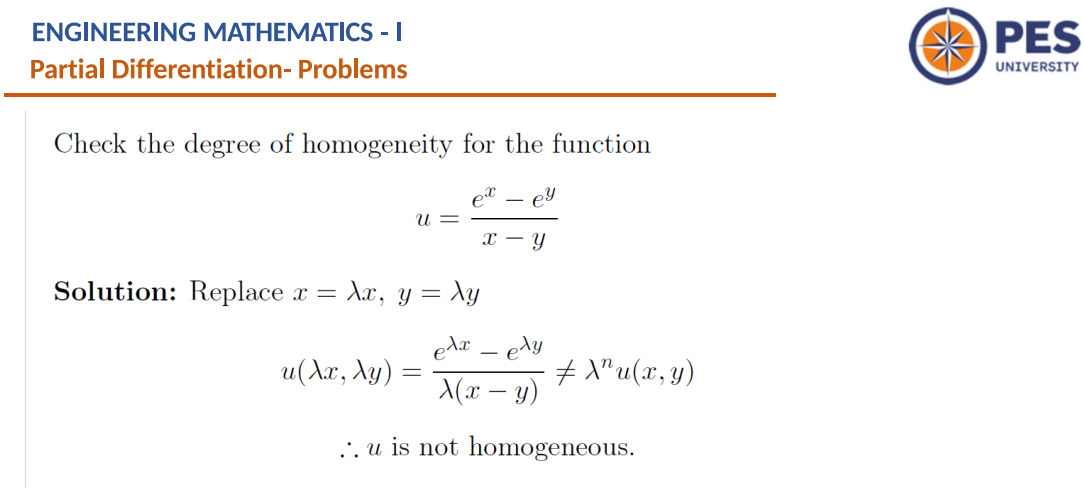

Figure G000091. Problems on Euler s theorem · source slide page 8.

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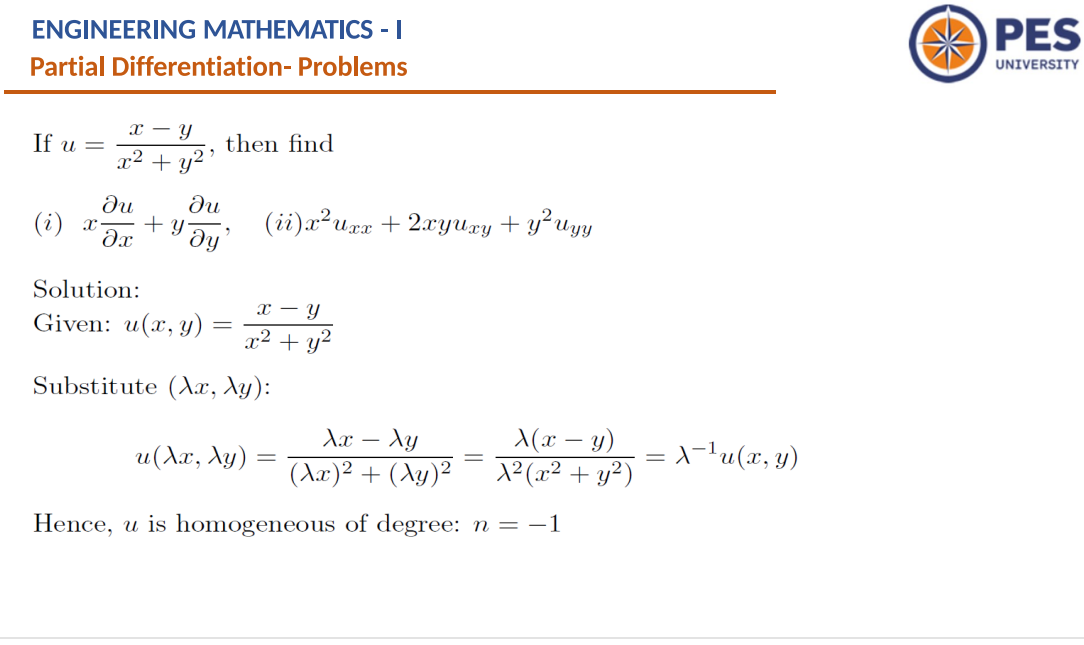

Figure G000092. Problems on Euler s theorem · source slide page 9.

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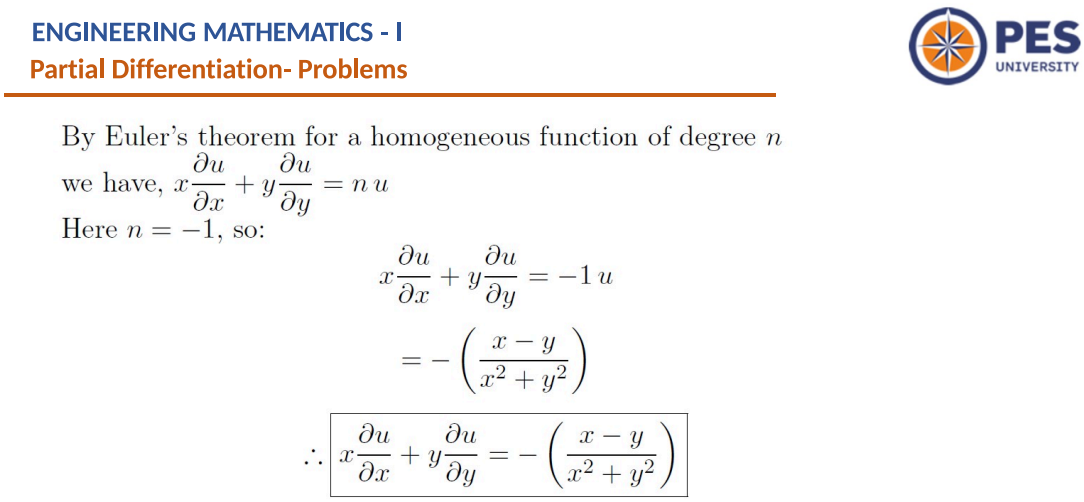

Figure G000093. Problems on Euler s theorem · source slide page 10.

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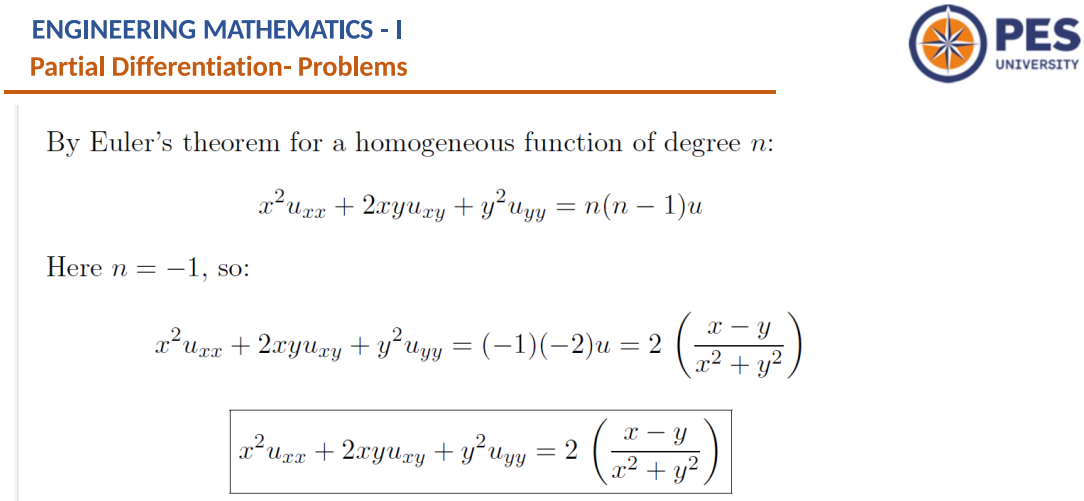

Figure G000094. Problems on Euler s theorem · source slide page 11.

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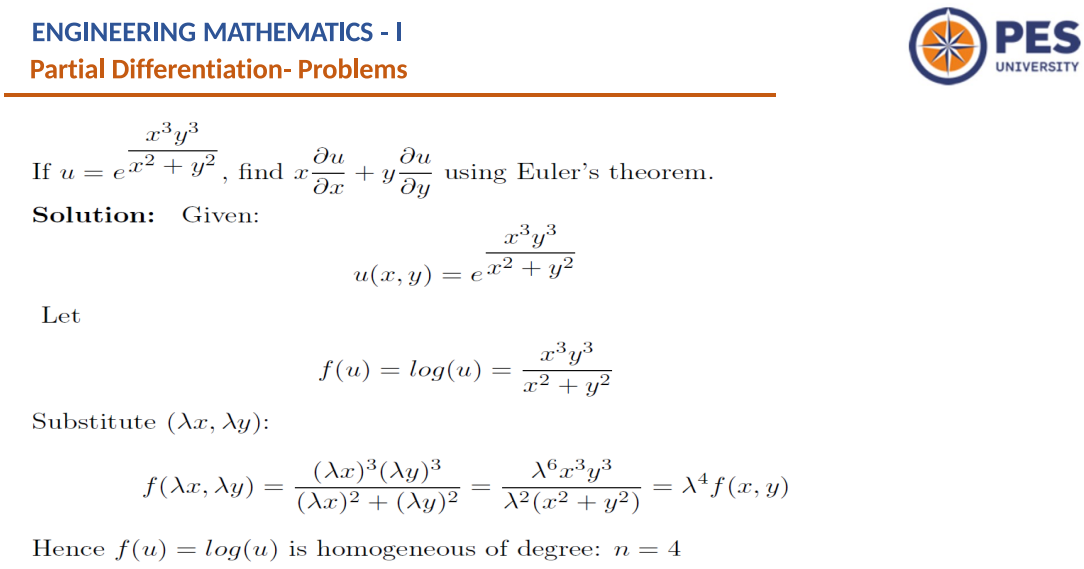

Figure G000095. Problems on Euler s theorem · source slide page 12.

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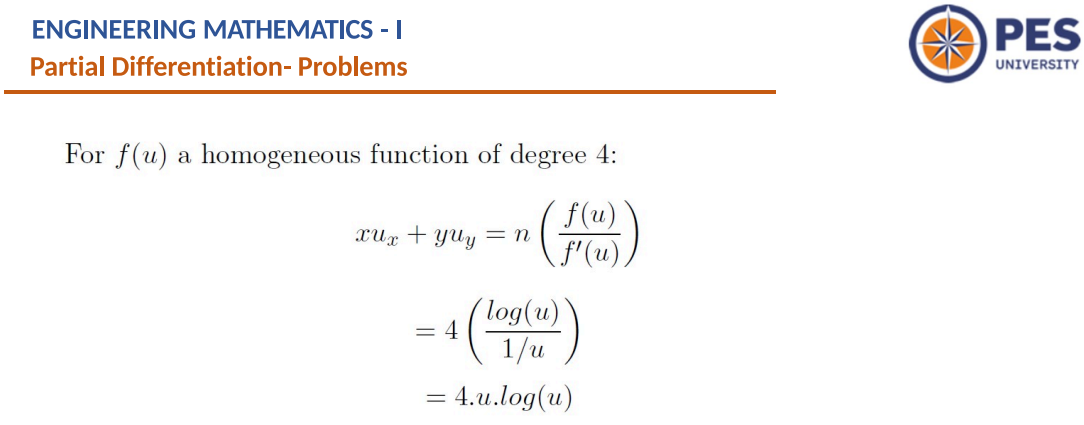

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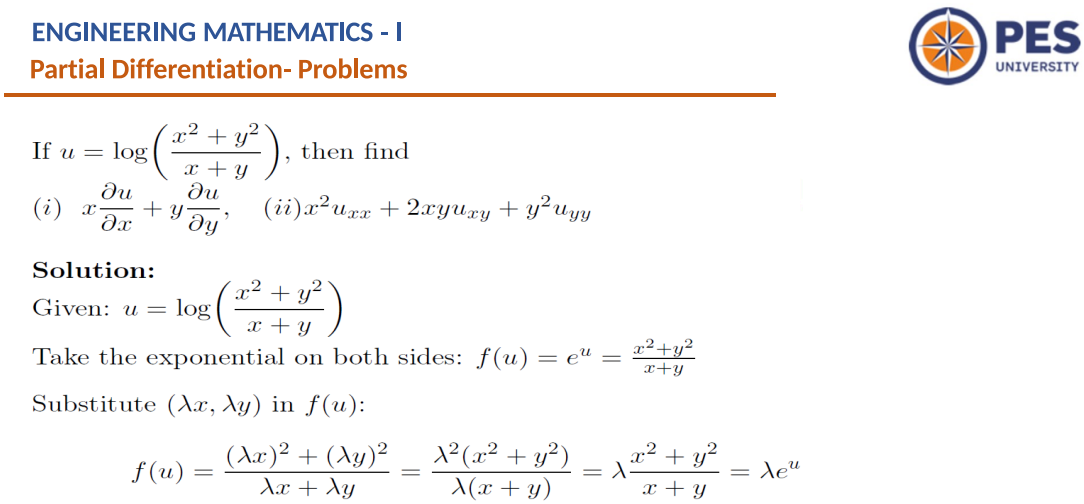

Figure G000097. Problems on Euler s theorem · source slide page 14.

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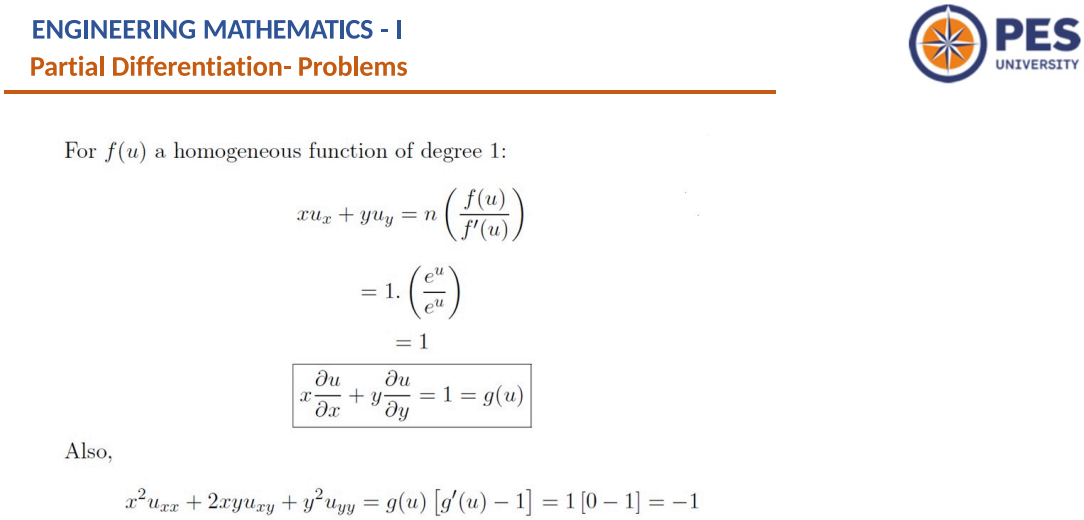

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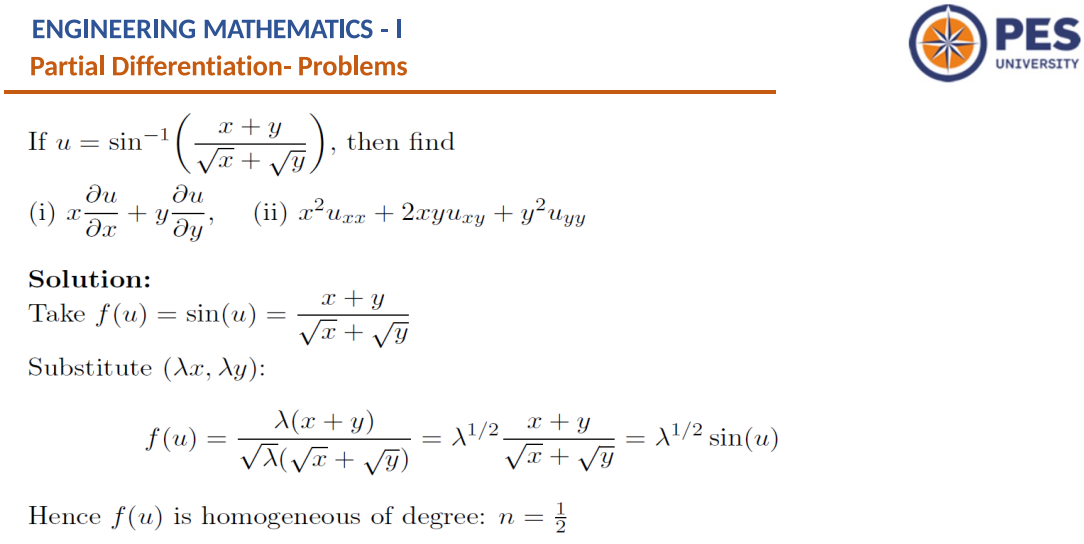

Figure G000099. Problems on Euler s theorem · source slide page 16.

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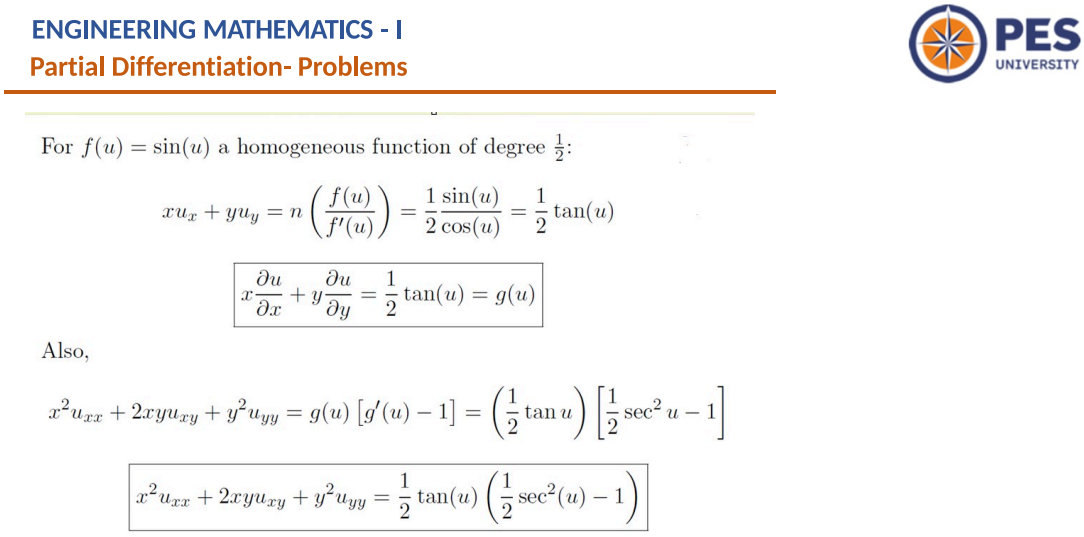

Figure G000100. Problems on Euler s theorem · source slide page 17.

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Figure G000101. Problems on Euler s theorem · source slide page 18.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000507 · EV0000508 · EV0000509
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 6 Sub Topic : Problems on Euler's Theorem EV0000510 · EV0000511 · EV0000512
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Problems on Euler's Theorem 1. Verify Euler's theorem for cos c+y Vc+√y c+y .2 1(1+) Solution: Z = cosu = =√f() √c+√y 1+1 Vc Vy Z is a homogeneous function of degree %2 Euler's theorem is given by c z +yy 0z Z √x+2y-y√ C 2(Vc+/g)2 2y√c+y√y-c√y y dy 2(vx+√g)2 EV0000513 · EV0000514 · EV0000515
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Problems on Euler's Theorem x+2y+3z 2. If u = sin−1 3$tanu$ = 0. Solution:$Sinu$ = , show that xu + yu + zu + x y z x8+y8+z8 =x−3 x+2y+3z y z g(,) 8 x x x8+y8+z $Sinu$ is a homogeneous function of degree -3. By Euler's theorem, ∂($sinu$) ∂($sinu$) ∂($sinu$) x + y + z = −3$sinu$ ∂x ∂y ∂z ∂u ∂u ∂u $xcosu$ + $ycosu$ + $zcosu$ = −3sinu ∂x ∂y ∂z ∂u ∂u ∂u x + y + z = −3$tanu$ ∂x ∂y ∂z EV0000516 · EV0000517 · EV0000518

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Check the degree of homogeneity for the function x4 + 2x2y2 + y4 u三 (c2 + y2)2 Solution: Replace = 入r, y = >y (入x)4 + 2(入x)²(入y)2 + (入y)4 u(Xx,Xy) = ((入c)2 + (入y)²)2 X4(x4 + 2x²y2 + y4) X4(x2 + y2)2 u(入x, Xy) = X°u(c, y) Degree = 0 →> 'u' is a homogeneous function of degree O. EV0000519 · EV0000520 · EV0000521

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Check the degree of homogeneity for the function sin-1(y/α) u= √x2 + y2 Solution: Replace = 入c, y = 入y sin- u(入c,入y) = (入x)2 +(入y)2 sin-1(y/α) =入-1u(x,y) 入Vx2 + y2 Degree = -1 → 'u' is a homogeneous function of degree - 1. EV0000522 · EV0000523 · EV0000524

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Check the degree of homogeneity for the function x²y + xy2 u= (x+ y)3 Solution: Replace = 入r, y = >y (入c + 入y)3 X3(x²y + cy²) =X0 u(x, y) X3(c + y)3 Degree = 0 → 'u' is a homogeneous function of degree O. EV0000525 · EV0000526 · EV0000527

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Check the degree of homogeneity for the function ed - ey u c-y Solution: Replace = Xc, y = Xy e^x - e^y u(入c, 入y) ≠ 入nu(c,y) X(x -y) .'. u is not homogeneous. EV0000528 · EV0000529 · EV0000530

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] c-y Ifu= du Ou +yy (ii)x²ucx + 2cyuxy + y²uyy (i) C- Solution: c-y Given: u(,y) = x2 + y2 Substitute (入c, Xy): 入c - 入y X(c -y) u(入x,>y)= =入-lu(c,y) (入x)2 +(入y)2 X2(x2 + y²) Hence, u is homogeneous of degree: n = -1 EV0000531 · EV0000532 · EV0000533

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] By Euler's theorem for a homogeneous function of degree n du Ou we have, - +y nu ϑc Oy Here n = -1, so: du du +y -1u Oy c-y 2 + y2 Ou Ou c-y +y Oy x2 + y2 EV0000534 · EV0000535 · EV0000536

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] By Euler's theorem for a homogeneous function of degree n: x²uex + 2xyury + y²uyy = n(n - 1)u Here n = -1, so: x²uee + 2ryury + y²uly = (-1)(-2)u = 2 c-y x2 + y2 c-y x²uex + 2xyuxy + y²ugy g=2 x2 + y2 EV0000537 · EV0000538 · EV0000539

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] du du +yy using Euler's theorem. Solution: Given: u(x,y) = ex2 + y2 Let f(u) = log(u)= 2²+y2 x3y3 Substitute (入x,\y): (入x)(入y)3 X6x3y3 f(入x,>y)= = X4f(x,y) (入x)2 + (入y)2 X2(x2 + y²) Hence f(u) = log(u) is homogeneous of degree: n = 4 EV0000540 · EV0000541 · EV0000542

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] For f(u) a homogeneous function of degree 4: f(u) cuc + yuy 三n f'(u) log(u) 4 1/u = 4.u.log(u) EV0000543 · EV0000544 · EV0000545

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Ifu = log then find c+y du Ou +y (ii)x²uxx + 2cyuxy + y²uyy (i) C 0c dy Solution: x² + y Given: u = log c+y c+y Substitute (入x,>y) in f(u): X²(x² +y²) x² + y2 = 入eu 入c + 入y 入(x +y) c+y Hence, f(u) is a homogeneous function of degree: n = 1 EV0000546 · EV0000547 · EV0000548

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] For f(u) a homogeneous function of degree 1: f(u) cur + yuy = n f'(u) = 1 du Ou +y0y = 1 = g(u) 0c Also, x²ucx + 2xyuxy + y²uyy = g(u) [g(u) - 1] = 1[0 - 1] = -1 EV0000549 · EV0000550 · EV0000551

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] c+y If u = sin . then find √c+√ ()0 du Ou +yy (ii) x²ucx + 2cyuxy + y²uyy Solution: c+y Substitute (入c, 入y): X(x + y) f(u)=Vx(√E+√) = A1/2_+y = X1/2 sin(u) √c+√y Hence f(u) is homogeneous of degree: n = 2 EV0000552 · EV0000553 · EV0000554

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] For f(u) = sin(u) a homogeneous function of degree 2: f(u) 1 sin(u) 1 cur + yuy = n tan(u) f'(u) 2 cos(u) 2 du du 1 +y tan(u) = g(u) 0c dy 2 Also, x²uee + 2xyuxy + y²uy = g(u) [g(u) - 1] = tan u sec²u- 1 2 x²ue + 2xyuey + y²uy = tan(u) 1 sec²(u) - 1 2 EV0000555 · EV0000556 · EV0000557

18. Source slide page 18 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000558 · EV0000559 · EV0000560

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] Solve the problem let's check the definition of Euler's theorem. If f of x y is homogeneous of degree n, then x fx plus y f y is equal to n into u. So for the given problem it's difficult for us to check if it is homogeneous readily. [00:00:18.084] So what we will do is I will take z to be cos inverse of x plus y by root x plus root y. So you I can take it as cos z. So I'll apply cos on both sides, I'll get x plus y by root x plus root y. [00:00:32.548] And it's uh we can check that this function is homogeneous conveniently by taking an x common and root x common from the denominator. [00:00:41.988] So we got u to be a homogeneous function of degree half. So x u x plus y u y is equal to half into u. That's a statement of Euler's theorem. Now we need to verify this, right? So let's find out that [00:00:56.560] The partial derivative of u with respect to x. So u we have x plus y by root x plus root y. [00:01:05.700] So, what will be u x then? Ux will be you can apply either quotient rule or product rule. So I will apply quotient rule to this root x plus root y into 1 minus x plus y into 1 by 2 root of x divided by [00:01:25.284] root x plus root y the whole square. That's u x. And similarly we can write u y to be root x plus root y minus x plus y into 1 over 2 root y'all. [00:01:46.148] whole divided by root x plus root y the whole square. So what we are supposed to do is now multiply this with x and this with y and then we have to combine them together. Right. So what will be x u x [00:02:05.188] Plus Y, U Y will look like [00:02:09.060] So if I multiply the first equation with x, I can write this as x by root x plus root y after some simplification, right? I can cancel this minus I'm multiply it with x. So I can cancel a root x with this, so it will be half. [00:02:27.364] x plus y into root x divided by root x plus root y the whole square. This is what I get from the first equation. So that is from u x plus [00:02:41.348] I can go with the second equation which is y into u y. So again if I do the similar thing I will get y by root x plus root y minus 1 half of x plus y divided by root x plus root y. [00:02:59.876] into root y for the second term. So if you observe carefully I can group these two terms because they have common denominator and I can group these two terms. So if I group these two I'll get x plus y [00:03:17.220] divided by root x plus root y that's for the first term minus 1 half into if I take out x plus y by root x plus root y the whole square I will be left with

root x in the first case [00:03:37.220] plus root y in the second case. Right? So if you observe little carefully, I can cancel this with one of the terms in the denominator. So that I'll be left with [00:03:52.068] x plus y divided by root x plus root y minus 1 half of x plus y divided by root x plus root y. And that will give me half of x plus y by root x plus root y. [00:04:11.972] And I know that this is nothing but u. So I will get half into u and that will be my right hand side. So I started with x u x plus y u y is equal to half u, and then I proved that I'll get the right hand side. Started with the left hand side. So uh this is the left hand side. [00:04:30.980] X UX plus Y Uy and whatever the resultant I got is the right hand. Let's try another problem on Euler's theorem. [00:04:40.836] Now the statement says if u is equal to sine inverse of x plus two y plus three z divided by square root of x par eight plus y par eight plus z par eight, show that x u x plus y u y plus z u z is minus three time. Again, [00:04:57.252] To apply Euler's theorem it's difficult. See just imagine differentiating this with respect to X, with respect to Y, with respect to Z, substituting here. So that's a very tedious task, right? Instead what we can do is we can take uh let's take uppercase z to be some sine u.

EV0000138 · EV0000139 · EV0000140 · EV0000141

Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.212] which will be x plus two y plus three z divided by square root of x par eight plus y par eight plus z to the power of eight. Right? [00:05:32.996] So if we take this uppercase z to be sin u then it's easier for us to check that this function here is homogeneous of degree, uh we'll check the degree. So I can take out x common from the numerator and write this as 1 plus uh [00:05:49.488] 2 y by x plus 3 z by x divided by root of x power 8 will be x to the power of 4 if I take out common, I'll get under root of 1 plus y by x to the power of 8 plus z by x to the power of 8. [00:06:10.084] Right. I don't need to worry about what is here because I can write this as x to the power of minus 3 into some function of y by x, comma z by x. So from the definition of homogeneous function, it's clear that sine u [00:06:28.388] Which is nothing but Z, this uppercase Z. [00:06:32.804] So we s got that z is equal to sin u, which is x to the power of minus three into some function of y by x and z by x. That means z is a homogeneous function of degree minus three. So that's what we got. So z is homogeneous of degree three minus three. [00:06:54.468] That means we can apply Euler's theorem to Z, which says that x into zx plus y into z y is equal to minus 3 into z. So now let's substitute for zx, z y and z. So all these are uppercase z. [00:07:13.828] So I have another term here x z plus y z y plus z uh z. So this is a uppercase z the function, this is lowercase z. Now what is this z uppercase z equal to sine u? So we'll find its partial derivative. So z is sine u. [00:07:33.988] That means ZX will be cos u into u x. Similarly, z y will be cos u into u y. [00:07:46.308] and Z will be cos u into U Z. If we substitute these things in this equation we will get x Z into cos u into U Z plus X into [00:08:04.708] Cos u into u x plus y into cos u into u y. This is nothing but minus 3 into z. Z is nothing but sine u. So this is minus 3 sin u. Now we can divide this throughout by cos u. [00:08:22.852] And let's make a small rearrangement of this. So I will write x first. So I will get x u x plus y u y plus z u z is equal to minus three tan u because I divide it through out by cos u I'll get this result which is the required result. [00:08:41.988] So you can see that using Euler's theorem things were very simple. We could just check if the function is homogeneous, apply the statement of Euler's theorem, and we establish [00:08:53.200] the result by taking sine. So whenever you have a complicated function, you can always use Euler's theorem if you feel that the function is homogeneous and obtain the required result. Next we will solve some problems approximation.

EV0000142 · EV0000143 · EV0000144 · EV0000145

Full transcript: 1382 normalized words, 34 timestamped segments, 546.602667 seconds duration. Low-confidence segments flagged: 0.
EV0000138 · EV0000139 · EV0000140 · EV0000141

11 Problems on Euler's theorem

This lesson is indexed from the source video titled Problems on Euler's theorem. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000146 · EV0000147

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.140] Hello everyone. Welcome back to the course on engineering mathematics 1. In this session, we will solve problems on homogeneous functions and Euler's theorem. [00:00:24.164] Let us solve the first problem u equals x power four plus two times x square into y square plus y power four divided by x square [00:00:43.012] plus y square one square. So this is a function given. [00:00:52.100] Now in order to test whether the given function is homogeneous or not, we need to prove this particular condition that is u of lambda x comma lambda y equals lambda power n times u of x comma y. It means [00:01:11.364] In the given function, we replace x by lambda x, y by lambda y. So that leads to lambda power n times u of x comma y. Remember, here n can be positive, negative. [00:01:29.924] Or even zero. So let us use this approach to test whether the given function is homogeneous or not. [00:01:41.156] So let us replace x by lambda x y by lambda y in the given function so that u of x comma y becomes u of lambda x comma lambda y on the right hand side becomes lambda x power plus [00:01:59.300] 2 times lambda x square times lambda y square plus lambda y power 4 in the denominator x square becomes lambda x whole square, y square becomes lambda y square and whole power two. [00:02:17.316] So from the numerator we can take lambda power 4 common. Then we are left out with x power 4 plus 2x square y square plus y power 4. In the denominator we get lambda power 4 of x square plus y square whole square. [00:02:37.924] So this is nothing but u of lambda x comma lambda y equals lambda power 0 into u of x comma y. After lambda power 4, lambda power 4, we can write 1 to be lambda power 0. [00:02:51.652] So this indicates that the given function u is the homogeneous of degree zero. [00:03:02.916] Let us move on to the second problem. U equals sine inverse of y by x divided by square root of x square plus y square. Let's check whether this function is homogeneous or not. [00:03:20.804] Replace x by lambda x, replace y by lambda y. So that the given function u of x comma y becomes u of lambda x comma lambda y equals [00:03:34.628] Sign in us of lambda y by lambda x. So after calc is of lambda, we can get sine in was of y by x. And in the denominate denominator, it is square root of lambda x whole square plus lambda y whole square. [00:03:48.900] So we can take out lambda outside. So that leads to lambda power minus 1 times u of x comma y this indicates that the given function u is a homogeneous of degree minus 1. [00:04:06.756] Let us look at the third problem. U equals x square y plus xy square divided by x plus y power 3. [00:04:18.788] So when we use x as lambda x and y

by lambda y, after we simplify we get lambda power 0 times u of x comma y. So this indicates that the given function u is also homogeneous of degree zero. [00:04:40.740] Let us look at uh one more function. U equals e power x minus e power minus y divided by x minus y. So when we use x as lambda x and y as lamb y. [00:04:59.012] This simplification doesn't give us lambda power n times u of x comma y. It means u of lambda x comma lambda y is not equal to lambda power n times u of x comma y. This indicates the fact that the given function is not homogeneous.

EV0000146 · EV0000147 · EV0000148 · EV0000149

Transcript block 2 (source interval 000300-000599 seconds): [00:05:18.884] I hope the concept of homogeneous function is clear to you. [00:05:27.556] After understanding the concept of homogeneous functions, let us solve problems on Euler's theorem. If u equal to x minus y divided by x square plus y square, then find the value of x into dou u by dou x. [00:05:46.756] plus y into dou u by dou y and x square into dou square u by dou x square plus [00:05:55.076] 2 x y into dou square u by dou x into dou y plus y square into dou square u by dou y square. As we know that this is nothing but extension of Euler's theorem. [00:06:11.556] So in order to use Euler's theorem, first we need to prove that u is a homogeneous function of degree n. So when we look at the function u e of x comma y equals x minus y divided by x square plus y square, after using this approach. [00:06:29.412] We get to know that u is a homogeneous function of degree minus 1 because u of lambda x comma lambda y equals lambda power minus 1 times u of x comma y. [00:06:45.092] After proving u to be homogeneous function, we can directly use Euler's formula, which is nothing but x into dou by dou x plus y into dou by dou y equals n u. [00:06:59.172] Because n is minus 1, we can get this particular answer that is x into dou by dou x plus y into dou by dou y equals minus u becomes minus half x minus y divided by x square plus y square. [00:07:23.268] So in order to solve the second part of the problem, we use extension of Euler's theorem, which is nothing but this LHS equals n into n minus 1 times u. Because n is minus 1, finally we get 2 times. [00:07:42.244] X minus y divided by x square plus y square. So finally this is the answer for the second part of the problem. [00:07:59.684] So let us solve this particular problem. It is given that u equals e power x power 3 times y power 3 divided by x square plus y square, then find the value of x into dou u by dou x plus y into dou u by dou y using euler's theorem. [00:08:20.292] Observe carefully, the given function is not homogeneous. So in this kind of situation, we use this kind of approach. Let us take log on both the sides, then [00:08:34.660] U of x comma y becomes log u, then this part becomes x power 3 times y power 3 divided by x square plus y square. Then check whether f of u is homogeneous or not. So when we put x as lambda x and y by lambda y. [00:08:53.476] After we simplify, we get lambda power x power 3 y power 3 divided by lambda square of x square plus y square. So finally this is same as lambda power 4 into f of x comma y. [00:09:08.772] So this clearly indicates that f of u equal to log u is homogeneous function of degree 4. So when the original function is not homogeneous, we can use this formula that is x into dou by dou x plus y into dou by dou y. [00:09:27.588] equals n times f of u by f dash of u. So that leads to 4 times u into log u. So this is the answer for the problem. [00:09:42.308] Let us look at this problem. If u equal to log of x square plus y square divided by x plus y, then find a value of these two. Again observe carefully. [00:09:55.652] The given origin function is not homogeneous. Again, we use a different approach because log is involved, we take exponential exponential on both the sides.

EV0000150 · EV0000151 · EV0000152 · EV0000153

Transcript block 3 (source interval 000600-000899 seconds): [00:10:07.396] Then that leads to f of u equals a power u equals x square plus y square divided by x plus y. Then check whether f of u is homogeneous or not. So when we use the normal process, we get to know that f of u is a homogeneous function. [00:10:26.244] or degree n equals to 1. So when we use this standard formula when the given function is not homogeneous, the final answer for the problem is going to be x into dou by doux plus y into dou by douy equals one. [00:10:45.316] Similarly, in order to prove the second part of the problem, we can directly use this formula. X square dou square u by dou x square plus 2xy dou square u by doux dou y. [00:10:59.332] Plus y square dou square u by dou y square equals g of u of g dash of u minus one. So when we use the substitution, the final answer becomes minus one. [00:11:16.804] So finally this answer for the problem. [00:11:22.084] Let us look at this particular problem. If u equal to sine inverse of x plus y divided by square root of x plus square root of y, then find the value of these two. Again observe carefully, the given function is not homogeneous. [00:11:40.548] So, in order to solve problems of this category, so the given function can be written as sin u equals x plus y divided by square root of x plus square root of y. So now we need to prove f of is homogeneous or not. So when we replace x by lambda x, y by lambda y. [00:12:00.676] After using simplification, we get f of u equals lambda power half times sin u. So this indicates that f of u is a homogeneous of degree n equals to plus 2. [00:12:19.652] Plus one by two. So after using the standard formula, we get x into dou by doux plus y into dou by dou y equals half times tau. [00:12:33.508] In a similar manner, when we use the extension of Euler's theorem, we get this particular step. The right hand side is going to be half times tanue. [00:12:42.468] And this part is going to be the derivative of half tan u, that is nothing but half second square u, and then we have minus one. So these are the problems on homogeneous functions as well as Euler's theorem. Thank you.

EV0000154 · EV0000155 · EV0000156 · EV0000157

Full transcript: 1657 normalized words, 48 timestamped segments, 783.637 seconds duration. Low-confidence segments flagged: 0.

EV0000146 · EV0000147 · EV0000148 · EV0000149

12 Taylor's and Maclaurin's expansion

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Graphs and diagrams

Figure G000102. Taylor and Maclaurin series expansion for a function of two variables · source slide page 1.

EV0000562

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

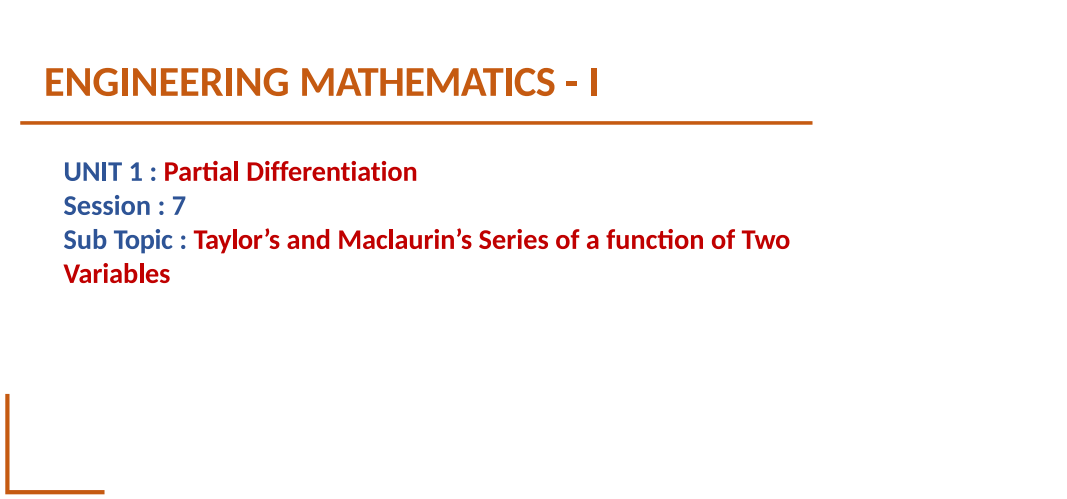

Figure G000103. Taylor and Maclaurin series expansion for a function of two variables · source slide page 2.

EV0000565

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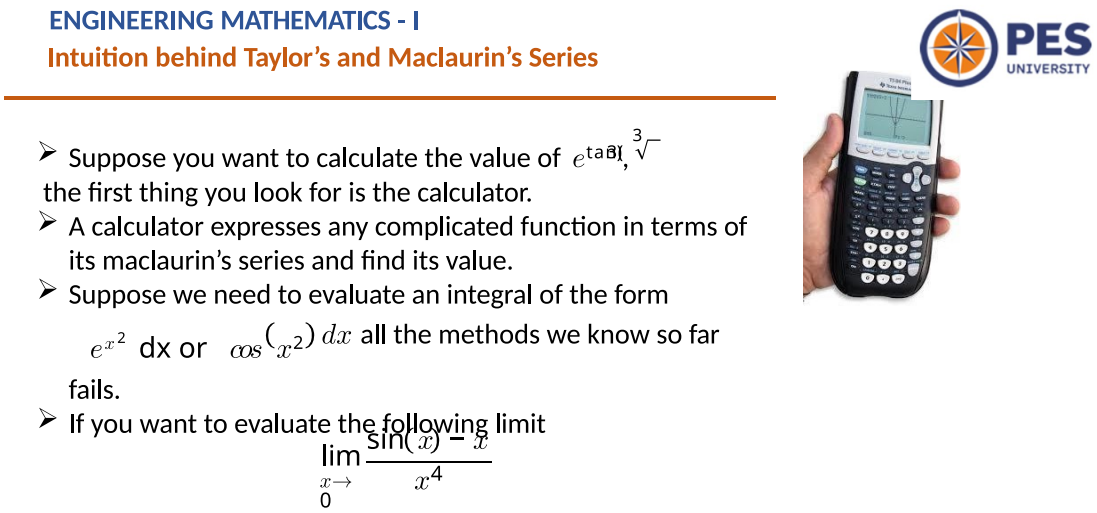

Figure G000104. Taylor and Maclaurin series expansion for a function of two variables · source slide page 3.

EV0000568

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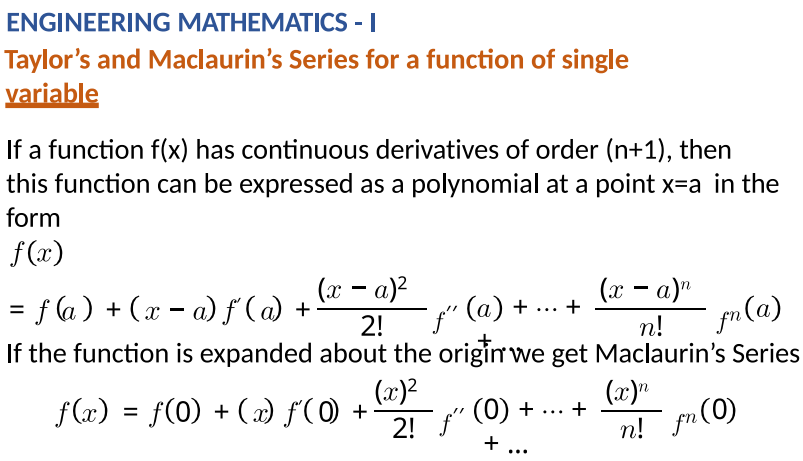

Figure G000105. Taylor and Maclaurin series expansion for a function of two variables · source slide page 4.

EV0000571

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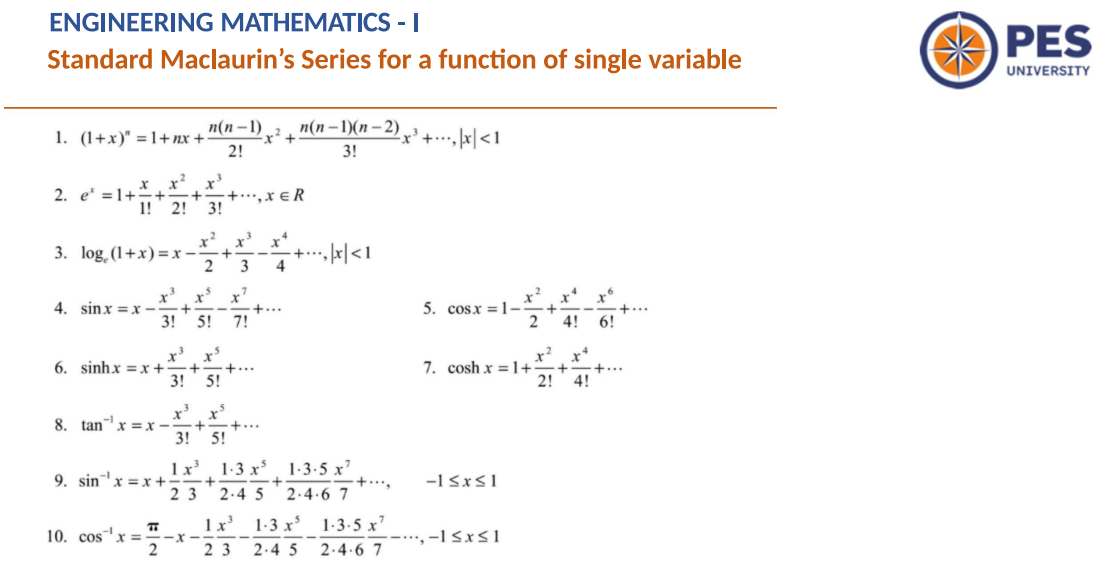

Figure G000106. Taylor and Maclaurin series expansion for a function of two variables · source slide page 5.

EV0000574

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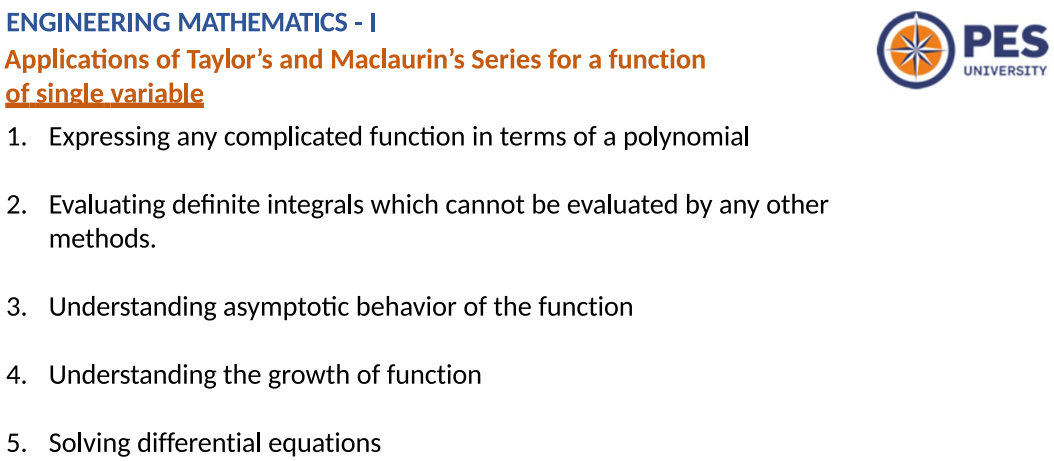

Figure G000107. Taylor and Maclaurin series expansion for a function of two variables · source slide page 6.

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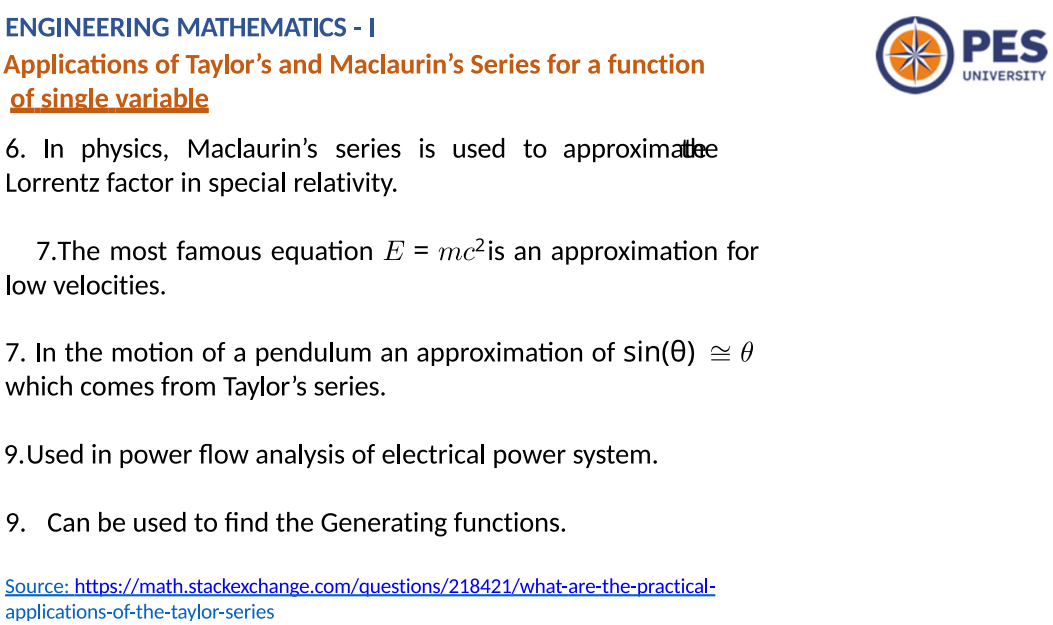

Figure G000108. Taylor and Maclaurin series expansion for a function of two variables · source slide page 7.

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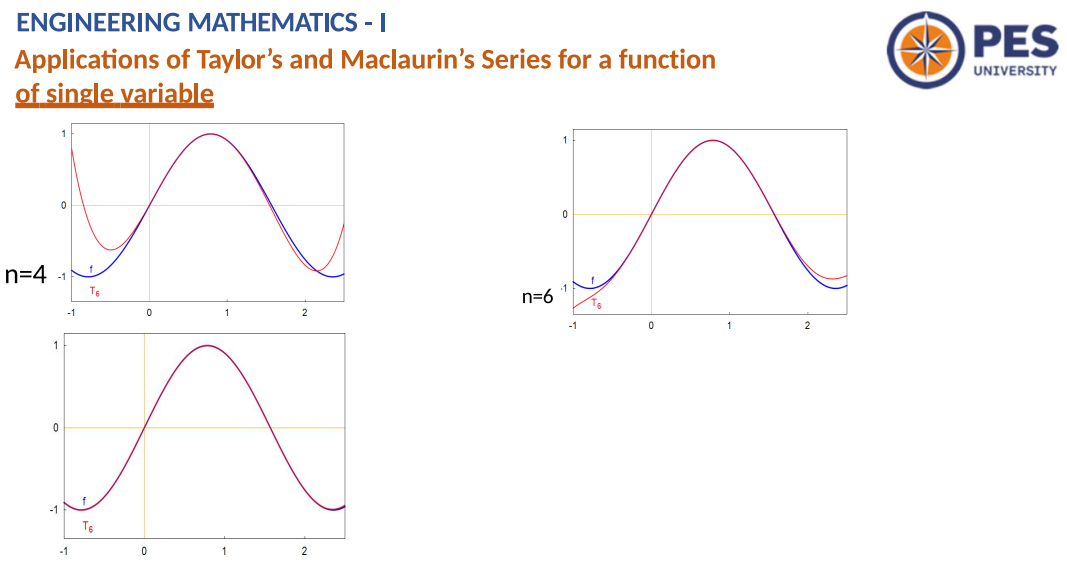

Figure G000109. Taylor and Maclaurin series expansion for a function of two variables · source slide page 8.

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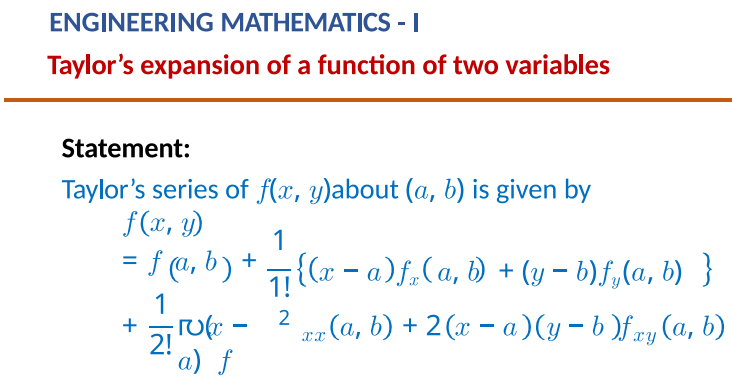

Figure G000110. Taylor and Maclaurin series expansion for a function of two variables · source slide page 9.

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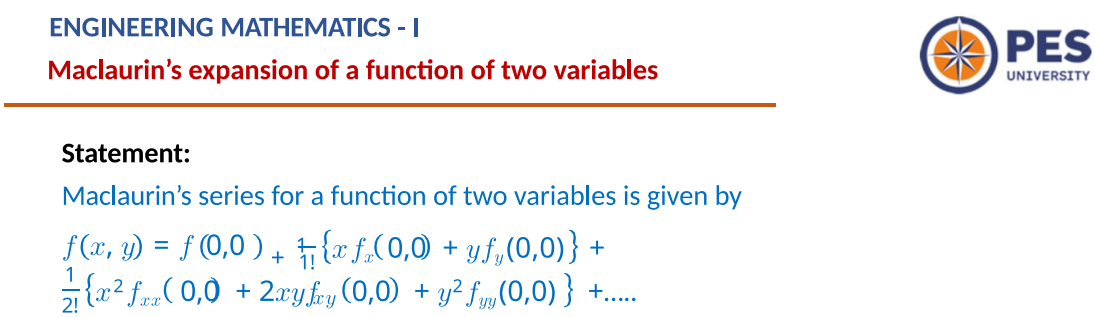

Figure G000111. Taylor and Maclaurin series expansion for a function of two variables · source slide page 10.

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ENGINEERING MATHEMATICS - I

Problems on Taylor's and Maclaurin's expansion

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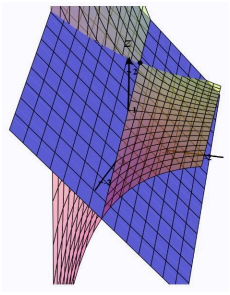

***Figure 3**: Graph of $f(x,y) = xe^y + 1$ and its 1st -degree Taylor polynomial, $L(x,y) = 1 + x + y$*

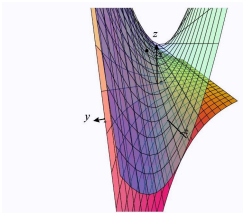

***Figure 4**: Graph of $f(x,y) = xe^y + 1$ and its 2nd -degree Taylor polynomial, $L(x,y) = 1 + x + xy + \frac{y^2}{2}$*

Figure G000112. Taylor and Maclaurin series expansion for a function of two variables · source slide page 11.

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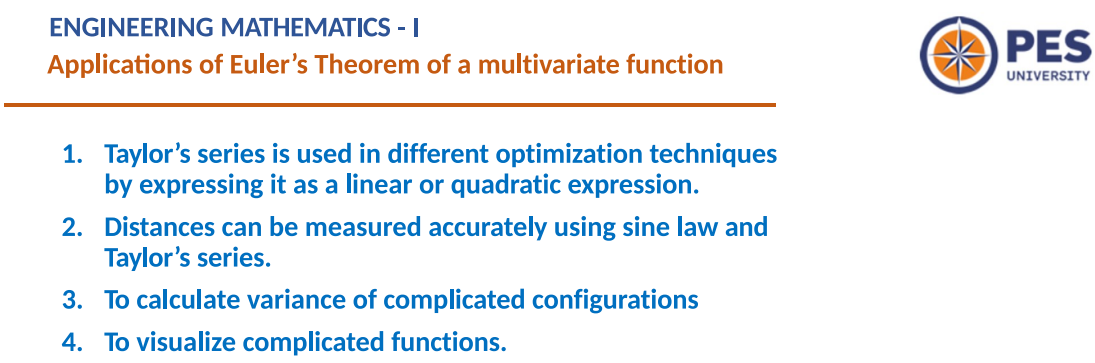

Figure G000113. Taylor and Maclaurin series expansion for a function of two variables · source slide page 12.

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Figure G000114. Taylor and Maclaurin series expansion for a function of two variables · source slide page 13.

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000561 · EV0000562 · EV0000563
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 7 Sub Topic : Taylor's and Maclaurin's Series of a function of Two Variables EV0000564 · EV0000565 · EV0000566
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Intuition behind Taylor's and Maclaurin's Series 3 Suppose you want to calculate the value of $e$tan(3), the first thing you look for is the calculator. A calculator expresses any complicated function in terms of its maclaurin's series and find its value. Suppose we need to evaluate an integral of the form ☒ $ex2$ dx or ☒$cos\ x2$ fails. If you want to evaluate the following limit dx all the methods we know so far sin $x - x$ lim $x\to0$ 4 x EV0000567 · EV0000568 · EV0000569
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Taylor's and Maclaurin's Series for a function of single variable If a function f(x) has continuous derivatives of order (n+1), then this function can be expressed as a polynomial at a point x=a in the form $f\ x\ (x - a)2\ (x - a)n\ f''\ fn = f\ a + x - a\ f'\ a + a + \cdots + a +$ ... 2! $n!$ If the function is expanded about the origin we get Maclaurin's Series $(x)2\ (x)n\ f''\ fn\ f\ x = f\ 0 + x\ f'\ 0 + 0 + \cdots + 0 +$ ... 2! $n!$ EV0000570 · EV0000571 · EV0000572

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Standard Maclaurin's Series for a function of single variable [Image text, OCR] n(n-1)(n-2)x3+., | x | <1 n(n-1) 1. (1+x)"=1+nx+ 2! 3! X X 2.e²=1+= ...,XER 1! 2! 3! 3 x X 3.log(1+x)=x <1 2 3 4 5 X x x 4.sinx=x- 5.cosx=1- 3! 5! 7! 2 4! 6! X 6.sinhx=x+ 7.coshx=1+ 3! 5! 2! 4! 3 X x 8.tan-x=x- 3! 5! 1 x3 1·3 xs 1·3.5x7 9.sinx=x+ -1≤x≤1 23 2.45 2.4.67 1x3 1.3 xs 1·3.5x7 T 10.cos-'x= -1 ≤x≤ 1 2 23 2.45 2.4.67 EV0000573 · EV0000574 · EV0000575
6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Applications of Taylor's and Maclaurin's Series for a function of single variable Expressing any complicated function in terms of a polynomial Evaluating definite integrals which cannot be evaluated by any other methods. Understanding asymptotic behavior of the function Understanding the growth of function Solving differential equations EV0000576 · EV0000577 · EV0000578
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Applications of Taylor's and Maclaurin's Series for a function of single variable In physics, Maclaurin's series is used to approximate the Lorrentz factor in special relativity. The most famous equation $E = mc2$ is an approximation for low velocities. In the motion of a pendulum an approximation of $\sin(\vartheta) \cong \theta$ which comes from Taylor's series. Used in power flow analysis of electrical power system. Can be used to find the Generating functions. Source: https://math.stackexchange.com/questions/218421/what-are-the-practical-applications-of-the-taylor-series EV0000579 · EV0000580 · EV0000581
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Applications of Taylor's and Maclaurin's Series for a function of single variable [Image text, OCR] 1 0 f 1 T6 2 [Image text, OCR] 0 1 T6 2 n=4 n=6 [Image text, OCR] 0 f 1 T6 2 EV0000582 · EV0000583 · EV0000584
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Taylor's expansion of a function of two variables Statement: Taylor's series of $f(x, y)$about (a, b) is given by $f\ x, y\ 1 = f\ a, b + 1!\ x - a\ f x\ a, b + (y - b) f\ y(a, b)\ 1\ 2 + ϖ(x - a)\ f\ a, b + 2\ x - a\ y - b\ f\ a, b\ xx\ xy\ 2!$ EV0000585 · EV0000586 · EV0000587
10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Maclaurin's expansion of a function of two variables Statement: Maclaurin's series for a function of two variables is given by $+ 1\ f\ x, y = f\ 0{,}0\ x f x\ 0{,}0 + y f\ y(0{,}0) + 1!\ 1\ 2!\ x2 f xx\ 0{,}0 + 2xy f\ 0{,}0 + y2 f\ yy(0{,}0) + ... xy$ EV0000588 · EV0000589 · EV0000590
11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion [Image text, OCR] Z y EV0000591 · EV0000592 · EV0000593
12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Applications of Euler's Theorem of a multivariate function Taylor's series is used in different optimization techniques by expressing it as a linear or quadratic expression. Distances can be measured accurately using sine law and Taylor's series. To calculate variance of complicated configurations To visualize complicated functions. EV0000594 · EV0000595 · EV0000596
13. Source slide page 13 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000597 · EV0000598 · EV0000599

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:02.372] Hello everyone. Welcome back to another session of partial differentiation. [00:00:07.748] Today you will be learning an important concept called the Taylor's and McLaren series, which are very very helpful in reducing any complicated problems in the form of a polynomial, and it has a lot of applications in engineering. [00:00:24.644] So, uh as usual, before I go to the topic, I would like to convince you why you need to learn Taylors and McLaren series. Um suppose I'll ask you to, you know, find the value of this function, e to the power of tan of cube root of three. [00:00:42.276] So most of you would um think that, you know, if I let you a calculator, you can just give me a jiffy. But if I don't let you use a calculator, you think that she has gone crazy and then you know, you don't know what's tan of cube root of three, and then on that I'm asking you to find exponential power of that. [00:01:02.692] So you might have wondered, but a calculator gives all these complicated expression in just the click of a button. So how does a calculator can calculate this? What goes behind that? So basically, calculator cannot understand [00:01:20.804] Anything that is complicated. It will not understand your differentiation of sine, cos, or you know, if you want to give a function that's like e to the power of tan of x square and then ask you ask the calculator to differentiate it. [00:01:35.040] What it does every time is it try to express any complex function as a series of polynomials. Few of them, I don't know, you might be aware because I think you are aware of binomial expansion. So I I will always try to [00:01:54.660] get any function in the form of a polynomial so that I can handle the function in a better way. Right? So what a calculator does it, it expresses any complicated functions in terms of a polynomial which is actually using Maclarin series. [00:02:14.820] So we should be grateful to two mathematicians, Taylors and McLevins. Another thing is suppose I'll ask you to evaluate this integral. Let's say I'll give you integral of sine of x. So everybody knows the answer for that. [00:02:32.164] But if I make a small change in this, I'll give sine x square dx. Then what is the integral of this? Maybe if you want, I'll give you definite integral. Let's make it 0 to pi by 2 or pi, whatever you want. [00:02:46.116] Try plugging it this in the calculator because anyway you cannot find the value of this directly. So if I plug this in in the calculator, [00:02:54.948] Then the calculator writes this sine of x square in the form of a polynomial up to few terms, depending on the configuration of the calculator. [00:03:06.244] And then it calculates once you get a polynomial, everything is easy, right? If I can write this as a polynomial, you integrate every term, apply the limits, simplify, you will get the answer. [00:03:17.508] That is what is happening. Or as simple as I have mentioned here, e to the power of x square, which you don't know how to integrate. Right? So like this, there are infinitely many functions which we cannot evaluate directly. [00:03:33.636] So this is a tool. So for example, you know uh about the indeterminate forms, right? Zero by zero, and then you use L'opitholes rule, you did learn mean value theorems like Cauchy's and Legrange and Rolls. [00:03:47.664] And you might be wondering how did they thought about those theorems? So the basis for all those theorems is the Taylor's theorem. And m even though we are not discussing Taylor's theorem, but Taylor's series is nothing but consequence of Taylor's theorem. [00:04:05.764] So I hope you pay little attention when you watch this video so that by uh end of this video you are in a position to answer so many unanswered questions. [00:04:19.780] So what is a Taylor series? So basically what happens is if I have a function which has continuous derivatives, that's very, very important. Why am I adding this continuous? Why am I prefixing this derivative with continuous? That's because I want to have [00:04:38.308] a derivative of n plus one th order. I'm not just finding the first derivative, the second derivative, or the third derivative. In fact, I'm finding the n plus one order

derivative. Then Taylor says that you can expand this function [00:04:56.772] around any point A using a polynomial. So this is not new. So that means it looks like f of x, I can write this as f of a plus x minus a into f dash of a

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:15.876] Plus x minus a the whole square divided by two factorial into f double dash of a and then this continues. [00:05:26.564] This is not new to you, right? Suppose I'll show you why it is not new. Suppose I'll erase all the higher degree terms, okay? So don't see this. This is an infinite series, of course, but don't see that. Aren't you familiar with this? What is this? [00:05:43.492] I can write rewrite this expression as f of x minus f of a divided by x minus a, which is nothing but f dash of a, right? [00:06:00.644] So the basis for the definition of the derivative comes from the t-shirts. That means given any function, I can write this as a polynomial of infinite order with infinitely many terms. [00:06:17.668] Instead of expanding it around a point A, if I expand the function about the origin, about the origin, then we get the McLaren series. So McLaren series is nothing but a particular case of Taylor series. Okay? [00:06:34.116] So, is the definition clear? So the whole idea or the intuition behind the Taylor series and McLaren series is you give me any complicated function, I will write that in the form of a polynomial. [00:06:50.276] And once I have a polynomial, I can do everything with it. I can differentiate, I can study what is the change that is happening, I can integrate, I can find the limit of that. Whatever you ask me to do, I can do it. [00:07:03.684] That's why these please remember that Taylor's in McLaren series always comes uh very handy for you when you have to handle complicated conditions or complicated functions. So these are some of the standard uh [00:07:19.396] Mechlerin series, which you might already know, right? So e power x is one thing which is very very important. So an exponential function can be written as 1 plus x by 1 factorial plus x square by 2 factorial and other things. Uh I to add a few, you can write [00:07:37.444] The series for this as well, which is 1 by 1 minus x. So this is actually 1 plus x plus x square plus x cube and it continues. Okay. [00:07:52.932] And one surprising thing is if they give me this series, I know that I'm getting a finite value. Right? For example, if they give one plus two plus two square plus two cube, one. [00:08:07.492] so on and so forth. So I'm replacing X with two, then I know that this entire even though uh it's uh you know changing a lot, you'll get minus one. And that's very strange. So [00:08:21.936] Actually Ramanujan proved so many such results. So you need to explore some of the convergence of series, which will really fascinate you. Because you know you are adding infinitely many positive terms and it's coming back to minus one, which is you know not that easy to accept. [00:08:39.664] what is happening there, right? But I want you to explore some of those things. So if e power x is this I can write e power minus x to be 1 minus x plus x square by 2 factorial and then minus x cube by 3 factorial and then plus etcetera. So this will continue. [00:08:59.108] I have log of 1 plus x, you can try log of 1 minus x, sine tan, hyperbolic, tan inverse. But how the question is, how are they getting this? Right? [00:09:10.692] It's very simple. I'll uh anyway, I won't be solving problems on Taylor's and McLaren series for one variable because I'll be extending it to uh bivariate function. But I can show you what happens. So if I want McLaren series for e par x, so this is my f of x, all I need to do is [00:09:28.464] is calculate its derivative as many as you want. So let's say you want four terms, you calculate up till the fourth derivative. And I know that derivative of e par x is e par x itself. No matter how many times I differentiate, I'll get [00:09:44.196] um e power x only. And I can substitute these, replace this all these x's with

0. So all these values will become 1. And that's not new to us, right? So that's why e to the power of x is this and e to the power of minus x will be that.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:01.060] So uh let me go back again so that you are well aware with the definition. So f of x, I can write any given function in terms of the Makelden series by using this series. All you need to do is find the value of the function at 0, provided f of 0 [00:10:18.512] does not tend to infinity. I cannot write equal to because we cannot reach infinity. So I'll say it cannot tend to infinity. That means if of suppose I want to find the McLaren series for 1 by X. [00:10:32.880] That's not defined because every at uh all the derivative at zero will tend to infinity. That's the only constraint. And here the same thing f of a cannot be infinite. Cannot tend to infinity or cannot be equal to infinite. So is the definition clear? [00:10:51.812] Okay. So what our uh idea is to express any given function, any complicated function in terms of a polynomial, which can be achieved using Taylor's and McLaren series. Okay so if I want to tell you uh what we are trying to do, so you give me a function. [00:11:11.332] And then I want to express it in terms of a polynomial. [00:11:17.700] If you want about a point x is equal to a, then you use Taylor series, and if you want about the origin, you use metogen series. [00:11:31.876] That's all you want, right? Now let's see what are the advantages of this Taylor's and McLaren series. As I said before, it you can use this to evaluate any complicated function by expressing it in terms of a polynomial. You can evaluate definite integrals, which I also discussed, right? 0 to 1 sine of x squared. [00:11:51.584] So now see I know the series for sine, right? So let me go back to my previous slide. So this is the series for sine. And if you want sine x square, just replace x with x square. That's all you need to do. So how does the series for sine x square look like? So sine of x square will look like x square. [00:12:11.364] Minus x to the power of 6 by 3 factorial plus x to the power of 10 by 5 factorial minus etc. [00:12:20.292] And the question says you can you have to integrate this from 0 to 1. So that's easy now for me because I know what is integral of x square, x par 6, x par 10, etc. I can integrate all these things with respect to x and then apply the limits. That's all I need to do. [00:12:37.348] So even calculator does the same thing. Okay so if you w if you give this in the calculator, it might not take infinitely many terms because mostly what happens is only if if we if we take a finite number of terms, that itself matches with the given function. [00:12:53.508] I'll show you that when we plot sign uh any function and its uh series, okay? [00:12:59.876] The next thing is uh understanding asymptottic behavior of the functions. So what do I mean by asymptottic behavior? Suppose I have a rational function here and I want to check what is happening to its asymptotes. You know that these are the asymptotes for the function, right? [00:13:13.636] So there is an asymptot here. So asymptotic behavior of the function, um you can understand the growth function using uh Taylor's and McLaren series, the most important would be to solve differential equations. So [00:13:29.060] In this course, you will be learning higher order differential equations and when you need to solve the higher order differential equation, we will be basically using the Taylor series. So I'll tell talk more about that when we do that unit. Okay. [00:13:44.420] And uh in physics, McLaren series used to approximate the Lorentz factor. So the Lorentz factor is nothing but this uh gamma uh in special relativity. [00:13:54.820] And E is equal to M C square. The most famous equation E is equal to M C square. It's an approximation on velocities, which is done again by the McLaren series. In the motion of a pendulum, we approximate sine theta to be theta, that itself is a that means in the series expansion of sine theta. [00:14:12.592] You're taking only one term. So this is theta minus theta

cube by 3 factorial. So you can actually leave all the higher degree terms and consider only one term. So we approximated this sine theta to be equal to theta. [00:14:27.268] And again, it's used in power flow analysis of electrical power systems and can be used to find generating functions. This is a very, very important concept for computer science uh students. Okay, but uh anyway, when we do uh four-year series in the next semester, we'll talk about uh [00:14:45.040] generating functions what it does and what do we mean by generating functions and other things. Okay. So those are the few applications. Now I want you to visualize what is happening. So here I have plotted as you can see this is a sign graph right the graph that is in blue is sign graph.

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:04.676] And the graph that is in red is the McLaren series for that. So what is sine x now? What is the series for sine x? Sine x is x by 1 factorial minus x cubed by 3 factorial and then plus x power 5 by 5 factorial and then etc. Right? So [00:15:24.452] This blue graph is the graph of sine x. And then you see I have taken only four terms here. I have taken the maybe one more I'll add. So x power 7 by 7 factorial. So for this four terms, this is how much it matches. So I am plotting this in red and sine x in blue. [00:15:44.100] Okay so that's what is happening. Um in fact um this is let me change this to blue so that you know that I'm plotting sin x in blue. The next graph that you see. [00:15:58.148] is again uh n is equal to 6. So I have added two more functions here. And see, this is maybe this is n is equal to 10. I'm just adding 10 uh [00:16:10.432] uh terms in that polynomial, which is not too many as compared to an infinite series, right? So just for ten uh terms in the polynomial, see the graph, how exactly it matches. It's overlapping. [00:16:24.996] So that's how close the approximation is. That's why it works so best. So you give me any function, there are two advantages, right? The first advantage is I'm getting everything in terms of a polynomial. That's the first advantage. So everything goes in terms of a polynomial. [00:16:45.252] I'm fine with that, I'm happy. The second thing is once it you are telling me to take infinite series, but most of the series converges for finitely many terms. So it's enough if I take finitely many terms to define my function. [00:17:03.268] Maybe not as m small as sinex, you might have to take say around 40 or 50 or whatever the number might be. That depends on the function. But definitely the function matches exactly with the given function in terms of tailors or in terms of material series. [00:17:20.260] Okay so in a similar fashion now we are going to define the Taylor series for a function of two independent variables. Again with the condition that this f of a comma b cannot go to infinity. [00:17:37.380] So um now that you already know, so what is the Taylor series for a function of single variable? f of x is equal to f of a plus x minus a into f dash of a. [00:17:51.940] plus this is f dash plus x minus a the whole square by two factorial into f double dash of n etc. I didn't talk much about this because you are already familiar with the binomial theorem, right? [00:18:07.428] you already know what is binomial theorem expansion is and then how do you get the coefficients and other things. So this is basically a series. So for a Taylor series for a function of two independent variables will look like this f of ab plus x minus a into f x of a b because there are two derivatives. [00:18:26.404] You have to take x minus a for fx, y minus b for f y. And there are four second order partial derivatives, fxx, fxy, and then f y. But you know that they are equal. F x y is the same as f yx. So it's okay, I have combined. That's why you got this two. [00:18:45.028] Right? So it is 1 by 2 factorial x minus a the whole square f x of ab, plus 2a into x minus a into y minus b of ab, plus y minus b the whole square f y of ab. And then of course the next term uh will continue. [00:19:01.636] So, uh can you take a guess on what could be the McLaren series for a

function of two variable will look like? Instead of AB I have to consider the origin. [00:19:12.388] So what I'll do is um I'll consider f of x y and I'll consider the value of the function at 0, 0. The only thing I need to worry is not just this, but any of these cannot tend to infinity. [00:19:26.692] Uh I did mention only this but not necessarily. None of the derivatives fx, f y and f x, f y y. So none of them can tend to infinity. [00:19:38.564] At zero, zero, of course. At zero, zero. So if you can take care of that, if it satisfies that, then you get the McLaren series for a function of two variables. And it's easy to remember because uh fx goes with x. [00:19:53.168] f y goes with y, f x x goes with x square, fx y goes with 2xy I'm getting this too because mixed second order partial derivatives are equal and then f y y goes with y square. Right?

EV0000170 · EV0000171 · EV0000172 · EV0000173

Transcript block 5 (source interval 001200-001499 seconds): [00:20:08.036] So, uh, before we solve some problems, this is the graph of uh [00:20:13.824] polynomial because it's two dimension two variable I have to go to three d so it's not as good as uh visualizing in uh just one dimension. But anyway I have done my best. So suppose I have a function which is x e to the power of y plus one. I have written a Taylor polynomial of this. [00:20:32.068] uh so of just the first degree. Um and that's what I get. Maybe they have used the McLaren series. So you see that it's not completely merging, but it's it's trying to but whereas the second function, um I have taken uh one more term extra here. So you see I took uh one plus X plus Y [00:20:51.104] And now I'm taking the second degree polynomial. So it is the same function x e to the power of y plus 1 and I'm going up to the second degree polynomial. So uh if I go up to second degree polynomial, this how much it matches. [00:21:05.796] Okay. So anyway, this is just to give you a visual um impact that you take any function or any surface, you can express it as a polynomial. And then uh we'll see how many terms we need to take so that they uh they're almost together. So approximately [00:21:22.912] We can express any given function in terms of its polynomial. These are some of the uh applications of Taylor series for two variables. So, Taylor series is used in different optimization techniques by expressing it as a linear or quadratic expression. So [00:21:42.084] Basically, what they do is they just take a they are going to make use of the linear and optimization process. [00:21:50.628] and distances can be measured accurately, variance of complicated configuration and as I said, visualization of complicated functions and uh um so geometrically uh it's just that. Okay? All right. So we will s [00:22:06.944] In the next session, we'll try to solve some problems on Taylor's and McLaren series for a function of two independent variables.

EV0000174 · EV0000175 · EV0000176 · EV0000177

Full transcript: 3627 normalized words, 86 timestamped segments, 1335.317333 seconds duration. Low-confidence segments flagged: 0.

EV0000158 · EV0000159 · EV0000160 · EV0000161

13 Problems on Taylor's and Maclaurin's series

This lesson is indexed from the source video titled Problems on Taylor's and Maclaurin's series. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json.

Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000178 · EV0000179

Graphs and diagrams

Figure G000115. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 1.

EV0000601

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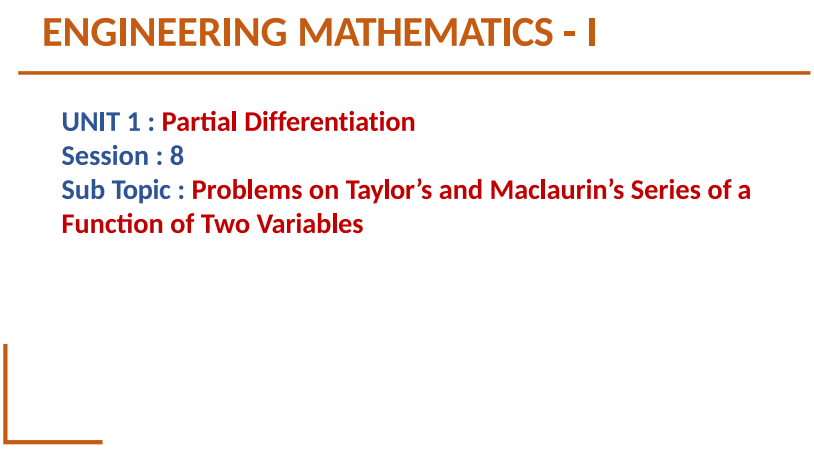

Figure G000116. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 2.

EV0000604

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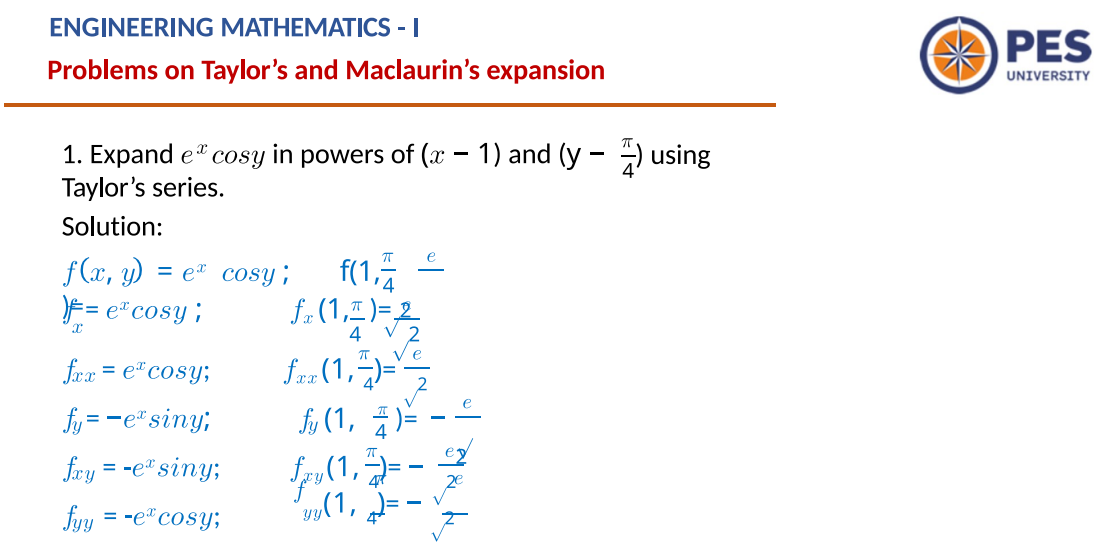

Figure G000117. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 3.

EV0000607

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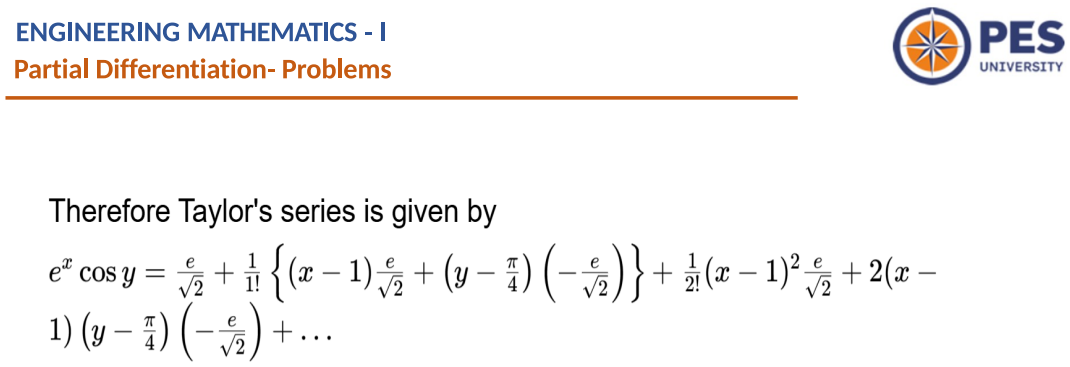

Figure G000118. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 4.

EV0000610

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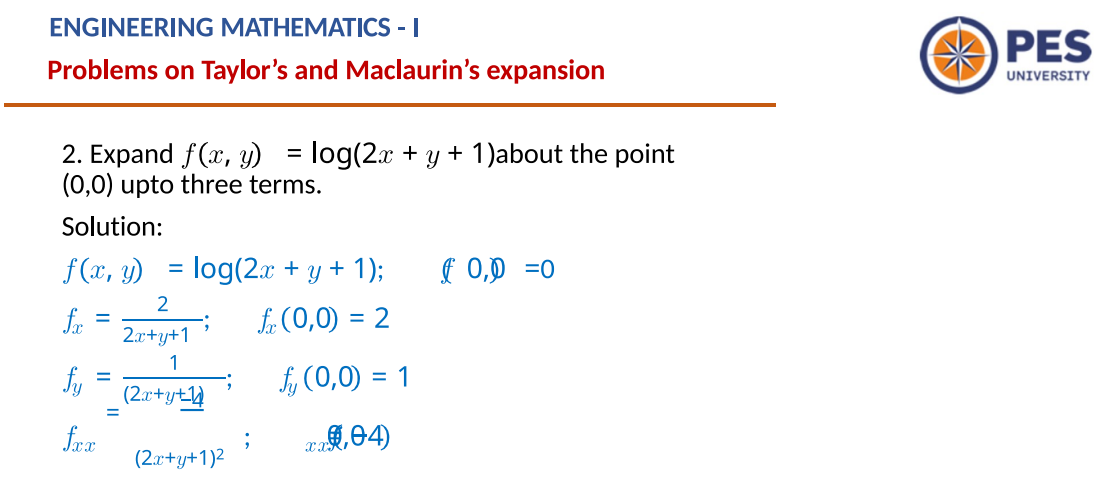

Figure G000119. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 5.

EV0000613

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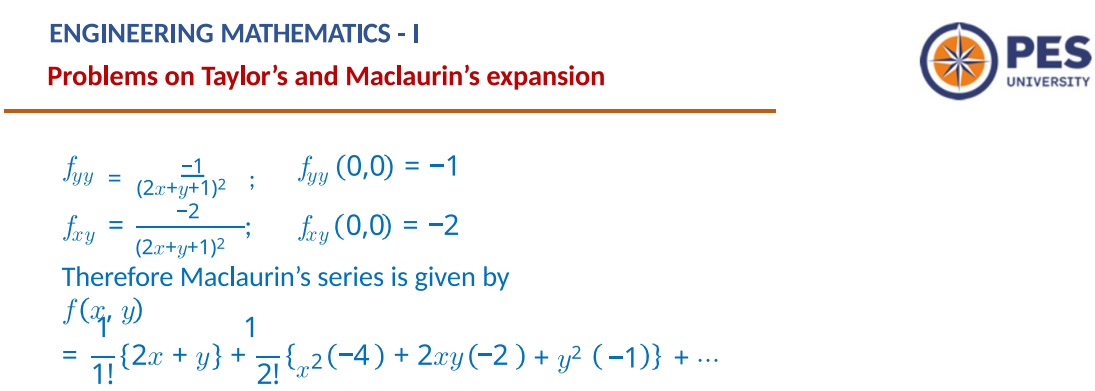

Figure G000120. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 6.

EV0000616

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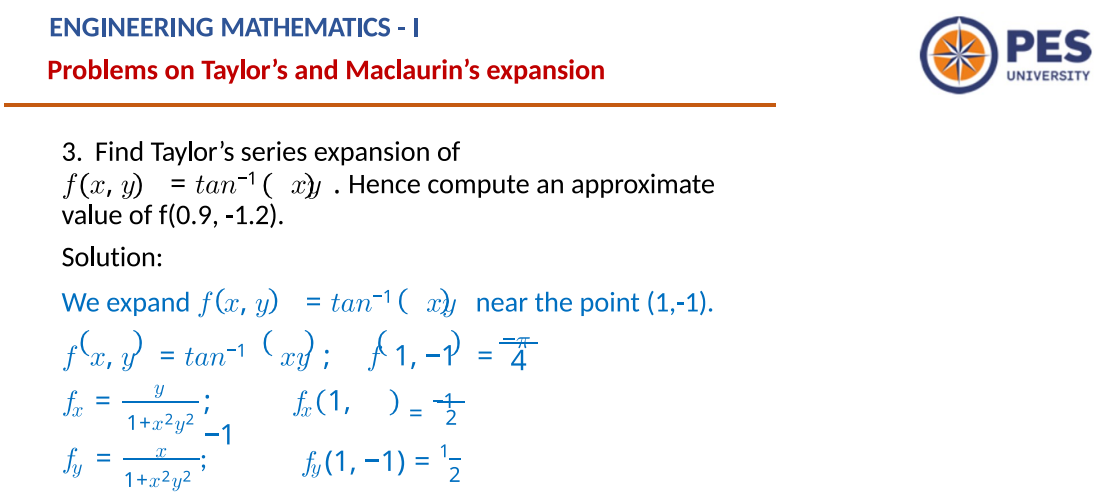

Figure G000121. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 7.

EV0000619

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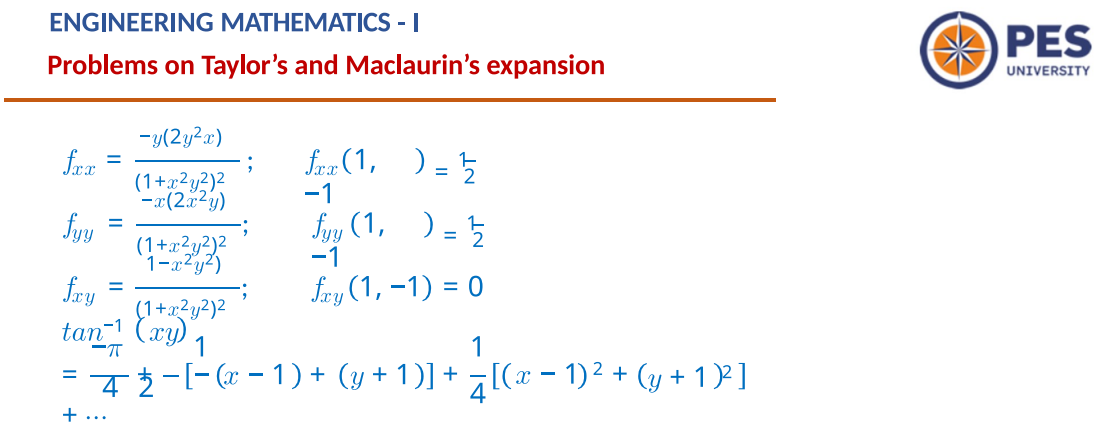

Figure G000122. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 8.

EV0000622

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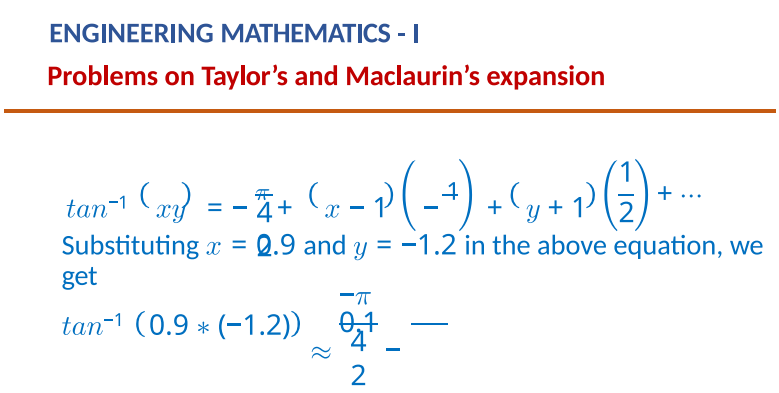

Figure G000123. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 9.

EV0000625

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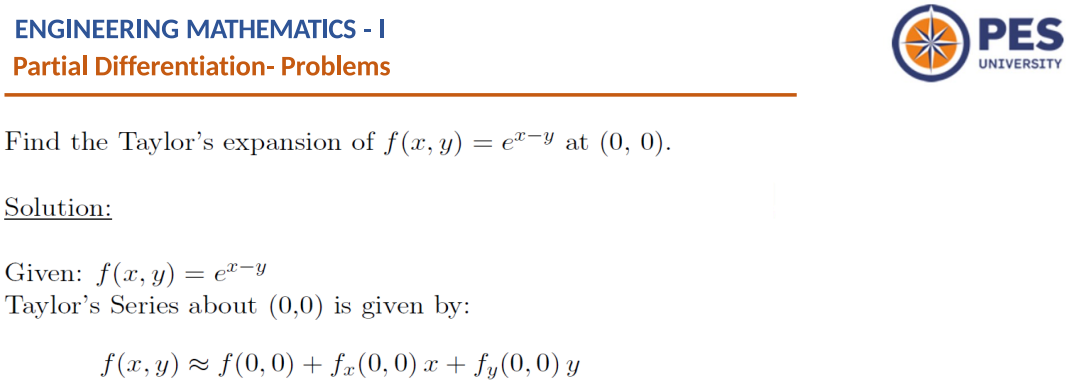

Find the Taylor's expansion of $f(x, y) = e^{x-y}$ at $(0, 0)$.

Solution:

Given: $f(x, y) = e^{x-y}$
Taylor's Series about (0,0) is given by:

$$\begin{aligned} f(x,y) \approx\ & f(0,0) + f_x(0,0)\,x + f_y(0,0)\,y \\ & + \frac{1}{2!}\left[f_{xx}(0,0)\,x^2 + 2f_{xy}(0,0)\,xy + f_{yy}(0,0)\,y^2\right] \\ & + \frac{1}{3!}[f_{xxx}(0,0)\,x^3 + 3f_{xxy}(0,0)\,x^2y \\ & + 3f_{xyy}(0,0)\,xy^2 + f_{yyy}(0,0)\,y^3] + \ldots \end{aligned}$$

Figure G000124. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 10.

EV0000628

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

PES UNIVERSITY

At (0,0):

$$f(0,0) = 1, \quad f_x(0,0) = 1, \quad f_y(0,0) = -1$$
$$f_{xx}(0,0) = 1, \quad f_{xy}(0,0) = -1, \quad f_{yy}(0,0) = 1$$
$$f_{xxx}(0,0) = 1, \quad f_{xxy}(0,0) = -1, \quad f_{xyy}(0,0) = 1,$$
$$f_{yyy}(0,0) = -1$$

Thus,

$$f(x,y) = e^{x-y}$$
$$\approx 1 + x - y + \frac{1}{2}(x^2 - 2xy + y^2) + \cdots$$

Figure G000125. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 11.

EV0000631

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

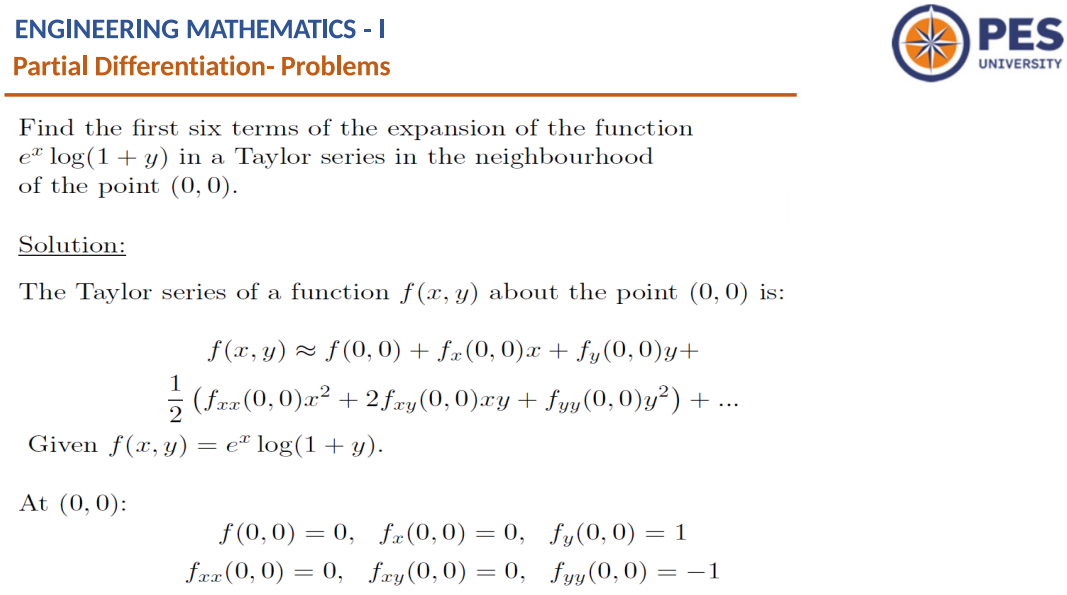

Figure G000126. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 12.

EV0000634

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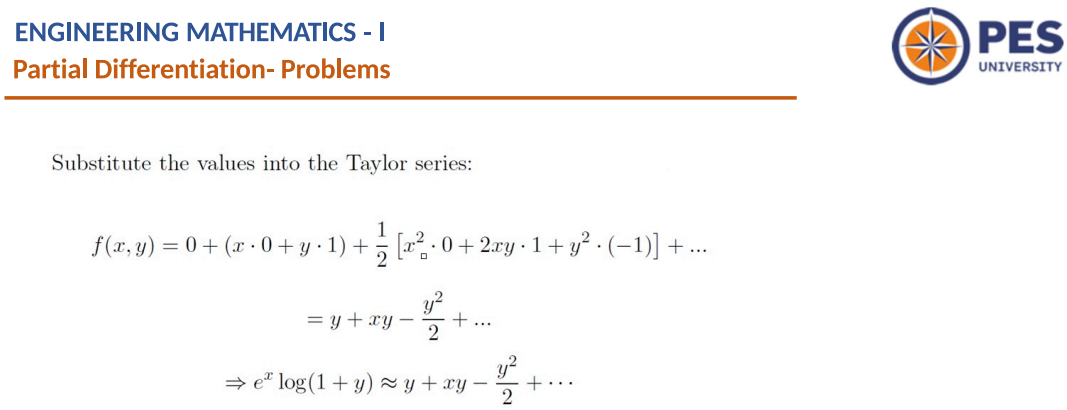

Figure G000127. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 13.

EV0000637

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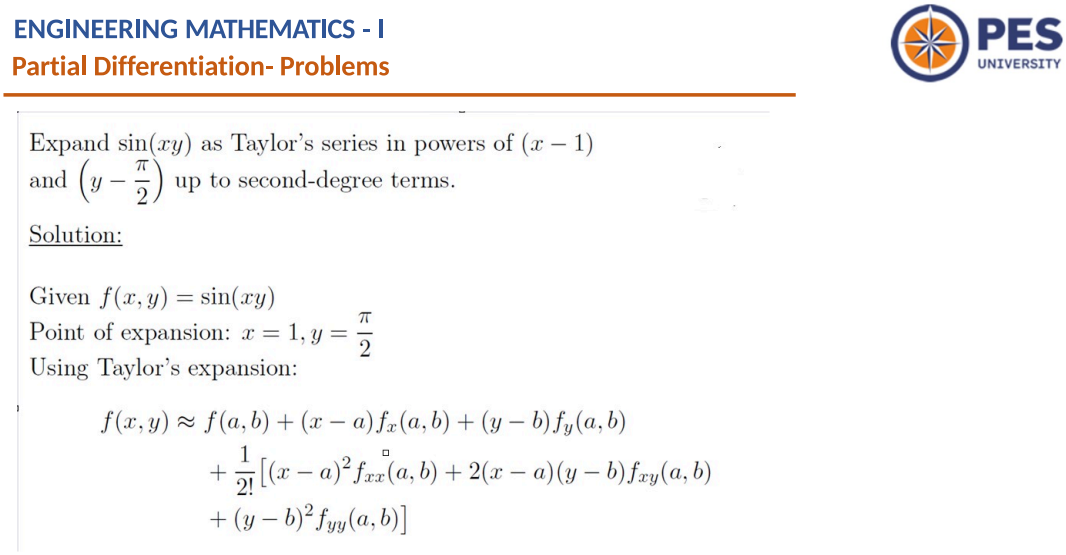

Figure G000128. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 14.

EV0000640

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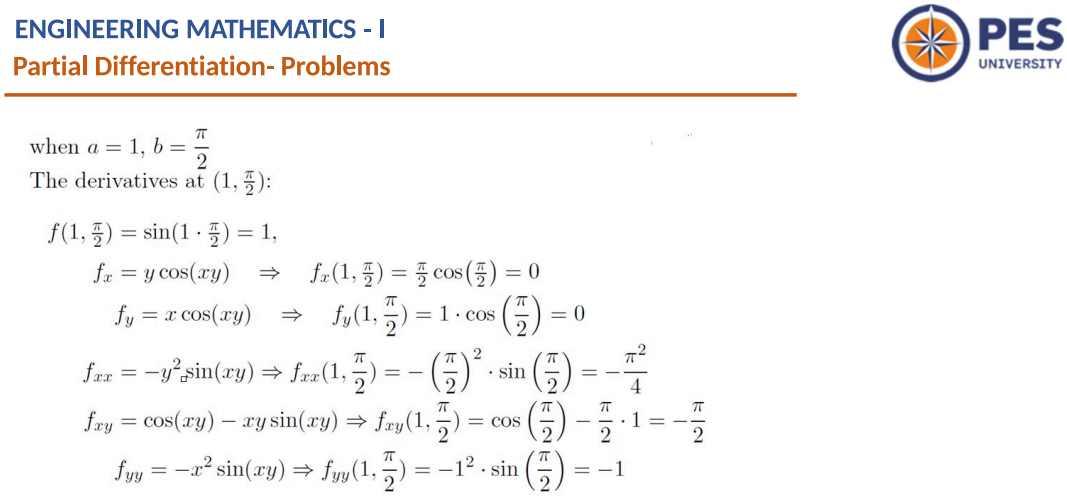

Figure G000129. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 15.

EV0000643

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

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$$f(x,y) \approx 1 + (x-1)\cdot 0 + \left(y-\frac{\pi}{2}\right)\cdot 0 + \frac{1}{2}\left[(x-1)^2\cdot\left(-\frac{\pi^2}{4}\right) + 2(x-1)\left(y-\frac{\pi}{2}\right)\cdot\left(-\frac{\pi}{2}\right) + \left(y-\frac{\pi}{2}\right)^2\cdot(-1)\right] + \ldots$$

$$sin(xy) \approx 1 - \frac{\pi^2}{8}(x-1)^2 - \frac{\pi}{2}(x-1)\left(y-\frac{\pi}{2}\right) - \frac{1}{2}\left(y-\frac{\pi}{2}\right)^2 + \ldots$$

Figure G000130. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 16.

EV0000646

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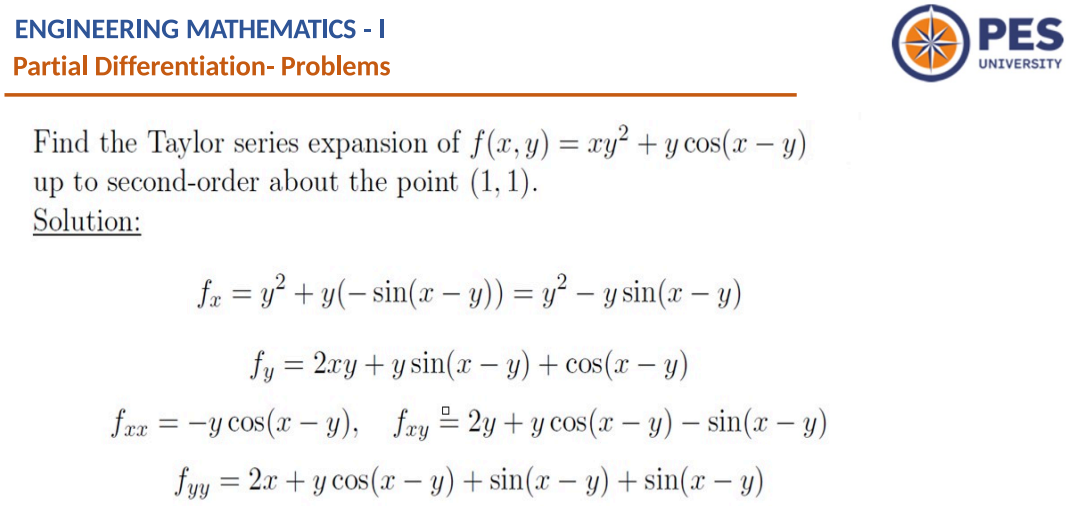

Figure G000131. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 17.

EV0000649

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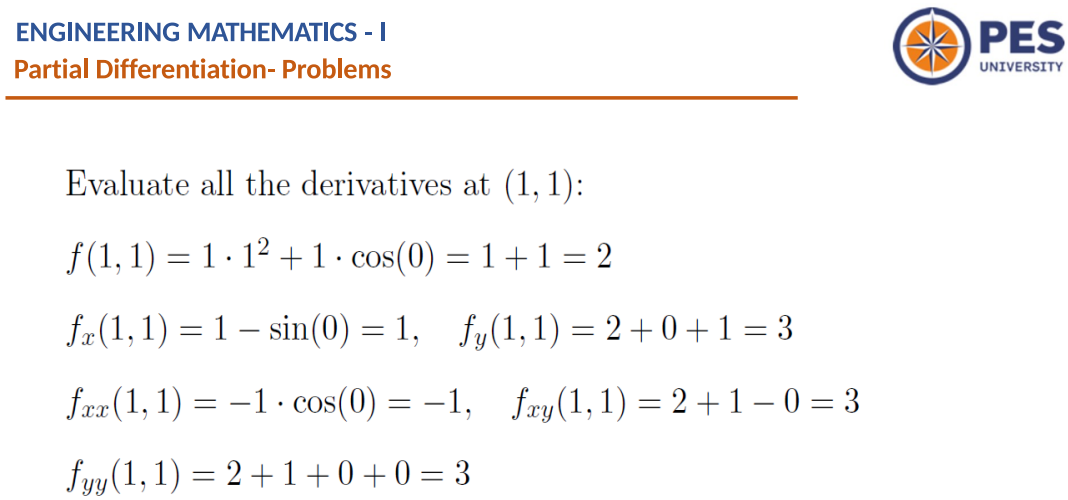

Figure G000132. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 18.

EV0000652

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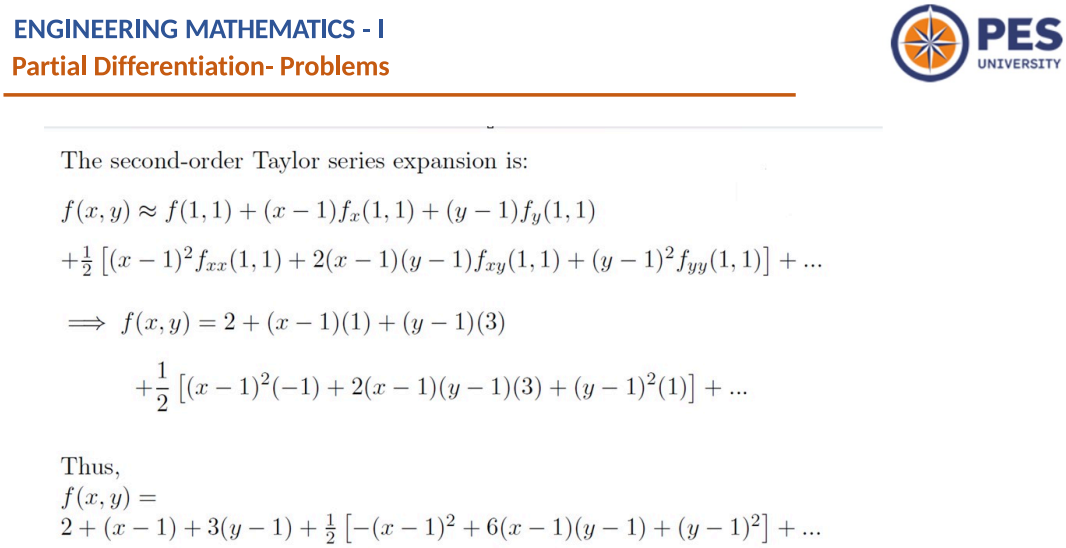

Figure G000133. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 19.

EV0000655

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000134. Taylor s and Maclaurin s expansion for a function of two variables continued · source slide page 20.

EV0000658

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000600 · EV0000601 · EV0000602
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 8 Sub Topic : Problems on Taylor's and Maclaurin's Series of a Function of Two Variables EV0000603 · EV0000604 · EV0000605
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion π 4 x 1. Expand $e\ cos y$ in powers of $(x - 1)$ and (y − Taylor's series. Solution:) using $\pi\ e\ f\ x, y = ex\ cos y$; f(1,)= 4 √2 $\pi\ e\ f = excos y$; $f x$ (1,)= 4 √2 $x\ \pi\ e\ f = excos y$; $f xx$(1, 4)= √2 $xx\ e\ f = -exsiny$; f (1, π)= − $y\ y$ 4 √2 $\pi\ e\ f$ = -$exsiny$; $f xy$(1, 4)= − √2 $xy\ f\ \pi\ e\ yy$(1, 4)= − √2 f = -$excos y$; yy EV0000606 · EV0000607 · EV0000608
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Problems on Taylor's and Maclaurin's expansion Therefore Taylor's series is given by {(x-1)+(u-)(-)}+(x-1)²+2(±- e cos y + 1)(µ-)(-) EV0000609 · EV0000610 · EV0000611
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion 2. Expand $f\ x, y = \log(2x + y + 1)$about the point (0,0) upto three terms. Solution: $f\ x, y = \log(2x + y + 1)$; f 0,0 =0 2 f = ; f 0,0 = 2 $x\ x$ 2x+y+1 1 f = ; f 0,0 = 1 $y\ y$ (2x+y+1) = −4 ; f 0,0 = −4 $f\ xx\ xx$ (2x+y+1)2 EV0000612 · EV0000613 · EV0000614

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion = −1 ; f f 0,0 = −1 yy yy (2x+y+1)2 −2 f = ; f 0,0 = −2 xy xy (2x+y+1)2 Therefore Maclaurin's series is given by f x, y 1 1 = 2x + y + x2 −4 + 2xy −2 + y2 −1 + ⋯ 1! 2! EV0000615 · EV0000616 · EV0000617

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion 3. Find Taylor's series expansion of f x, y = tan−1 xy . Hence compute an approximate value of f(0.9, -1.2). Solution: We expand f x, y = tan−1 xy near the point (1,-1). f x, y = tan−1 xy ; f 1, −1 = −π 4 = −1 y f = ; f 1, −1 x x 1+x2y2 x 2 f = ; f (1, −1) = 1 y y 1+x2y2 2 EV0000618 · EV0000619 · EV0000620

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion = 1 −y(2y2x) ; f = f 1, −1 xx xx (1+x2y2)2 2 = 1 −x(2x2y) ; f = f 1, −1 yy yy (1+x2y2)2 2 1−x2y2) ; f = f 1, −1 = 0 xy xy (1+x2y2)2 tan−1 xy −π 1 = + − x − 1 1 + x − 1 2 + + y + 1 y + 1 2 4 2 + ⋯ 4 EV0000621 · EV0000622 · EV0000623

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Problems on Taylor's and Maclaurin's expansion 1 2 tan−1 xy = − π + x − 1 − 1 + y + 1 + ⋯ 4 2 Substituting x = 0.9 and y = −1.2 in the above equation, we get −π 0.1 ≈ − tan−1 0.9 ∗ (−1.2) 4 2 EV0000624 · EV0000625 · EV0000626

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Find the Taylor's expansion of f(c, y) = e-y at (O, 0). Solution: Given: f(x,y) = er-y Taylor's Series about (O,0) is given by: f(x,y) f(0, 0) + fx(0,0) + fy(0,0) y 1-2 [fcx(0, 0) x² + 2fay(0, 0) xy + fyy(0, 0) y2 ┼ 1 [fcxx(0, 0) x² + 3fcxy(0, 0) α²y -3 + 3fryy(0, 0) cy² + fyyy(0, 0) y²] + .. EV0000627 · EV0000628 · EV0000629

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] At (0,0): f(0,0) = 1, fx(0,0) = 1, fμy(0,0) = -1 fcx(0,0) = 1, fxy(0,0) = -1, fyy(0,0) = 1 fccx(0,0) = 1, fcxy(0,0) = -1, fryy(0,0) = 1, fyyy(0,0) = -1 Thus, f(x,y) = er-y 1 ≈1+c-y+ 2 EV0000630 · EV0000631 · EV0000632

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Find the first six terms of the expansion of the function e* log(1 + y) in a Taylor series in the neighbourhood of the point (O, 0). Solution: The Taylor series of a function f(c,y) about the point (O,O) is: f(α,y) ≈ f(0, 0) + fx(0,0)c + fy(0,0)y+ (fcx(0, 0)x² + 2fey(0, 0)xy + fyg(0, 0)y²) + . 2 Given f(x,y) = e" log(1 + y). At (0, 0): f(0,0) = 0, fx(0,0) = 0, fμ(0,0) = 1 fcx(0,0) = 0, fcy(0,0) = 0, fyy(0,0) = -1 EV0000633 · EV0000634 · EV0000635

13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Substitute the values into the Taylor series: [x2. 0 + 2cy · 1 + y² .(-1)] + .. f(c,y) = 0+(c·0+y·1) + y = y + cy 2 → e" log(1 + y) ≈ y + cy G2 2 EV0000636 · EV0000637 · EV0000638

14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Expand sin(cy) as Taylor's series in powers of (c - 1) π and up to second-degree terms. Solution: Given f(c,y) = sin(cy) T Point of expansion: = 1, y = 2 Using Taylor's expansion: f(x,y) f(a,b) + (c - a)fx(a,b) + (y - b)fy(a,b) + (y - b)² fyy(a,b)] EV0000639 · EV0000640 · EV0000641

15. Source slide page 15 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] T when α = 1, b = 2 The derivatives at (1,): f(1,)= sin(1·)= 1, fx = y cos(cy) →fx(1,)=cos()= 0 fy =xcos(xy) → fy(1,2))= 1·cos π 2 fax = -y²sin(ry) ≥ faa(1,)=-()·sin T T TT fcy = cos(cy)- cy sin(cy) → fey(1, = COS 1 2 2 2 fyy = -αx² sin(cy) → fy(1,)) = -12.sin EV0000642 · EV0000643 · EV0000644

16. Source slide page 16 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] (-)·0 f(,y) ≈1+ (- 1)·0+ —# 2[(x -1)2. +2(x-1)(μ-≤)·(-) 2 2 sin(cy) ≈ 1 (c - 1)² EV0000645 · EV0000646 · EV0000647

17. Source slide page 17 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Find the Taylor series expansion of f(x,y) = cy² + y cos(c - y) up to second-order about the point t (1,1). Solution: fc = y² + y(- sin(x -y)) =y² - ysin(c -y) fy = 2cy + ysin(x - y) + cos(x - y) fcx = -ycos(c - y), fxy = 2y + ycos(c - y)- sin(c - y) fyy = 2x + y cos(x - y) + sin(c - y) + sin(c - y) EV0000648 · EV0000649 · EV0000650

18. Source slide page 18 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Evaluate all the derivatives at (1, 1) : f(1,1) = 1· 12 + 1· cos(0) = 1 + 1 = 2 fx(1,1) =1 - sin(0) = 1, fy(1,1) = 2+0+ 1 =3 fcx(1,1) = -1· cos(0) = -1, fry(1,1) = 2 + 1 - 0 = 3 fyy(1,1) = 2+ 1+ 0 + 0 = 3 EV0000651 · EV0000652 · EV0000653

19. Source slide page 19 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] The second-order Taylor series expansion is: f(,y) ≈ f(1,1) + (α - 1)fx(1,1) + (y - 1)fy(1,1) + [(c - 1)²fcx(1,1) + 2(x - 1)(y - 1)fxy(1,1) + (y - 1)²fyy(1, 1)] + . > f(c,y) = 2 + (c - 1)(1) + (y -1)(3) Thus, f(c, y) = 2 + (c - 1) + 3(y - 1) + 2 [-(x - 1)² + 6(x - 1)(y - 1) + (y - 1)²] + EV0000654 · EV0000655 · EV0000656

20. Source slide page 20 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000657 · EV0000658 · EV0000659

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:01.220] Alright, now let's try some problems on Taylors and McLaren series. As I was mentioning in the lecture, we are not going to solve problems on the Taylors and McLaren series for a function of single variable. But we'll try [00:00:14.896] And find out the uh Taylor series and the McLaren series for a function of two independent variables. [00:00:21.860] So the first question is expand e power x cos y in powers of x minus 1 and y minus 4 up to second degree term. So that's what it says, right? So first let's write out what is the Taylor series expansion. [00:00:37.472] So it says we can express any given function of two independent variables at a point. So the point is 1 comma pi by 4 because we want to express it in that form. So this will be f of 1 comma pi by 4 plus x minus 1. [00:00:57.316] x minus 1 into f x of 1 comma pi by four plus y minus pi by four into [00:01:06.916] f y of 1 comma 5 by 4. So this is only the first degree term. The question says up to the second degree term. So the next one will be plus uh x minus 1 the whole square. So [00:01:25.988] x minus 1 the whole square by two factorial into f x x of one comma five by four plus two into [00:01:40.068] x minus 1 into y minus pi by four divided by two factorial into f x y of one comma pi by four plus [00:01:54.980] 2 into or sorry, y minus pi by four the whole square divided by two factorial into f y y of one comma pi four. [00:02:07.104] And then plus etc. Of course, it's an infinite series. But what we are required

to find is I need to calculate what is the value of the function. The function is e power x into cos y. So I will take f of x y as e to the power of x into cos y. [00:02:25.796] And then we will find all the partial derivatives of this, or till whatever we require and the corresponding values. So what will be f of 1 comma pi by 4? F of 1 is e and a cos of pi by 4 is root 2. So this will be e divided by root 2. [00:02:45.508] Then what is f x of x y? F x of x y is you have to treat cos y as constant. So this is e power x into cos y. So f x at 1 comma pi by 4 is again e bar root. [00:03:05.380] What will be f y of x y now? I need to differentiate cos y by treating x as constant. So this will be minus e par x sine y. [00:03:17.348] Right. So this is F y of XY. So F y at one comma pi by four will be again sin pi by four is also one by root two. So this is minus e by root. [00:03:31.648] Now we are done with the first order partial derivatives. Now let's try the second order partial derivative. So f xx of x comma y. F xx of x comma y will be I need to differentiate this again with respect to x. [00:03:47.136] So I get e to the power of x into cos y again. So f x at one comma pi by four will be e divided by two. [00:03:59.140] And then now let's find the mixed second order partial derivative fxy of x y. So what will be f x y? I need to differentiate fx with respect to y. So this will be minus e to the power of x sine y. [00:04:16.036] So the value of f x y at one comma five by four will be again minus e divided by two. And the last one that I need is f y y f y y of uh x comma y. [00:04:34.052] So that that means I need to differentiate f y with respect to y again. So this will be minus e to the power of x into cos y. So I'll get f y y at 1 comma y by four will be minus e divided by. [00:04:54.436] If you observe carefully, all these values are same and only thing they differ is because of the sign. Okay? That's because of the point that they have given about which we need to expand the function. So now if I substitute all these partial derivatives in the given problem, I can write

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:11.936] So let me just write the final answer. I can write f of x comma y as [00:05:19.268] f of 1 comma pi by 4 is e pi root 2. So in fact I can take out e by root two common from all the terms because every term here has e by root 2. So I'll get e by root 2 which is 1 plus. Next one is x minus 1 [00:05:36.164] Okay, and I am done. Plus, uh for y it is minus, so y minus y by four. This is the second degree term. Okay. Plus I have x minus one the whole square by two factorial. F x x is positive. [00:05:53.284] and minus x minus one of course into two into y minus pi by four divided by two factorial and the last term is also minus I'll get y minus pi by four the whole square divided by two factorial. [00:06:10.672] So I can stop here for now because they have asked me to expand only up till the second degree term. So but sometimes if it is uh a third degree term we might have to go for a third order partial derivatives also. [00:06:23.696] So maybe in the next problem we'll see an expansion for Taylor series or McLaren series wherein we go up to third degree term. [00:06:34.084] Okay. Now let's try another problem. Expand. F of x y is equal to log of two x plus y plus one about the point zero comma zero up to three terms. That's a question. Okay so about zero comma zero means we are expanding my cleric series. [00:06:52.356] Right. So let's write down what is the McLaren series for the function f of x y. [00:06:58.436] So the McLaren series will be f of zero comma zero plus x f x of zero comma zero plus y f y of zero comma zero plus uh x square [00:07:14.468] F x at zero coma zero by two factorial plus two x y. [00:07:22.564] f x y 0 comma zero by two factorial plus f uh y y at zero comma zero into y square by two factorial and then of course plus etcetera other terms will continue. [00:07:38.564] Okay. Now when if the question says find up to three terms, make sure that you will get three non zero terms. Okay. So uh let's start by finding the value. So uh I want f of x y is log of [00:07:57.892] 2x plus y plus 1. Right? So what is the value of this function at 0, 0? I'll get

log of 1, which is 0 for me. Okay. So first term, the constant term is not there for this particular function. So what is fx of xy? [00:08:18.596] f x will be 1 by 2x plus y plus 1 into 2. Because I am differentiating with respect to x, I need to treat the other two variables as constant. So f x at 0 0 will be uh this is 0, this is 0, so I will get 2. [00:08:38.596] What will be FY of X, comma Y [00:08:42.468] If y is again 1 by 2x plus y plus 1 and the derivative of y is just 1. So f y of x comma y or 0 0 will be just 1 because these two terms goes to 0. So I got uh [00:09:00.228] two terms which are non zero. Now we differentiate this again with respect to x. Okay. So derivative will be minus two by two x plus y plus one the whole square into two. [00:09:17.188] So that means f x at 0 comma 0 will be minus 4. So, okay. So if they say up to three terms, it's enough if you find three non-zero coefficients. But let me find the two more uh terms, which is fx y. [00:09:37.220] of x y so that I can complete all the second degree term. So f x y is differentiate this with respect to y. So it will be minus 2 by 2x plus y plus 1 the whole square. So f x y [00:09:53.348] at zero zero will be minus two. [00:09:58.308] And the last one would be to calculate f y y of x y. So f y y of x y will be differentiating this again with respect to y, which will be minus 1 by 2x plus y plus 1 the whole square.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:16.164] Okay. Now what is the value of this at 0 0? So f y y at 0 0 will be again you put 0 here, so this will be minus 1. So now I got all the coefficients, so I can write the function. So log of 2x plus y plus 1. [00:10:35.812] Will be the first term is 0. So I'll get 2x and then plus y. Okay, that's the first term. Minus 4x square by 2 factorial. Minus 2xy by 2 factorial. [00:10:55.428] Okay, and then the last term will be minus y square by two factorial plus x. Actually these two terms were not necessary because the question says up to three terms. So if I need to simplify this little more, I'll get log of [00:11:11.280] Two x plus y plus one will be two x minus or plus y minus two uh x square, okay, minus x y minus y square by two plus x. [00:11:28.064] So this is how you can express any given complicated functions in terms of a polynomial. So that polynomial it will be in terms of X and Y. So the next problem we'll see it as an application of Taylor series. [00:11:47.140] Now in the next problem we'll see how to apply this Taylor series to evaluate a value. [00:11:53.988] Let's try another problem on find Taylor series expansion of f of x y equal to tan inverse of x y and find the approximate value of f of zero point nine comma minus one point two. This is the question, right? [00:12:10.384] So that means uh just observe these values. Because I want the approximate value, can I say that f of 0.9 and minus 1.2 is approximately equal to f of 1 comma minus 1. [00:12:28.772] Right. So if I find the Taylor series expansion of uh this particular function above this point I'll get my value. So what is tan inverse of uh so let's let's write down the function f of x y is tan inverse of [00:12:48.324] x y. So, we will expand this as a Taylor series expansion about the point 1 comma minus 1 and if we simplify that maybe we will go up to the second degree term and if we simplify we get an approximate value of the function at that particular point. [00:13:05.968] Right. So if f of x y then f of one comma minus one will be tan inverse of minus one, which is minus pi by four. Okay. So and then [00:13:20.612] if x sub x comma y if I differentiate this partially with respect to x, I will get 1 by 1 plus x y, the whole square. [00:13:31.140] Okay. So derivative of tan inverse x is 1 by 1 plus x square and derivative of x y with respect to x is y. So I will get a y here. And then so f x at 1 comma minus 1 will be you substitute y as minus 1 and this is square which is 2. So this is minus half. [00:13:51.524] Right. Next is we'll find out what is f y of x comma y. So you differentiate this again with respect to y, which is similar to this, so you get x divided by 1 plus xy, the whole

square. [00:14:07.972] So now if y at 1 comma minus 1 will be if I substitute 1 and minus 1 here, I'll get half. [00:14:18.148] So more terms you take you will get a better approximation of the answer. Okay. But for now I think because we have three non zero terms, this should be sufficient. That means we got tan inverse of x y. I can write tan inverse of x y as [00:14:35.908] uh minus pi by four plus I'm expanding about the point uh x minus one so x minus one into minus half plus y plus one into plus half. [00:14:53.348] Plus, etcetera of course. So if you want a better approximation of the value, then you have to expand for a few more terms. But this also can give a better uh you know close approximation of this. Uh now the question is I want to calculate what is f of 0.9 and minus 1.2. So tan inverse of

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:13.028] 0.9 and minus 1.2 will be minus 5 by 4. And if you substitute here for x is 0.9, minus 1 will be just 0.1 with a minus into minus half. [00:15:28.228] Okay. Plus y is minus one point two plus one will be again minus zero point uh one minus zero point two in fact. Okay, right, minus zero point two into plus half. [00:15:42.852] So, an approximate value of the function would be minus pi by four and then plus one by two and then minus 0.1 or this is minus pi by four and then minus point one by two. [00:16:02.020] Minus 0.1 by 2 is just the 1 by 20. So uh anyway, so this this is if you can substitute the value of pi by four, you can further simplify this. [00:16:11.344] So I hope the problems on Taylor's and McLaren series is clear. This is just an application problem of how we can use the Taylor series to evaluate a function at a particular point. Okay.

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Full transcript: 2399 normalized words, 64 timestamped segments, 983.744 seconds duration. Low-confidence segments flagged: 0.

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14 Extremas of functions

This lesson is indexed from the source video titled Extremas of functions. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000194 · EV0000195

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.836] Hello everyone. Welcome to another session on partial differentiation. Today we'll be doing applications of partial differentiation, which is extremes of functions. So uh before I go to the definition of what what do you mean by a maximum of a function. [00:00:16.836] So the graph here shows the local maxima of a graph, right? So these points you call them as the local maxima. [00:00:26.660] I hope you remember how you calculated maxima for a function of single variable, right? So uh suppose y is equal to f of x was a function of single variable, then uh how did we calculate the maxima or the extrema for this curve? So you had a curve, something like this. [00:00:46.116] And then we wanted to know what are these points, right? That's what we did. [00:00:50.724] So what is the procedure

that we followed for uh to calculate the local extremas? It could be either maximum or minimum. The first thing I did was to calculate the uh critical points. So how do we calculate the critical point? You calculate the first derivative and equate them to zero. [00:01:10.276] What is the practical advantage of this? You know that if you have a local maxima, then the tangent at these points will be parallel to the x axis. Right? Suppose that's the case. Can you visualize that here? See k you can see that here. So if you have [00:01:27.652] uh a plane that is parallel to the x axis or a tangent that is parallel to the x axis then its slope will be zero. So we get the critical point by solving this equation that's the first step. [00:01:40.100] The second step would be you've got to find the second order partial derivative. Second order not partial derivative, second order derivative. And if the value of this is less than zero, [00:01:54.468] At the critical point, we say that the function is maximum, right? And if it is greater than zero we say that the function has a minimum at that point. And if t square y by dx square is equal to zero, then it's neither maximum nor minimum. [00:02:11.780] So we'll follow a similar procedure to find the extremas for a function of two independent variables. So uh we'll see how does a minima look, right? So suppose uh I have minima, so you ha you can see these points are maximum and these points are the minimum points. [00:02:31.972] And the condition holds for both because the tangents as you see, the tangents at these points are also parallel to the x axis. So basically what we did you calculate the first order derivative, equate it to zero to get the [00:02:47.428] Critical points. Critical points means the points at which the curve might have a extrema, either it could be a maxima or a minima. [00:02:57.476] Then we need to decide whether that critical point represents a maximum point or a minimum point. So to do that we'll take the second derivative, substitute the value of the critical point in the second derivative, and we'll check if that point gives you negative or positive. [00:03:14.560] If it gives zero then we cannot conclude anything. Such points are called saddle point. I hope you remember that, right so a saddle point looks something like this where so if saddle point is this where the function is uh neither maximum nor minimum. So it's in between. That means the test fails basically. [00:03:31.972] So, if we extend the same thing to a function of two independent variables, then uh we'll see what are the conditions, okay? But before that when do we say that a function of two independent variables has a maximum? [00:03:46.692] Or uh you know suppose suppose I have a curve or a surface something like this. And let's say I have a point here and I call that as f of x comma y. I can decide that this point is the maximum for this surface at a point A, comma b if [00:04:06.724] Any neighborhood point the value of the function is less than the value of the function at A B. [00:04:12.868] That means this condition. If f of ab is greater than f of a plus h, comma b plus k. So you take any point in the neighborhood of this point, very close to this point, okay. Neighborhood means very close to this. So the value of the function at any point in the neighborhood of A B [00:04:30.640] must be less than the value of the function at eight. Then we say that the function has a relative maximum. [00:04:40.580] Why the word relative? It's because I'm considering the points that are closer to A B, I'm in a finite range, right? I'm not taking the uh surface in the entire uh real plane from minus infinity to infinity. That's why the term relative maximum. Because in that neighborhood that's the maximum point. [00:04:58.052] Similarly, I can say that the sum point here is the minimum if f of a b is less than f of a plus h, comma b plus k. So, the value of the function is less than the value of the function in any neighborhood of that particular point.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:17.444] Then we say that that's a relative minimum. So um that means [00:05:23.968] In other words, you can have another definition if delta of A plus H comma B plus K minus F of A B is of the same sign, that's what I mean. So you just look at this condition here. So if f of a plus h comma b plus k minus f of ab is of the same sign [00:05:41.648] For all small h and k and if sin is negative then f of ab is maximum. If it's positive we say that f of ab is minimum. That's the definition, okay. So you don't need to worry too much about it because you know what what we call it as a maximum point and what we define it as a minimum point. [00:06:00.228] Now, how do I check if the function has an extrema at a given point for a function of two independent variables. Okay. So if we want to compare that with a function of single variable, let me write down for reference what all we did. So what we did first is we got dy by dx equal to 0. [00:06:19.748] So, corresponding to this, what we have for a function of two independent variables, you know that there are two first order partial derivatives. So, that is fx of a b which is equal to 0 and f y of ab which is equal to 0. [00:06:35.748] So the first one to find the critical point I'm going to equate the first order partial derivatives to 0. So f x of ab to 0 and f y of ab to zero. Okay? And the second condition, what is the second condition here? I need to find [00:06:53.124] The second order derivative, substitute the point that you got here and check if the value is negative or if it's positive or if it's equal to C. [00:07:04.804] Now I'll do a similar thing for this curve as well. That is, I am going to find because you know that there are uh four mixed order partial derivatives, right? R T minus a s square. Where what is r? R is nothing but f x. So this is r f x y is uh s [00:07:22.276] And so F y is also y x is also S. You know that mixed second order partial derivatives are always equal and FYY will be T. And this determinant is called a Hesian matrix. [00:07:37.060] So, basically we are expanding the determinant of the Hesian matrix and if I expand this determinant I will get F x x into f y y minus f x y the whole square is nothing but [00:07:51.556] Is that's a condition I want. So if this is greater than zero, only then that point would represent an extreme. Okay? So remember here we got it directly, but here we need to check another condition. So the first [00:08:07.524] preliminary condition that should be satisfied for a point to represent an extrema, it could be a maxima or a minima is f x x into f y minus f x y the whole square must be greater than zero. [00:08:23.812] If this R T minus S square is less than 0, then f of a b is not an extrema value at all. You cannot conclude anything from that. [00:08:33.232] Okay. So this has to be greater than zero. Just in case if it becomes equal to zero, then the test completely fails. You don't know whether you don't know it the you don't get any information about the nature of that point, whether it is a maxima or a minima or a saddle point, nothing we get. [00:08:50.020] Okay. So there are two important conditions that you have to remember. The first thing is find out both the first order partial derivatives, equate them to zero. [00:09:00.580] Secondly, find R T minus S square, that is a combination of the second order partial derivative. So you calculate f x into f y minus f x y the whole square should be greater than 0. So this will decide whether that [00:09:16.544] uh I'll have an extrema or not. Okay. At some point if this is zero, how do I decide that that point is a maximum or a minimum [00:09:26.500] In that case, we have few conditions. Okay. So what you need to do, we got this R T minus a square. Along with this condition, if R is less than 0 at any point AB, then f of A B has a maximum value. [00:09:43.076] R means what then? F x x. So you need to check f x x f y y minus f x y the whole square. This must be greater than 0. Along with this you need to check that f x must be less than 0.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:02.596] which is R. So f x x must be less than 0. So if this condition is satisfied then it will have a minimum or maximum. This is similar to the condition, see you remember this d square y by d x square has to be less than 0. [00:10:20.100] So this is similar to you can consider that as it's similar to Fxx. And the second condition is so if R T minus S square this has to hold good for it to be an extrema. Along with that I have F Y Y now to be um [00:10:38.468] Uh sorry F x x only. So this is F xx if it is greater than zero. So let me just erase that so that it doesn't confuse you. [00:10:47.056] So, for a condition for it to be minimum now, right? Condition for it to be minimum is f x must be greater than 0 and that's what you concluded, right? In the case of single variable. If d square y by dx square is greater than 0, then the function will have a minimum at that point. [00:11:06.564] So, even here I got the same thing. And then lastly if I have R T minus S square to be less than 0, then it's it represents a saddle point. It's it it it doesn't sway to the left, neither it sway to the right. So it represents neither mix maximum nor a minimum point. [00:11:24.368] And of course if this is zero then I cannot conclude anything, so you have to say that no conclusion can be made. [00:11:32.420] So is the working rule clear? Now um let me just write all of them together. The first thing is uh you find out f x equated to zero, find f y equated to zero. Solve these two equations to get critical points. [00:11:51.300] Okay? So solve them. Next is you calculate the second order partial derivative. So you calculate f x x f y y minus minus f x y the whole square. [00:12:06.948] Okay? So this has to be positive. This has to be positive. Along with this, if f x is less than 0, I'll get maximum. That means that point represents maximum. And if x f x is greater than 0, [00:12:24.256] So uh then it represents a minimum point. So let me just uh write that properly. [00:12:33.828] So if xx is greater than 0 then this would represent a minimum point, right? And if f x x into f y minus fxy the whole square, if it is less than 0, then it's a saddle point. [00:12:53.380] And if just in case if this is equal to zero then no conclusion. So you can write uh no conclusion. Is it clear? So anyway we'll solve the problem maybe so that makes things little more uh clear for you.

EV0000202 · EV0000203 · EV0000204 · EV0000205

Full transcript: 2186 normalized words, 52 timestamped segments, 788.416 seconds duration. Low-confidence segments flagged: 0.

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15 Extremas of functions continued

This lesson is indexed from the source video titled Extremas of functions continued. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. No slide deck is assigned to this video; candidate slide sources remain separate in the unit master. EV0000206 · EV0000207

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.000] So the question says the temperature T at any point XYZ in space is 400 XYZ square, so that's the temperature function.

Find the highest temperature on the surface of the unit sphere. [00:00:14.884] Now there are two things that you need to observe. First thing is so that means they have given a sphere which might look something like that. [00:00:24.708] because it's already in the xy plane. So that's a sphere. And they have given the temperature function t of x y z as 400 x y z square which varies with the par with the spatial coordinates. So x y and z square. [00:00:40.388] But do we know how to find the extremas for a function of three independent variables? No, right? That's a problem I have. So what I will do is so this is a constraint that is given to me, that means you need to find the temperature in space, find the highest temperature on the surface of the sphere. [00:01:00.164] So I can solve for z square from the sphere. I can write z square as 1 minus y square minus x square. So what will be my t now the temperature function? So t of xyz will be 400. [00:01:15.088] x y into for z square I can substitute 1 minus x square minus y square. Right? Now this is a function of only two independent variables x and y. Now I can differentiate. [00:01:31.844] Right. So here I have taken it as f, so f is this function. You differentiate that with respect to x partially and equate it to 0. So what do I get if I differentiate t with respect to x partially? [00:01:44.292] I will get 400 y is a constant, I do not need to worry, derivative of x is 1, derivative of x cube is 3 x square and derivative of x y square is y square. If I equate this to 0, [00:01:57.604] Then I get 1 minus 3x square minus y square to be 0. Similarly, if I take f y, which is t y. [00:02:05.956] So, what is T y now four hundred x is a constant, I'll get one minus x square minus three y square. That's a derivative. And if you equate this to zero you will get this equation. So, can I solve these two equations for x and y? [00:02:24.420] They are like simultaneous equations. So uh solving shouldn't be a problem. Um you subtract them together. If you subtract I'll get minus two x square and then plus two y square equal to zero. [00:02:42.180] Sorry. So if I solve this then I get x square to be y square or x will be plus or minus 1. If x is plus or minus y, [00:02:56.868] What w what is the value of x and y that I get if I substitute as x is equal to plus or minus y there. So after all the simplification I got x is equal to plus or minus y. Let me substitute in either this equation or this equation, it doesn't matter. So one minus three x square minus x square is zero. [00:03:17.156] Or I get 1 is equal to 4 x square or x square equal to 1 by 4. So x will be plus or minus half. Right? [00:03:27.204] But also what we got is y equal to 0 or this. So f y we got x equal to 0. Actually this should be x equal to 0, not y equal to 0. So we have x is 0, x is equal to half, y is equal to half. [00:03:46.724] x is equal to minus half and y can be minus half. What we need to do is we just need to uh c get a combination of these things. So how many points you will get? You get these many points. Either it could be 0 0 [00:04:01.924] uh half comma half, half comma minus half, minus half comma half and minus half comma minus half. Now what these are the critical points or the stationary points, okay? So [00:04:13.668] That means these are the points at which there is a possibility for the function to have an extrema. I don't know whether it will be a maximum or a minimum or a saddle point, I don't know. Right? So next thing what we need to do is we need to find the second order partial derivatives. [00:04:31.652] So you need to find f x f y y minus f x y the whole square. So for this problem I have already calculated that and for 0 0 at the point 0 0 I get minus 1600. So if it's negative you know that it's a saddle point. [00:04:51.044] And for half comma half it is positive, for minus half comma half also it is positive. And uh so you can calculate what is the maximum value that it can attain and minimum value it can attain by substituting these values, right. So we have

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.100] maximum at two points. Can we conclude just by taking this? Just by taking this condition? No. [00:05:16.228] I need to check along with that that R is less than zero. So let me quickly go back to the previous slide. Okay? So what is uh the function for me now? Uh I can erase the things that are there here. The function is this. So what is F x now? [00:05:32.592] We got F x already as four hundred y into one minus y square minus three x square. Right? What is f x? [00:05:45.092] So 400y is again a constant. I need to differentiate this which will give me minus 6x. So at any given point because of this negative, okay, if both x and y are positive, [00:05:59.728] or both x and y are negative, I'll get f of x to be negative and that's what I want, right? So that's what we get here. So at half comma half and at minus half comma half r t minus s square is definitely greater than zero and r is less than zero. [00:06:17.988] Because it satisfies this condition, we say that the T is maximum at these two points on the surface of a sphere. And if you want to calculate what is that maximum value, substituting four hundred uh so we have four hundred into x. [00:06:33.572] into y into z square. But z square is actually 1 minus x square minus y square. Right? So this is basically hundred and um this will be hundred into one minus half which is half again. So fifty. [00:06:51.428] That means the maximum temperature that is attained on the surface of a sphere is fifty. So this is one method wherein we actually eliminate and then we substitute. [00:07:06.308] Let's do another problem. This is more of a practical problem, but I felt like you will appreciate this because you will understand how to solve a problem. It says a container with an open top, let me underline whatever you need to be careful about is to have a ten meter cube capacity. [00:07:24.804] and be made of thin sheet of metal. Calculate the dimensions of the box if it is to use the minimum possible amount of metal. [00:07:38.596] So it's an open top means what? Because I am worried about the surface area. So can I write this as a is equal to 2xy plus 2yz plus zx. [00:07:51.716] I hope it's clear why didn't I prefix this with two is because my top is open here as you can see this is not closed so because it's an open top box. And of course it's a rectangular box, so they have given XYZ to be ten. That's a wall. Again. So [00:08:10.948] I hope you understood how I wrote this. This is because I need to minimize the possible amount of metal that is required for me to made it is made out of thin. So basically I am looking for the surface area. Correct? And then they have given me that the volume of that container is 10 meter cube. [00:08:27.684] Now the question is how do I eliminate a variable? I cannot handle three independent variables, right? So suppose the x y z is equal to ten, can I write z as ten by x y? [00:08:41.508] Like we did in the previous case, we use the equation of the sphere. We are going to eliminate z. So what will happen to area now if I substitute z I get 2 x y plus 2y into z, z will be 10 by x y [00:08:57.520] plus zx. So this is 10 by x y into x. Right? So this x gets cancelled. I I am doing little simplification here. So I'll get area to be 2 x y [00:09:10.436] plus twenty by x plus ten by y. So this is the function which I need to minimize because I want the you know minimum possible amount of metal to be used. Now what are the condition for this to be minimum? So let me just uh write that again. [00:09:30.756] So what I have is area is twenty uh xy, right? So it is two x y, sorry. [00:09:43.076] area is equal to 2xy plus 20 divided by x plus twenty divided by y and that's what we got. Right? [00:09:56.004] What are the conditions for us to minimize this? So the first thing is I need at what points it can be minimum. So I have to find a x. Ax will be 2 y minus 20 by x square, right? Equate this to 0. What do I get if I equate this to zero?

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:15.012] I can um take out uh uh y comma two comma and I can divide two out through out by two. So I'll get y is equal to ten by x square. Is that fine? So this is the first equity that I get y is equal to twenty by or ten by x square. [00:10:33.892] Similarly, I will find a y. What's a y? I will get 2x again minus 20 by y square equal to 0. If I equate this to 0, I will get x is equal to 10 by y square. [00:10:50.020] So y is equal to ten by x square or x is equal to ten by y square. Now I need to solve them of course, but I need some kind of equation. Where should I put that to solve, right? I need to solve for x and then y and then z to get the critical points. So where can we substitute that? [00:11:09.476] The only equation that I know is XYZ equal to ten. So from where I eliminated this. I got x y is equal to ten by x square and x is equal to ten by y square. What is the next thing I know? [00:11:26.820] I know that I have to find a x a y minus a x y the whole square must be greater than zero. Right. So [00:11:43.524] Alright, so I have these two equations now y is equal to 10 by x square and x is equal to 10 by y square. Can I substitute one in another? I have y is equal to 10 by x square. So let me substitute for x here 10 by uh x is 10 by [00:12:02.372] y square the whole square. Right? So y will be 10 y power 4 divided by 10 square. [00:12:14.852] Or I can eliminate this. So I get y is equal to y to the power of four divided by ten. Can we solve this? I'll get y minus y to the power of four by ten to be equal to zero. Is it fine? [00:12:32.708] So that means I can take out y common and then I get 1 minus y cube by 10 equal to 0. What do we get when we solve this equation? I'll get y equal to 0. [00:12:46.916] as the first value or so the other equation that I have is 1 minus 1 minus y cube by 10 equal to 0. So if I solve this I will get 1 equal to y cube by 10 or y is equal to cube root of 10. Right. [00:13:07.876] So I got y equal to zero um and y is equal to cube root of ten. Y is equal to zero doesn't make sense for me. And so how do I find x now? I know that x is equal to ten by y squared. [00:13:22.724] So that's the equation that I have from the a y equal to 0. So if I substitute that here I will get 10 divided by y square which is 10 to the power of 2 by 3. So this is 10 to the power of 1 minus 2 by 3, which is again cube root of 10. [00:13:40.612] Right. So I got x as cube root of ten and y as cube root of ten. [00:13:48.996] And of course the y equal to 0 doesn't work because if y becomes 0 x tends to infinity. That's another way to eliminate that. And how do I get z? Z is equal to 10 by x y. So ten by cube root of 10 into cube root of 10. [00:14:05.892] So, can we simplify this little bit? This is 10 divided by 10 to the power of 2 by 3 or this is 10 to the power of 1 minus 2 by 3 which is 10 to the power of 1 by 3 again. Is it fine? So we have um z is equal to 10 by x y. [00:14:25.412] And the we just substituted that. [00:14:32.420] We got x is equal to y is equal to z. So if you want to minimize that's the value of uh z that I'll get. Okay. So we got the value of x is equal to cube root of ten, y is equal to cube root of ten and z is also cube root of ten. [00:14:48.868] But how do we check that this will m this is the minimum value or the minimum amount of metal that I can get, right? So what I have is I have a x and a y. So can you find what is a x from this from this equation? So I'll get a x as um

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Transcript block 4 (source interval 000900-001199 seconds): [00:15:08.548] 20 by x square and derivative of 1 by x square will be 2 by x cube with a minus. So this will be 40 by x cube. Right? And a y will be same thing, which is 40 by y cube. And then I need another axy. So what will be a x y is just 2. [00:15:27.428] So, the condition that we got we got the critical points now let's check the condition. What are the condition we have? Ax x into ay minus a x y the whole square. [00:15:39.332] And what happens to this? A x we got the value right. What is the value that we got? 40 by x cube. So this is 40 divided by x cube and this is 40 divided by y cube minus a x y we

got it as just 2. So this is just 4. Okay. [00:15:59.332] So this will be 40 into 40 divided by I already know that x is cube root of 10. So this will be just 10 into 10 minus 4. So that means this is greater than 0. [00:16:16.868] And what is Ax? You can observe that for all the points that we got for x, y and z, a x is always greater than 0. So if a x is greater than 0 then we get the minimum value. [00:16:33.060] That means the dimensions of the box so that the minimum of metal sheet is used will be x is equal to cube root of ten. It's a basically a cube and y is equal to cube root of ten and z is equal to cube root of ten. [00:16:49.156] I hope the problem is clear. Uh in the next session we will see another method wherein we don't have to eliminate. See the problem with us we had like we have to eliminate one of the variables to make it into a function of two independent variables. It might not be as simple as this in every case, right. [00:17:07.872] So we'll see another method which will help you to uh minimize that so that you know how to solve the problem.

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Full transcript: 2781 normalized words, 66 timestamped segments, 1034.837333 seconds duration. Low-confidence segments flagged: 0.

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16 Problems on Maxima and Minima for a function of two variables

This lesson is indexed from the source video titled Problems on Maxima and Minima for a function of two variables. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000222 · EV0000223

Graphs and diagrams

Figure G000145. Problems on Maxima and Minima for a function of two variables · source slide page 1.

EV0000691

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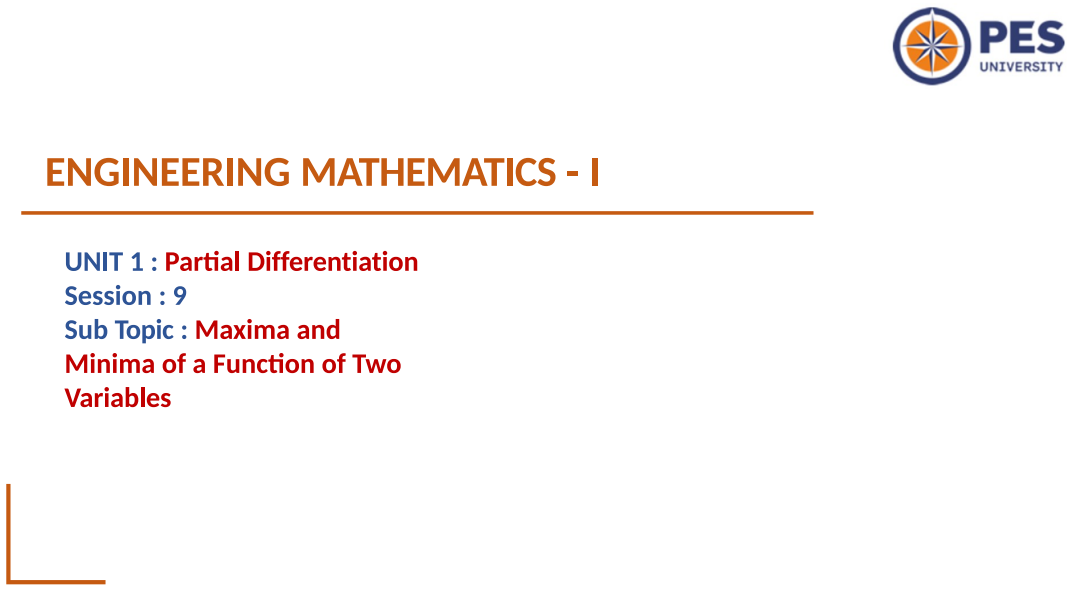

Figure G000146. Problems on Maxima and Minima for a function of two variables · source slide page 2.

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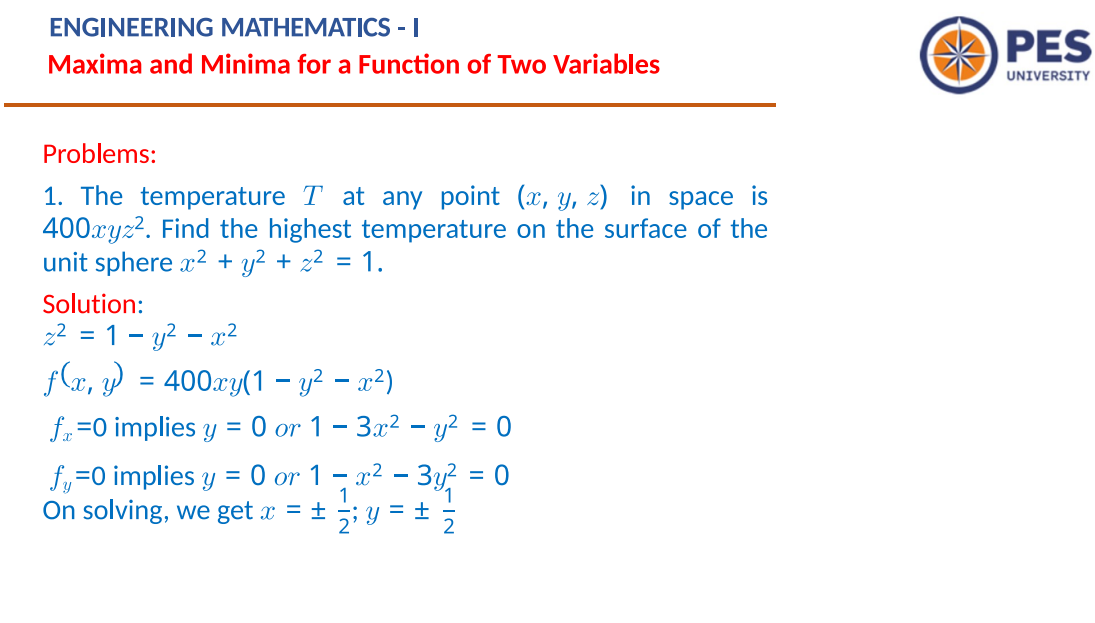

Figure G000147. Problems on Maxima and Minima for a function of two variables · source slide page 3.

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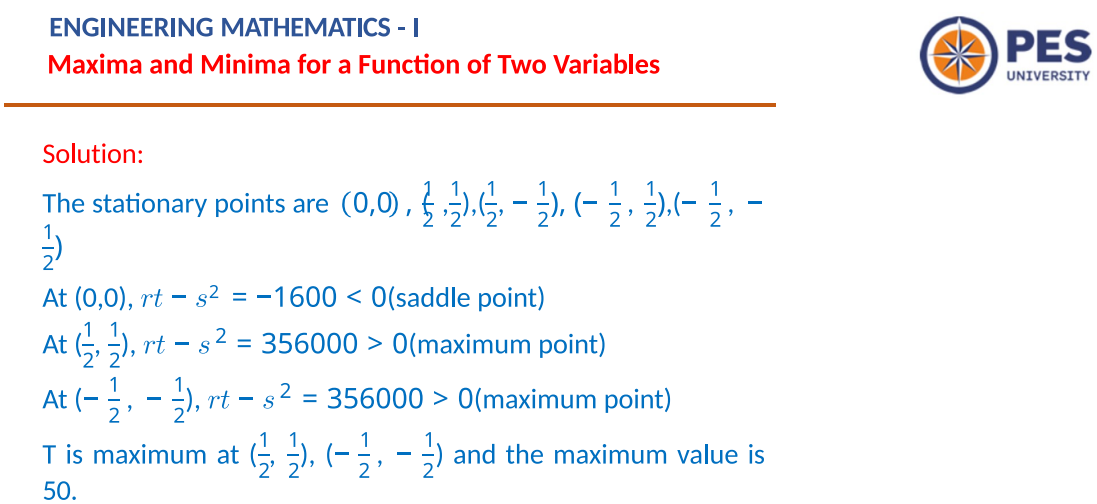

Figure G000148. Problems on Maxima and Minima for a function of two variables · source slide page 4.

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ENGINEERING MATHEMATICS - I

Maxima and Minima for a Function of Two Variables

2. A Container with an open top is to have 10 m^3 capacity and be made of thin sheet metal. Calculate the dimensions of the box if it is to use the minimum possible amount of metal.

Solution:

Let x, y and z ft. be the edges of the box and A be its surface. Then $A = 2xy + 2yz + xz$.....(1)

$xyz = 10$.....(2)

Substitute (2) in (1)

$A = 2xy + 10\left(\frac{2}{x} + \frac{2}{y}\right)$

$A_x = 0\ gives\ 2y - \frac{20}{x^2} = 0\ and\ y = \frac{10}{x^2}$

Figure G000149. Problems on Maxima and Minima for a function of two variables · source slide page 5.

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ENGINEERING MATHEMATICS - I

Maxima and Minima for a Function of Two Variables

Contd.....

$A_y = 0$ *gives* $2x - \frac{20}{y^2} = 0$ *and* $x = \frac{10}{y^2}$

Solving we get $x = \sqrt[3]{10}, y = \sqrt[3]{10}, z = \sqrt[3]{10}$

$A_{xx} = \frac{40}{x^3}$

$A_{yy} = \frac{40}{y^3}$

$A_{xy} = 2$

$A_{xx}A_{yy} - (A_{xy})^2 > 0$ *and* $A_{xx} > 0$

Therefore it is minimum and the dimensions of the box are

$x = \sqrt[3]{10}, y = \sqrt[3]{10}, z = \sqrt[3]{10}$

Figure G000150. Problems on Maxima and Minima for a function of two variables · source slide page 6.

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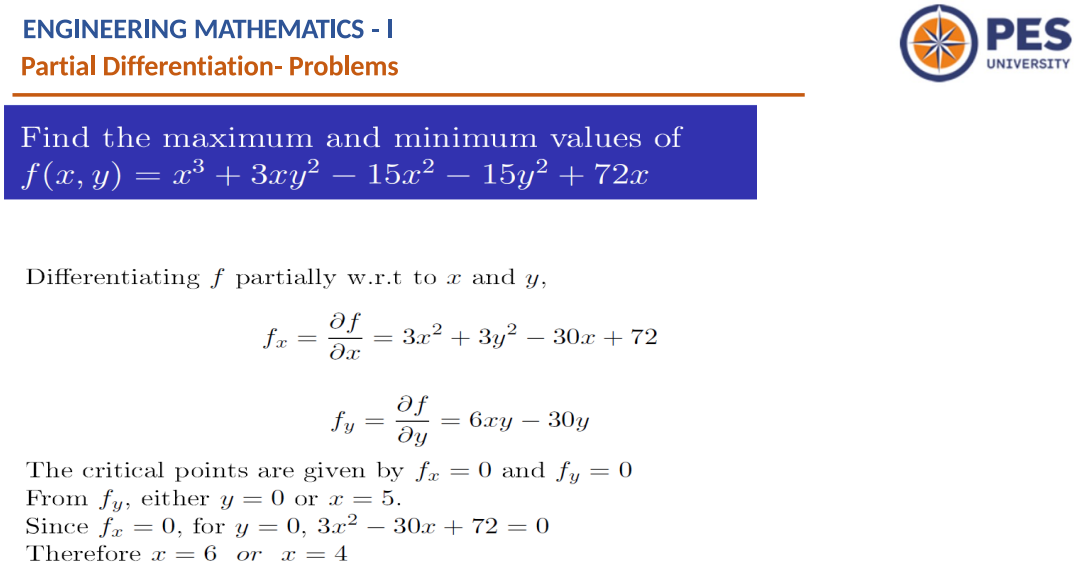

Figure G000151. Problems on Maxima and Minima for a function of two variables · source slide page 7.

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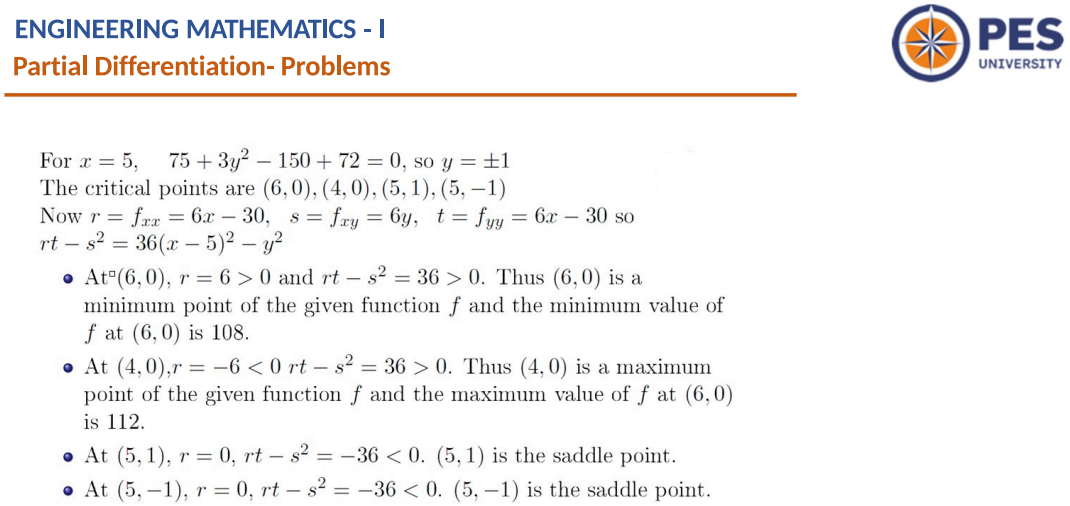

Figure G000152. Problems on Maxima and Minima for a function of two variables · source slide page 8.

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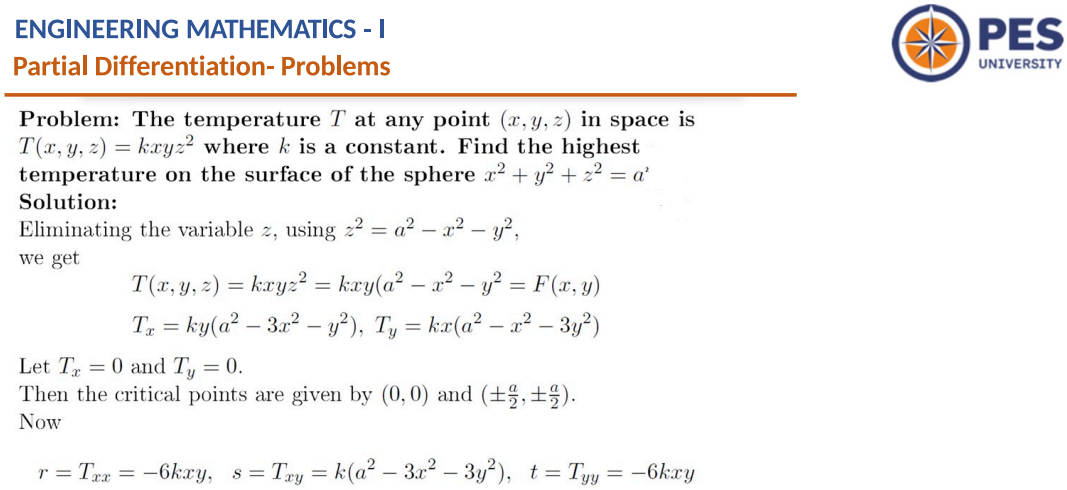

Figure G000153. Problems on Maxima and Minima for a function of two variables · source slide page 9.

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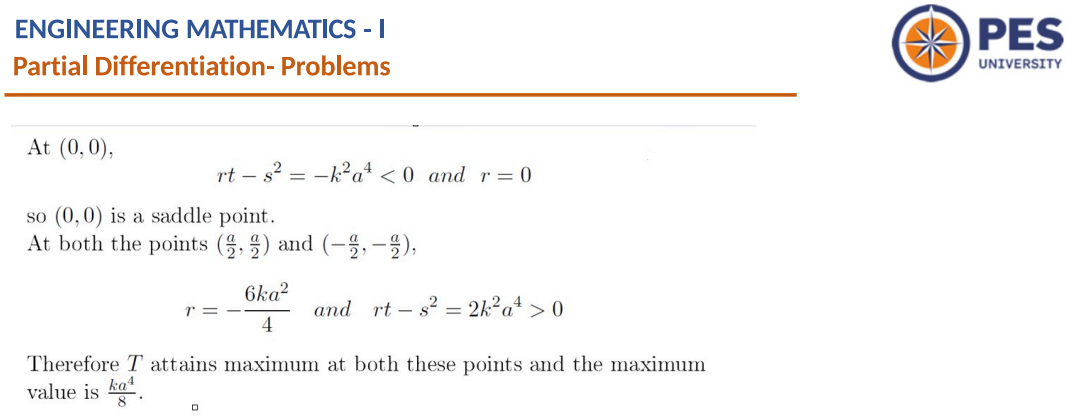

Figure G000154. Problems on Maxima and Minima for a function of two variables · source slide page 10.

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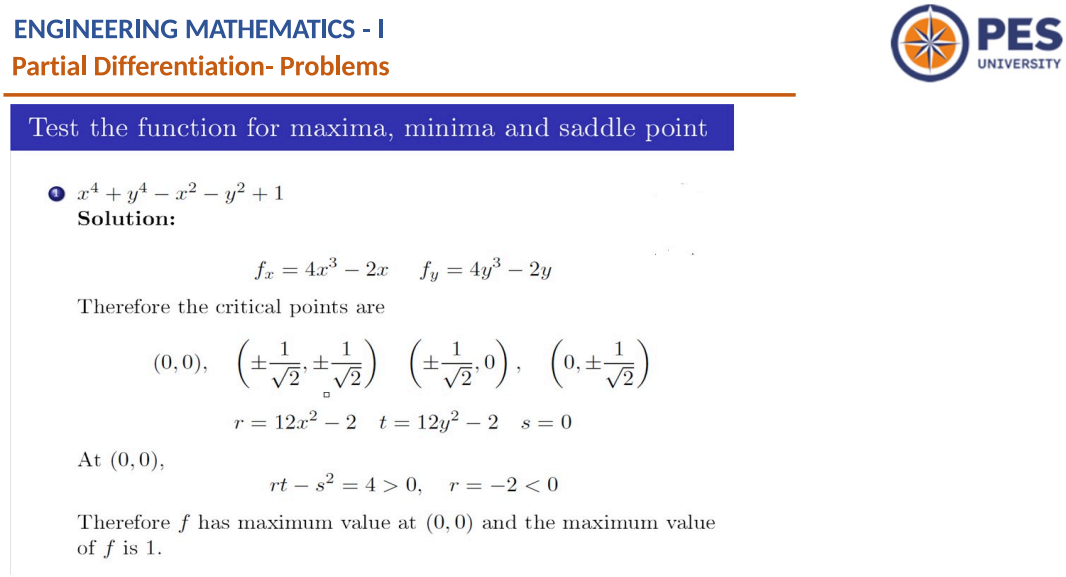

Figure G000155. Problems on Maxima and Minima for a function of two variables · source slide page 11.

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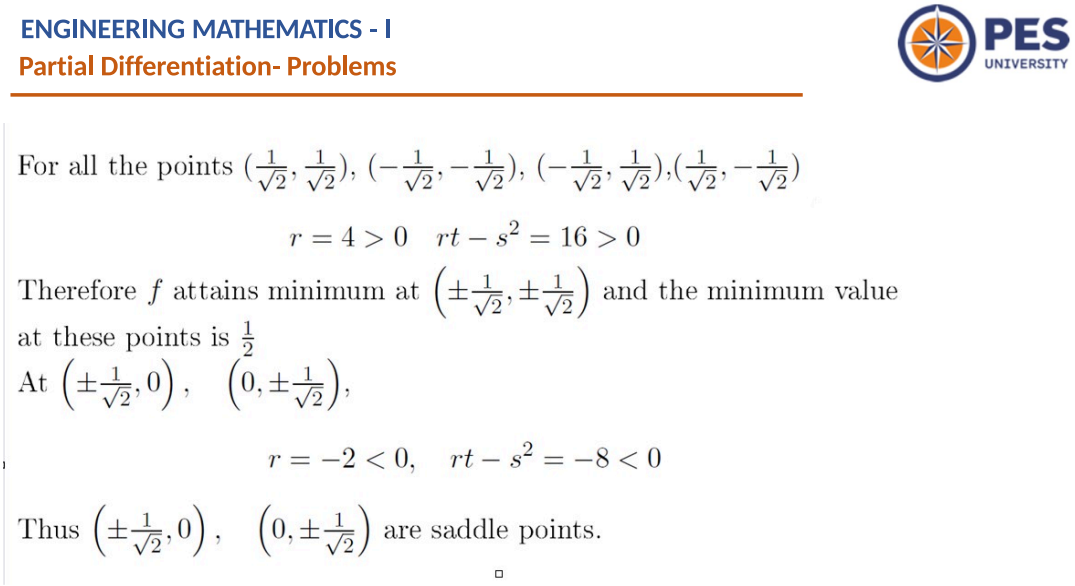

Figure G000156. Problems on Maxima and Minima for a function of two variables · source slide page 12.

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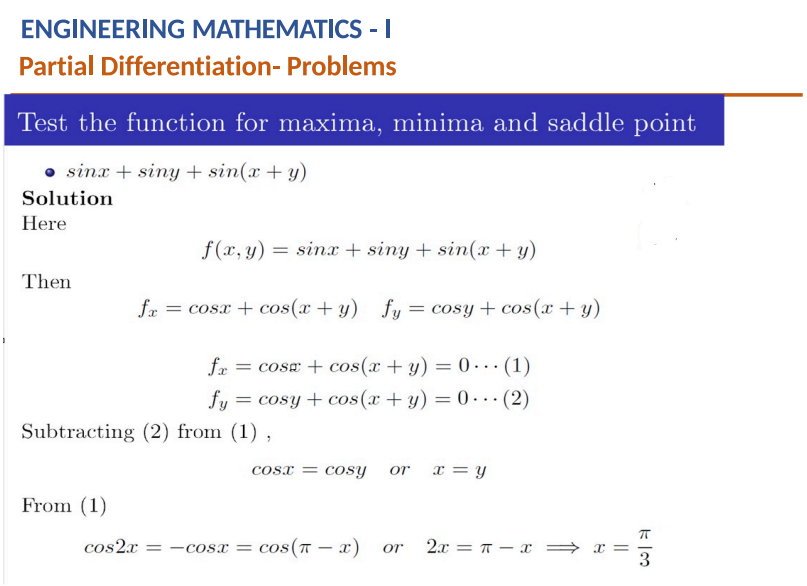

Figure G000157. Problems on Maxima and Minima for a function of two variables · source slide page 13.

EV0000727

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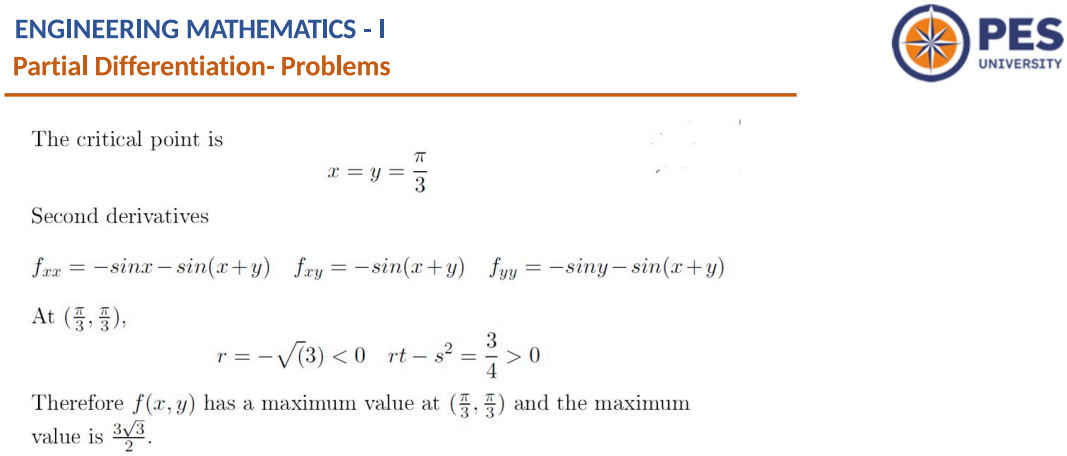

Figure G000158. Problems on Maxima and Minima for a function of two variables · source slide page 14.

EV0000730

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Figure G000159. Problems on Maxima and Minima for a function of two variables · source slide page 15.

EV0000733

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000690 · EV0000691 · EV0000692
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 9 Sub Topic : Maxima and Minima of a Function of Two Variables EV0000693 · EV0000694 · EV0000695
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Problems: 1. The temperature T at any point (x, y, z) in space is $400xyz2$. Find the highest temperature on the surface of the unit sphere $x2 + y2 + z2 = 1$. Solution: $z2 = 1 - y2 - x2$ $f\ x, y = 400xy(1 - y2 - x2)$ fx=0 implies $y = 0\ or\ 1 - 3x2 - y2 = 0$ fy=0 implies $y = 0\ or\ 1 - x2 - 3y2 = 0$ 1 2 1 On solving, we get $x = \pm$; $y = \pm$ 2 EV0000696 · EV0000697 · EV0000698
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Solution: 1 1 1 1 1 1 1 The stationary points are 0,0 , (,),(, −), (− ,),(− , − 2 2 2 2 2 2 2 1) 2 At (0,0), $rt - s2 = -1600 < 0$(saddle point) 1 1 2 At (,), $rt - s =$ $356000 > 0$(maximum point) 2 2 1 1 2 At (− , −), $rt - s = 356000 > 0$(maximum point) 2 2 1 1 1 1 T is maximum at (,), (− , −) and the maximum value is 2 2 2 2 50. EV0000699 · EV0000700 · EV0000701

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables 2. A Container with an open top is to have 10 m^3 capacity and be made of thin sheet metal. Calculate the dimensions of the box if it is to use the minimum possible amount of metal. Solution: Let x, y and z ft. be the edges of the box and A be its surface. Then $A = 2xy + 2yz + xz$.....(1) $xyz = 10$.....(2) Substitute (2) in (1) 2 1 x y $A = 2xy + 10 + 20$ 10 $Ax = 0$ $gives$ $2y - x2 = 0$ and $y = x2$ EV0000702 · EV0000703 · EV0000704

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Contd..... 20 10 $Ay = 0$ $gives$ $2x - y2 = 0$ and $x = y2$ 3 3 $Solving\ we\ get\ x = 10,\ y = 10,\ z = 3$ 10 = 40 $A\ xx\ x3 = 40\ A\ yy\ y3\ Axy = 2\ AxxAyy - Axy$ $2 > 0$ and $Axx > 0$ Therefore it is minimum and the dimensions of the box are 3 $x = 3$ 10, $y = 10$, $z = 3$ 10 EV0000705 · EV0000706 · EV0000707

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Find the maximum and minimum values of f(c,y) = x² + 3cy² - 15x² - 15y² + 72c Differentiating f partially w.r.t to and y, fx = Oc af fy=y = 6cy - 30y The critical points are given by fx = O and fy = 0 From fy, either y = 0 or = 5. Since f = 0, for y = 0, 3x² - 30 + 72 = 0 Therefore c = 6 or = 4 EV0000708 · EV0000709 · EV0000710

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] For c = 5, 75 + 3y² - 150 + 72 = 0, s0 y = ±1 The critical points are (6,0), (4, 0), (5, 1), (5, -1) Now r = fcx = 6x - 30, s = fcy = 6y, t = fyy = 6c - 30 s0 rt - s² = 36(c - 5)² - y² At(6,0), r = 6 > 0 and rt - s² = 36 > 0. Thus (6,0) is a minimum point of the given function f and the minimum value of f at (6, 0) is 108. At (4,0),r = -6 < 0 rt - s² = 36 > 0. Thus (4,0) is a maximum point of the given function f and the maximum value of f at (6,0) is 112. At (5,1), r = 0, rt - s2 = -36 < 0. (5,1) is the saddle point. ● At (5,-1), r = 0, rt - s² = -36 < 0. (5, -1) is the saddle point. EV0000711 · EV0000712 · EV0000713

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Problem: The temperature T at any point (,y,z) in space is T(c,y,2) = kryz2 where k is a constant. Find the highest temperature on the surface of the sphere r2 + y2 + z2 = a* Solution: Eliminating the variable z, using 2 = a² - x² - y², weget T(x,y,z) = kxy2² = kxy(a² - x² - y² = F(x,y) Tx = ky(a² - 3x² - y²), Ty = kx(a² - x² - 3y²) Let Tr = 0 and Ty = 0. Then the critical points are given by (O, 0) and (±,±). Now r = Trx = -6kcy, s = Try = k(a² - 3x² - 3y²), t = Tyy = -6kcy EV0000714 · EV0000715 · EV0000716

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] At (0, 0), rt - s² =-k²a < 0 and r = 0 so (O, O) is a saddle point. At both the points (2,%) and (-,-) 6ka2 and rt - s² = 2k²a4 > 0 4 Therefore T attains maximum at both these points and the maximum value is ka4 EV0000717 · EV0000718 · EV0000719

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Test the function for maxima, minima and saddle point x4 + y4 - x² - y² + 1 Solution: fc = 4x3 - 2c fy = 4y² - 2y Therefore the critical points are (), (0,± (0,0), r = 12α² -2 t = 12y² - 2 s = 0 At (0,0), rt - s² = 4 > 0, r = -2 < 0 Therefore f has maximum value at (O, O) and the maximum value of f is 1. EV0000720 · EV0000721 · EV0000722

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] For all the points (), (——), (—)() r =4>0 rt-s² =16 >0 Therefore f attains minimum at (and the minimum value at these points is 2 0,± At ±, 0) r=-2<0, rt-s²=-8< 0 ±,0) (0,± Thus are saddle points. EV0000723 · EV0000724 · EV0000725
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Test the function for maxima, minima and saddle point sinc + siny + sin(c + y) Solution Here f(x,y) = sinx + siny + sin(x + y) Then fx = cosc + cos(c + y) fy = cosy + cos(c + y) fc = cosc + cos(c +y) = 0..· (1) fy = cosy + cos(c + y) = 0..· (2) Subtracting (2) from (1); cosr = cosy or=y From (1) T cos2=-cos = cos(π-) or 2=π- =→ 3 EV0000726 · EV0000727 · EV0000728
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] The critical point is T c=y= 3 Second derivatives fcx = -sinc -sin(c+y)) fcy =-sin(c+y)) fyy = -siny- sin(c+y) At (,), 3 r=-V(3)<0 rt-s2 √ 4 Therefore f(c,y) has a maximum value at (,) and the maximum value is 3v3 2 EV0000729 · EV0000730 · EV0000731
15. Source slide page 15 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000732 · EV0000733 · EV0000734

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.172] Hello everyone. Welcome back to the course on Engineering Mathematics 1. In this session, we solve problems on maxima and minima for a function of two variables. F of x comma y. [00:00:23.972] Let's start with this problem. Find the maximum and minimum values of function x power 3 plus 3xy square minus 15x square minus 15y square. [00:00:43.172] plus 72x. As per the procedure, we first find out the partial derivative of f with respect to x and y, which we call them as p as well as q. [00:01:00.676] Now for the given f of x comma y, p becomes three x square plus three y square minus thirty x plus seventy p. Similarly, q becomes six x y minus thirty y. [00:01:23.748] Next, in order to find out critical points, we need to equate fx as well as f y to zero. So [00:01:37.060] Then [00:01:40.644] From f y equal to zero, we see that either y equal to zero or x equal to five. It means when we equate 6x minus 30y to 0, we see that either y 0 or x equal to 5. [00:01:59.204] So when y is 0, this expression 1 becomes 3x square minus 30x plus 72 equals 0. So when we solve this, we see that x is 6 or x is 4. [00:02:16.196] So for this particular case the critical points are 6,0 as well as 4 comma 0. We have one more case left out. In a similar manner, we have to put x equal to 5 in expression number 1. [00:02:34.148] So when x is phi from equation one, we see that 75 plus 3y square minus 115 plus 72 equals 0. So when we solve this, we get y equal to plus R minus 1. [00:02:51.780] So the critical points for this particular case phi comma 1 as well as phi comma minus 1. So finally we have these four critical points, namely 6 comma 0, 4 comma 0, 5 comma 1 and 5 comma minus 1. [00:03:12.836] Then we need to compute R S t. R is nothing but dou square f divided by dou x square, which is 6x minus 30. Then S is nothing but dou square f divided by dou x dou y which is nothing but 6y. [00:03:30.564] Similarly, T is dou square f divided by dou y square, which is nothing but 6x minus 30. So therefore R T minus s square becomes 36 of x minus 5 whole square minus pi square. [00:03:48.804] As we see that there are four critical points. Now we need to compute R T minus S square at each of these critical points. So we can think of four cases because of four critical points. [00:04:02.980] At 0, 6 comma 0 we see that R t minus s square is positive, but R is positive. Therefore, we can say that at 6 comma 0 the function is minimum. So the

minimum value is given by 108. [00:04:24.612] At the critical point 4 comma zero, what happens? At 4 comma zero, R T minus S square is 36, which is positive, but R is negative. Therefore, we say that at the point 4 comma 0, the function is maximum. [00:04:42.020] So the maximum value at 6 comero is 12. So at 5 comma 1 as 5 comma minus 1 we see that R T minus S square is negative. Therefore we say that we say that these are the two stattle points.

EV0000222 · EV0000223 · EV0000224 · EV0000225

Transcript block 2 (source interval 000300-000599 seconds): [00:05:00.644] So this is the complete answer for this problem. We first found out PQ, then we equated P and Q to 0. Then we found out critical points. Then we computed RT minus S square. [00:05:16.740] Then evaluated RT when C square at each of those critical points then arrived at this conclusion. [00:05:25.028] Let's look at another problem. The temperature capital T at any point XYZ in space is given by this expression KXY Z square where K is a constant. [00:05:40.548] Find the highest temperature on the surface of the sphere x square plus y square plus z square equal to a square. Note that because we are looking for maximum temperature, we assume that k to be a positive constant. Also, [00:06:00.388] We need to convert this function of three variables to a function of two variables by using z value. So from the given expression, we solve for z square, it is going to be a square minus x square minus y square. So finally [00:06:18.304] P of x y z becomes k x y of a square minus x square minus y square. So for this function we find out p and q. So p is going to be ky of a square minus 3x square minus y square. [00:06:37.028] Similarly, Ty is going to be kx of a square minus x square minus 3y square. So t x and t y are nothing but the partial derivative of t with respect to x and y. [00:06:52.868] Now to compute critical points we need to equate tx and t y to zero. So when we do this procedure we see that 0 comma 0 plus r minus a by 2 plus r minus a by 2 are the three critical points. Then [00:07:10.244] The value of R is going to be minus 6K XY. The value of S is going to be K of A square minus 3x square minus 3y square. The value of T is going to be minus 6 Kxy. Let's evaluate R T minus S square at each of these critical points. [00:07:30.500] So at 0 com we see that R T minus S square is negative. Therefore, we say that 0.0 is a saddle point. In a similar manner, we see that at the points a by 2, a by 2, as well as minus a by 2, and minus a by 2, R T [00:07:50.020] Minus S square is positive, but R is negative because we are assuming K to be positive. Therefore, we say that P attains maximum at both these points and the maximum value is given by k a power 4 divided by 8. [00:08:12.484] Let us take another example. Test the function for maxima, minima and settle point. The function is given by x power 4 plus y power 4 minus x square y square plus 1. [00:08:29.348] Now for this function, fx is going to be 4x power 3 minus 2x. Fy is going to be 4y power 3 minus 2y. So these are nothing but the partial derivative of f with respect to x and y. [00:08:45.604] Now when we equate P and Q to 0 after we saw we see that these are the critical points. So at 0 comma 0 we see that R T minus s square is positive, but R is negative. [00:09:00.580] Therefore, we say that the function is maximum at the point zero comma zero and the maximum value is one. [00:09:09.188] In a similar manner, it is easy to see that at all these four points, R T minus S square is positive, but R is also positive, therefore we say that F is minimum at these points. And the minimum value is given by plus 1 by 2. [00:09:26.564] Lastly, at plus r minus 1 by root 2 0, and 0, plus r minus 1 by root 2, we can see that r t minus s square is minus 8, which is negative, but r is also negative. Therefore, we say that these are the two saddle points. [00:09:50.212] Let us take one more example. F of x comma y to be sine x plus sine of y plus sine of x plus y. So for this particular example, a p is going to be cos x plus cos of x plus y.

EV0000226 · EV0000227 · EV0000228 · EV0000229

Transcript block 3 (source interval 000600-000899 seconds): [00:10:08.196] Fy is going to be cos y plus cos of x plus y. So when we equate f x and f y to 0, we get expression 1 and 2. So we solve this in a different manner to find out the critical point. [00:10:23.300] Subtracting 2 from 2, we see that x equals y after cancellation of plus of x plus y and plus of x plus y. Then substitute x as y in expression 1, we see that x equal to 5 by 3. So because x is y. [00:10:41.284] the only critical point is pi by three comma pi by three so at pi by three comma pi by three we see that after evaluat r t minus s square is positive but r is negative therefore we say that [00:10:57.060] The function is maximum at the point pi by three, comma pi by three, the maximum value is given by three root three divided by two. Thank you.

EV0000230 · EV0000231 · EV0000232 · EV0000233

Full transcript: 1445 normalized words, 39 timestamped segments, 670.485 seconds duration. Low-confidence segments flagged: 0.

EV0000222 · EV0000223 · EV0000224 · EV0000225

17 Lagrange's method of undetermined coefficient

This lesson is indexed from the source video titled Lagrange's method of undetermined coefficient. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000234 · EV0000235

Graphs and diagrams

Figure G000160. Lagrange s method of un determined multipliers for two variables · source slide page 1.

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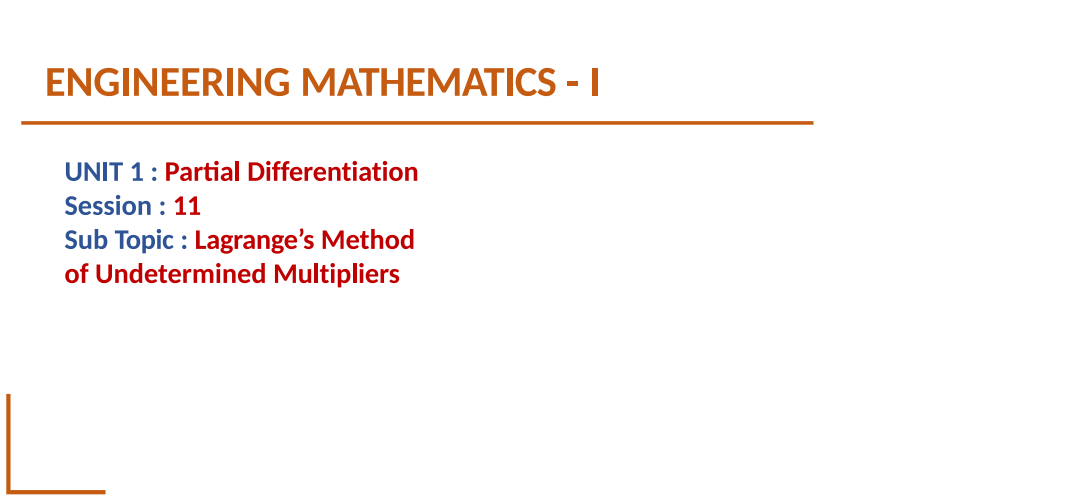

Figure G000161. Lagrange s method of un determined multipliers for two variables · source slide page 2.

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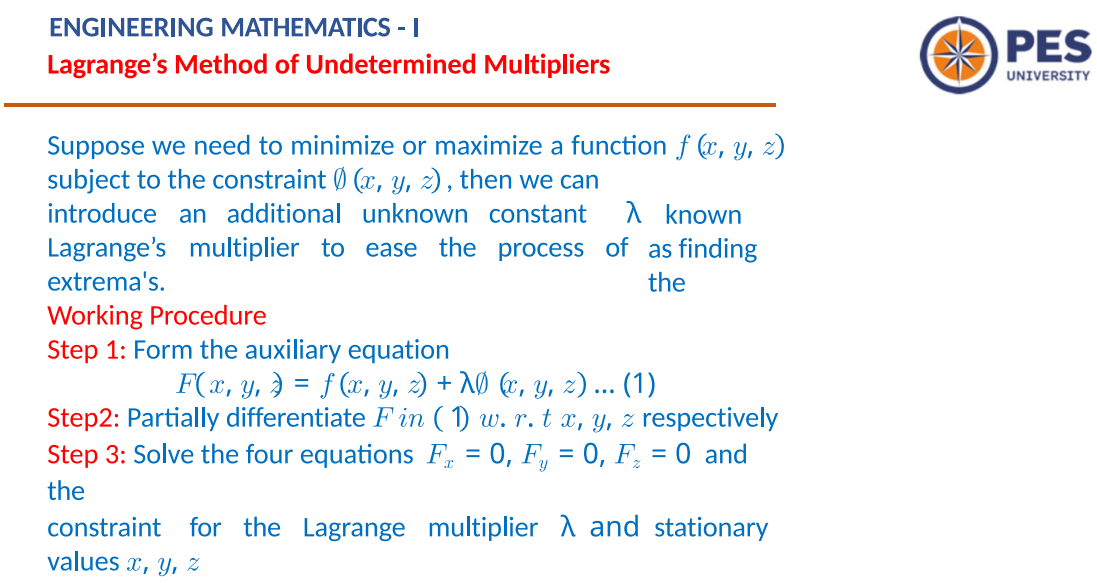

Figure G000162. Lagrange s method of un determined multipliers for two variables · source slide page 3.

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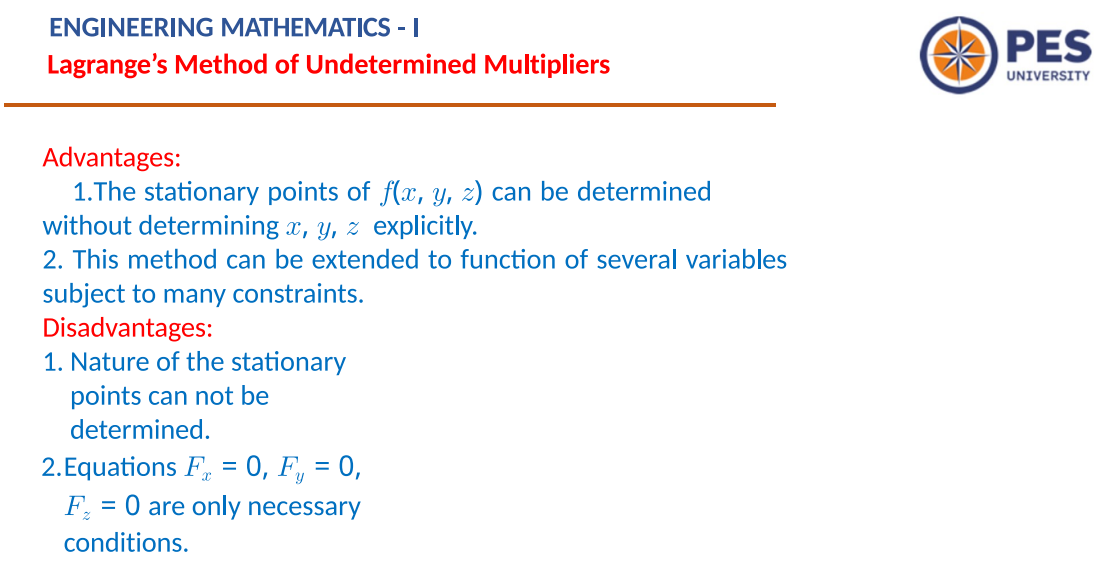

Figure G000163. Lagrange s method of un determined multipliers for two variables · source slide page 4.

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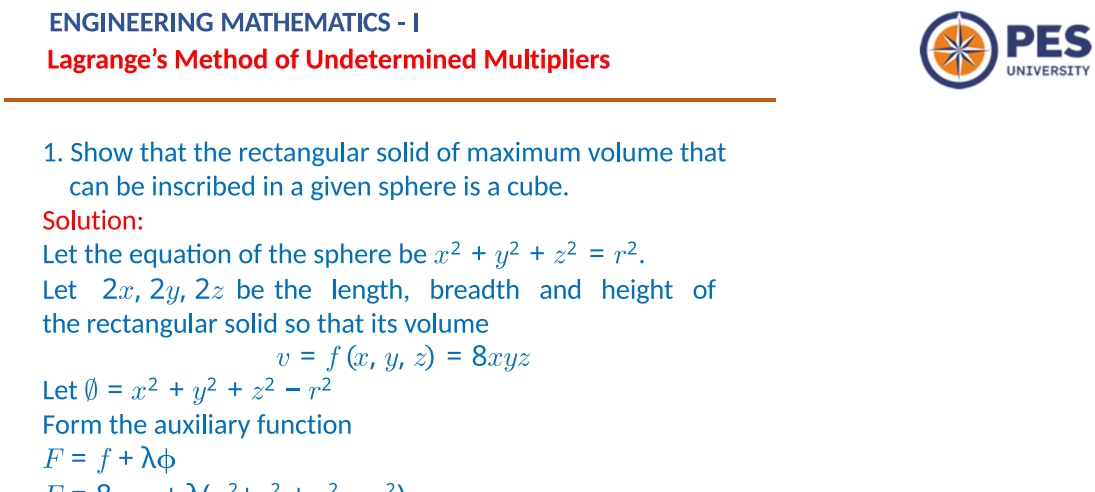

Figure G000164. Lagrange s method of un determined multipliers for two variables · source slide page 5.

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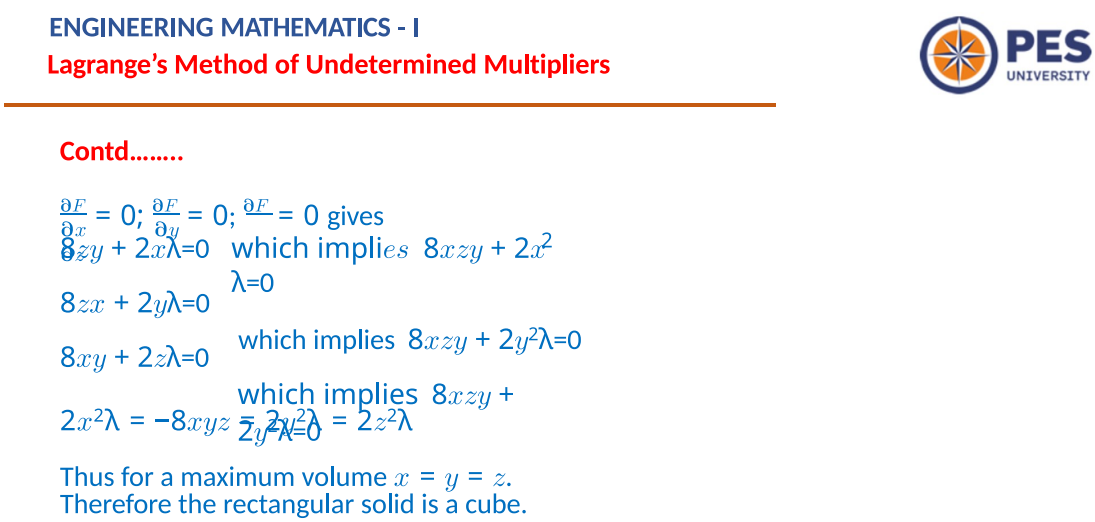

Figure G000165. Lagrange s method of un determined multipliers for two variables · source slide page 6.

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ENGINEERING MATHEMATICS - I

Lagrange's Method of Undetermined Multipliers

2. Find the points closest to the origin on the hyperbolic cylinder $x^2 - z^2 = 1$.

Solution:

To find the points on the cylinder closest to the origin, imagine a small sphere centered at the origin expanding like a soap bubble until it just touches the cylinder $x^2 - z^2 - 1 = 0$.

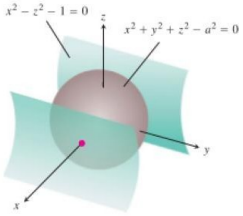

Figure G000166. Lagrange s method of un determined multipliers for two variables · source slide page 7.

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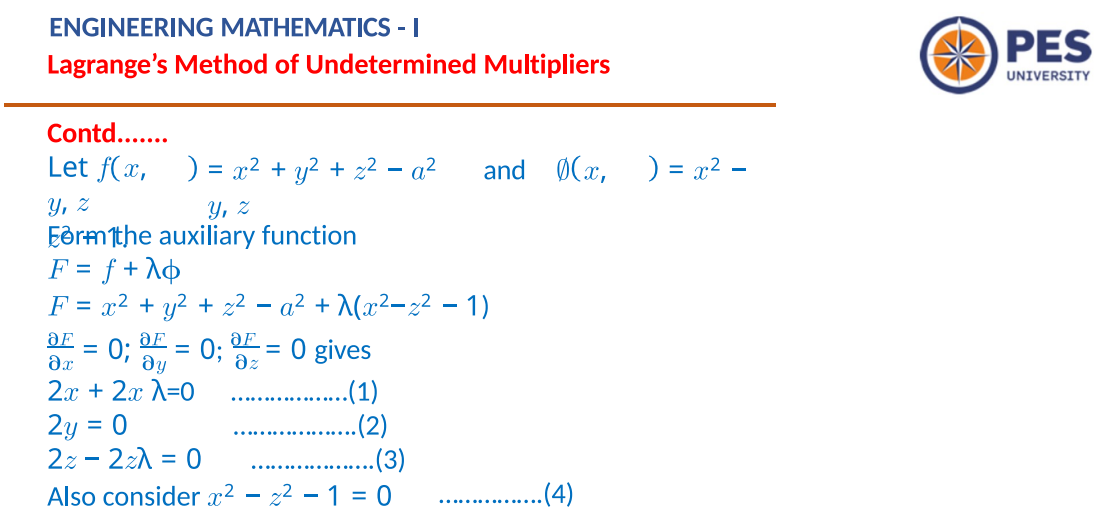

Figure G000167. Lagrange s method of un determined multipliers for two variables · source slide page 8.

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ENGINEERING MATHEMATICS - I

Lagrange's Method of Undetermined Multipliers

Contd.....

From (2), $y = 0$

Form (1),

$2x(1 + \lambda)=0$ which gives $x = 0$ or $\lambda = -1$

But x cannot be equal to 0 and hence $\lambda = -1$ From (3), we get $z = 0$, $\lambda = 1$

Substitute $z = 0$ in (4), we get $x = \pm 1$.

Therefore the points closest to the origin on the hyperbolic cylinder are (±1,0,0).

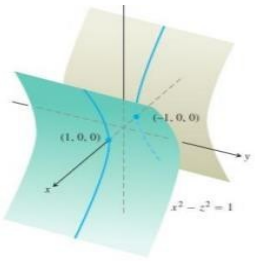

Figure G000168. Lagrange s method of un determined multipliers for two variables · source slide page 9.

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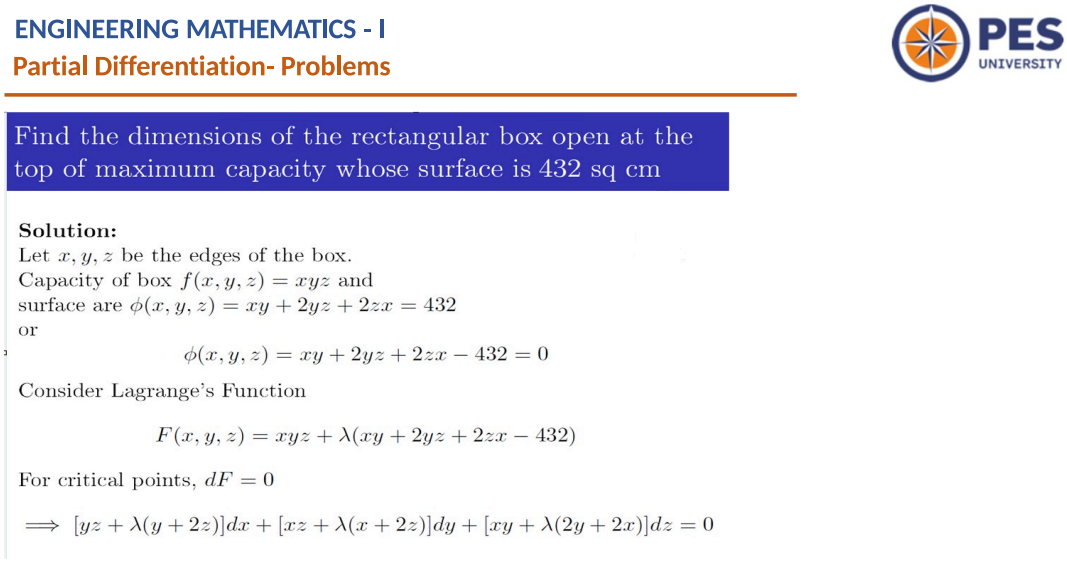

Figure G000169. Lagrange s method of un determined multipliers for two variables · source slide page 10.

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

PES UNIVERSITY

$$yz + \lambda(y + 2z) = 0 \cdots (1)$$
$$xz + \lambda(x + 2z) = 0 \cdots (2)$$
$$xy + \lambda(2y + 2x) = 0 \cdots (3)$$

Multiplying (1) by x, (2) by y and subtracting, we get

$$2z\lambda(x - y) = 0 \quad \Longrightarrow \quad x = y$$

Multiplying (2) by y, (3) by z and subtracting, we get

$$x\lambda(y - 2z) = 0 \quad \Longrightarrow \quad y = 2z$$

Therefore $x = y = 2z$
From $\phi(x, y, z)$

$$3x^2 = 432 \quad or \quad x = 12$$

Hence

$$y = 12 \quad z = 6$$

Hence the dimension of the box are $12cm$,$12cm$,$6cm$.

Figure G000170. Lagrange s method of un determined multipliers for two variables · source slide page 11.

EV0000766

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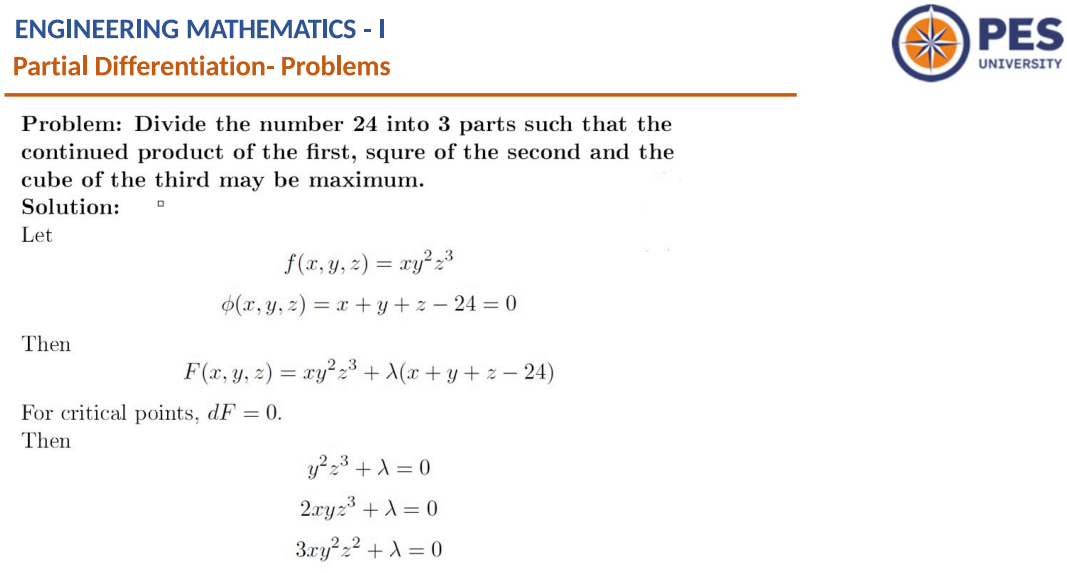

ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

PES UNIVERSITY

Problem: Divide the number 24 into 3 parts such that the continued product of the first, squre of the second and the cube of the third may be maximum.

Solution:

Let

$$f(x,y,z) = xy^2z^3$$

$$\phi(x,y,z) = x + y + z - 24 = 0$$

Then

$$F(x,y,z) = xy^2z^3 + \lambda(x + y + z - 24)$$

For critical points, $dF = 0$.

Then

$$y^2z^3 + \lambda = 0$$

$$2xyz^3 + \lambda = 0$$

$$3xy^2z^2 + \lambda = 0$$

Figure G000171. Lagrange s method of un determined multipliers for two variables · source slide page 12.

EV0000769

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ENGINEERING MATHEMATICS - I
Partial Differentiation- Problems

PES UNIVERSITY

In each case

$$\lambda = -y^2z^3 \quad \lambda = -2xyz^3 \quad \lambda = -3xy^2z^2$$

Equating

$$-y^2z^3 = -2xyz^3 \implies y = 2x$$

$$-2xyz^3 = -3xy^2z^2 \implies z = \frac{3}{2}y \implies z = 3x$$

Therefore from $\phi(x, y, z)$,

$$x + 2x + 3x = 24 \implies x = 4$$

Hence

$$x = 4, \quad y = 8, \quad z = 12$$

Figure G000172. Lagrange s method of un determined multipliers for two variables · source slide page 13.

EV0000772

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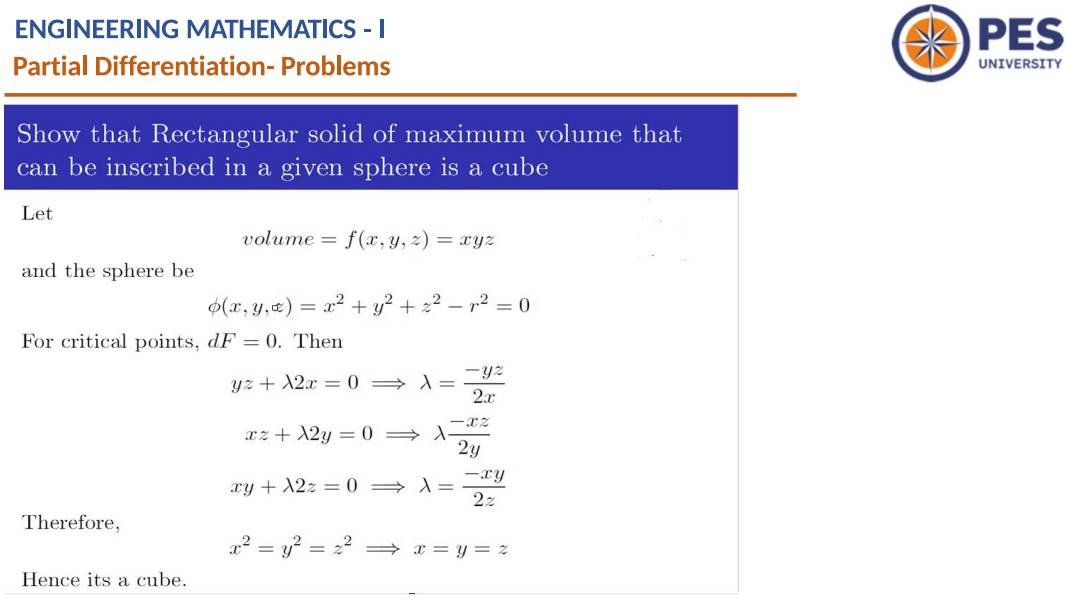

Figure G000173. Lagrange s method of un determined multipliers for two variables · source slide page 14.

EV0000775

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Figure G000174. Lagrange s method of un determined multipliers for two variables · source slide page 15.

EV0000778

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Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000735 · EV0000736 · EV0000737
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 11 Sub Topic : Lagrange's Method of Undetermined Multipliers EV0000738 · EV0000739 · EV0000740
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers Suppose we need to minimize or maximize a function $f\ x, y, z$ subject to the constraint $\varnothing\ x, y, z$, then we can introduce an additional unknown constant λ known as finding the Lagrange's multiplier to ease the process of extrema's. Working Procedure Step 1: Form the auxiliary equation $F\ x, y, z = f\ x, y, z + \lambda\varnothing\ x, y, z$ ... (1) Step2: Partially differentiate $F\ in\ 1\ w.r.t\ x, y, z$ respectively Step 3: Solve the four equations $Fx = 0$, $Fy = 0$, $Fz = 0$ and the constraint for the Lagrange multiplier λ and stationary values x, y, z EV0000741 · EV0000742 · EV0000743
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers Advantages: The stationary points of $f(x, y, z)$ can be determined without determining x, y, z explicitly. This method can be extended to function of several variables subject to many constraints. Disadvantages: Nature of the stationary points can not be determined. Equations $Fx = 0$, $Fy = 0$, $Fz = 0$ are only necessary conditions. EV0000744 · EV0000745 · EV0000746

5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers 1. Show that the rectangular solid of maximum volume that can be inscribed in a given sphere is a cube. Solution: Let the equation of the sphere be $x2 + y2 + z2 = r2$. Let $2x$, $2y$, $2z$ be the length, breadth and height of the rectangular solid so that its volume $v = f\ x,\ y,\ z = 8xyz$ Let $\varnothing = x2 + y2 + z2 - r2$ Form the auxiliary function $F = f + \lambda\phi\ F = 8xyz + \lambda(x2+y2 + z2 - r2)$ EV0000747 · EV0000748 · EV0000749

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers Contd..... $\partial F = 0$; $\partial F = 0$; $\partial F = 0$ gives $\partial x\ \partial y\ \partial z$ 2 $8zy + 2x\lambda=0$ which impli$es\ 8xzy + 2x\ \lambda=0$ which implies $8xzy + 2y2\lambda=0$ which implies $8xzy + 2y2\lambda=0$ $8zx + 2y\lambda=0$ $8xy + 2z\lambda=0$ $2x2\lambda = -8xyz = 2y2\lambda = 2z2\lambda$ Thus for a maximum volume $x = y = z$. Therefore the rectangular solid is a cube. EV0000750 · EV0000751 · EV0000752

7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers 2. Find the points closest to the origin on the hyperbolic cylinder $x2 - z2 = 1$. Solution: To find the points on the cylinder closest to the origin, imagine a small sphere centered at the origin expanding like a soap bubble until it just touches the cylinder $x2 - z2 - 1 = 0$. [Image text, OCR] x²-2²-1=0 x²+y²+z²-a²=0 EV0000753 · EV0000754 · EV0000755

8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers Contd..... Let $f\ x,\ y,\ z\ z2 - 1. = x2 + y2 + z2 - a2$ and $\varnothing\ x,\ y,\ z = x2 -$ Form the auxiliary function $F = f + \lambda\phi\ F = x2 + y2 + z2 - a2 + \lambda(x2-z2 - 1)$ $\partial F = 0$; $\partial F = 0$; $\partial F = 0$ gives $\partial x\ \partial y\ \partial z$ .....(1) .....(2) .....(3) $2x + 2x\ \lambda=0$ $2y = 0$ $2z - 2z\lambda = 0$ Also consider $x2 - z2 - 1 = 0$ .....(4) EV0000756 · EV0000757 · EV0000758

9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Lagrange's Method of Undetermined Multipliers Contd..... From (2), $y = 0$ Form (1), $2x(1 + \lambda)=0$ [Image text, OCR] (-1.0.0) (1.0.0) which gives $x = 0$ or $\lambda = -1$ But x cannot be equal to 0 and hence $\lambda = -1$ From (3), we get $z = 0$, $\lambda = 1$ Substitute $z = 0$ in (4), we get $x = \pm1$. Therefore the points closest to the origin on the hyperbolic hyperbolic cylinder are (±1,0,0). EV0000759 · EV0000760 · EV0000761

10. Source slide page 10 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Find the dimensions of the rectangular box open at the top of maximum capacity whose surface is 432 sq cm Solution: Let , y, z be the edges of the box. Capacity of box f(x,y,≥) = cyz and surface are Φ(c,y,z) = cy + 2yz + 2z = 432 Or Φ(,y,z) = cy + 2yz + 2z - 432 = 0 Consider Lagrange's Function F(c,y,z) = cyz + 入(cy + 2yz + 2zc - 432) For critical points, dF = 0 →[yz + X(y + 2z)]d + [cz + X(x + 2z)]dy + [cy + 入(2y + 2x)]dz = 0 EV0000762 · EV0000763 · EV0000764

11. Source slide page 11 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] yz +入(y+ 2z)=0..·(1) +X(+22)=0..·(2) cy + 入(2y + 2x) = 0..·(3) Multiplying (1) by x, (2) by y and subtracting, we get 2z入(c-y) = 0 =→=y Multiplying (2) by y, (3) by z and subtracting, we get >(y - 2z) = 0 =→ y= 2z Therefore = y = 2z From Φ(,y,2) $3c^2$=432 or = 12 Hence y=12 =6 Hence the dimension of the box are 12cm,12cm,6cm. EV0000765 · EV0000766 · EV0000767

12. Source slide page 12 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Problem:Divide the number 24 into 3 parts suchthat the continued product of the first, squre of the second and the cube of the third may be maximum. Solution: 口 Let f(x,y,z) = xy²x3 (x,y,z) = + y + - 24 = 0 Then F(x,y,z) = xy²x3 + X(x + y + - 24) For critical points, dF = 0. Then y²x²+入= 0 2cyz² + 入 = 0 3cy²² + 入 = 0 EV0000768 · EV0000769 · EV0000770
13. Source slide page 13 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] In each case =-y²3 入= -3xy²22 入= -2cyz 3 Equating -y²x3 = -2xyz3 y = 2c 3 -2xyz3 = -3cy²x2 =→→= 2y → 2 =3c Therefore from Φ(x, y,z), + 2 +3 =24 =→ = 4 Hence = 4， y=8， = 12 EV0000771 · EV0000772 · EV0000773
14. Source slide page 14 (extracted text): ENGINEERING MATHEMATICS - I Partial Differentiation-Problems [Image text, OCR] Show that Rectangular solid of maximum volume that can be inscribed in a given sphere is a cube Let volume = f(x,y,z) = cyz and the sphere be Φ(x,y,) = x² +y² + x2 -r² = 0 For critical points, dF = O. Then -yz y≈+ X2c = 0 => 入= 2c 2y -cy cy + 入2 =0 =→ 入= 2z Therefore, x² = y² =x² => c =y= 2 Hence its a cube. EV0000774 · EV0000775 · EV0000776
15. Source slide page 15 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000777 · EV0000778 · EV0000779

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:00.420] Welcome back. In the previous session we discussed about the extrema of a function of two independent variables, right? And we did solve some problems also. Now we are going to study another method which is called the Lagrangius method of undetermined multipliers. [00:00:17.956] So, in the first problem if you recollect what we did is we studied the temperature if the temperature function was XYZ was 400 XYZ square, this was given to us, and then we had to calculate what is the maximum temperature on the surface of a sphere. [00:00:36.100] Right. So we had some kind of constraint that I need to walk only on the surface of a sphere. So such problems wherein we get a val the function along with the constraint can be solved by using Lagrange's method of undetermined multiplier. [00:00:53.060] So, the definition goes suppose we need to maximize or minimize a function f of x y z subject to the constraint phi of x y z then we can introduce an additional parameter or unknown which is called lambda. So he used lambda [00:01:08.996] and we can write the auxiliary equation. So we can write f of x y z to be the given function plus lambda time its constraint. [00:01:18.244] The next thing that is you need to do is you solve the first order partial derivatives. So f x equal to 0, f y equal to 0, f z is also equal to 0. This is an advantage. You remember what we did here? We had x square plus y square plus z square equal to 1. So we had to eliminate the z. [00:01:36.016] from the given function. So we substituted z square as 1 minus x square minus y square. So this z square we substituted back here and then eliminated that z that can be avoided if we use this method. [00:01:51.012] Okay. So what you have to do is you calculate this new function f which is which is obtained by introducing the parameter lambda. That's why it's called an multiplier because this is like a scaling of the constraint, right? [00:02:04.036] And it is undetermined, so we don't need to worry about what is the value of that lambda, but we'll calculate the value, okay? So calculate the first order partial derivatives, equate it to zero and use the constraint. [00:02:17.188] Constraint is uh phi of x y z. Okay. Um for to to solve lambda x y and z. So you remember like if you have three unknowns, you need three equity to solve. [00:02:30.372] Right. So here I have four

unknowns, I have lambda x y and z. So to solve four unknowns I need four equations. Three of them I got here f x 0, f y 0 and f z 0. So the fourth one I can use phi of xyz equal to z. [00:02:48.260] I'll I'll use that constraint and then I'll s use that to solve the lambda value. Let's try a problem, maybe things will be little more clear, right? But before I go to problems, let's discuss what are the advantages of this method and what are the disadvantages. [00:03:04.452] So here I can find the stationary points. The stationary points are nothing but the critical points, right? So I can find the critical points without actually determining XYZ explicitly. [00:03:17.764] You remember what we did in the previous case? We reduced the given situation to a function of two independent variables where then we found the first order partial derivatives, we equate it to zero and then from there it started, right? So here we don't have to do that. [00:03:33.104] You can directly find Fx, FY and F Z, equate them to zero and also consider the constraints so all uh the stationary points can be obtained uh explicitly. [00:03:45.380] Ah and second advantage is this method can be extended to n parameters, right? If I have x of x y z w, then you write the equation by introducing lambda, that's all you have to do. So this can be extended to a several variables subject to the constraints, subject to as many constraints as they have given. [00:04:05.924] But the disadvantage is you will not be able to know what is the nature of the stationary point or what is the nature of the critical point. That means you will not be able to decide whether that will represent a maxima point or a minimum point, that is the disadvantage. [00:04:23.428] All you can do is get the stationary points. You know that it will represent either maximum or minimum. But I cannot verify if I have to verify this R T minus S s square to be greater than 0 and R is less than 0 or R is greater than 0, I have to go back to the previous method only. [00:04:41.200] That's the disadvantage of this. But definitely to find the critical point you can make use of this. So let's try a problem. [00:04:49.892] So the problem is show that the rectangular solid of maximum volume that can be inscribed in a s given sphere is a Q. So that means that might look something like let me try to sketch that. Um I don't know how well I can sketch. So suppose this is the sphere that we have.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:08.132] Okay. Now you try to inscribe a uh rectangle inside this. So let me change the [00:05:21.476] This so I have the sphere. [00:05:26.244] And then now what I want is um I have to inscribe a rectangle in it. So maybe a blue. So what you need to do is suppose you're inscribing a rectangle inside a sphere, the question says [00:05:41.988] If this has to have a maximum value, then it has to be a cube. Let's prove that mathematically, okay. So maybe geometrically it's difficult to visualize, but you can do that. So what is the equation of a sphere? It's x square plus y square plus z square equal to r square with the center of course at the origin and radius r. [00:06:01.796] And what what is the volume of the um rectangular uh solid, F of X Y Z? Is [00:06:08.580] Two eight x. That's because I have taken two x and then two y and two z. So um uh so that the rectangle is solid of maximum volume that can be inscribed in a given sphere is a cube. So I will keep that uh [00:06:24.112] as it is, then the volume is this 8 of XYZ, that's the constraint that is given to me. Which one I need to maximize or minimize, I need to worry about this. What is the constraint inside the sphere. [00:06:38.436] Right. So the axillary equation would look like eight of x y z plus lambda x square plus y square plus z square minus r square. [00:06:50.340] So [00:06:53.092] So th for the one thing that we have to do is we have to consider you can take you know if you don't want this eight, you can just take xy z also. The volume is like you know length into breadth into height, so you you can take that as

x y z. Now what happens if I calculate f x f x will be y z [00:07:12.644] plus lambda into two z. Right? And what is FY? F y will be x z plus lambda into um f y so I have to I'll get two y. Similarly f z will be x y [00:07:30.820] Plus lambda into two set. Right? [00:07:34.916] Now what do I get if I equate these to zero? I can take out z common and then I get lambda y plus uh two lambda plus y equal to zero. So z is zero or two lambda equal to minus y. So I'll get something like that, right. So I'll get these equipment [00:07:51.360] Don't take this 8 also it is absolutely fine. But what you can do is if you multiply this first equation with x, I will get x y z plus um 2 x square lambda is 0. I will do the same thing for the next equation. I will take x y z plus [00:08:11.012] to y square lambda equal to 0. What do I get if I solve these two equations? I will subtract them. I will get 2x square lambda minus 2y square lambda equal to 0. This will give me lambda equal to 0 or [00:08:28.804] x is equal to plus or minus 1, right. Because it's a measure minus y does not matter for me, so I will get x is equal to y. Similarly, if you use these two equations you will get lambda equal to 0 or y is equal to z. [00:08:44.836] So from these two I can conclude that x is equal to y is equal to z. So if you prove that you have a rectangular solid with you know different sides and you just prove that all the three sides are equal so it will become a q. [00:08:59.396] So, you understood the advantage? We never use the fact that I have to take R T minus S square and then you know find out R greater than zero or less than zero. That was not necessary because the question said that maximum volume, they have already given that it is maximum. [00:09:15.940] So all I need to show that that maximum volume is nothing but a Q. In such problems the Slagrange's method of undetermined multipliers are very helpful. Okay? So but with the constraint. So the most important thing is you identify the function and the constraint. Let's try another problem. [00:09:35.588] The question is find the points closest to the origin of the hyperbolic cylinder. So whenever they say a point in a space, can I consider that point can be x square plus y square plus z square equal to one maybe? So that point can be represented as [00:09:55.332] X square plus Y square plus Z square because uh a sphere can be taken as a point. Maybe if I take this as R instead of that. As R becomes R square maybe. Okay, let me let me just do it.

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:08.644] So I'll take the point to be x square plus y square plus z square. [00:10:17.092] If R becomes zero, I will get the s uh smallest a point which is nothing but the origin. So here I want to find the point on uh point that is closest to the origin. On the hyperbolic cylinder, x square minus z square equal to [00:10:36.484] So what would be my axillary equation? My axillary equation would look like f of x y z is equal to uh this is the given function x I want to uh make this as small as possible with the given condition that x square minus z square minus [00:10:57.508] Okay, this is the given constraint. Or if you want to call g of x y z, g of x y z is x square minus z square minus 1. [00:11:07.364] Now what I need to do? I need to find XY Z and that lambda. So I need four equations. So what are those equations? First let me calculate what's f x is. F x will be two x plus two x lambda, right? [00:11:21.764] and f y will be again two y plus a zero, there is no y term there, and f z will be two z minus two lambda z. [00:11:35.076] Now I have these three equations but I have four unknowns. But I have another condition right which is x square minus z square minus one equal to zero. Now I have to make use of all these four conditions to solve for x, y, z and lambda. [00:11:51.268] Now just observe the second equation. I have to equate them to zero, right? That's that's what is the condition, f x is zero, f y is zero and f z is zero. So from the second equation, isn't it evident that I'll get y is equal to zero? So I already got the value of [00:12:07.844] What about the first equation if you observe

carefully the first equation is 2x plus 2x lambda is 0. So I can take out 2x common, I will get 1 plus lambda is 0. This gives me x equal to 0 or lambda is equal to minus 1. [00:12:26.820] But can x be zero? There is a problem with that. If x becomes zero, if x is zero, then what do I get? I'll get z square to be minus one. So that means my z will not be in the real plane at all. That's a problem I have, right? [00:12:44.484] So that means x is equal to zero is not possible. So here this is a contradiction. So I cannot consider this. But I can take definitely lambda equal to minus one. Now what is the next equation I have? I have two z [00:13:01.220] Ah 2 z minus 2 lambda z equal to 0. This will give me z equal to 0 or lambda equal to 1. [00:13:10.820] Correct. So z equal to zero or lambda equal to one. That is fine for me. So I got so far I got lambda to be plus or minus one, I got y to be zero, I got z to be zero. [00:13:24.932] But if z is 0 what is the value of x that I get? I have x square minus z square minus 1 is 0. So if z becomes 0 I will get x square as 1, so x will be plus or minus 1. [00:13:41.412] Right. That means the point that will be very very close to the origin will be plus or minus 1, 0 and 0. [00:13:50.192] So, there are two points that lie on a hyperbolic cylinder which is x square minus z square is equal to one, which is closest to the origin are these two points. [00:14:01.572] Is it clear? So you remember if you are using Lagrangius method of multiplier, the condition will also be mentioned. That means that point is minimum only for you. Here it is closest. So I am calculating that the point to be at a minimum distance from the origin. [00:14:16.900] In the previous example it was maximum right. What is the rectangular solid of maximum volume that can be inscribed in a sphere. So whenever you have a constraint then you can definitely make use of Lagrange's method of undetermined multiplier, that's one thing. [00:14:33.744] The second thing is whenever they have already given the condition that you have to maximize the temperature. Similarly the first problem that we did in the previous session can also be solved using Lagrangian's method of undetermined multipliers. You practice some more problems then concept should be more clear to you, right?

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Full transcript: 2506 normalized words, 60 timestamped segments, 893.909333 seconds duration. Low-confidence segments flagged: 0.

EV0000234 · EV0000235 · EV0000236 · EV0000237

18 Problems on Lagrange's method of undetermined multipliers

This lesson is indexed from the source video titled Problems on Lagrange's method of undetermined multipliers. Its complete timestamped transcript is preserved below in segment order. The original video path and content hash are recorded in datasets/lecture-corpus.json. Transcript text is source wording; it has not been silently corrected or rewritten. A candidate slide deck is grouped here according to the current matching score; the exact path and page locators are listed in the source evidence. EV0000246 · EV0000247

SHORT-NOTE RECAP

Transcript block 1 (source interval 000000-000299 seconds): [00:00:03.524] Hello everyone. Welcome back to the course on engineering mathematics one. In this session, we will small problems on

Lagrange Method. [00:00:19.812] So let's look at the first problem. Find the dimensions of the rectangle box open at the top of maximum capacity whose surface is 432 square centimeter. So in this problem we need to find out the dimensions of the rectangle box. [00:00:40.452] So let us observe this particular diagram. [00:00:50.596] So this is a rectangular box. So that X, Y, and Z edges of the box. So the total surface area for this rectangular box in general is [00:01:09.636] Two XY plus two Y Z plus two Z because the boxes open at the top. [00:01:26.532] So this total surface area reduces to XY plus 2yz plus 2 ZX. Because we need to find out the dimensions that is XYZ of this rectangular box. [00:01:46.020] This volume xyz becomes function for the problem and xy plus 2yz plus 2 z equals 432 becomes constraint for the problem. So phi of xyz can be written as xy plus 2y z. [00:02:06.212] plus 2 zx minus 432 equals 0. Let us use the grandiose function which is given by f of x y z equals f of x y z plus lambda times phi. [00:02:21.732] Well lambda is a Lagrange multiplier. So this equation reduces to f of xyz equals where f of x y z is going to be our xyz plus lambda times phi. That is xy plus two yz plus two zx minus four thirty two. [00:02:40.932] Next, as per the procedure, we need to compute the partial derivative of f with respect to x, y, as well as z. So that is the f x, of y, then f z. Then we equate fx to zero. [00:02:56.388] That leads to yz lambda of y plus two z equals zero. Similarly, partial derivative of y. [00:03:05.540] of f with respect to y equals zero gives x plus lambda x plus two z equals zero. Similarly the partial derivative f with respect to z equals zero gives x y plus lambda of two y plus two x equals zero. [00:03:24.644] Now, the task is to find out the values of xyz from these three expressions. So let us solve these three expressions one by one. So when we multiply 1 by x and multiply 2 by y and subtracting. [00:03:40.452] We see that 2z lambda of x minus y equals 0. Because 2z lambda is not zero, we can equate x minus y equals 0, then that gives x equals y. Similarly, multiply 2 by y [00:03:59.332] and 3 by z and subtracting we get x x lambda of y minus 2 z t equals 0 because x lambda not zero we can equate y minus 2 z to 0 that gives y equal to 2 z [00:04:18.596] So finally we arrive at a conclusion x equals y equals to z. Let us use this in the constraint. Substitute x equals y, y equals to z in the constant and for you have x y z. We see that. [00:04:34.948] 3x square equals 432 or x equals 12. Because y is x, so y is 12. Because z is y by 2, z becomes 6. [00:04:47.396] So finally, the dimensions of the box are 12 centimeter, 12 centimeter, and 6 centimeter. So this is how we can solve problem using Lagrange method.

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Transcript block 2 (source interval 000300-000599 seconds): [00:05:03.204] Let's look at the second problem. Divide the number 24 into three parts such that the continued product of the first square of the second and the cube of the third may be maximum. [00:05:23.492] Before we solve this problem, let's understand what is the meaning of continued product. Continued product means the product we get by multiplying together in succession. [00:05:37.668] It means if we have three parts, let us say x comma y comma z, then the continued product of the first square of the second and the cube of the third is going to be x times y square times z power three. [00:05:57.604] Then divide the number 24 into three three parts. It means we want to split 24 into three non-negative parts. Let's say xyz, that is x plus y plus z equals 24. [00:06:16.900] So as to maximize this x y square z power 3. So based on this idea for this problem, x y square z power 3 is a function, x plus y plus z equals 24 is a constraint. [00:06:33.540] So let us write the grandest equation that is capital F equals f of lambda phi. So we get f of x y z equals x y square z power three plus lambda of [00:06:50.308] X plus Y plus Z minus 24. Let us find out the partial derivative of f with respect to XYZ as earlier and equate them to 0. So f x to 0 gives y square z power 3 plus lambda equals 0.

F [00:07:09.636] to zero gives two x y z power three plus lambda zero. Similarly F Z zero gives three x y square z square plus lambda equals zero. [00:07:24.228] Observe carefully in all these three expressions, we can use this idea. In each case, lambda equals minus y square z power 3, minus 2x y z power 3, minus 3xy square z square. On equate [00:07:43.620] We see that minus y square z power 3 and minus 2x y z power 3. It means from the first two terms we can get y equals to 2x. Similarly, from the second and third we get z equal to 3x. [00:07:59.588] Let us substitute y as 2x and z as 3x in the constraints that leads to x as it is 2x plus 3x equals 24. So we get x equal to 4. [00:08:11.876] So because y is 2x, y becomes 8, and because z is 3x, z is 12. So x equal to 4, y equals to 8, and z equals to 12. It means the three parts are 4 and 8 and 12. [00:08:30.372] So this choice indeed makes x y square z power three as large as possible under the constraint x plus y plus z equals twenty-four. So this is the answer for this particular problem. [00:08:50.788] Let us look at one more problem. Show that the rectangular solid of maximum volume that can be inscribed in the given sphere is a q. So for this problem we need to prove that x equals y equal to z. [00:09:08.388] Because this is the only condition under which the inscribed rectangular solid achieves maximum volume inside the sphere. So for this problem, xyz is going to be the function and x square plus y square plus z square equal to r square is going to be the constraint. [00:09:28.516] So we want to maximize xyz subject to constraint x square plus y square plus z square equal to r square. So as usual, after writing Lagrangus equation, after equating fx to 0 and f y to 0. [00:09:46.372] And f z to 0, so we see that lambda equal to minus y z by 2x, lambda equal to minus x z by two y, lambda equals minus x by 2z. When we solve all these in ratios, we see that x equals y equals a

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Transcript block 3 (source interval 000600-000899 seconds): [00:10:05.732] So this completes the problem and prove that it is a Q. Thank you.

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Full transcript: 1215 normalized words, 36 timestamped segments, 613.376 seconds duration. Low-confidence segments flagged: 0.

EV0000246 · EV0000247 · EV0000248 · EV0000249

19 Source slide deck: Introduction to Maxima and Minima for a function of two variables

This entry preserves the complete slide deck titled Introduction to Maxima and Minima for a function of two variables as searchable extracted page text and page-faithful rendered images. Its association with a particular video is not confirmed; it remains a separate topic under Engineering Mathematics - I, Unit_01, to avoid an unsupported match. EV0000660 · EV0000661

Graphs and diagrams

Figure G000135. Introduction to Maxima and Minima for a function of two variables · source slide page 1.

EV0000661

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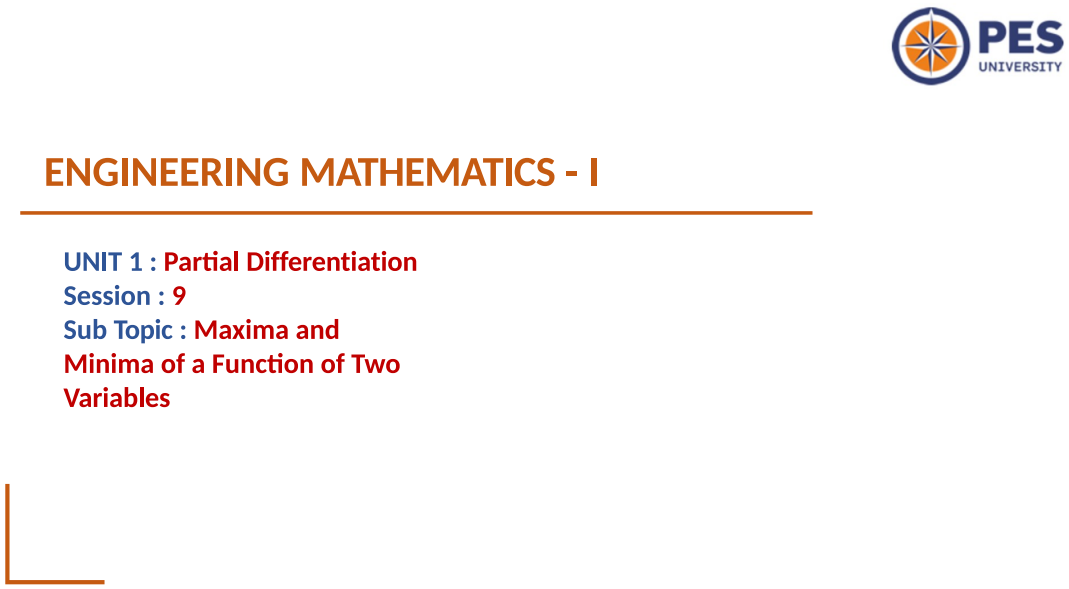

Figure G000136. Introduction to Maxima and Minima for a function of two variables · source slide page 2.

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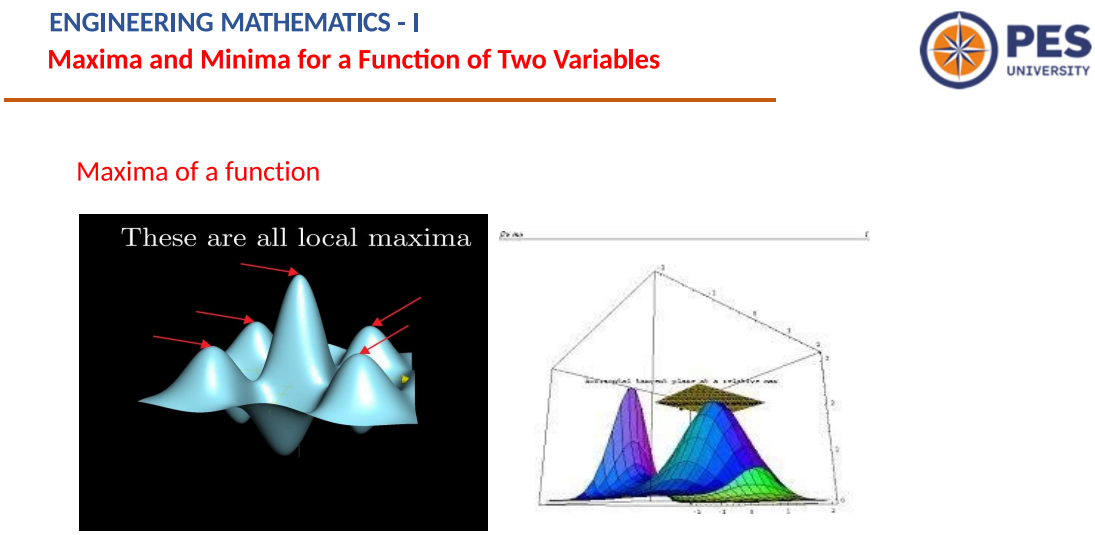

Figure G000137. Introduction to Maxima and Minima for a function of two variables · source slide page 3.

EV0000667

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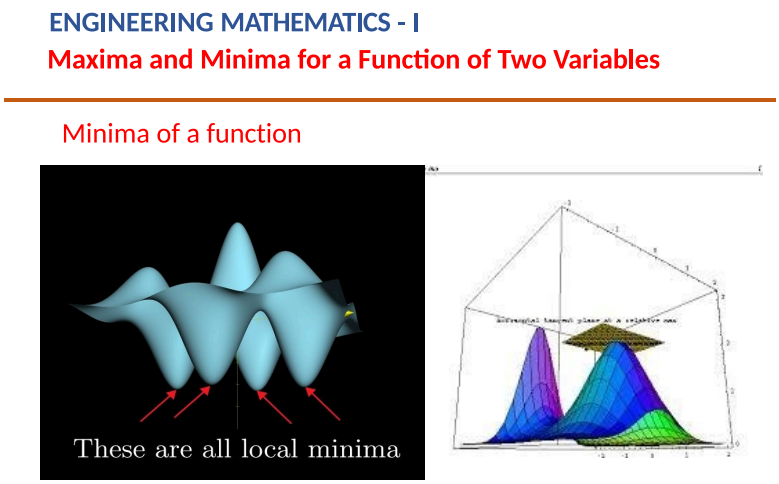

Figure G000138. Introduction to Maxima and Minima for a function of two variables · source slide page 4.

EV0000670

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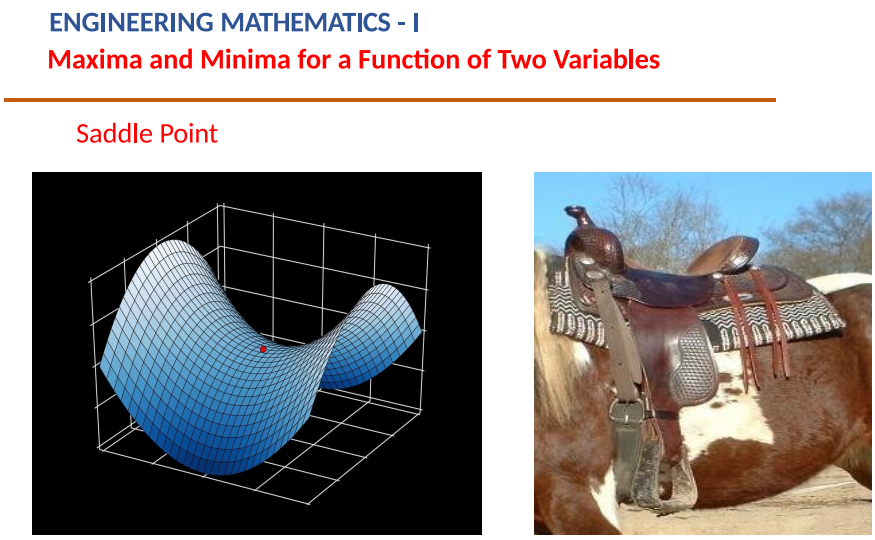

Figure G000139. Introduction to Maxima and Minima for a function of two variables · source slide page 5.

EV0000673

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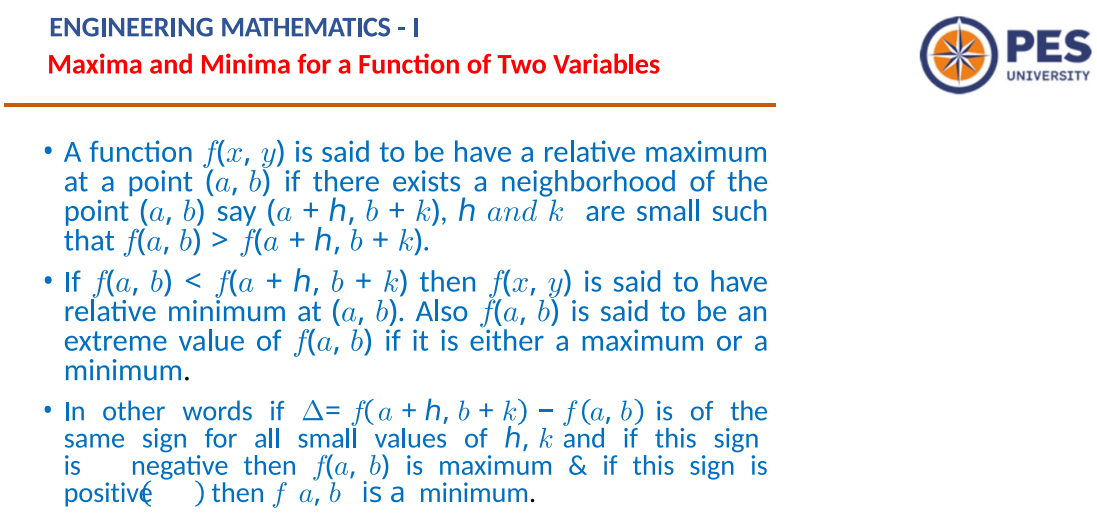

Figure G000140. Introduction to Maxima and Minima for a function of two variables · source slide page 6.

EV0000676

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

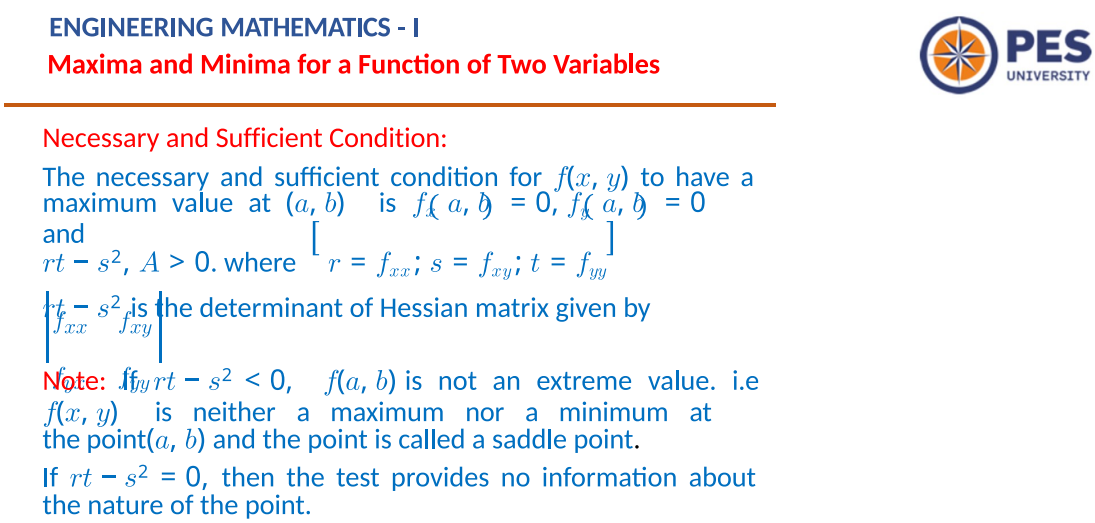

Figure G000141. Introduction to Maxima and Minima for a function of two variables · source slide page 7.

EV0000679

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

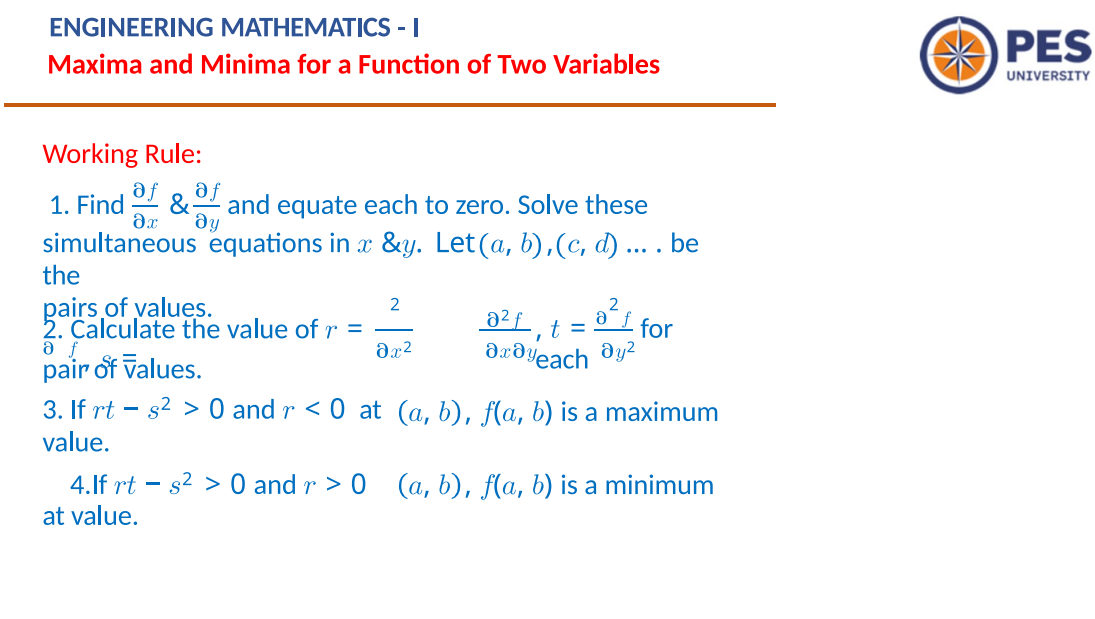

Figure G000142. Introduction to Maxima and Minima for a function of two variables · source slide page 8.

EV0000682

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

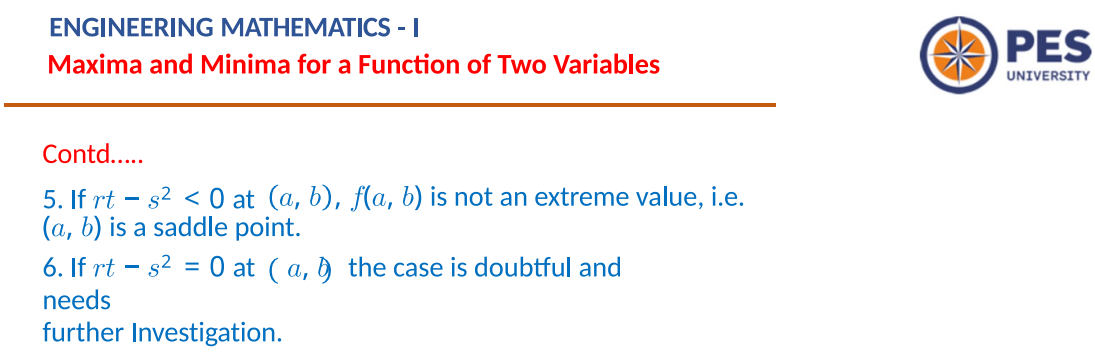

Figure G000143. Introduction to Maxima and Minima for a function of two variables · source slide page 9.

EV0000685

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Figure G000144. Introduction to Maxima and Minima for a function of two variables · source slide page 10.

EV0000688

How to read it. Use this original slide page to inspect formulas, diagrams, layout and any text extraction gaps. The original deck path and hash are recorded in its source metadata.

Important points and why

1. Source slide page 1 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS ENGINEERING MATHEMATICS - I UE26MA141A Department of Science and Humanities EV0000660 · EV0000661 · EV0000662
2. Source slide page 2 (extracted text): ENGINEERING MATHEMATICS - I UNIT 1 : Partial Differentiation Session : 9 Sub Topic : Maxima and Minima of a Function of Two Variables EV0000663 · EV0000664 · EV0000665
3. Source slide page 3 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Maxima of a function [Image text, OCR] These are all local maxima EV0000666 · EV0000667 · EV0000668
4. Source slide page 4 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Minima of a function [Image text, OCR] These are all local minima EV0000669 · EV0000670 · EV0000671
5. Source slide page 5 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Saddle Point [Image text, OCR] 09998 EV0000672 · EV0000673 · EV0000674

6. Source slide page 6 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables A function $f(x, y)$ is said to be have a relative maximum at a point (a, b) if there exists a neighborhood of the point (a, b) say $(a + h, b + k)$, h and k are small such that $f(a, b) > f(a + h, b + k)$. If $f(a, b) < f(a + h, b + k)$ then $f(x, y)$ is said to have relative minimum at (a, b). Also $f(a, b)$ is said to be an extreme value of $f(a, b)$ if it is either a maximum or a minimum. In other words if $\Delta = f\ a + h, b + k - f\ a, b$ is of the same sign for all small values of h, k and if this sign is negative then $f(a, b)$ is maximum & if this sign is positive then $f\ a, b$ is a minimum. EV0000675 · EV0000676 · EV0000677
7. Source slide page 7 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Necessary and Sufficient Condition: The necessary and sufficient condition for $f(x, y)$ to have a maximum value at (a, b) is $fx\ a, b = 0$, $fy\ a, b = 0$ and $rt - s2$, $A > 0$. where $r = fxx$; $s = fxy$; $t = fyy\ rt - s2$ is the determinant of Hessian matrix given by $fxx\ fyx\ fxy\ fyy$ Note: If $rt - s2 < 0$, $f(a, b)$ is not an extreme value. i.e $f(x, y)$ is neither a maximum nor a minimum at the point(a, b) and the point is called a saddle point. If $rt - s2 = 0$, then the test provides no information about the nature of the point. EV0000678 · EV0000679 · EV0000680
8. Source slide page 8 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Working Rule: $\partial f\ \partial f$ 1. Find & and equate each to zero. Solve these $\partial x\ \partial y$ simultaneous equations in x &y. Let a, b, c, d ... . be the pairs of values. 2 2. Calculate the value of $r = \partial\ f$, $s = 2$, $t = \partial\ f$ for each $\partial 2f\ \partial x2\ \partial x\partial y\ \partial y2$ pair of values. If $rt - s2 > 0$ and $r < 0$ at value. If $rt - s2 > 0$ and $r > 0$ at value. a, b , $f(a, b)$ is a maximum a, b , $f(a, b)$ is a minimum EV0000681 · EV0000682 · EV0000683
9. Source slide page 9 (extracted text): ENGINEERING MATHEMATICS - I Maxima and Minima for a Function of Two Variables Contd..... 5. If $rt - s2 < 0$ at a, b , $f(a, b)$ is not an extreme value, i.e. (a, b) is a saddle point. 6. If $rt - s2 = 0$ at a, b the case is doubtful and needs further Investigation. EV0000684 · EV0000685 · EV0000686
10. Source slide page 10 (extracted text): [Image text, OCR] PES UNIVERSITY CELEBRATING5OYEARS Thank you Department of Science and Humanities EV0000687 · EV0000688 · EV0000689

SHORT-NOTE RECAP

Source deck preserved: Introduction to Maxima and Minima for a function of two variables. Its extracted page text is listed as searchable points and all rendered source pages are attached as figures. Match to a specific video is score-based; this index does not claim a manual content review. EV0000660

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Official unit catalogue	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/units.json	Official course/unit title and index.
Continuity and differentiability of a function of two independent variable transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Continuity and differentiability of a function of two independent variable transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Continuity_and_differentiability_of_a_function_of_two_independent_variable__543f4d65e9b6ed52.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Introduction to Partial Differentiation transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Introduction to Partial Differentiation transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/001_Introduction_to_Partial_Differentiation__7d6d5767ec830adb.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Geometrical interpretation of Partial derivatives transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9990; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Geometrical interpretation of Partial derivatives transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/002_Geometrical_interpretation_of_Partial_derivatives__374c7bb76e445af3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9981; only the JSON segmentation supplies the displayed timestamps.
Problems on Total derivatives, Chain rule transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Problems_on_Total_derivatives_Chain_rule__4d669ee31cdd4a30.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Total derivative, Chain rule and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Total derivative, Chain rule and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/003_Total_derivative_Chain_rule_and_Implicit_functions__d194dd0f27d571b2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Implicit functions and problems transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9995; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Implicit functions and problems transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Implicit_functions_and_problems__28ebf6864cd576d3.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Composite and Implicit functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/004_Problems_on_Composite_and_Implicit_functions__d1ea652fce3206d2.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Homogeneous functions and Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9991; only the JSON segmentation supplies the displayed timestamps.
Homogeneous functions and Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/005_Homogeneous_functions_and_Euler_s_theorem__a0c2c43d28f6ec69.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Euler's Theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9982; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's Theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_Theorem__6c4797f9ed9a5b61.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9985; only the JSON segmentation supplies the displayed timestamps.
Problems on Euler's theorem transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/006_Problems_on_Euler_s_theorem__0d726228252093d7.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Taylor's and Maclaurin's expansion transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9992; only the JSON segmentation supplies the displayed timestamps.
Taylor's and Maclaurin's expansion transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/007_Taylor_s_and_Maclaurin_s_expansion__ff2506f0c8105451.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Taylor's and Maclaurin's series transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9983; only the JSON segmentation supplies the displayed timestamps.
Problems on Taylor's and Maclaurin's series transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/008_Problems_on_Taylor_s_and_Maclaurin_s_series__4b73a1bc5fc83388.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/009_Extremas_of_functions__5ebc2dd46f817380.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Extremas of functions continued transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9993; only the JSON segmentation supplies the displayed timestamps.
Extremas of functions continued transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Extremas_of_functions_continued__5cf502cc80940c87.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Problems on Maxima and Minima for a function of two variables transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9962; only the JSON segmentation supplies the displayed timestamps.
Problems on Maxima and Minima for a function of two variables transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__e276ddb59439e6ec.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Lagrange's method of undetermined coefficient transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9988; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Lagrange's method of undetermined coefficient transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Lagrange_s_method_of_undetermined_coefficient__6812c905c391978e.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9967; only the JSON segmentation supplies the displayed timestamps.
Problems on Lagrange's method of undetermined multipliers transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/AV_Summaries/011_Problems_on_Lagrange_s_method_of_undetermined_multipliers__94e58d0103d5b799.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Source	Path	Use
Slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Partial differentiation	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Introduction to Partial differentiation (converted PDF; original hash a1c216d29db4461b50640f770f898294307495d0f4ecabc6d0c546abd126a64d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Geometrical interpretation of Partial derivatives Problems on partial derivatives	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Geometrical interpretation of Partial derivatives Problems on partial derivatives	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Geometrical interpretation of Partial derivatives Problems on partial derivatives (converted PDF; original hash 4df6156d5226cd03f8889623b0ef2a5bde035e033d9b3ce84064ba9360944efc)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Total derivative Chain rule problems	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Total derivative Chain rule problems	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Total derivative Chain rule problems (converted PDF; original hash 8fb90d8c86fdd0e6672954f6874fa5df34e8e2988f0f7bf6085fd3b0b27ca5c6)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Implicit functions Problems on Implicit and Composite functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Implicit functions Problems on Implicit and Composite functions	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Implicit functions Problems on Implicit and Composite functions (converted PDF; original hash 33599a90d72d93b72a632725fa279ebbdd8da8def5fd3e5c8e8524224ba3373d)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Homogeneous functions and Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Homogeneous functions and Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Homogeneous functions and Euler s theorem (converted PDF; original hash 5e9df5fbab859c24989026e80d8ec451101874e569137c4a4fb0789f4a16fd53)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Problems on Euler s theorem	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Euler s theorem (converted PDF; original hash 84ce9a0df0c6f69428b12df247d69c4d2609eecefdb0ef07aca0cba7ab50bfa7)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Taylor and Maclaurin series expansion for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Taylor and Maclaurin series expansion for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Taylor and Maclaurin series expansion for a function of two variables (converted PDF; original hash 90064a7ee2a213d4017032a6ebaf80ebc3d8d90aaa7ffd21b26db378c85946ef)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Source	Path	Use
Slide page text: Taylor s and Maclaurin s expansion for a function of two variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Taylor s and Maclaurin s expansion for a function of two variables continued	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Taylor s and Maclaurin s expansion for a function of two variables continued (converted PDF; original hash 9c39f5b326ed0806d84b6995f4f749e9afaf10afd48666cd3bcb52027f2ea4b2)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Introduction to Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Introduction to Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.

Source	Path	Use
Page-render source for Introduction to Maxima and Minima for a function of two variables (converted PDF; original hash e02458424ddafb4c04372fdf8ab6714366a7cf827e1ca9a819bbcc4c16e56cbe)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Problems on Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.
Alternate extracted slide page text: Problems on Maxima and Minima for a function of two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Problems on Maxima and Minima for a function of two variables (converted PDF; original hash 13b0c81bb2e7cdd3ac55cdaaa4466dad1e67d1b8b7d9b89827aac3734fffb57f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.
Slide page text: Lagrange s method of un determined multipliers for two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md	Complete extracted page text from the bucket's .slides.md sidecar.

Source	Path	Use
Alternate extracted slide page text: Lagrange s method of un determined multipliers for two variables	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.txt	The bucket's alternate .slides.txt text sidecar; content differences are kept as separate page points.
Page-render source for Lagrange s method of un determined multipliers for two variables (converted PDF; original hash 80306ba85a4b394f5259f1d9bcbc45bd8dd670f103fbdbc479dd0d1504523e5f)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx	Page inventory and visual source for the rendered original deck; figure pages are derived from this local PDF. The original bucket path/hash are retained in datasets/content-index.json.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. No historical archive question page met the unit-match candidate threshold; all subject-level and ambiguous pages remain in datasets/question-archive.json.
2. 2 video-to-slide associations were not promoted to a match. Candidate rankings are listed in the transcript corpus and video-slide-matches.csv; the slide decks appear separately.
3. 1 transcript segments have low ASR confidence; they are flagged with timestamps in the transcript blocks and indexed in the corpus.
4. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
5. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
6. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.

Coverage

Every inventoried source unit has an explicit coverage decision. Covered means reviewed and mapped to content; exclusions and gaps remain visible.

Evidence	Source unit	Status	Decision
EV0000001	SRC00001/catalogue	covered	Unit title and index copied from the official units.json catalogue; human content review is pending.

Evidence	Source unit	Status	Decision
EV000002	SRC00002/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000003	SRC00003/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000004	SRC00004/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000005	SRC00005/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000006	SRC00002/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000007	SRC00003/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV000008	SRC00004/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000009	SRC00005/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000010	SRC00002/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000011	SRC00003/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000012	SRC00004/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000013	SRC00005/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000014	SRC00002/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000015	SRC00003/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000016	SRC00004/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000017	SRC00005/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0061 to s0081. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000018	SRC00002/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000019	SRC00003/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000020	SRC00004/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000021	SRC00005/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0082 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000022	SRC00006/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000023	SRC00007/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000024	SRC00008/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000025	SRC00009/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000026	SRC00006/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000027	SRC00007/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000028	SRC00008/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000029	SRC00009/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000030	SRC00006/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000031	SRC00007/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000032	SRC00008/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000033	SRC00009/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0061. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000034	SRC00006/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000035	SRC00007/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000036	SRC00008/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000037	SRC00009/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0062 to s0080. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000038	SRC00010/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000039	SRC00011/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000040	SRC00012/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000041	SRC00013/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000042	SRC00010/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000043	SRC00011/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000044	SRC00012/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000045	SRC00013/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000046	SRC00010/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000047	SRC00011/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000048	SRC00012/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000049	SRC00013/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0043 to s0063. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000050	SRC00010/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000051	SRC00011/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000052	SRC00012/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0066. Sidecar variants compare at similarity 0.9990; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000053	SRC00013/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0064 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000054	SRC00014/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000055	SRC00015/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000056	SRC00016/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000057	SRC00017/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000058	SRC00014/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000059	SRC00015/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000060	SRC00016/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000061	SRC00017/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000062	SRC00014/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000063	SRC00015/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000064	SRC00016/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare at similarity 0.9981; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000065	SRC00017/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0045. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000066	SRC00018/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000067	SRC00019/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000068	SRC00020/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000069	SRC00021/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000070	SRC00018/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000071	SRC00019/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000072	SRC00020/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000073	SRC00021/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000074	SRC00018/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000075	SRC00019/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000076	SRC00020/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000077	SRC00021/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000078	SRC00018/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000079	SRC00019/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000080	SRC00020/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000081	SRC00021/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000082	SRC00018/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000083	SRC00019/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000084	SRC00020/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0098. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000085	SRC00021/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0098. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00000086	SRC00018/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0099 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000087	SRC00019/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0099 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000088	SRC00020/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0099 to s0104. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000089	SRC00021/t001500-001799	covered	Timestamp interval t001500-001799; canonical segment IDs s0099 to s0104. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000090	SRC00022/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000091	SRC00023/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00000092	SRC00024/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000093	SRC00025/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000094	SRC00022/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000095	SRC00023/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000096	SRC00024/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000097	SRC00025/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000098	SRC00022/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000099	SRC00023/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 00	SRC00024/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 01	SRC00025/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 02	SRC00022/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 03	SRC00023/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 04	SRC00024/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 05	SRC00025/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0079. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 06	SRC00022/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000107	SRC00023/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000108	SRC00024/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0090. Sidecar variants compare at similarity 0.9995; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000109	SRC00025/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0080 to s0090. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000110	SRC00026/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000111	SRC00027/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000112	SRC00028/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000113	SRC00029/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 14	SRC00026/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 15	SRC00027/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 16	SRC00028/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 17	SRC00029/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 18	SRC00026/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 19	SRC00027/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 20	SRC00028/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001121	SRC00029/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0037 to s0042. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001122	SRC00030/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001123	SRC00031/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001124	SRC00032/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001125	SRC00033/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001126	SRC00030/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001127	SRC00031/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000128	SRC00032/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000129	SRC00033/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000130	SRC00030/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000131	SRC00031/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000132	SRC00032/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000133	SRC00033/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0040 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000134	SRC00030/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000135	SRC00031/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000136	SRC00032/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 0.9991; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000137	SRC00033/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000138	SRC00034/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000139	SRC00035/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000140	SRC00036/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000141	SRC00037/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001142	SRC00034/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001143	SRC00035/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001144	SRC00036/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare at similarity 0.9982; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001145	SRC00037/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0034. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001146	SRC00038/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001147	SRC00039/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001148	SRC00040/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001149	SRC00041/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0019. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001150	SRC00038/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001151	SRC00039/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001152	SRC00040/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001153	SRC00041/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0020 to s0037. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001154	SRC00038/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001155	SRC00039/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000156	SRC00040/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare at similarity 0.9985; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000157	SRC00041/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0038 to s0048. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000158	SRC00042/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000159	SRC00043/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000160	SRC00044/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000161	SRC00045/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000162	SRC00042/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000163	SRC00043/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000164	SRC00044/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000165	SRC00045/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0038. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000166	SRC00042/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000167	SRC00043/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000168	SRC00044/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000169	SRC00045/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0039 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000170	SRC00042/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000171	SRC00043/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000172	SRC00044/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000173	SRC00045/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0077. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000174	SRC00042/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000175	SRC00043/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000176	SRC00044/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0086. Sidecar variants compare at similarity 0.9992; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000177	SRC00045/t001200-001499	covered	Timestamp interval t001200-001499; canonical segment IDs s0078 to s0086. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000178	SRC00046/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000179	SRC00047/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000180	SRC00048/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000181	SRC00049/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000182	SRC00046/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000183	SRC00047/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001184	SRC00048/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001185	SRC00049/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001186	SRC00046/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001187	SRC00047/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001188	SRC00048/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001189	SRC00049/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0059. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001190	SRC00046/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000191	SRC00047/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000192	SRC00048/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0064. Sidecar variants compare at similarity 0.9983; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000193	SRC00049/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0060 to s0064. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000194	SRC00050/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000195	SRC00051/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000196	SRC00052/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000197	SRC00053/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0021. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00001 98	SRC00050/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00001 99	SRC00051/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 00	SRC00052/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 01	SRC00053/t00030 0-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0022 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 02	SRC00050/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 03	SRC00051/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 04	SRC00052/t00060 0-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 05	SRC00053/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0052. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 06	SRC00054/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 07	SRC00055/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 08	SRC00056/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 09	SRC00057/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 10	SRC00054/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 11	SRC00055/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000212	SRC00056/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000213	SRC00057/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0040. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000214	SRC00054/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000215	SRC00055/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000216	SRC00056/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000217	SRC00057/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0041 to s0058. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000218	SRC00054/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000219	SRC00055/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000220	SRC00056/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 0.9993; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000221	SRC00057/t000900-001199	covered	Timestamp interval t000900-001199; canonical segment IDs s0059 to s0066. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000222	SRC00058/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000223	SRC00059/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000224	SRC00060/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 0.9962; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000225	SRC00061/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0017. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002026	SRC00058/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002027	SRC00059/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002028	SRC00060/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 0.9962; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002029	SRC00061/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0018 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002030	SRC00058/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002031	SRC00059/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002032	SRC00060/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare at similarity 0.9962; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000233	SRC00061/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0039. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000234	SRC00062/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000235	SRC00063/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000236	SRC00064/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000237	SRC00065/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000238	SRC00062/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000239	SRC00063/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV00002 40	SRC00064/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 41	SRC00065/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0021 to s0041. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 42	SRC00062/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 43	SRC00063/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 44	SRC00064/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 0.9988; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 45	SRC00065/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0042 to s0060. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV00002 46	SRC00066/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000247	SRC00067/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000248	SRC00068/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000249	SRC00069/t000000-000299	covered	Timestamp interval t000000-000299; canonical segment IDs s0001 to s0018. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000250	SRC00066/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000251	SRC00067/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000252	SRC00068/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000253	SRC00069/t000300-000599	covered	Timestamp interval t000300-000599; canonical segment IDs s0019 to s0035. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.

Evidence	Source unit	Status	Decision
EV0000254	SRC00066/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000255	SRC00067/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000256	SRC00068/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare at similarity 0.9967; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000257	SRC00069/t000600-000899	covered	Timestamp interval t000600-000899; canonical segment IDs s0036 to s0036. Sidecar variants compare at similarity 1.0000; verbatim transcript text is included in the linked note. This is a source-coverage check, not an audiovisual fact check.
EV0000258	SRC00070/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000259	SRC00072/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000260	SRC00071/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000261	SRC00070/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000262	SRC00072/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000263	SRC00071/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000264	SRC00070/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000265	SRC00072/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000266	SRC00071/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 67	SRC00070/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 68	SRC00072/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 69	SRC00071/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 70	SRC00070/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 71	SRC00072/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 72	SRC00071/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 73	SRC00070/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 74	SRC00072/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 75	SRC00071/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 76	SRC00070/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 77	SRC00072/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 78	SRC00071/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002279	SRC00070/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002280	SRC00072/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002281	SRC00071/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00002282	SRC00070/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002283	SRC00072/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002284	SRC00071/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 85	SRC00070/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 86	SRC00072/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 87	SRC00071/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV00002 88	SRC00070/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 89	SRC00072/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 90	SRC00071/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000291	SRC00070/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000292	SRC00072/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000293	SRC00071/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000294	SRC00070/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000295	SRC00072/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000296	SRC00071/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00002 97	SRC00070/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00002 98	SRC00072/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00002 99	SRC00071/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 00	SRC00070/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 01	SRC00072/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 02	SRC00071/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 03	SRC00070/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 04	SRC00072/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 05	SRC00071/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 06	SRC00070/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 07	SRC00072/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 08	SRC00071/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000309	SRC00070/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000310	SRC00072/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/001_Introduction_to_Partial_differentiation__8f4f825e9c46150f.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000311	SRC00071/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000312	SRC00073/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000313	SRC00075/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000314	SRC00074/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003315	SRC00073/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003316	SRC00075/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003317	SRC00074/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV00003318	SRC00073/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003319	SRC00075/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003320	SRC00074/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003321	SRC00073/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003322	SRC00075/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003323	SRC00074/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV00003324	SRC00073/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003325	SRC00075/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003326	SRC00074/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003327	SRC00073/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003328	SRC00075/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003329	SRC00074/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00003330	SRC00073/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003331	SRC00075/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003332	SRC00074/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000333	SRC00073/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000334	SRC00075/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000335	SRC00074/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000336	SRC00073/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000337	SRC00075/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000338	SRC00074/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000339	SRC00073/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000340	SRC00075/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000341	SRC00074/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000342	SRC00073/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000343	SRC00075/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/002_Geometrical_interpretation_of_Partial_derivatives_Problems_on_partial_derivatives__714d38d7a06bef55.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000344	SRC00074/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 45	SRC00076/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 46	SRC00078/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 47	SRC00077/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 48	SRC00076/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 49	SRC00078/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 50	SRC00077/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 51	SRC00076/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 52	SRC00078/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 53	SRC00077/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00003 54	SRC00076/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003 55	SRC00078/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003 56	SRC00077/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000357	SRC00076/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000358	SRC00078/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000359	SRC00077/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000360	SRC00076/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000361	SRC00078/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000362	SRC00077/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003363	SRC00076/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003364	SRC00078/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003365	SRC00077/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00003366	SRC00076/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003367	SRC00078/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003368	SRC00077/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003369	SRC00076/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003370	SRC00078/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003371	SRC00077/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV00003372	SRC00076/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003373	SRC00078/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003374	SRC00077/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003375	SRC00076/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003376	SRC00078/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003377	SRC00077/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV00003378	SRC00076/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00003379	SRC00078/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00003380	SRC00077/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000381	SRC00076/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000382	SRC00078/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000383	SRC00077/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000384	SRC00076/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000385	SRC00078/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000386	SRC00077/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000387	SRC00076/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000388	SRC00078/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000389	SRC00077/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000390	SRC00076/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000391	SRC00078/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000392	SRC00077/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000393	SRC00076/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000394	SRC00078/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000395	SRC00077/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000396	SRC00076/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000397	SRC00078/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000398	SRC00077/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00003 99	SRC00076/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 00	SRC00078/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 01	SRC00077/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 02	SRC00076/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 03	SRC00078/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 04	SRC00077/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 05	SRC00076/p21	covered	Extracted slide page 21 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 06	SRC00078/p21	covered	Original slide page 21 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 07	SRC00077/p21	covered	Alternate extracted slide text page 21; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 08	SRC00076/p22	covered	Extracted slide page 22 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 09	SRC00078/p22	covered	Original slide page 22 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 10	SRC00077/p22	covered	Alternate extracted slide text page 22; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 11	SRC00076/p23	covered	Extracted slide page 23 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 12	SRC00078/p23	covered	Original slide page 23 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 13	SRC00077/p23	covered	Alternate extracted slide text page 23; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 14	SRC00076/p24	covered	Extracted slide page 24 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 15	SRC00078/p24	covered	Original slide page 24 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 16	SRC00077/p24	covered	Alternate extracted slide text page 24; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000417	SRC00076/p25	covered	Extracted slide page 25 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000418	SRC00078/p25	covered	Original slide page 25 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000419	SRC00077/p25	covered	Alternate extracted slide text page 25; normalized text matches the .slides.md page. Human comparison is pending.
EV0000420	SRC00076/p26	covered	Extracted slide page 26 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000421	SRC00078/p26	covered	Original slide page 26 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000422	SRC00077/p26	covered	Alternate extracted slide text page 26; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000423	SRC00076/p27	covered	Extracted slide page 27 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000424	SRC00078/p27	covered	Original slide page 27 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000425	SRC00077/p27	covered	Alternate extracted slide text page 27; normalized text matches the .slides.md page. Human comparison is pending.
EV0000426	SRC00076/p28	covered	Extracted slide page 28 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000427	SRC00078/p28	covered	Original slide page 28 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/003_Total_derivative_Chain_rule_problems__74d776a2caa69540.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000428	SRC00077/p28	covered	Alternate extracted slide text page 28; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000429	SRC00079/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000430	SRC00081/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000431	SRC00080/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000432	SRC00079/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000433	SRC00081/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000434	SRC00080/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000435	SRC00079/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000436	SRC00081/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000437	SRC00080/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000438	SRC00079/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000439	SRC00081/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000440	SRC00080/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000441	SRC00079/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000442	SRC00081/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000443	SRC00080/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000444	SRC00079/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000445	SRC00081/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000446	SRC00080/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000447	SRC00079/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000448	SRC00081/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000449	SRC00080/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000450	SRC00079/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000451	SRC00081/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000452	SRC00080/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000453	SRC00079/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000454	SRC00081/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000455	SRC00080/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000456	SRC00079/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000457	SRC00081/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000458	SRC00080/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000459	SRC00079/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000460	SRC00081/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000461	SRC00080/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000462	SRC00079/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000463	SRC00081/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000464	SRC00080/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000465	SRC00079/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000466	SRC00081/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000467	SRC00080/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000468	SRC00079/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000469	SRC00081/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000470	SRC00080/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000471	SRC00079/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000472	SRC00081/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000473	SRC00080/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000474	SRC00079/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000475	SRC00081/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000476	SRC00080/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000477	SRC00079/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000478	SRC00081/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000479	SRC00080/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000480	SRC00079/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000481	SRC00081/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/004_Implicit_functions_Problems_on_Implicit_and_Composite_functions__8dea1ab57d4cc9f6.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000482	SRC00080/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000483	SRC00082/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000484	SRC00084/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000485	SRC00083/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000486	SRC00082/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000487	SRC00084/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000488	SRC00083/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000489	SRC00082/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000490	SRC00084/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000491	SRC00083/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000492	SRC00082/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000493	SRC00084/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000494	SRC00083/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00004 95	SRC00082/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 96	SRC00084/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00004 97	SRC00083/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV00004 98	SRC00082/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00004 99	SRC00084/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 00	SRC00083/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 01	SRC00082/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 02	SRC00084/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 03	SRC00083/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 04	SRC00082/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 05	SRC00084/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/005_Homogeneous_functions_and_Euler_s_theorem__b933003cc540ac04.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 06	SRC00083/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000507	SRC00085/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000508	SRC00087/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000509	SRC00086/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000510	SRC00085/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000511	SRC00087/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000512	SRC00086/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000513	SRC00085/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000514	SRC00087/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000515	SRC00086/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000516	SRC00085/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000517	SRC00087/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000518	SRC00086/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000519	SRC00085/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000520	SRC00087/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000521	SRC00086/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000522	SRC00085/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000523	SRC00087/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000524	SRC00086/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000525	SRC00085/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000526	SRC00087/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000527	SRC00086/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000528	SRC00085/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000529	SRC00087/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000530	SRC00086/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000531	SRC00085/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000532	SRC00087/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000533	SRC00086/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000534	SRC00085/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000535	SRC00087/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000536	SRC00086/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000537	SRC00085/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000538	SRC00087/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000539	SRC00086/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000540	SRC00085/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000541	SRC00087/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000542	SRC00086/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000543	SRC00085/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000544	SRC00087/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000545	SRC00086/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000546	SRC00085/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000547	SRC00087/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000548	SRC00086/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000549	SRC00085/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000550	SRC00087/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000551	SRC00086/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.
EV0000552	SRC00085/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000553	SRC00087/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000554	SRC00086/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000555	SRC00085/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000556	SRC00087/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000557	SRC00086/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.
EV0000558	SRC00085/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000559	SRC00087/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/006_Problems_on_Euler_s_theorem__1dc65a95c3b42beb.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000560	SRC00086/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000561	SRC00088/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000562	SRC00090/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000563	SRC00089/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000564	SRC00088/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000565	SRC00090/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000566	SRC00089/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005567	SRC00088/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005568	SRC00090/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005569	SRC00089/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV00005570	SRC00088/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005571	SRC00090/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005572	SRC00089/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000573	SRC00088/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000574	SRC00090/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000575	SRC00089/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000576	SRC00088/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000577	SRC00090/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000578	SRC00089/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00005 79	SRC00088/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 80	SRC00090/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 81	SRC00089/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV00005 82	SRC00088/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00005 83	SRC00090/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00005 84	SRC00089/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000585	SRC00088/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000586	SRC00090/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000587	SRC00089/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000588	SRC00088/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000589	SRC00090/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000590	SRC00089/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000591	SRC00088/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000592	SRC00090/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000593	SRC00089/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000594	SRC00088/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000595	SRC00090/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000596	SRC00089/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000597	SRC00088/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000598	SRC00090/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/007_Taylor_and_Maclaurin_series_expansion_for_a_function_of_two_variables__20e30802511eb2ac.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000599	SRC00089/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000600	SRC00091/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000601	SRC00093/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000602	SRC00092/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000603	SRC00091/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000604	SRC00093/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000605	SRC00092/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000606	SRC00091/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000607	SRC00093/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000608	SRC00092/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000609	SRC00091/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000610	SRC00093/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000611	SRC00092/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000612	SRC00091/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000613	SRC00093/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000614	SRC00092/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000615	SRC00091/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000616	SRC00093/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000617	SRC00092/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000618	SRC00091/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000619	SRC00093/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000620	SRC00092/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000621	SRC00091/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000622	SRC00093/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000623	SRC00092/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000624	SRC00091/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000625	SRC00093/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000626	SRC00092/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000627	SRC00091/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000628	SRC00093/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000629	SRC00092/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000630	SRC00091/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000631	SRC00093/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000632	SRC00092/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000633	SRC00091/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000634	SRC00093/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000635	SRC00092/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000636	SRC00091/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000637	SRC00093/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000638	SRC00092/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000639	SRC00091/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000640	SRC00093/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000641	SRC00092/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000642	SRC00091/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000643	SRC00093/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000644	SRC00092/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000645	SRC00091/p16	covered	Extracted slide page 16 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000646	SRC00093/p16	covered	Original slide page 16 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000647	SRC00092/p16	covered	Alternate extracted slide text page 16; normalized text matches the .slides.md page. Human comparison is pending.
EV0000648	SRC00091/p17	covered	Extracted slide page 17 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000649	SRC00093/p17	covered	Original slide page 17 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000650	SRC00092/p17	covered	Alternate extracted slide text page 17; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000651	SRC00091/p18	covered	Extracted slide page 18 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000652	SRC00093/p18	covered	Original slide page 18 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000653	SRC00092/p18	covered	Alternate extracted slide text page 18; normalized text matches the .slides.md page. Human comparison is pending.
EV0000654	SRC00091/p19	covered	Extracted slide page 19 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000655	SRC00093/p19	covered	Original slide page 19 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000656	SRC00092/p19	covered	Alternate extracted slide text page 19; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000657	SRC00091/p20	covered	Extracted slide page 20 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000658	SRC00093/p20	covered	Original slide page 20 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/008_Taylor_s_and_Maclaurin_s_expansion_for_a_function_of_two_variables_-_continued__8431d55b4539aacd.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000659	SRC00092/p20	covered	Alternate extracted slide text page 20; normalized text matches the .slides.md page. Human comparison is pending.
EV0000660	SRC00094/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000661	SRC00096/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000662	SRC00095/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000663	SRC00094/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000664	SRC00096/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000665	SRC00095/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000666	SRC00094/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000667	SRC00096/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000668	SRC00095/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000669	SRC00094/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000670	SRC00096/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000671	SRC00095/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000672	SRC00094/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000673	SRC00096/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000674	SRC00095/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00006775	SRC00094/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006776	SRC00096/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006777	SRC00095/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV00006778	SRC00094/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00006779	SRC00096/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00006780	SRC00095/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000681	SRC00094/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000682	SRC00096/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000683	SRC00095/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV0000684	SRC00094/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000685	SRC00096/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000686	SRC00095/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000687	SRC00094/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000688	SRC00096/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/009_Introduction_to_Maxima_and_Minima_for_a_function_of_two_variables__4eceea203b0b8f6d.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000689	SRC00095/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000690	SRC00097/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000691	SRC00099/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000692	SRC00098/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000693	SRC00097/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000694	SRC00099/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000695	SRC00098/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.
EV0000696	SRC00097/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000697	SRC00099/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000698	SRC00098/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000699	SRC00097/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000700	SRC00099/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000701	SRC00098/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.
EV0000702	SRC00097/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000703	SRC00099/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000704	SRC00098/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000705	SRC00097/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000706	SRC00099/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000707	SRC00098/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.
EV0000708	SRC00097/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000709	SRC00099/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000710	SRC00098/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007711	SRC00097/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007712	SRC00099/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007713	SRC00098/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.
EV00007714	SRC00097/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007715	SRC00099/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007716	SRC00098/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000717	SRC00097/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000718	SRC00099/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000719	SRC00098/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.
EV0000720	SRC00097/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000721	SRC00099/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000722	SRC00098/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000723	SRC00097/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000724	SRC00099/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000725	SRC00098/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.
EV0000726	SRC00097/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000727	SRC00099/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000728	SRC00098/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000729	SRC00097/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000730	SRC00099/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000731	SRC00098/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.
EV0000732	SRC00097/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000733	SRC00099/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/010_Problems_on_Maxima_and_Minima_for_a_function_of_two_variables__901fa082eda4bf29.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000734	SRC00098/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000735	SRC00100/p1	covered	Extracted slide page 1 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000736	SRC00102/p1	covered	Original slide page 1 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000737	SRC00101/p1	covered	Alternate extracted slide text page 1; normalized text matches the .slides.md page. Human comparison is pending.
EV0000738	SRC00100/p2	covered	Extracted slide page 2 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000739	SRC00102/p2	covered	Original slide page 2 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000740	SRC00101/p2	covered	Alternate extracted slide text page 2; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000741	SRC00100/p3	covered	Extracted slide page 3 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000742	SRC00102/p3	covered	Original slide page 3 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000743	SRC00101/p3	covered	Alternate extracted slide text page 3; normalized text matches the .slides.md page. Human comparison is pending.
EV0000744	SRC00100/p4	covered	Extracted slide page 4 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000745	SRC00102/p4	covered	Original slide page 4 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000746	SRC00101/p4	covered	Alternate extracted slide text page 4; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000747	SRC00100/p5	covered	Extracted slide page 5 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000748	SRC00102/p5	covered	Original slide page 5 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000749	SRC00101/p5	covered	Alternate extracted slide text page 5; normalized text matches the .slides.md page. Human comparison is pending.
EV0000750	SRC00100/p6	covered	Extracted slide page 6 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000751	SRC00102/p6	covered	Original slide page 6 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000752	SRC00101/p6	covered	Alternate extracted slide text page 6; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000753	SRC00100/p7	covered	Extracted slide page 7 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000754	SRC00102/p7	covered	Original slide page 7 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000755	SRC00101/p7	covered	Alternate extracted slide text page 7; normalized text matches the .slides.md page. Human comparison is pending.
EV0000756	SRC00100/p8	covered	Extracted slide page 8 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000757	SRC00102/p8	covered	Original slide page 8 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000758	SRC00101/p8	covered	Alternate extracted slide text page 8; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000759	SRC00100/p9	covered	Extracted slide page 9 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000760	SRC00102/p9	covered	Original slide page 9 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000761	SRC00101/p9	covered	Alternate extracted slide text page 9; normalized text matches the .slides.md page. Human comparison is pending.
EV0000762	SRC00100/p10	covered	Extracted slide page 10 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000763	SRC00102/p10	covered	Original slide page 10 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000764	SRC00101/p10	covered	Alternate extracted slide text page 10; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000765	SRC00100/p11	covered	Extracted slide page 11 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000766	SRC00102/p11	covered	Original slide page 11 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000767	SRC00101/p11	covered	Alternate extracted slide text page 11; normalized text matches the .slides.md page. Human comparison is pending.
EV0000768	SRC00100/p12	covered	Extracted slide page 12 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000769	SRC00102/p12	covered	Original slide page 12 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000770	SRC00101/p12	covered	Alternate extracted slide text page 12; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV0000771	SRC00100/p13	covered	Extracted slide page 13 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000772	SRC00102/p13	covered	Original slide page 13 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000773	SRC00101/p13	covered	Alternate extracted slide text page 13; normalized text matches the .slides.md page. Human comparison is pending.
EV0000774	SRC00100/p14	covered	Extracted slide page 14 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV0000775	SRC00102/p14	covered	Original slide page 14 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV0000776	SRC00101/p14	covered	Alternate extracted slide text page 14; normalized text matches the .slides.md page. Human comparison is pending.

Evidence	Source unit	Status	Decision
EV00007777	SRC00100/p15	covered	Extracted slide page 15 from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.slides.md; original page is also embedded when the source render is available. Human subject review is pending.
EV00007778	SRC00102/p15	covered	Original slide page 15 rendered from pesuc/PESU_Materials/Sem-1/UE26MA141A_Engineering_Mathematics_-_I/Unit_01/Slides/011_Lagrange_s_method_of_un_determined_multipliers_for_two_variables__238e903a20cec7ad.pptx; visual page included as vector figure. Human layout/content review is pending.
EV00007779	SRC00101/p15	covered	Alternate extracted slide text page 15; normalized text matches the .slides.md page. Human comparison is pending.

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